Silicon Quantum Dot Quantum Computing

Research Notes

Preface

These notes ask how electrons confined in a solid can store, manipulate, and reveal quantum information. We begin with the physical system that holds the electrons, determine which of their states can be used, and then develop the interactions and measurements needed to operate on those states. The main text gives the results and conditions needed for this explanation; the appendices develop their derivations and the more specialized calculations.

Spin and Pauli operators. Spin states are labelled by the total-spin quantum number S and the magnetic quantum number m. For a physical electron spin, we use the ordered basis $(|\uparrow\rangle, |\downarrow\rangle)$ and Pauli operators $\sigma_x, \sigma_y, \sigma_z$, with $\sigma_z|\uparrow\rangle = |\uparrow\rangle$ and $\sigma_z|\downarrow\rangle = -|\downarrow\rangle$. The dimensionless spin operator is $\mathbf{s} = \boldsymbol{\sigma}/2$; its angular momentum is $\hbar\mathbf{s}$. Spin and Pauli operators are written without hats.

For a logical qubit, the Pauli operators X, Y, Z refer to the ordered basis $(|0\rangle, |1\rangle)$:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{Pref.1}$$

The physical-spin operators $\sigma_x, \sigma_y, \sigma_z$ have these same matrices in the physical-spin basis above. Their relation to X, Y, Z depends on the encoding, whose logical states are defined in the main text. Subscripts identify the spin or logical qubit on which an operator acts; identity operators on other factors are implicit. Two-qubit matrices use the basis order $(|00\rangle, |01\rangle, |10\rangle, |11\rangle)$ unless stated otherwise.

Logical rotations. We use $R_k(\vartheta) = \exp(-i\vartheta K/2)$, with $(k, K) = (x, X), (y, Y), (z, Z)$:

$$\begin{aligned} R_x(\vartheta) &= \begin{pmatrix} \cos(\vartheta/2) & -i\sin(\vartheta/2) \\ -i\sin(\vartheta/2) & \cos(\vartheta/2) \end{pmatrix}, \\ R_y(\vartheta) &= \begin{pmatrix} \cos(\vartheta/2) & -\sin(\vartheta/2) \\ \sin(\vartheta/2) & \cos(\vartheta/2) \end{pmatrix}, \\ R_z(\vartheta) &= \begin{pmatrix} e^{-i\vartheta/2} & 0 \\ 0 & e^{i\vartheta/2} \end{pmatrix}. \end{aligned} \tag{Pref.2}$$

Encoding subscripts identify the logical states on which these operators act. Products act from right to left.

Evolution and phases. For a Hamiltonian $H(t)$ applied over an interval τ, the evolution operator is

$$U(\tau) = \mathcal{T} \exp\left[-\frac{i}{\hbar}\int_0^\tau H(t)\, dt\right]. \tag{Pref.3}$$

The time-ordering operator $\mathcal{T}$ places later-time operators to the left. When specifying gates on an isolated computational space, an overall phase does not change their physical action. A scalar term $c(t)I$ in the Hamiltonian contributes only the phase $e^{-i\int_0^\tau c(t)\,dt/\hbar}$ and may therefore be omitted. Relative phases between computational basis states are part of the gate and must be retained.

Pauli and Hadamard gates. The operators X, Y, Z also define the Pauli gates. The Hadamard gate is

$$H_{\mathrm{H}} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \tag{Pref.4}$$

Their decompositions into rotations are

$$X = iR_x(\pi), \qquad Y = iR_y(\pi), \qquad Z = iR_z(\pi), \qquad H_{\mathrm{H}} = iR_y(\pi/2)R_z(\pi). \tag{Pref.5}$$

More generally, the Euler product $R_z(\vartheta_1)R_y(\vartheta_2)R_z(\vartheta_3)$ gives an arbitrary single-qubit unitary up to global phase.

SWAP and $\sqrt{\mathrm{SWAP}}$. SWAP exchanges the two logical states, $\mathrm{SWAP}|ab\rangle = |ba\rangle$ for $a, b \in \{0, 1\}$. We use the following choice of its square root:

$$\mathrm{SWAP} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad \sqrt{\mathrm{SWAP}} = \frac{1}{2}\begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 1+i & 1-i & 0 \\ 0 & 1-i & 1+i & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}. \tag{Pref.6}$$

CZ and CNOT gates. CZ adds a minus sign to $|11\rangle$. CNOT flips the target qubit when the control is $|1\rangle$. With the first qubit as control,

$$\mathrm{CZ} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \qquad \mathrm{CNOT}_{1\to 2} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \tag{Pref.7}$$

For logical qubits A, B, the same CNOT is

$$\mathrm{CNOT}_{A\to B} = (|0\rangle\langle 0|)_A \otimes I_B + (|1\rangle\langle 1|)_A \otimes X_B. \tag{Pref.8}$$

It maps $|ab\rangle$ to $|a, b \oplus a\rangle$, where $\oplus$ denotes addition modulo two. Hadamard gates on the target convert CZ into CNOT:

$$\mathrm{CNOT}_{A\to B} = (I_A \otimes H_{\mathrm{H},B})\,\mathrm{CZ}_{AB}\,(I_A \otimes H_{\mathrm{H},B}). \tag{Pref.9}$$

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1 What is a quantum dot?

A quantum dot is a small region in a material that confines electrons in all three spatial directions. A potential well provides a useful picture: an electron has lower potential energy inside the region and must cross an energy barrier to leave it. The localized spatial states of the electron are its *orbitals*, with discrete energies set by the size and shape of the well. A dot can contain one electron or several; confinement makes a small electronic system with its own states and energy levels. This is the origin of the description of quantum dots as artificial atoms [1].

These notes consider quantum dots in crystalline *silicon*, a semiconductor. Metal electrodes held at chosen voltages create electric fields in the silicon and shape the potential energy of its mobile electrons. These electrodes are called *gates*, and the resulting localized electronic region is a *gate-defined quantum dot*. The well occupies a small part of a continuous piece of silicon. Changing the gate voltages changes its energy levels and the barriers around it, allowing us to control which electrons it holds and how strongly they couple to electrons elsewhere in the device.

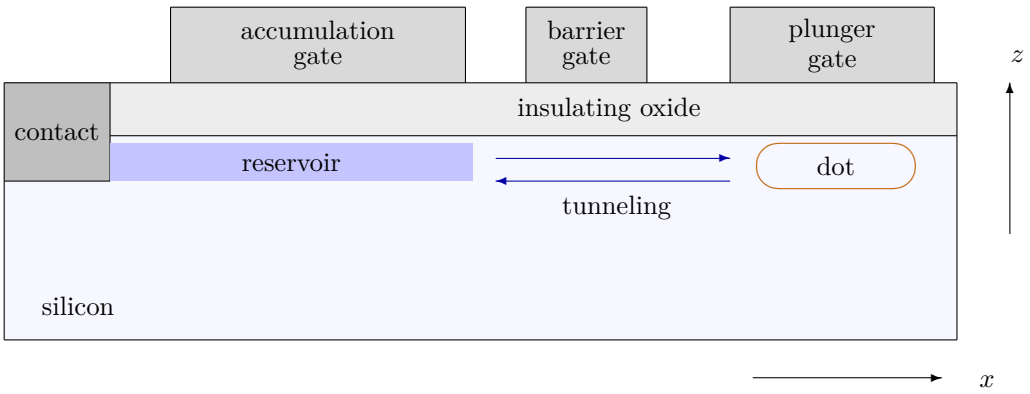
1.1 How gates form a dot

Figure 1.1 shows a silicon surface covered by an insulating layer of silicon dioxide, with metal gates above it. We take z perpendicular to the interface and x, y along it. To make a dot, the device must first confine electrons near the interface in z and then restrict them to a small region in the x, y plane.

The two parts of this confinement have different physical origins. An electron in the silicon encounters a high material energy barrier at the oxide. A suitable gate voltage lowers the electron's energy near the interface relative to the silicon farther below it. The oxide barrier on one side and the rising energy into the silicon on the other form a well in z. The shapes and voltages of the gates also make the potential vary along the interface. A lateral minimum surrounded by barriers then confines the electron in x and y, completing the three-dimensional well [2].

The voltage of each gate is set by an external source. Through the electric field, this voltage changes the electric potential Φ in the silicon and hence the potential energy $-e\Phi$ of an electron, whose charge is $-e$ with $e > 0$. The insulating oxide separates the gate from the electrons. Electrons are supplied through the contact on the left of the figure, which connects to a much larger populated region of silicon. This region serves as an *electron reservoir*: it can give up or accept one electron with negligible change in its macroscopic state. The accumulation gate helps populate this region by lowering the electron potential energy beneath it.

(a) Material cross-section



(b) Electron potential energy along the layer

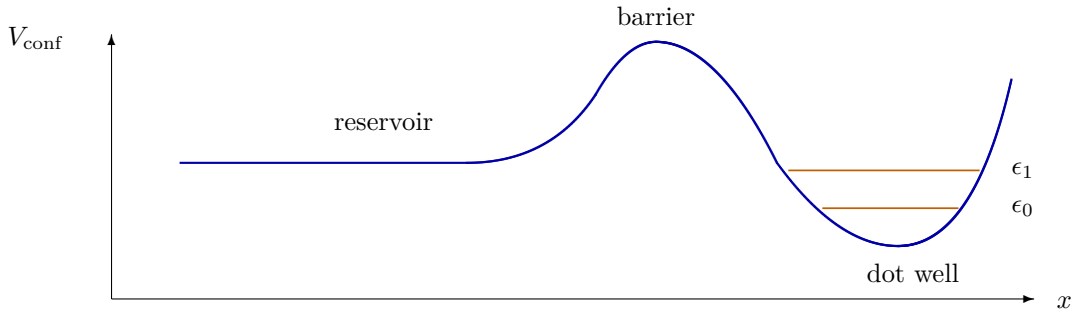


Figure 1.1: Confinement and electron access in a silicon metal–oxide–semiconductor device. (a) The gates act through the insulating oxide to shape the electron potential in the silicon. Electrons reach the reservoir through the contact and the dot through a tunable tunneling barrier. The transverse y direction is omitted. (b) A lateral cut through the confining potential shows the reservoir, barrier, and dot well. The lines ϵ_0 and ϵ_1 indicate discrete orbital energies schematically; the actual levels depend on confinement in all three directions.

Electrons pass between the reservoir and the dot by tunneling through the intervening barrier within the silicon. The *barrier gate* primarily changes the height of this barrier and therefore the tunneling rate. The *plunger gate* primarily shifts the dot's energy levels, changing the energy cost of putting an electron there. These roles explain how the same device can admit an electron and then retain it: adjust the dot levels to allow loading, and raise the reservoir barrier to suppress escape. Section 2 gives the energy conditions that select a particular electron number. Gate fields overlap, so the names describe the main intended effects rather than perfectly independent controls.

The stack of metal, oxide, and semiconductor in the figure gives this structure its name, *MOS*. Another common structure uses a thin silicon layer between silicon–germanium layers. An electron has lower energy in the silicon layer, which therefore supplies a vertical well; gates again provide lateral confinement [3]. In either case, we describe the total confining potential energy by $V_{\text{conf}}(\mathbf{r})$, combining the material energy offsets with the electrostatic contribution $-e\Phi(\mathbf{r})$ (Appendix A).

1.2 The states of one confined electron

The well determines where an electron is localized. To calculate its orbitals and their energies, we also need its kinetic energy in the silicon crystal. The periodic lattice changes the relation between energy and wave vector from that of an electron in vacuum. For the mobile electrons of interest, this relation is the conduction-band dispersion $E_c(\mathbf{k})$. In silicon it has several minima at different wave vectors. The states near each minimum form a *valley*; different valleys thus describe different crystal components available to an electron in the same dot (Fig. 1.2).

Figure 1.2(a) shows calculated bands of bulk silicon along a path in wave-vector space. The point Γ is $\mathbf{k} = 0$; X and L lie at the boundary of the first Brillouin zone along the crystal directions [100] and [111], respectively. The lowest conduction band has a minimum between Γ and X. In unstrained bulk silicon, crystal symmetry gives six equivalent minima along the positive and negative [100], [010], and [001] directions [4]. Panel (b) locates these six valleys; the Γ–X segment in panel (a) passes through one of them. These are crystal directions and need not coincide with the device axes x, y, z.

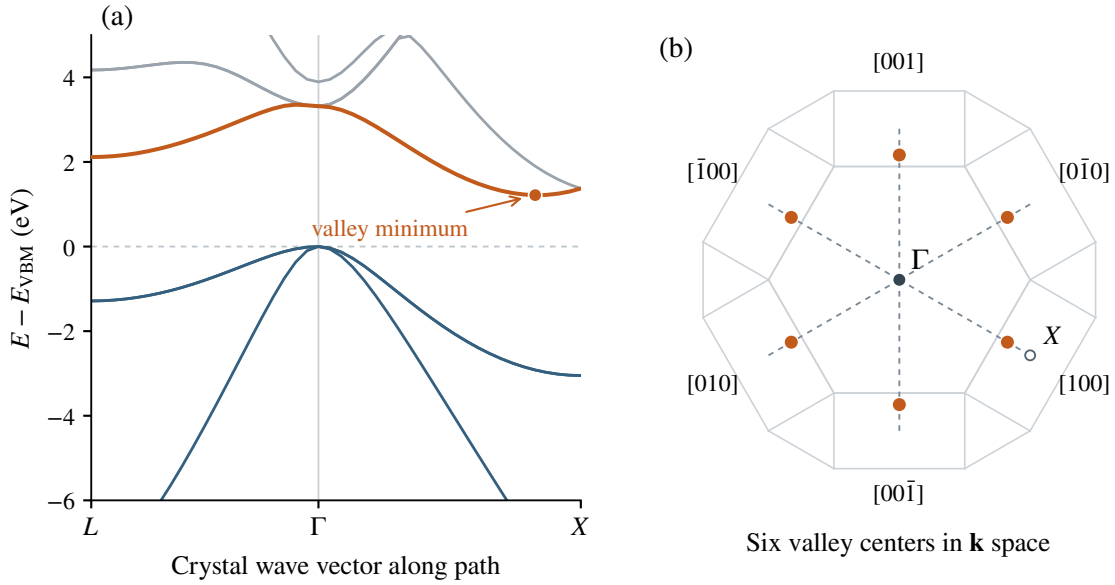


Figure 1.2: Silicon bands and valleys. (a) HSE06 band energies for bulk diamond silicon, replotted from NIST JARVIS-DFT record JVASP-1002 [5]. The valence-band maximum is the energy zero. Blue curves show valence bands; the lowest conduction band is orange and higher conduction bands are gray. The dot marks the sampled minimum between Γ and X. (b) Positions of the six symmetry-equivalent valley centers (orange dots) in crystal wave-vector space. The wireframe outlines the first Brillouin zone; dashed lines point along the six crystal directions. This panel shows positions, not energy surfaces.

First consider states near one minimum, at $\mathbf{k}_0$. The dispersion is approximately quadratic there, so its curvature defines an inverse effective-mass tensor (Appendix B):

$$(\mathbf{M}^{-1})_{\alpha\beta} = \frac{1}{\hbar^2} \left. \frac{\partial^2 E_c}{\partial k_\alpha \partial k_\beta} \right|_{\mathbf{k}_0}, \qquad \alpha, \beta = x, y, z. \tag{1.1}$$

Here $\mathbf{M}$ is the effective-mass tensor of the chosen valley; it is constant in this uniform-crystal model. Its components are written in the device coordinates x, y, z defined above and need

not form a diagonal matrix.[1]

For confinement that varies slowly on the scale of a crystal unit cell, we can separate the rapid spatial variation of the crystal state from the variation across the dot. To leading order, the spatial wave function of a confined electron has the form

$$\psi_n(\mathbf{r}) \simeq \chi_c(\mathbf{r})\phi_n(\mathbf{r}).$$

Here χ_c is the extended crystal state at the chosen band minimum, while ϕ_n is a slowly varying, localized *envelope*; the index n labels the dot orbitals. The envelope sets the extent and shape of the electron's probability distribution across the dot, retaining the crystal structure in χ_c.

Within this valley, for energies near the minimum and weak admixture of other bands, the effective-mass reduction derived in Appendix B gives the envelope equation

$$\underbrace{\left[-\frac{\hbar^2}{2}\sum_{\alpha,\beta}(\mathbf{M}^{-1})_{\alpha\beta}\partial_{r_\alpha}\partial_{r_\beta} + V_{\text{conf}}(\mathbf{r})\right]}_{h_0}\phi_n = \epsilon_n\phi_n. \tag{1.2}$$

Thus the orbital problem is a three-dimensional Schrödinger equation with the effective-mass tensor and the device potential. The energy of the reference crystal minimum is chosen as the zero; material offsets and voltage shifts remain in V_{conf}. We will use the same reference for the dot and reservoir energies. The lowest solution ϕ_0 has energy ϵ_0, and higher solutions describe excited orbitals such as the level ϵ_1 in Fig. 1.1(b). Their spacing is set by confinement; for example, reducing the size of a well increases the kinetic energy required for spatial variation and typically increases the orbital gaps. Weak tunneling to the reservoir allows us to treat these localized levels as states of the dot.

The silicon–oxide or silicon–germanium interfaces that provide vertical confinement also affect the valley states. Their potential varies on the atomic scale and can couple crystal components near different band minima [6]. Thus the single-valley orbital equation supplies one part of the electron's spatial description. To determine the low-energy states, we must include the interface coupling between the retained valleys.

For N_v retained valleys, collect the spatial envelopes into $\mathbf{F} = (F_1, \ldots, F_{N_v})^{\mathsf{T}}$. The multi-valley model in Appendix C gives the block Hamiltonian acting on these components:

$$h_{\text{mv}} = \begin{pmatrix} h_{0,1} & \hat{v}_{12} & \cdots & \hat{v}_{1,N_v} \\ \hat{v}_{21} & h_{0,2} & \cdots & \hat{v}_{2,N_v} \\ \vdots & \vdots & \ddots & \vdots \\ \hat{v}_{N_v,1} & \hat{v}_{N_v,2} & \cdots & h_{0,N_v} \end{pmatrix}. \tag{1.3}$$

Each diagonal block $h_{0,\nu}$ is the orbital Hamiltonian of Eq. (1.2) for valley ν, with its own effective-mass tensor and confining potential. The off-diagonal block $\hat{v}_{\nu\nu'}$ transfers amplitude from valley ν' to valley ν, with $\hat{v}_{\nu'\nu} = \hat{v}^\dagger_{\nu\nu'}$. In the local model used here, it multiplies the spatial envelope by a coupling function $v_{\nu\nu'}(\mathbf{r})$. These functions encode the microscopic interface structure and are additional inputs to the effective model. The eigenstates of h_{mv} are combined orbital–valley states; we denote the lowest one by $|\chi_0\rangle$.

The consequence is particularly simple when the two lowest valleys are equivalent and share a ground orbital ϕ_0 of energy ϵ_0. Assume that all other orbital–valley states lie higher in energy and that coupling to them is small compared with their separation from this pair.

[1] The principal axes are the eigenvector directions of $\mathbf{M}$ and need not coincide with the device axes. Isotropy is the special case $\mathbf{M} = m^*\mathbf{I}$, where $\mathbf{I}$ is the three-dimensional identity matrix.

Projection onto this orbital gives the valley Hamiltonian (Appendix C)

$$H_{\mathrm{v}} = \epsilon_0 I + \begin{pmatrix} 0 & v_0 \\ v_0^* & 0 \end{pmatrix} \quad \text{in the basis } (|\nu_1\rangle, |\nu_2\rangle), \tag{1.4}$$

where v_0 is the interface coupling matrix element within ϕ_0. For $v_0 = |v_0|e^{i\theta_{\mathrm{v}}} \neq 0$, the eigenstates, energies, and valley gap are

$$\begin{aligned} |v_\pm\rangle &= \frac{|\nu_1\rangle \pm e^{-i\theta_{\mathrm{v}}}|\nu_2\rangle}{\sqrt{2}}, \quad \epsilon_\pm = \epsilon_0 \pm |v_0|, \\ \Delta_{\mathrm{val}} &= \epsilon_+ - \epsilon_- = 2|v_0|. \end{aligned} \tag{1.5}$$

In this limit, the lowest spatial state is $|\chi_0\rangle = |\phi_0\rangle \otimes |v_-\rangle$. The coupling can be small compared with an orbital excitation energy while still determining the valley eigenstates and their entire splitting. If mixing with excited orbitals is appreciable, the full Hamiltonian (1.3) determines $|\chi_0\rangle$ and the excitation gaps without assuming this product form.

Each spatial state also carries the electron's spin. When a common orbital basis can be used for the retained valleys and spin-dependent mixing is weak, we can organize the one-electron states as

$$\mathcal{H}_{N=1} \simeq \mathcal{H}_{\mathrm{orb}} \otimes \mathcal{H}_{\mathrm{val}} \otimes \mathbb{C}^2_{\mathrm{spin}}. \tag{1.6}$$

The first factor describes orbital envelopes, the second the retained valley components, and the third the spin. The coupled eigenstates need not factor into these parts. To use a single electron as a spin qubit, we retain the lowest combined spatial state $|\chi_0\rangle$ and use its two spin states. Section 4.1 gives the conditions under which thermal excitation and control pulses leave the electron in that spatial state.

1.3 When the dot contains several electrons

Several electrons in the same well share the available one-electron states and repel one another. Their allowed states must also satisfy fermionic antisymmetry. The energy of an N-electron dot therefore depends both on its one-electron spectrum and on the interaction between each pair of electrons.

Let h_1 be the one-electron Hamiltonian acting on the orbital, valley, and spin states retained in the description, including any relevant valley mixing and spin-dependent energies. With spin-dependent terms omitted, $h_1 = h_{\mathrm{mv}} \otimes I_{\mathrm{spin}}$, where h_{mv} is the coupled Hamiltonian in Eq. (1.3) and I_{spin} is the identity on the two spin states. The pair interaction $W(\mathbf{r}, \mathbf{r}')$ includes the dielectric and conductor response to the electron charges (Appendix D). For N electrons, this screened repulsion gives the many-electron Hamiltonian

$$H_N = \sum_{a=1}^{N} h_1^{(a)} + \sum_{a<b} W(\mathbf{r}_a, \mathbf{r}_b), \tag{1.7}$$

where $h_1^{(a)}$ acts on electron a, and the second sum counts each interacting pair once.

At fixed gate voltages and other external fields, let E_N denote the ground-state energy of H_N in the antisymmetric N-electron state space. The resulting sequence $E_0, E_1, E_2, \ldots$ describes the lowest energy available at each electron number. Which number is actually occupied depends on the energy gained or lost when electrons are exchanged with the reservoir. That comparison is developed in Section 2.

1.4 Several coupled quantum dots

Gate electrodes can create several potential minima, each surrounded by barriers, in the same silicon region. We label the resulting dots by i. Orthonormal localized orbitals provide a basis for specifying how many electrons occupy each dot; several orbital–valley states and their spin states can be retained on each dot. The barriers control electron tunneling between dots, while Coulomb interactions act both within and between them.

For an array isolated from reservoirs, fix the total electron number N, and write N_i for the number on dot i, so that $N_1 + N_2 + \cdots = N$. The occupation tuple $\mathbf{N} = (N_1, N_2, \ldots)$ defines the subspace $\mathcal{H}_{(N_1,N_2,\ldots)}$. The single index in $\mathcal{H}_N$ fixes only the total number, while the tuple specifies the number on every dot. For example, the two-electron space $\mathcal{H}_2$ of two dots splits into $\mathcal{H}_{(1,1)}$, $\mathcal{H}_{(2,0)}$, and $\mathcal{H}_{(0,2)}$. Each subspace contains the allowed orbital, valley, and spin states at those occupations. In this local basis, grouping the many-electron Hamiltonian by charge configuration (Appendix E.2) gives

$$\begin{aligned} \mathcal{H}_N &= \bigoplus_{N_1+N_2+\cdots=N} \mathcal{H}_{(N_1,N_2,\ldots)}, \\ H_N^{(0)} &= \bigoplus_{N_1+N_2+\cdots=N} H_{(N_1,N_2,\ldots)}, \qquad H_N = H_N^{(0)} + \mathcal{T}. \end{aligned} \tag{1.8}$$

Both direct sums run over the allowed occupation tuples with total N. Each block $H_{(N_1,N_2,\ldots)}$ acts within the subspace with that same tuple, and $\mathcal{T}$ transfers charge between different subspaces. The superscript (0) denotes the reference with these transfers omitted. Each block then has its own ground energy $E_{\mathbf{N},0}$ and excited energies $E_{\mathbf{N},a}$; the first energy index labels the occupation tuple, and a labels a level within that subspace. These energies include the interactions between electrons on different dots. Thus even when tunneling is suppressed, the array need not behave as independent dots.

Restoring tunneling couples states in different charge subspaces while preserving N. For two such states with uncoupled energies $E_{\mathbf{N},a}$ and $E_{\mathbf{N}',b}$, mixing is small when their coupling matrix element is much smaller than $|E_{\mathbf{N},a} - E_{\mathbf{N}',b}|$. This is the regime used to keep local occupations nearly fixed when the relevant charge-transfer channels are weakly mixed. Near a degeneracy, even a small nonzero tunnel matrix element can strongly mix the two states. In a general orbital model, interaction terms can also transfer charge; they belong to $\mathcal{T}$ along with single-electron hopping, as detailed in Appendix E.2.

For a simple example, take one electron in two dots. Retain only the lowest localized spatial state on each dot and a common fixed spin state, assuming that coupling to other orbital–valley and spin states can be neglected. Denote the retained states by $|1,0\rangle$ and $|0,1\rangle$. The local-basis matrix elements in Appendix E.2 give

$$h_{\text{pair}} = \begin{pmatrix} \epsilon_1 & -t_{12} \\ -t_{12}^* & \epsilon_2 \end{pmatrix} \quad \text{in the basis } (|1,0\rangle, |0,1\rangle). \tag{1.9}$$

The diagonal entries are the energies of the two localized states, and t_{12} is their tunnel coupling, with units of energy. Gate voltages tune both the detuning $\epsilon_1 - \epsilon_2$ and the barrier that controls t_{12}. When $|\epsilon_1 - \epsilon_2| \gg |t_{12}|$, each eigenstate lies mostly in one charge subspace. When the detuning becomes comparable to or smaller than $|t_{12}|$, both occupations contribute appreciably. At $\epsilon_1 = \epsilon_2$ and $t_{12} \neq 0$, the two eigenstates have equal weights in $(1,0)$ and $(0,1)$, with energy separation $2|t_{12}|$ (Appendix E.2). Small tunneling can therefore coexist with strong mixing near resonance; its magnitude still sets the small splitting and the associated slow coherent transfer.

2 Controlling the electron number

To use confined electrons as spin qubits, we must prepare the charge configuration assumed by the spin model and retain it during control. The single-spin qubit considered below requires one electron in a dot; our two-dot spin model starts with one electron in each dot. A reservoir supplies these electrons through tunneling. Gate voltages change the dot energies to select the desired occupations, while the barriers control how readily electrons can enter or leave. Preparing a useful spin system therefore requires control of both its energetically preferred electron number and its exchange of electrons with the reservoir.

To compare different electron numbers, we account for the energy of both the dot and the reservoir that supplies its electrons. Adding an electron to an $(N-1)$-electron dot costs $E_N - E_{N-1}$. At zero temperature, the large reservoir loses one electron, changing its energy by $-\mu_{\mathrm{R}}$, where μ_{R} is its chemical potential. The transfer lowers the combined energy when $E_N - E_{N-1} - \mu_{\mathrm{R}} < 0$. For N electrons transferred from the reservoir into the dot, the corresponding energy to compare is $E_N - \mu_{\mathrm{R}} N$, up to a common reservoir reference.

We collect these charge sectors in the full dot Hamiltonian H, which acts as H_N in each N-electron sector. At zero temperature, a dot in equilibrium with a weakly coupled reservoir of chemical potential μ_{R} occupies a ground state of

$$\widehat{\Omega} = H - \mu_{\mathrm{R}} \widehat{N}, \tag{2.1}$$

where $\widehat{N}$ counts electrons. In the N-electron sector, this operator is $H_N - \mu_{\mathrm{R}} N$, whose ground-state energy is $E_N - \mu_{\mathrm{R}} N$. The equilibrium electron number is therefore

$$\boxed{N_{\mathrm{eq}}(\mu_{\mathrm{R}}) \in \underset{N=0,1,\ldots}{\arg\min} [E_N - \mu_{\mathrm{R}} N]}. \tag{2.2}$$

This determines the relation between chemical potential and electron number from the energies E_N, at fixed dot parameters [7].

Addition thresholds and charge gaps. Equating $E_N - \mu_{\mathrm{R}} N$ and $E_{N-1} - \mu_{\mathrm{R}}(N-1)$ gives the threshold for adding the Nth electron:

$$\mu_N = E_N - E_{N-1}. \tag{2.3}$$

The spacing between successive thresholds is

$$E_{\mathrm{add}}(N) = \mu_{N+1} - \mu_N = E_{N+1} - 2E_N + E_{N-1}. \tag{2.4}$$

If E_N is strictly convex, meaning $E_{\mathrm{add}}(N) > 0$ for every $N \geq 1$, these thresholds increase in order. Minimizing $E_N - \mu_{\mathrm{R}} N$ then selects charge $N \geq 1$ throughout the interval

$$\mu_N < \mu_{\mathrm{R}} < \mu_{N+1}. \tag{2.5}$$

Each plateau has width $E_{\mathrm{add}}(N)$, so electrons enter one at a time as μ_{R} increases. At a boundary, neighboring charges are degenerate; the empty dot is selected for $\mu_{\mathrm{R}} < \mu_1$. Without strict convexity, some plateaus can disappear, and the original minimization remains the general rule.

Within a plateau, adding or removing one electron changes the energy of $\widehat{\Omega}$ by

$$\Delta_+ = E_{N+1} - E_N - \mu_{\mathrm{R}} = \mu_{N+1} - \mu_{\mathrm{R}}, \tag{2.6}$$

$$\Delta_- = E_{N-1} - E_N + \mu_{\mathrm{R}} = \mu_{\mathrm{R}} - \mu_N. \tag{2.7}$$

Both costs are positive, suppressing low-energy sequential charge transfer: this is the Coulomb-blockade regime [7].

Selecting a charge state with gate voltages. Gate voltages control the electron number through their effect on the electric potential $\Phi(\mathbf{r})$ in the dot. For an electron of charge $-e$, a potential change $\delta\Phi(\mathbf{r})$ gives a potential energy shift $-e\,\delta\Phi(\mathbf{r})$. This alters the many-electron energies E_N and the addition thresholds $\mu_N = E_N - E_{N-1}$. For a uniform shift $\delta\Phi$ over the dot, $\delta E_N = -Ne\,\delta\Phi$, so $\delta\mu_N = -e\,\delta\Phi$. With the reservoir chemical potential μ_R held fixed, moving a threshold across it changes the equilibrium electron number according to Eq. (2.2). Electrical transport or charge sensing can detect this change (Appendix G). For a given gate geometry, a first-order expansion about a reference voltage setting $\mathbf{V}_0$ relates the threshold shift to the applied voltage changes:

$$\delta\mu_N \simeq -e\sum_g \alpha_{Ng}\,\delta V_g, \qquad \alpha_{Ng} = -\frac{1}{e}\left.\frac{\partial\mu_N}{\partial V_g}\right|_{\mathbf{V}_0}. \tag{2.8}$$

Here α_{Ng} is the dimensionless *lever arm* of gate g for the Nth addition threshold: $e\alpha_{Ng}$ converts a change in gate voltage into a shift of that threshold. A physical gate generally alters both the dot levels and the tunneling barriers, so both effects must be considered when choosing the operating voltages. Appendix F derives the corresponding voltage responses.

2.1 Loading a single dot

For the single-spin qubit, the two logical states are spin states of the same confined electron. The required charge state is therefore $N = 1$. *Loading* prepares this occupation by bringing an electron from the reservoir into the dot through tunneling.

Figure 2.1 shows the charge regions along a gate-voltage scan at fixed μ_R. To prepare $N = 1$, first empty the dot at a setting such as E: set $\mu_1 - \mu_\mathrm{R} \gg k_BT$ while allowing tunneling to the reservoir, so that an electron already in the dot can escape. Then change the gate voltage to a setting such as L, where $\mu_1 < \mu_\mathrm{R} < \mu_2$, keeping the reservoir barrier low enough to permit tunneling. An electron from an occupied reservoir state can now enter the dot. Once it has entered, adding a second electron costs $\mu_2 - \mu_\mathrm{R} > 0$, and removing the first costs $\mu_\mathrm{R} - \mu_1 > 0$. Choosing both costs large compared with k_BT suppresses these further charge changes and selects single occupation.

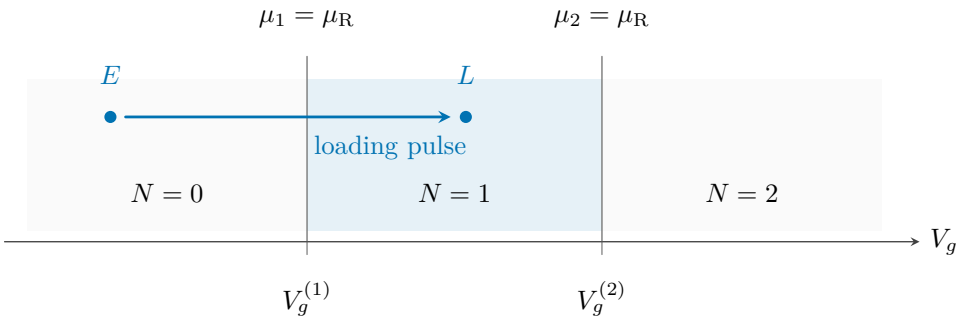


Figure 2.1: Single-dot charge regions at fixed reservoir chemical potential μ_R. For the gate response illustrated here, increasing V_g lowers the addition thresholds. The voltages $V_g^{(1)}$ and $V_g^{(2)}$ satisfy $\mu_1 = \mu_\mathrm{R}$ and $\mu_2 = \mu_\mathrm{R}$, respectively. The loading pulse moves from the empty-dot setting E into the $N = 1$ region at L, while tunneling to the reservoir remains possible. Raising the reservoir barrier after loading is a separate step.

Crossing the addition threshold makes entry energetically favorable; the dot still has to wait for an electron to tunnel through the barrier. The loading voltages are therefore held for

a finite interval, long compared with the typical entry waiting time, which depends on the barrier transparency. After this loading interval, raising the reservoir barrier reduces electron exchange and retains the electron while its spin is manipulated. For an unwanted escape rate Γ_{escape} and a control interval τ, the condition $\Gamma_{\text{escape}}\tau \ll 1$ makes loss during that interval unlikely. Energetic stability suppresses unwanted charge changes through their energy costs; a high barrier additionally suppresses the tunneling that would carry them out.

2.2 Loading two coupled dots

The two-dot spin model used below requires one localized spin on each dot. Its starting charge configuration is therefore $(N_1, N_2) = (1, 1)$. Preparing it requires selecting the occupation of each dot, since two electrons in the pair could also occupy $(2, 0)$ or $(0, 2)$.

The plunger-gate voltages V_1, V_2 tune the dot energies. Repulsion between dots makes each addition threshold depend on the occupation of the other dot. With the reservoirs at a common chemical potential μ_{R} and interdot mixing neglected, the equilibrium condition partitions the (V_1, V_2) plane into regions of fixed (N_1, N_2) (Appendix H). Figure 2.2 shows the resulting *charge stability diagram* around $(1, 1)$. Crossing a blue boundary exchanges an electron with a reservoir; for example, $(1, 0) \to (1, 1)$ loads dot 2. Crossing an orange boundary transfers an electron between dots while preserving their total number; for example, $(1, 1) \to (2, 0)$ moves an electron from dot 2 to dot 1.

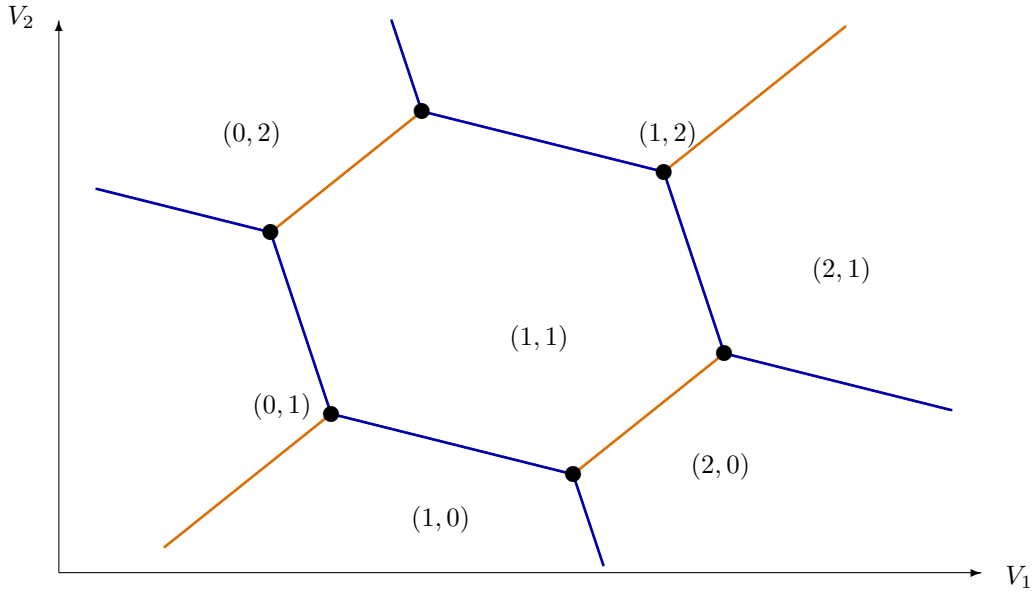


Figure 2.2: Schematic charge regions of two dots with negligible interdot mixing and a common reservoir chemical potential. Blue boundaries change the total electron number by one; orange boundaries transfer an electron between dots. Black points, called triple points, mark where three charge regions meet. The central $(1, 1)$ region provides the charge configuration for one localized spin on each dot. The axes are plunger voltages; offsets, scales, and cross-couplings are unspecified.

To prepare $(1, 1)$, choose voltages inside its region, sufficiently far from the boundaries that unwanted charge changes cost much more than $k_B T$. Allow exchange with the reservoirs long enough for the occupations to equilibrate, then raise the reservoir barriers to retain the two electrons. The interdot tunnel coupling must also remain small compared with the relevant

charge-transfer gaps to keep the occupations close to one electron per dot. Appendix H derives the boundary conditions and explains how measured charge transitions locate these regions.

3 Qubit architectures and encodings

After selecting and retaining the $N = 1$ charge state, we can keep the electron in the lowest orbital–valley state $|\chi_0\rangle$ and use its two spin states to store a qubit. With several confined electrons, we can instead select two joint spin states and use them as one logical qubit. The choice of this two-dimensional *computational subspace* is the encoding; transitions out of it are leakage. Different encodings use the available physical interactions differently: a field acts on an individual spin, while an interaction between spins can act within a jointly encoded space.

We retain one orbital–valley state per dot and assume predominantly one electron per occupied dot.

3.1 Single-spin qubits

For one electron, we choose the logical basis

$$|0\rangle = |\chi_0\rangle \otimes |\downarrow\rangle, \qquad |1\rangle = |\chi_0\rangle \otimes |\uparrow\rangle. \tag{3.1}$$

The computational space is $\mathcal{H}_{\mathrm{q}} = \mathrm{span}\{|0\rangle, |1\rangle\}$, and a pure qubit state is $|\psi\rangle = a|0\rangle + b|1\rangle$, with $|a|^2 + |b|^2 = 1$. Leakage involves charge, orbital, or valley excitations.

Charge preparation alone does not fix the spin state. For example, the two-dot $(1, 1)$ charge configuration introduced in Section 1.4, with one orbital–valley state per dot, contains four spin states:

$$|\downarrow\downarrow\rangle, \quad |\downarrow\uparrow\rangle, \quad |\uparrow\downarrow\rangle, \quad |\uparrow\uparrow\rangle. \tag{3.2}$$

The arrows refer to dots 1 and 2. A charge stability diagram identifies charge occupations and transitions, not the spin state within a charge sector; its boundaries are derived in Appendix H.1. The four states can represent two single-spin qubits or supply a subspace for one encoded qubit [8].

3.2 Singlet–triplet qubits

Two spins decompose as $\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$. We encode one qubit in the $m = 0$ states: the singlet $(S = 0)$ and the triplet $(S = 1)$,[2]

$$\begin{aligned} |0_{\mathrm{ST}}\rangle = |S\rangle &= \frac{|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle}{\sqrt{2}}, \\ |1_{\mathrm{ST}}\rangle = |T_0\rangle &= \frac{|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle}{\sqrt{2}}. \end{aligned} \tag{3.3}$$

The remaining triplets $|T_+\rangle = |\uparrow\uparrow\rangle$ and $|T_-\rangle = |\downarrow\downarrow\rangle$ are spin-leakage states.

3.3 Exchange-only qubits

Three spin-1/2 electrons can encode one logical qubit. Their spin space decomposes as

$$\tfrac{1}{2} \otimes \tfrac{1}{2} \otimes \tfrac{1}{2} = \tfrac{1}{2} \oplus \tfrac{1}{2} \oplus \tfrac{3}{2}.$$

[2]The label S in $|S\rangle$ stands for "singlet"; the total-spin quantum number of this state is $S = 0$.

Taking the $m = +1/2$ state from each of the two $S = 1/2$ representations gives a two-dimensional computational space. We choose the basis

$$|0_{\mathrm{E}}\rangle = \frac{|\uparrow\downarrow\uparrow\rangle - |\downarrow\uparrow\uparrow\rangle}{\sqrt{2}},$$
$$|1_{\mathrm{E}}\rangle = \frac{2|\uparrow\uparrow\downarrow\rangle - |\uparrow\downarrow\uparrow\rangle - |\downarrow\uparrow\uparrow\rangle}{\sqrt{6}}. \tag{3.4}$$

The arrows refer to spins 1, 2, and 3. Spins 1 and 2 form a singlet in $|0_{\mathrm{E}}\rangle$ and a triplet in $|1_{\mathrm{E}}\rangle$, distinguishing the two $S = 1/2$ representations. The spin-rotation-invariant interaction derived in Section 4.3 is called isotropic exchange. It preserves S, m but can mix these two states, allowing logical rotations using exchange alone (Section 6.3).

4 Engineering spin control and exchange interactions

An encoding specifies which states store quantum information. To manipulate that information, we need a Hamiltonian that changes relative phases or populations within the chosen spin space. A static magnetic field splits the two levels of an individual spin and makes their relative phase evolve. A transverse drive mixes those levels, while tunneling between dots produces an exchange interaction that couples different spins. These physical terms supply the controls whose action on each encoding is worked out in Section 6.

The spin description assumes that charge, orbital, and valley excitations can be excluded as independent computational states. This restriction must remain valid at the operating field and under the applied pulses, even when virtual excitations contribute to the effective spin Hamiltonian [8].

4.1 Zeeman splitting and the two-level approximation

Consider a singly occupied dot with the electron in the spatial state $|\chi_0\rangle$. A static magnetic field along $+z$ lifts the degeneracy of its two spin states. For a positive isotropic effective g factor, the Zeeman Hamiltonian in the physical-spin basis $(|\uparrow\rangle, |\downarrow\rangle)$ is

$$H_Z = \frac{E_Z}{2}\sigma_z, \qquad E_Z = g\mu_B B > 0. \tag{4.1}$$

Here μ_B is the Bohr magneton and B the field magnitude. With the logical basis in Eq. (3.1), $|0\rangle$ is the lower level and $|1\rangle$ the upper level. In this basis, $H_Z|_{\mathcal{H}_{\mathrm{q}}} = -E_Z Z/2$. The minus sign follows from assigning the lower spin level to $|0\rangle$. This diagonal Hamiltonian changes the relative phase of a spin superposition while leaving the two populations unchanged.

The two-level approximation requires the electron to remain in the retained orbital–valley state and charge sector. Let Δ_{orb} and Δ_{val} denote the relevant gaps to excluded orbital and valley states, measured at the operating field (Appendix I). The condition $k_B T \ll \min(\Delta_{\mathrm{orb}}, \Delta_{\mathrm{val}})$ suppresses their thermal occupation. If electron exchange with a reservoir remains possible, every accessible charge-transfer process out of either retained spin state must have a positive energy cost large compared with $k_B T$. The costs $\Delta_{\pm}$ in Eq. (2.7) refer to the ground state of the occupied charge sector, with E_N evaluated including the Zeeman energies at the operating field. Starting in its upper Zeeman state reduces a transfer cost by E_Z for the same final state. Alternatively, reservoir exchange can be made negligible on the operation timescale by raising its tunnel barrier. Low temperature alone does not prevent control-induced leakage: a pulse that couples to an excluded state can populate it if its frequency approaches the corresponding transition. The coupling matrix elements, excitation gaps, and pulse spectrum must therefore be checked together [8].

Both Zeeman states belong to the retained qubit space. Keeping the electron in that space does not require it to occupy only the lower spin level, and does not by itself require $k_BT \ll E_Z$. That stronger spin-polarization condition is useful when thermal relaxation is used to prepare $|0\rangle$. During coherent operations, the intended state may instead be any superposition of the two levels. Spin relaxation and loss of phase coherence constrain how long that state can be retained even when charge, orbital, and valley leakage remain negligible.

4.2 Transverse spin control

Our aim is to control both the relative phase and the populations of the two spin states. A suitable Hamiltonian is

$$H_{\text{target}}(t) = a(t)\sigma_z + b(t)\sigma_x, \tag{4.2}$$

where a, b are real energy coefficients. The σ_z term changes the relative phase of $|\uparrow\rangle$ and $|\downarrow\rangle$, while σ_x mixes them. A static Zeeman field already supplies the longitudinal term. We need to generate a useful transverse term, either directly in the laboratory or in an effective rotating frame.

A constant transverse coupling is the simplest candidate. To see its limitation, keep the Zeeman splitting fixed and switch on a term $\hbar\Omega_x\sigma_x/2$ at $t = 0$:

$$H_{\text{static}} = \frac{\hbar}{2}(\omega_Z\sigma_z + \Omega_x\sigma_x), \qquad \omega_Z = \frac{E_Z}{\hbar}. \tag{4.3}$$

Since $(\omega_Z\sigma_z + \Omega_x\sigma_x)^2 = (\omega_Z^2 + \Omega_x^2)I$, a spin initially in $|\downarrow\rangle$ has transition probability

$$P^{\text{static}}_{\downarrow\to\uparrow}(t) = \frac{\Omega_x^2}{\omega_Z^2 + \Omega_x^2}\sin^2\left(\frac{t}{2}\sqrt{\omega_Z^2 + \Omega_x^2}\right), \qquad P^{\text{static}}_{\max} = \frac{\Omega_x^2}{\omega_Z^2 + \Omega_x^2}. \tag{4.4}$$

For $|\Omega_x| \ll \omega_Z$, the maximum is approximately $(\Omega_x/\omega_Z)^2 \ll 1$, regardless of pulse duration. The spin rotates about $(\Omega_x, 0, \omega_Z)$, an axis close to z, so it cannot reach the opposite Zeeman state. With zero longitudinal splitting, a full flip would take $\pi/|\Omega_x|$; the bound applies to a single constant pulse with fixed ω_Z.

Resonant driving avoids this limitation by making the transverse coupling oscillate near the Zeeman frequency.

Resonant driving and Rabi oscillations. Keep the Zeeman splitting fixed and take an oscillating transverse term with real drive coefficient Ω, angular frequency ω and phase ϕ (Appendix M):

$$H(t) = \frac{\hbar\omega_Z}{2}\sigma_z + \hbar\Omega\cos(\omega t + \phi)\sigma_x, \qquad \omega_Z = \frac{E_Z}{\hbar}. \tag{4.5}$$

The drive couples the Zeeman states because $\langle\uparrow|\sigma_x|\downarrow\rangle = 1$. In a frame rotating at the drive frequency, $|\psi(t)\rangle = e^{-i\omega t\sigma_z/2}|\psi_{\text{r}}(t)\rangle$, define the detuning $\delta = \omega_Z - \omega$. For $|\delta|, |\Omega| \ll \omega_Z$, the rotating-wave approximation (RWA) gives the effective Hamiltonian and evolution (Appendix L)

$$\begin{aligned} H_{\text{RWA}} &= \frac{\hbar}{2}\left[\delta\sigma_z + \Omega(\cos\phi\,\sigma_x + \sin\phi\,\sigma_y)\right], \\ U_{\text{r}}(t) &= \cos\frac{\Omega_{\text{eff}}t}{2}\,I - i\sin\frac{\Omega_{\text{eff}}t}{2}\,\frac{\delta\sigma_z + \Omega(\cos\phi\,\sigma_x + \sin\phi\,\sigma_y)}{\Omega_{\text{eff}}}, \end{aligned} \tag{4.6}$$

where $\Omega_{\text{eff}} = \sqrt{\delta^2 + \Omega^2}$ is the rotation frequency. The phase sets the transverse direction of the rotation axis $(\Omega\cos\phi, \Omega\sin\phi, \delta)$. For $\phi = 0$, the target coefficients are $a = \hbar\delta/2$ and $b = \hbar\Omega/2$. To find the pulse duration for a spin flip, start in $|\downarrow\rangle$ and evaluate the up-spin probability:

$$P_{\downarrow\to\uparrow}(t) = \frac{\Omega^2}{\delta^2 + \Omega^2}\sin^2\left(\frac{t}{2}\sqrt{\delta^2 + \Omega^2}\right). \tag{4.7}$$

This population-transfer probability is independent of ϕ. Tune the drive frequency to the measured spin resonance to set $\delta = 0$ (Appendix N.1.1). The measured rotation rate at the chosen drive amplitude then sets the spin-flip duration $t_\pi = \pi/|\Omega|$, even for $|\Omega| \ll \omega_Z$. Detuning tilts the effective rotation axis and reduces the maximum inversion. High-fidelity inversion therefore requires $|\delta| \ll |\Omega|$ and a pulse shorter than the coherence time.

4.3 Exchange from virtual tunneling

For electron spins confined in neighboring quantum dots, tunneling through the interdot barrier provides an exchange interaction whose strength can be controlled electrically. Together with the single-spin controls described above, this interaction supplies the means to implement two-spin gates. In the regime of interest, the electron pair remains predominantly in the $(1, 1)$ configuration: double occupation is energetically costly, but its weak admixture can still affect the spin dynamics.

Virtual tunneling and exchange. Consider a double dot isolated from its reservoirs, so that its total electron number remains fixed at two. Tunneling connects the $(1, 1)$ configuration to $(2, 0)$ and $(0, 2)$ by transferring an electron onto the occupied neighboring dot. To describe these processes, we retain one orbital–valley state on each dot, with both spin states, and assume that tunneling conserves spin. The Hubbard approximation developed in Appendix J.1 gives

$$\begin{aligned} H_{\mathrm{Hub}} = &\sum_{i=1}^{2} \epsilon_i n_i + U \sum_{i=1}^{2} n_{i\uparrow} n_{i\downarrow} + V n_1 n_2 \\ &- t_c \sum_{\sigma=\uparrow,\downarrow} (c_{1\sigma}^\dagger c_{2\sigma} + c_{2\sigma}^\dagger c_{1\sigma}). \end{aligned} \tag{4.8}$$

Here $c_{i\sigma}$ annihilates an electron with spin σ on dot i. The occupation operators are $n_{i\sigma} = c_{i\sigma}^\dagger c_{i\sigma}$ and $n_i = n_{i\uparrow} + n_{i\downarrow}$. Each retained orbital has energy ϵ_i, while the Coulomb repulsion depends on whether the electrons occupy the same dot or different dots. These two cases contribute the energies U and V, respectively; for simplicity, we take the same U on both dots. The final term transfers an electron between the dots with real tunnel amplitude t_c. This model neglects direct Coulomb exchange, and we leave out the local Zeeman terms when calculating the exchange contribution.

The energetic cost of a tunneling process follows from both the orbital energies and the repulsion. Moving an electron onto the occupied dot replaces the interdot repulsion V with the on-site repulsion U. It also changes the orbital energy by $+\varepsilon$ or $-\varepsilon$, depending on the direction of transfer, where $\varepsilon = \epsilon_1 - \epsilon_2$. The two double-occupation costs, measured from the energy of $(1, 1)$ in the absence of tunneling, are therefore

$$\Delta_{20} = U - V + \varepsilon, \qquad \Delta_{02} = U - V - \varepsilon. \tag{4.9}$$

We choose an operating point at which both costs are positive and much larger than $|t_c|$. Tunneling then mixes only small components of the doubly occupied states into the low-energy states associated with $(1, 1)$. Provided thermal and drive-induced charge excitation are also negligible, the doubly occupied states can be eliminated from the low-energy description while retaining the energy shifts that their virtual mixing produces.

The spin dependence of these shifts follows from fermionic antisymmetry. Within the retained orbital–valley state, a doubly occupied dot can support only a spin singlet. The $(1, 1)$ singlet can therefore mix with both higher-energy charge states, whereas the triplets have no such tunneling channel. This mixing lowers the singlet energy and leaves the triplet energy

unchanged. Eliminating the two charge states to second order in t_c gives the singlet–triplet energy difference (Appendix J.2)

$$J \equiv E_T - E_S = 2t_c^2 \left(\frac{1}{\Delta_{20}} + \frac{1}{\Delta_{02}} \right). \tag{4.10}$$

This energy difference is the exchange coupling that enters the spin Hamiltonian. After the higher-energy charge states have been eliminated, the interaction within the $(1, 1)$ spin space takes the Heisenberg form [9]

$$H_{\text{ex}} = J\, \mathbf{s}_1 \cdot \mathbf{s}_2. \tag{4.11}$$

A common energy shift has been omitted. The charge degrees of freedom thus enter the spin dynamics through the coefficient J, whose value is set by the tunnel amplitude and the two double-occupation costs.

Tuning exchange with gate voltages. Gate voltages control this interaction by changing the interdot tunneling and the relative orbital energies. Substituting the charge costs into the expression for J makes these dependencies explicit:

$$J(\varepsilon) = \frac{4t_c^2(U - V)}{(U - V)^2 - \varepsilon^2}, \qquad J(0) = \frac{4t_c^2}{U - V}. \tag{4.12}$$

Adjusting the interdot barrier changes the tunnel amplitude t_c, while shifting the orbital energy of one dot relative to the other changes the detuning ε. The tunnel amplitude at the chosen barrier setting must be measured (Appendix K); the exchange strength used to set a spin pulse is calibrated at the corresponding compensated voltages (Appendix N.2.2). In either case, the resulting exchange pulse must remain within the regime used to obtain the spin Hamiltonian. If a double-occupation cost becomes comparable to $|t_c|$, the associated charge state mixes strongly with $(1, 1)$ and must be included explicitly in the dynamics. The apparent divergence of the expression for J near a charge degeneracy is therefore outside its range of validity.

In a device, changing the barrier voltage generally changes the tunnel amplitude and shifts the dot levels at the same time. A pulse intended to tune exchange can consequently move the addition thresholds that define the chosen charge operating point. These shifts can be compensated by adjusting the two plunger voltages together with the barrier voltage.

For a small barrier-voltage change δV_{B}, let $\delta \mathbf{V}_{\text{P}}$ denote the two plunger adjustments and $\delta \boldsymbol{\mu}$ the resulting shifts of the two addition thresholds. The matrix G_{P} contains the measured derivatives of these thresholds with respect to the two plunger voltages at the operating point, while the vector $\mathbf{g}_{\text{B}}$ contains their measured derivatives with respect to the barrier voltage. With independent plunger responses, G_{P} is invertible, and the voltage response gives the compensation required to keep both thresholds fixed (Appendix F.4):

$$\delta \boldsymbol{\mu} = G_{\text{P}} \delta \mathbf{V}_{\text{P}} + \mathbf{g}_{\text{B}} \delta V_{\text{B}}, \qquad \delta \mathbf{V}_{\text{P}}|_{\delta \boldsymbol{\mu} = 0} = -G_{\text{P}}^{-1} \mathbf{g}_{\text{B}} \delta V_{\text{B}}. \tag{4.13}$$

Coordinated voltage combinations chosen to control specified quantities are called *virtual gates*. This combination cancels the changes in the two chosen thresholds to first order around the operating point, while allowing the tunnel amplitude and other Hamiltonian parameters to vary.

5 Initialization and spin readout

Initialization prepares a known logical input; readout converts the logical basis populations into distinguishable electrical signals. The readout schemes considered here first convert the spin distinction into different charge configurations or tunneling records. A nearby charge sensor detects these through their electric effect on its current (Appendix G.2).

5.1 Single-spin qubits

The initialization target is $|0\rangle = |\downarrow\rangle$, and readout distinguishes $|\downarrow\rangle$ from $|\uparrow\rangle$. We suppress the common orbital–valley state in the notation. The lower spin state can be prepared either by relaxation after loading or by selecting it during loading.

Initialization by relaxation. First load one electron from a reservoir. Its spin may initially occupy either Zeeman level. Raising the reservoir barrier then suppresses electron exchange and keeps the dot occupied. The spin remains coupled to the thermal environment, so an excited $|\uparrow\rangle$ can release energy and relax to the lower $|\downarrow\rangle$ state. Waiting for this dissipative process prepares the input state; a unitary rotation alone cannot reset an arbitrary unknown spin.

The achievable ground-state population and required waiting time follow from the relaxation dynamics. Let $p_\uparrow(t)$ be the probability of occupying the upper spin level. For a bath at temperature T, its equilibrium value is $p_\uparrow^{\mathrm{eq}} = [1 + \exp(E_Z/k_BT)]^{-1}$, where k_B is Boltzmann's constant. A simple relaxation model is

$$\dot{p}_\uparrow = -\frac{p_\uparrow - p_\uparrow^{\mathrm{eq}}}{T_1}, \qquad p_\uparrow(t) = p_\uparrow^{\mathrm{eq}} + [p_\uparrow(0) - p_\uparrow^{\mathrm{eq}}]e^{-t/T_1}. \tag{5.1}$$

Here T_1 is the longitudinal relaxation time. The initial-state dependence decays on the scale T_1, but the remaining excited-state population is set by the bath temperature. High-probability ground-state preparation therefore requires both a wait much longer than T_1 and $E_Z \gg k_BT$. Use the measured T_1 and long-time excited-state population at the holding voltages to choose the waiting time and determine the attainable preparation probability (Appendix U.1.2). The loading and isolation settings are specified in Appendix R.1.

Spin-selective loading. The second method prepares the lower spin state as the electron enters the dot, without relying on subsequent spin relaxation. Start from an empty dot. The energy required to add the first electron is $\mu_\sigma = E(1,\sigma) - E(0)$, where $E(0)$ and $E(1,\sigma)$ are the empty-dot and one-electron energies. Its spin dependence comes from the Zeeman energy of the final state. If ϵ_d is the addition energy without the Zeeman term,

$$\mu_\downarrow = \epsilon_d - \frac{E_Z}{2}, \qquad \mu_\uparrow = \epsilon_d + \frac{E_Z}{2}. \tag{5.2}$$

Thus the two loading thresholds differ by E_Z even for the first electron, without an electron–electron interaction. Gate voltages shift their common position ϵ_d relative to the reservoir chemical potential μ_{R}. To allow down-spin loading while suppressing up-spin loading, set

$$\mu_\downarrow < \mu_{\mathrm{R}} < \mu_\uparrow. \tag{5.3}$$

The spin-resolved addition thresholds must be measured to select the plunger voltage that satisfies this window (Appendix S.1). At low temperature, reservoir states at $\mu_\downarrow$ are occupied and can supply an electron, while states at $\mu_\uparrow$ are mostly empty. Charging energy must also prevent a second electron from entering.

The level alignment determines which process is allowed; the tunnel rates determine how long to keep the loading window open. For weak spin-conserving tunneling, let Γ_σ be the bare tunnel rate and T_e the reservoir electron temperature. The Fermi occupation factor gives the conditional rates

$$\begin{aligned} f_{\mathrm{R}}(E) &= \frac{1}{1 + \exp[(E - \mu_{\mathrm{R}})/(k_BT_e)]}, \\ \Gamma_{\mathrm{in},\sigma} &= \Gamma_\sigma f_{\mathrm{R}}(\mu_\sigma), \qquad \Gamma_{\mathrm{out},\sigma} = \Gamma_\sigma[1 - f_{\mathrm{R}}(\mu_\sigma)]. \end{aligned} \tag{5.4}$$

The factors f_{R} and $1 - f_{\mathrm{R}}$ require an occupied source state and an empty destination state, respectively. Appendix Q.1 derives these reservoir rates and their waiting-time law from tunneling amplitudes. Both margins in Eq. (5.3) should exceed $k_B T_e$ and tunnel broadening so that loading and unloading remain spin-selective.

Neglecting other events, an initially empty dot remains empty with probability $e^{-\Gamma_{\mathrm{in},\downarrow} t}$. Measure the loading rate at the chosen reservoir-barrier setting (Appendix R.1) and use it to choose a loading interval much longer than $\Gamma_{\mathrm{in},\downarrow}^{-1}$. Then raise the reservoir barrier to retain the electron during the gate sequence. Spin-selective loading has been demonstrated in Si/SiGe dots [10]. Excited orbitals and energy-dependent barriers can modify the rates.

Readout by energy-selective tunneling. The readout goal is to determine whether the occupied dot contains an up-spin or down-spin electron after the gates. We convert that distinction into a charge transient by allowing only the upper spin level to empty into the reservoir.

Reopen the reservoir barrier and tune the addition energies to Eq. (5.3). An up-spin electron can tunnel into an empty reservoir state; a down-spin electron remains because the corresponding states are occupied. If the dot empties, it can reload a down-spin electron. The two initial spin states therefore give different charge records:

$$\begin{aligned} |\uparrow\rangle: &\quad N = 1 \longrightarrow 0 \longrightarrow 1, \\ |\downarrow\rangle: &\quad N = 1 \quad \text{throughout the readout interval.} \end{aligned} \tag{5.5}$$

A sensor dot or quantum point contact detects the temporary empty state through a current or radio-frequency response [11]. The evidence for an initial up spin is the $1 \to 0 \to 1$ transient, not the final occupation, which can be one for either spin.

The electron must escape before it relaxes: a spin that relaxes first leaves no up-spin charge record. To quantify this competition, take constant escape and relaxation rates $\Gamma_{\mathrm{out},\uparrow}$ and T_1^{-1} during a low-temperature readout interval τ_{r}. The probability of neither event by time t is $e^{-(\Gamma_{\mathrm{out},\uparrow} + T_1^{-1})t}$. Multiplying by the escape rate and integrating gives the probability of producing the desired charge transient:

$$P_{\mathrm{escape}|\uparrow}(\tau_{\mathrm{r}}) = \frac{\Gamma_{\mathrm{out},\uparrow}}{\Gamma_{\mathrm{out},\uparrow} + T_1^{-1}} \left[1 - e^{-(\Gamma_{\mathrm{out},\uparrow} + T_1^{-1})\tau_{\mathrm{r}}}\right]. \tag{5.6}$$

Even when escape occurs, the detector must resolve the empty interval. Fast reloading or sensor noise can hide it, while thermal down-spin escape can produce a false up-spin signal. Thus readout fidelity depends on the charge-conversion probability above and on the sensor response. The measured escape rates and T_1 at the readout voltages determine the reservoir-barrier setting and readout interval: up-spin escape should be likely before relaxation, while down-spin escape remains unlikely (Appendix S.4).

For $a|\downarrow\rangle + b|\uparrow\rangle$, ideal readout gives the up-spin outcome with probability $|b|^2$. Measurements after calibrated rotations on repeated preparations access other spin components. Escape and reloading make this protocol destructive. Appendix S.1 specifies the conversion window, and Appendix S.4 derives the competing-rate probability and duration constraints.

5.2 Singlet–triplet qubits

Here initialization prepares $|0_{\mathrm{ST}}\rangle = |S_{11}\rangle$, and readout distinguishes it from $|1_{\mathrm{ST}}\rangle = |T_{0,11}\rangle$. Both tasks use the singlet's access to double occupation.

Initialization. A convenient route first puts both electrons on one dot, where the lowest orbital supports a unique spin singlet. Tune to a region where $|D_{02}\rangle$ is the nondegenerate ground state, load two electrons, and allow relaxation. Then sweep across the singlet charge anticrossing to separate them into $(1,1)$ while preserving the singlet spin state. The sweep must follow the charge anticrossing adiabatically. The measured tunnel coupling and detuning response determine the allowed voltage-ramp speed through the anticrossing. Once exchange becomes small, use the measured Zeeman difference to limit the time available for singlet–triplet rotation, or compensate that rotation (Appendix R.2). Accessible orbital or valley excitations can spoil this preparation.

Readout by Pauli spin blockade. The goal is to map the singlet and triplet to different charge states. Start in $(1,1)$ and pulse toward $(0,2)$. The singlet couples to $|D_{02}\rangle$ through the tunnel matrix element in Eq. (J.10). A doubly occupied triplet needs an excited orbital–valley state; keeping that state outside the readout window blocks triplet transfer. Choose the readout detuning from the measured singlet and triplet charge-transfer thresholds so that only the singlet transition is allowed (Appendix S.2). With negligible spin mixing and charge-transfer error,

$$|S_{11}\rangle \longrightarrow |D_{02}\rangle, \qquad |T_{11}\rangle \longrightarrow |T_{11}\rangle, \tag{5.7}$$

Here $|T_{11}\rangle$ is a triplet within the blocked window. A charge sensor distinguishes the resulting $(0,2)$ and $(1,1)$ occupations, implementing singlet–triplet readout through Pauli spin blockade [12]. Because the two configurations have the same total electron number, the sensor must have different sensitivity to charge on the two dots (Appendix G.2). This pulse deliberately permits real charge transfer, so the weak-admixture approximation used to derive exchange does not describe the transfer itself.

The measurement tests whether the input has a singlet component. For an initial two-spin density operator ρ, the probabilities of the two charge outcomes are

$$p_{02} = \mathrm{Tr}(\rho P_S), \qquad p_{11} = \mathrm{Tr}[\rho(I - P_S)], \qquad P_S = |S_{11}\rangle\langle S_{11}|. \tag{5.8}$$

Within the singlet–triplet encoding, these probabilities are the populations of $|0_{\mathrm{ST}}\rangle$ and $|1_{\mathrm{ST}}\rangle$. A blocked outcome outside that encoding cannot distinguish T_0 from $T_\pm$. Relaxation or field-gradient-induced mixing can turn a blocked state into one that transfers, so the sensor must resolve the charge outcome before those processes erase the distinction. Appendix S.2 develops the readout window and the conditions for resolving the two charge outcomes.

5.3 Exchange-only qubits

For the fixed-m encoding in Eq. (3.4), initialization prepares $|0_{\mathrm{E}}\rangle$, and readout must distinguish it from $|1_{\mathrm{E}}\rangle$. The useful distinction is the spin of pair 1–2: a singlet in $|0_{\mathrm{E}}\rangle$ and a triplet in $|1_{\mathrm{E}}\rangle$.

Initialization. Let $|S_{12}\rangle = (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)/\sqrt{2}$ denote the singlet of spins 1 and 2. The chosen logical zero factorizes as

$$|0_{\mathrm{E}}\rangle = |S_{12}\rangle \otimes |\uparrow\rangle_3 = \frac{|\uparrow\downarrow\uparrow\rangle - |\downarrow\uparrow\uparrow\rangle}{\sqrt{2}}. \tag{5.9}$$

We therefore prepare a singlet on dots 1–2 and an up spin on dot 3, with tunneling between dots 2 and 3 suppressed during preparation. Use the double-occupation preparation described above for the pair: load its singlet in $(0,2)$, then separate the electrons without mixing singlet and triplet. Keeping one electron on dot 3, the charge sequence is $(0,2,1) \to (1,1,1)$. The

singlet-loading and charge-transfer steps have the same temperature, excited-state, and ramp constraints as for the singlet–triplet qubit.

The up spin on dot 3 is an additional requirement of our choice $m = +1/2$. With the Zeeman convention used here, relaxation prepares the opposite spin. If individually addressable transverse control is available, one route is to load $|\downarrow\rangle_3$ as in Section 5.1, then apply a resonant π pulse to spin 3 before the encoded gates. Its frequency and duration are set by the measured resonance and rotation rate of spin 3 (Appendix N.1.1). This preparation uses a control beyond exchange; the subsequent logical rotations use exchange alone. An exchange pulse cannot set m, because it conserves the total spin projection along z.

Polarizing the third spin is not required by the broader exchange-only subsystem encoding used in silicon experiments [13]. Preparing only the pair singlet gives $\rho_{\text{init}} = |S_{12}\rangle\langle S_{12}| \otimes \rho_3$, where ρ_3 is the third spin's density operator. For example, an unpolarized $\rho_3 = I_3/2$ populates both $m = \pm 1/2$ sectors. Both correspond to logical zero because pair 1–2 is a singlet; the value of m is left unspecified. This is a larger encoding than the fixed-m subspace used above. Using it for two-qubit gates requires a sequence that implements the same logical gate for either value of each qubit's m, as in the Fong–Wandzura construction [13, 14]. Appendix R.3 gives the preparation steps and distinguishes fixed-m from subsystem initialization.

Readout of pair 1–2. To measure the logical basis, suppress J_{23} and pulse dots 1–2 toward double occupation while dot 3 remains singly occupied. A pair singlet can enter the lowest $(0, 2)$ state; a pair triplet is blocked when the excited doubly occupied triplets lie outside the readout window. The sensor therefore distinguishes $(0, 2, 1)$ from $(1, 1, 1)$, without measuring spin 3. Pair singlet preparation and Pauli-blockade readout are used in silicon triple-dot qubits [15].

To see why this measures the encoded basis, write the singlet projector on the three-spin space as

$$Q_S = |S_{12}\rangle\langle S_{12}| \otimes I_3 = \frac{1}{4} I - \mathbf{s}_1 \cdot \mathbf{s}_2. \tag{5.10}$$

Here I_3 acts on spin 3 and I on all three spins. Pair exchange has eigenvalue $-3/4$ on the pair singlet and $1/4$ on any pair triplet, so

$$Q_S|0_{\text{E}}\rangle = |0_{\text{E}}\rangle, \qquad Q_S|1_{\text{E}}\rangle = 0, \qquad P_{\text{E}} Q_S P_{\text{E}}|_{\text{E}} = \frac{I_{\text{E}} + Z_{\text{E}}}{2}, \tag{5.11}$$

where P_{E} projects onto the two-dimensional encoded space, I_{E} is its identity, and $Z_{\text{E}} = |0_{\text{E}}\rangle\langle 0_{\text{E}}| - |1_{\text{E}}\rangle\langle 1_{\text{E}}|$. Consequently, ideal charge conversion for $|\psi\rangle = a|0_{\text{E}}\rangle + b|1_{\text{E}}\rangle$ gives

$$p_{021} = |a|^2, \qquad p_{111} = |b|^2. \tag{5.12}$$

The subscripts label the three-dot occupations. A preceding encoded rotation changes the measured logical axis.

This assignment assumes that the input is in the computational space. States with total spin $S = 3/2$ also have a triplet on pair 1–2 and produce the blocked outcome; pair readout alone cannot distinguish that leakage from logical one. It also does not determine m. Thus the same singlet test works for the subsystem encoding with an unpolarized third spin, but does not certify preparation in our fixed-m subspace. Residual J_{23}, magnetic-field gradients, or relaxation during conversion can change the pair spin before it is resolved. Their timescales must be long compared with conversion and sensing. Appendix S.3 develops this pair-readout protocol and its limitations.

5.4 From repeated records to logical-state probabilities

Each readout returns one electrical record, such as an empty-dot transient or a charge configuration after Pauli-blockade conversion. The conversion rules above assign that record a logical outcome. For an input density operator ρ within the chosen qubit space, ideal logical-basis measurement has probabilities

$$p_0 = \mathrm{Tr}(\rho|0\rangle\langle 0|), \qquad p_1 = 1 - p_0, \qquad \widehat{p}_0 = \frac{n_0}{M}. \tag{5.13}$$

Here $|0\rangle, |1\rangle$ denote the logical basis of the encoding under study, and n_0 is the number of zero outcomes in M repetitions. Each repetition prepares the input again and applies the same control sequence before readout. A single outcome is random; the fraction n_0/M estimates its probability. For independent repetitions with fixed p_0, its variance is $p_0(1-p_0)/M$. Readout errors change the relation between the observed frequency and the logical population. If logical 0 is reported as 1 with probability ϵ_0, and logical 1 as 0 with probability ϵ_1, fixed classical assignment errors give

$$p_{1,\mathrm{obs}} = \epsilon_0 + (1 - \epsilon_0 - \epsilon_1)p_1. \tag{5.14}$$

Readout thus introduces an offset and reduces contrast. Recovering p_1 requires nonzero contrast and independently established error probabilities. Appendix S.5 derives this assignment map and its inverse in Eq. (S.10).

Logical populations alone do not determine a quantum state. For example, $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ and the mixed state $I/2$ both give $p_0 = p_1 = 1/2$. Their difference appears if a known logical rotation U_a precedes the same readout. The probability after an analysis rotation is then

$$p_0(U_\mathrm{a}) = \mathrm{Tr}\Big(\rho\, U_\mathrm{a}^\dagger|0\rangle\langle 0|U_\mathrm{a}\Big), \qquad p_0\big(R_y(-\pi/2)\big) = \frac{1 + \mathrm{Tr}(\rho X)}{2}. \tag{5.15}$$

The rotation $R_y(-\pi/2)$ maps $|+\rangle$ to $|0\rangle$, giving a zero outcome with probability one, whereas $I/2$ remains unchanged and still gives probability 1/2. Similarly, $R_x(\pi/2)$ before readout measures $(1 + \mathrm{Tr}(\rho Y))/2$. The pulse implementations of these rotations for each encoding are developed in Section 6. Repeated readout with different analysis rotations can therefore determine the relative phase information as well as the logical populations. Testing a gate requires its action on independent input states; the populations of one output cannot establish its action on an arbitrary superposition. Appendices T.1 and T.2 develop state and gate reconstruction; Appendix T.3 defines the quantitative comparison with a target gate.

6 Quantum gates

Initialization and readout provide a known input and a way to measure the final spin populations. Between these stages, the Hamiltonians in Section 4 control the coherent evolution of the qubits. Implementing a particular gate requires choosing the pulse amplitudes, durations, phases, and ordering so that the resulting evolution has the desired action on the computational states. For a pulse described by $H(t)$ over a time interval τ, the evolution is $U(\tau)$ in Eq. (Pref.3).

To implement a unitary gate, the pulse must return every initial computational state to the chosen computational space. If P_q projects onto that space, this condition is $(I - P_\mathrm{q})U(\tau)P_\mathrm{q} = 0$; the restriction of $P_\mathrm{q}U(\tau)P_\mathrm{q}$ then gives the gate operator. Intermediate admixture of charge or other spin states is allowed, provided that no population remains outside the computational space at the end of the pulse. A calculation based on an effective spin Hamiltonian must account for any such intermediate excursions through the approximations used to obtain it.

6.1 Gates for single-spin qubits

Transverse rotation gates: R_x and R_y. We want to choose the axis and angle of a logical spin rotation. The drive phase sets the transverse axis, while its amplitude and duration set the angle. In the logical basis $(|0\rangle, |1\rangle) = (|\downarrow\rangle, |\uparrow\rangle)$, the Pauli operators obey $\sigma_x = X$, $\sigma_y = -Y$, and $\sigma_z = -Z$.

Writing Eq. (4.6) in this basis and allowing a time-dependent drive envelope $\Omega(t)$ gives

$$H_{\mathrm{rot,q}}(t) = \frac{\hbar}{2}\left[-\delta Z + \Omega(t)(\cos\phi\, X - \sin\phi\, Y)\right]. \tag{6.1}$$

To reach resonance, tune the drive frequency to the measured spin resonance ω_Z (Appendix N.1.1). On resonance ($\delta = 0$), fixing ϕ keeps the rotation axis constant even if the envelope varies. The angle depends on the full pulse area, including its rise and fall, $\vartheta = \int_0^\tau \Omega(t)\, dt$:

$$U_\phi(\vartheta) = \exp\left[-\frac{i\vartheta}{2}(\cos\phi\, X - \sin\phi\, Y)\right]. \tag{6.2}$$

For a constant-amplitude pulse, the rotation rate Ω at that amplitude is measured (Appendix N.1.1) and the duration is chosen so that $\Omega\tau = \vartheta$. For a pulse with finite rising and falling edges, choose the amplitude and hold time from the calibrated area of the complete waveform (Appendix N.1.2).

To select the transverse axis, calibrate the relative phase of successive pulses against a common reference (Appendix N.1.3). Taking $\phi = 0$ as the x reference, the two phase settings give

$$\begin{aligned} \delta = 0, \quad \phi = 0: &\qquad U_0(\vartheta) = R_x(\vartheta), \\ \delta = 0, \quad \phi = -\pi/2: &\qquad U_{-\pi/2}(\vartheta) = R_y(\vartheta), \end{aligned} \qquad \vartheta = \int_0^\tau \Omega(t)\, dt. \tag{6.3}$$

Phase rotation gate: R_z. A logical z rotation changes the relative phase without moving population between the basis states. At the chosen holding voltages, measure the detuning from the control reference (Appendix U.1.1). With the transverse drive off, choose the waiting time to accumulate the required angle:

$$U_z(\tau) = \exp\left[\frac{iZ}{2}\int_0^\tau \delta(t)\, dt\right] = R_z\left(-\int_0^\tau \delta(t)\, dt\right). \tag{6.4}$$

For a constant measured detuning, the condition is $\vartheta_z = -\delta\tau$. Phase-coherent control can also track a z rotation by updating the programmed phases of later pulses; those phases use the same calibrated reference as the transverse gates above.

The calibrated rotations implement the Pauli and Hadamard gates through the sequences in Eq. (Pref.5).

Exchange gates: SWAP and $\sqrt{\mathrm{SWAP}}$. We next use exchange to entangle two single-spin qubits and combine it with local rotations to obtain CNOT. Keep the dots in $(1,1)$ and remove common Zeeman evolution. The local Zeeman-energy difference must be calibrated for the intended exchange operating point (Appendix N.2.2). Choose holding and exchange voltages for which this difference has negligible action during the pulse. If no such setting is available, the measured difference must be included in the pulse Hamiltonian instead of using the exchange-only form below. The exchange pulse gives

$$\begin{aligned} H_{\mathrm{ex}}(t) &= \frac{J(t)}{4}(X_1X_2 + Y_1Y_2 + Z_1Z_2), \\ U_{\mathrm{ex}}(\theta) &= e^{-i\theta(X_1X_2+Y_1Y_2+Z_1Z_2)/4}, \qquad \theta = \frac{1}{\hbar}\int_0^\tau J(t)\, dt. \end{aligned} \tag{6.5}$$

For the SWAP operator in Eq. (Pref.6), $\mathrm{SWAP} = (I + X_1X_2 + Y_1Y_2 + Z_1Z_2)/2$ and $\mathrm{SWAP}^2 = I$, so

$$U_{\mathrm{ex}}(\theta) = e^{i\theta/4}\left[\cos(\theta/2)I - i\sin(\theta/2)\mathrm{SWAP}\right]. \tag{6.6}$$

To set θ, calibrate the exchange strength and the phase accumulated on the pulse edges along the compensated barrier-voltage path (Appendix N.2.2). The barrier pulse amplitude and hold time are then chosen to reach the target area (Appendix N.2.3). The pulse areas $\theta = \pi$ and $\theta = \pi/2$ implement the two gates in Eq. (Pref.6), up to global phase:

$$\mathrm{SWAP} = e^{i\pi/4}U_{\mathrm{ex}}(\pi), \qquad \sqrt{\mathrm{SWAP}} = e^{i\pi/8}U_{\mathrm{ex}}(\pi/2). \tag{6.7}$$

SWAP exchanges the qubits without entangling a product state. The half-area pulse can entangle them; for example,

$$U_{\mathrm{ex}}(\pi/2)|01\rangle = \frac{e^{i\pi/8}}{\sqrt{2}}(|01\rangle - i|10\rangle). \tag{6.8}$$

The displayed output is maximally entangled [9].

Controlled gates: CZ and CNOT. To construct CZ from partial exchange, we need to cancel the X_1X_2 and Y_1Y_2 contributions while retaining Z_1Z_2. Set $G = U_{\mathrm{ex}}(\pi/2)$. Conjugation by the local phase flip Z_1 reverses the first two terms but leaves the third unchanged. Because the three terms commute, two exchange pulses give

$$U_{ZZ} \equiv GZ_1GZ_1 = \exp(-i\pi Z_1Z_2/4). \tag{6.9}$$

The phase flip Z_1 is supplied by a physical local π rotation [Eq. (Pref.5)], with exchange suppressed during that rotation. Each G pulse uses the calibrated exchange area $\pi/2$. The resulting U_{ZZ} assigns phases to all four basis states. Local z rotations remove the unwanted single-qubit phases, leaving a minus sign only on $|11\rangle$:

$$\mathrm{CZ} = e^{-i\pi/4}R_{z,1}(-\pi/2)R_{z,2}(-\pi/2)U_{ZZ}. \tag{6.10}$$

For the complete sequence, the conditional phase and the two local phases must be measured (Appendix N.2.4). Adjust the exchange hold times until the conditional phase is π modulo 2π, and set the local z corrections from the measured local phases. Applying the calibrated Hadamard gate to the target before and after this CZ gives CNOT [Eq. (Pref.9)]. This sequence requires separately addressable single-spin pulses, with charge leakage and unwanted Zeeman terms suppressed during exchange. It provides the entangling gate that, together with arbitrary single-qubit rotations, forms a universal gate set [9].

6.2 Gates for singlet–triplet qubits

For the basis $(|S\rangle, |T_0\rangle)$, we again need controls that change both relative phase and population. Exchange changes the singlet–triplet energy splitting and hence their relative phase. A difference between the longitudinal fields on the two dots couples the two states. For local Zeeman energies $E_{Z,i} = g_i\mu_B B_{z,i}$, with effective factors g_i and fields $B_{z,i}$, the physical Hamiltonian is

$$H = J\,\mathbf{s}_1 \cdot \mathbf{s}_2 + E_{Z,1}s_{1z} + E_{Z,2}s_{2z}. \tag{6.11}$$

This Hamiltonian preserves the $m = 0$ computational space. Exchange gives energies $-3J/4$ and $J/4$ to $|S\rangle$ and $|T_0\rangle$, while $\Delta E_Z = E_{Z,1} - E_{Z,2}$ couples them. With $Z_{\mathrm{ST}} = |S\rangle\langle S| - |T_0\rangle\langle T_0|$ and $X_{\mathrm{ST}} = |S\rangle\langle T_0| + |T_0\rangle\langle S|$, the logical Hamiltonian, after dropping $-J/4$, is

$$H_{\mathrm{ST}} = -\frac{J}{2}Z_{\mathrm{ST}} + \frac{\Delta E_Z}{2}X_{\mathrm{ST}}, \tag{6.12}$$

A uniform longitudinal field vanishes in the $m = 0$ space; the difference ΔE_Z, rather than the common field, sets the mixing term. Fluctuations of either J or ΔE_Z therefore change the logical evolution.

Logical R_x and R_z gates. For constant controls held for a time τ, the two pure-axis rotations are

$$\begin{aligned} J = 0: &\quad U_{\mathrm{ST}} = R_{x,\mathrm{ST}}(\Delta E_Z \tau/\hbar), \\ \Delta E_Z = 0: &\quad U_{\mathrm{ST}} = R_{z,\mathrm{ST}}(-J\tau/\hbar). \end{aligned} \tag{6.13}$$

To obtain $R_{x,\mathrm{ST}}$, set the compensated barrier and detuning voltages so that exchange is negligible, measure ΔE_Z there, and choose τ for the required angle (Appendix O.1). A direct $R_{z,\mathrm{ST}}$ requires a setting with negligible ΔE_Z; at that setting, the measured exchange strength determines the pulse duration for the required angle.

Rotations with a fixed field gradient. With a fixed nonzero ΔE_Z, turning on J tilts the axis rather than giving a pure z rotation. The values of J and ΔE_Z must be measured at each selected voltage setting (Appendix O.1). Holding both controls constant for time τ gives the gate below, with $E_{\mathrm{ST}} = \sqrt{J^2 + (\Delta E_Z)^2}$:

$$U_{\mathrm{ST}}(\tau) = \cos\left(\frac{E_{\mathrm{ST}}\tau}{2\hbar}\right) I - i \sin\left(\frac{E_{\mathrm{ST}}\tau}{2\hbar}\right) \frac{\Delta E_Z X_{\mathrm{ST}} - J Z_{\mathrm{ST}}}{E_{\mathrm{ST}}}. \tag{6.14}$$

The rotation axis is proportional to $(\Delta E_Z, 0, -J)$. Choose voltage settings whose measured J and ΔE_Z give nonparallel axes. The durations of these voltage holds can then be chosen to compose the desired logical rotation. Such control has been demonstrated in a silicon double dot with a micromagnet [12]. If $J/\Delta E_Z$ changes within a pulse, the axis also changes and time ordering is generally required.

Encoded CZ and CNOT gates. To entangle two singlet–triplet qubits, couple two double dots A, B containing four electrons in total. Capacitive coupling makes the phase acquired by one qubit depend on the state of the other, adding an interaction to the separate single-qubit Hamiltonians:

$$H_{\mathrm{logical}} = H_A \otimes I_B + I_A \otimes H_B + H_{\mathrm{cap}}. \tag{6.15}$$

Here H_A, H_B include local energy shifts and H_{cap} is the additional two-qubit interaction. After eliminating charge excitations, assume this coupling is diagonal in the product singlet–triplet basis. Here $|S\rangle$ denotes the charge-hybridized singlet. Let $E_{00}, E_{01}, E_{10}, E_{11}$ be the energies of $|SS\rangle$, $|ST_0\rangle$, $|T_0S\rangle$, and $|T_0T_0\rangle$, respectively. Its conditional part is fixed by $\kappa = E_{00} - E_{01} - E_{10} + E_{11}$: independent shifts of either qubit cancel in this combination. Absorbing those shifts into H_A, H_B, and writing $P_{S,q} = (I + Z_q)/2$ with Z_q the encoded Pauli operator on $q = A, B$, gives the interaction (Appendix O.2.1)

$$H_{\mathrm{cap}} = \kappa P_{S,A} P_{S,B} = \frac{\kappa}{4}(I + Z_A + Z_B + Z_A Z_B). \tag{6.16}$$

The Zeeman differences at the joint interaction setting must also be measured. Select the interaction voltages and duration so that the resulting transverse evolution is negligible over the pulse. If this condition cannot be reached, the full coupled Hamiltonian is needed (Appendix O.2.4). Under this condition and with negligible charge leakage, the nonlocal factor of the interaction pulse is

$$U_{\mathrm{ent}}(\Phi) = \exp(-i\Phi Z_A Z_B/4), \qquad \Phi = \frac{1}{\hbar}\int_0^\tau \kappa(t)\, dt. \tag{6.17}$$

To reach $\Phi = \pi$, measure the conditional phase of the full joint voltage pulse and adjust its barrier and detuning settings or its duration (Appendix O.2.2). The local phases accumulated in the same pulse must also be measured; they determine the net local corrections that convert the full pulse into CZ. In terms of the isolated nonlocal factor, the ideal conversion is

$$\mathrm{CZ}_{AB} = e^{-i\pi/4} R_{z,A}(-\pi/2) R_{z,B}(-\pi/2) U_{\mathrm{ent}}(\pi), \tag{6.18}$$

Using the calibrated local rotations to implement Hadamard gates on B before and after CZ gives the encoded CNOT [Eq. (Pref.9)], with $|0\rangle = |S\rangle$ and $|1\rangle = |T_0\rangle$ in each double dot. The physical gate acts on four spins. Uncompensated local X terms do not commute with $Z_A Z_B$ and require the full time-ordered evolution. Capacitive singlet–triplet entanglement was demonstrated in GaAs [16]; silicon requires its own coupling calibration.

Charge admixture sets the conditional energy. The conditional energy κ comes from the Coulomb interaction between the two double dots. In a density-interaction model without inter-block tunneling,

$$H_{\mathrm{micro}} = H_{\mathrm{Hub},A} + H_{\mathrm{Hub},B} + H_C^{AB}, \qquad H_C^{AB} = \sum_{i,j=1}^{2} W_{ij} n_{A,i} n_{B,j}, \tag{6.19}$$

where $n_{A,i}$ counts electrons on dot i in A, and W_{ij} is its Coulomb coupling to dot j in B, in energy units. These cross-qubit terms are additional to each double dot's U, V. Coulomb interaction acts on charge, so it can distinguish the logical spin states only when they have different charge distributions. Near a $(1,1)$–$(0,2)$ anticrossing, tunneling mixes $|S(1,1)\rangle$ with $|S(0,2)\rangle$, while $|T_0\rangle$ remains predominantly in $(1,1)$. The resulting dipole difference makes the inter-qubit energy depend on both logical states.

For a singlet with probability p_q of occupying $(0,2)$ and a triplet confined to $(1,1)$, the weak inter-block density interaction gives

$$\kappa = p_A p_B (W_{11} - W_{12} - W_{21} + W_{22}). \tag{6.20}$$

This assumes that the coupling does not appreciably change the separate charge eigenstates. Gate voltages change p_A, p_B through detuning and tunneling, thereby controlling the conditional energy. Increasing charge admixture can strengthen coupling but also increase charge-noise sensitivity [8, 16]. Appendix O.2.1 obtains the admixtures from the charge Hamiltonian and separates the energy shifts in Eq. (O.15).

Local gates in the presence of residual coupling. The same interaction is present during a nominally local gate if κ remains nonzero. In the diagonal limit above, two local gates act independently only when the residual Φ is negligible or is cancelled, modulo 2π and local phases. With simultaneous local X terms, independence must instead be checked from the full two-qubit evolution. Reducing charge admixture suppresses κ but also changes each qubit's exchange; these controls cannot be assumed to be independent. In the diagonal limit, measure the local and residual conditional phases of the proposed simultaneous pulses and adjust their voltages, durations, or scheduling to retain the target local angles while canceling the unwanted conditional phase (Appendix O.2.4). An echoed entangling pulse is constructed in Appendix O.2.3.

6.3 Gates for exchange-only qubits

Single-pair exchange gates. Here the goal is to generate logical rotations using exchange pulses alone. In the basis of Eq. (3.4), exchange on spins 1 and 2 distinguishes their singlet

and triplet, whereas exchange on spins 2 and 3 mixes the encoded states. We determine the two rotation axes from the physical Hamiltonian

$$H_{\mathrm{E}} = J_{12}\,\mathbf{s}_1 \cdot \mathbf{s}_2 + J_{23}\,\mathbf{s}_2 \cdot \mathbf{s}_3, \tag{6.21}$$

Isotropic pair exchange conserves total spin and its z component, so it preserves the $(S, m) = (1/2, +1/2)$ computational space even when its strength changes in time. Nonuniform magnetic fields need not preserve that space. In the encoded basis, and after dropping a common energy, the Hamiltonian is

$$H_{\mathrm{E,q}} = \frac{\sqrt{3}J_{23}}{4} X_{\mathrm{E}} + \left(-\frac{J_{12}}{2} + \frac{J_{23}}{4}\right) Z_{\mathrm{E}}. \tag{6.22}$$

The two projected exchange matrices are evaluated in Appendix P.1.1, Eq. (P.1). For a pulse on one pair with the other exchange negligible, let $\theta_{ij} = \int_0^\tau J_{ij}(t)\,dt/\hbar$. The active pair's exchange area and the other pair's residual exchange must be measured at the selected voltage settings (Appendix P.1.2). Choose the compensated barrier waveform and its duration for the required θ_{ij}, with the inactive pair held at its calibrated low-exchange setting. Up to global phase,

$$\begin{aligned} U_{12,\mathrm{E}}(\theta_{12}) &= \exp(i\theta_{12} Z_{\mathrm{E}}/2) = R_{z,\mathrm{E}}(-\theta_{12}), \\ U_{23,\mathrm{E}}(\theta_{23}) &= \exp\left[-\frac{i\theta_{23}}{2}\left(\frac{\sqrt{3}}{2} X_{\mathrm{E}} + \frac{1}{2} Z_{\mathrm{E}}\right)\right]. \end{aligned} \tag{6.23}$$

The two pulses rotate about $-\hat{z}$ and $(\sqrt{3}/2, 0, 1/2)$. Because these axes are nonparallel, their generators and commutators span the single-qubit rotation algebra. Choosing pulse areas and alternating the two couplings therefore gives arbitrary encoded rotations [17].

A direct logical R_x gate. For constant simultaneous couplings, $J_{12} = J_{23}/2$ cancels the Z_{E} term. Both couplings must be calibrated with the two barrier controls applied together; the simultaneous voltages are adjusted until the measured couplings satisfy this ratio (Appendix P.1.3). Holding these couplings for time τ gives

$$H_{\mathrm{E,q}} = \frac{\sqrt{3}J_{23}}{4} X_{\mathrm{E}}, \qquad U_{\mathrm{E}}(\tau) = R_{x,\mathrm{E}}\left(\frac{\sqrt{3}J_{23}\tau}{2\hbar}\right). \tag{6.24}$$

For time-dependent pulses, the same gate follows if the ratio is maintained at every instant, including the ramps:

$$J_{12}(t) = \frac{1}{2} J_{23}(t) \quad \Longrightarrow \quad U_{\mathrm{E}}(\tau) = R_{x,\mathrm{E}}(\vartheta_x), \qquad \vartheta_x = \frac{\sqrt{3}}{2\hbar} \int_0^\tau J_{23}(t)\,dt. \tag{6.25}$$

A ratio imposed only during the constant hold does not make the entire gate a pure x rotation. The complete gate angle must therefore be calibrated for the chosen pair of waveforms. If the ratio cannot be maintained on their edges, use a sequence of the calibrated single-pair rotations instead (Appendix P.1.3).

Encoded CNOT: target gate and pulse requirements. To implement CNOT [Eq. (Pref.8)] between two triple-dot qubits $A = (1, 2, 3)$ and $B = (4, 5, 6)$, we need exchange between the blocks and a pulse sequence that returns both blocks to their computational spaces. The logical states in the target gate are $|0_{\mathrm{E}}\rangle, |1_{\mathrm{E}}\rangle$ in each block. For a linear six-dot chain, the available Hamiltonian is

$$H_6(t) = \sum_{i=1}^{5} J_{i,i+1}(t) \mathbf{s}_i \cdot \mathbf{s}_{i+1}. \tag{6.26}$$

The boundary term J_{34} couples the blocks, but it also creates a leakage problem absent from the one-block gate. In particular, $[\mathbf{s}_3 \cdot \mathbf{s}_4, \mathbf{s}_A^2] \neq 0$, where $\mathbf{s}_A = \mathbf{s}_1 + \mathbf{s}_2 + \mathbf{s}_3$. It can change either block's total spin or its z component despite conserving total six-spin angular momentum. Thus a physical $\sqrt{\text{SWAP}}$ on spins 3 and 4 is not an encoded gate by itself.

The pulse sequence must cancel any final leakage while retaining an entangling action on the code. To test both requirements, calculate the product of M pair pulses in the full six-spin space:

$$U_6 = U_{i_M j_M}(\theta_M) \cdots U_{i_1 j_1}(\theta_1), \qquad U_{ij}(\theta) = \exp[-i\theta\, \mathbf{s}_i \cdot \mathbf{s}_j]. \tag{6.27}$$

Let P_{AB} project onto the product of the two code spaces, each with $S = 1/2, m = +1/2$. An encoded CNOT, including local corrections, must satisfy

$$\begin{aligned} (I - P_{AB})U_6 P_{AB} &= 0, \\ P_{AB}U_6 P_{AB}|_{\mathrm{q},A\otimes\mathrm{q},B} &= e^{i\chi}\mathrm{CNOT}_{A\to B}, \end{aligned} \tag{6.28}$$

Here χ is a global phase. The first condition excludes final leakage; the second fixes the logical gate. Intermediate leakage is allowed, so exponentiating only the projected boundary exchange is insufficient.

Fong and Wandzura give exact six-spin exchange-only CNOT sequences [14]. Their pulse angles and ordering are not reproduced here. For a chosen sequence, each target exchange area must be converted to a barrier waveform and duration using the measured response of that pair, including the boundary pair 3–4. The final leakage and logical action of the complete sequence, including relative phases, must then be measured (Appendix P.2). Together with one-block rotations, such a gate gives universal encoded computation.

7 Coherence and loss of quantum information

Keeping an electron in its dot does not by itself preserve a spin superposition. Its relative phase depends on the energy splitting throughout the wait or gate. If that splitting fluctuates in a way we do not know, repetitions of the same control sequence need not produce the same phase. Other processes change spin populations or populate states outside the chosen qubit space. These mechanisms have different consequences for the operations derived above.

7.1 Phase accumulation, dephasing, and spin echo

Consider a single-spin qubit with the drive switched off. In a frame rotating at a fixed reference frequency, write the longitudinal Hamiltonian in our logical basis as

$$H_{\mathrm{wait}}(t) = -\frac{\hbar}{2}[\delta_0 + \xi(t)]Z, \qquad \phi(t) = \int_0^t \xi(t')\, dt'. \tag{7.1}$$

Here δ_0 is the mean spin frequency minus the reference frequency, and $\xi(t)$ is a classical fluctuation of that difference, with units of angular frequency. Magnetic fluctuations can change the local field; electric fluctuations can change the effective g factor or the position of the electron in a field gradient. The model retains only the resulting longitudinal frequency noise. It does not include spin flips or excitation of other electronic states.

For a given realization of the noise, the two logical amplitudes acquire opposite phases. Consequently the density-matrix element $\rho_{01} = \langle 0|\rho|1\rangle$ is multiplied by $e^{i\delta_0 t + i\phi(t)}$, while both populations stay unchanged. Averaging over repetitions gives

$$\overline{\rho}_{01}(t) = e^{i\delta_0 t} W(t) \rho_{01}(0), \qquad W(t) = \left\langle e^{i\phi(t)} \right\rangle. \tag{7.2}$$

The brackets denote an average over noise realizations sampled in the experiment. The measured mean detuning determines the correction to the reference frequency, or the phase tracked during the wait (Appendix U.1). A spread of unknown phases reduces $|W|$ and hence the coherence of the averaged state. This is dephasing, even though every realization of this particular model evolves unitarily.

The analysis rotation in Eq. (5.15) makes this reduction observable. Prepare $|+\rangle$, wait for time t, apply $R_y(-\pi/2)$, and measure in the logical basis. Since $\mathrm{Tr}(\rho X) = 2\,\mathrm{Re}\,\rho_{01}$, the zero probability is

$$p_0(t) = \frac{1 + \mathrm{Re}[e^{i\delta_0 t} W(t)]}{2}. \tag{7.3}$$

This is a Ramsey measurement. The known detuning produces fringes; phase fluctuations reduce their contrast. In a simple limit, $\xi(t) = \xi$ is constant during each wait but varies between repetitions with a zero-mean Gaussian distribution of variance σ_ω^2. The coherence and measured probability are then

$$\begin{aligned} W(t) &= e^{-\sigma_\omega^2 t^2/2}, \\ p_0(t) &= \frac{1 + e^{-(t/T_2^*)^2}\cos(\delta_0 t)}{2}, \qquad T_2^* = \frac{\sqrt{2}}{\sigma_\omega}. \end{aligned} \tag{7.4}$$

Thus T_2^* is the $1/e$ time of the free-evolution contrast in this model. Other noise statistics give other envelopes, and noise slower than the duration of the data acquisition may appear as a fixed offset instead of a sampled distribution.

A midpoint π rotation about X exchanges $|0\rangle$ and $|1\rangle$. Its frequency, phase, and pulse area use the calibrated transverse controls in Section 6.1. A component that accumulated one sign of phase in the first half therefore accumulates the opposite sign in the second half. In a frame that tracks this known pulse, the random phase becomes

$$\phi_{\mathrm{e}}(t) = \int_0^{t/2} \xi(t')\,dt' - \int_{t/2}^{t} \xi(t')\,dt', \qquad W_{\mathrm{e}}(t) = \left\langle e^{i\phi_{\mathrm{e}}(t)} \right\rangle. \tag{7.5}$$

A constant offset contributes equally to the two integrals and cancels, including δ_0. Slowly varying noise is partly cancelled; noise that changes substantially between the two halves need not be. This spin-echo sequence can therefore preserve contrast longer than free evolution. Its measured decay time depends on the pulse sequence and the noise, so it must be distinguished from T_2^*. Appendix U.1.1 develops the pulse phases and measured probabilities for Ramsey and echo experiments.

7.2 Relaxation and leakage

Longitudinal phase noise is only one way of losing quantum information. Coupling to the environment can transfer energy and change the populations of $|0\rangle$ and $|1\rangle$. The relaxation law in Eq. (5.1) describes their approach to thermal equilibrium on the timescale T_1. The same process that helps prepare the ground spin state destroys an arbitrary stored superposition. Exchanging the two states halfway through a wait does not reverse an uncontrolled energy transfer to the environment, so spin echo does not in general remove this relaxation.

A separate possibility is population outside the computational space. For the projector P_{q} defined in the introduction to Section 6, the population outside that space is

$$L(t) = 1 - \mathrm{Tr}[P_{\mathrm{q}}\rho(t)], \qquad \mathrm{Tr}\,\rho(t) = 1. \tag{7.6}$$

Examples include unwanted orbital or valley excitation, or occupation of the $S = 3/2$ sector of a three-spin exchange-only qubit. Pure dephasing and relaxation between the two retained

logical states both have $L = 0$. Conversely, a leaked state may produce the same charge-readout outcome as a logical state, as the pair-singlet test above illustrates. Leakage must therefore be specified relative to the chosen computational space and cannot always be inferred from a single readout population.

These distinctions also explain why one coherence time is not a complete measure of gate quality. A gate can have a reproducible angle error without appreciable idle dephasing, or drive leakage even when the idle spin is long lived. The relevant error depends on the state, control pulse, and observation being considered. For exchange control, Eq. (4.12) gives one useful consequence: at $\varepsilon = 0$, the first derivative $\partial J/\partial\varepsilon$ vanishes when t_c, U, V are fixed. Small detuning fluctuations then enter only at higher order, while barrier fluctuations still change t_c and hence the rotation angle. Locate the low-sensitivity setting from the measured exchange response along the compensated detuning-voltage path, then tune the barrier to obtain the required exchange strength (Appendix U). Avoiding leakage additionally requires limiting unwanted transitions across charge, orbital, valley, or encoding gaps during the pulse. Appendix U quantifies the angle sensitivity in Eq. (U.7) and the unwanted transition amplitudes in Eq. (U.12).

8 Selecting and connecting qubits in an array

The gates above specify an operation on one spin or a chosen group of spins. In an array, the applied electric and magnetic fields can also reach other dots, and neighboring exchange couplings may remain finite. *Addressing* means making the intended qubit or pair respond selectively to a control. Frequency separation, spatially concentrated driving, and control of individual exchange links provide complementary ways to do this.

8.1 Selecting a spin by frequency and drive amplitude

For approximately parallel local quantization axes, spin i has resonance frequency $\omega_i = g_i\mu_B B_i/\hbar$, with B_i the local static field magnitude and g_i its effective g factor. A spatially varying static field can separate these frequencies. A gate voltage can also shift a resonance by changing g_i or the electron's position in the field. Electrical tuning of the g factor has been demonstrated in silicon quantum dots [18]. One can set the drive frequency to the target resonance, or tune the target into resonance with a chosen drive frequency.

Suppose a constant pulse is resonant with spin i and gives it Rabi frequency $\Omega_i > 0$. Its π-pulse time is $\tau_\pi = \pi/\Omega_i$. If the same pulse drives another spin j with Rabi frequency Ω_j, that spin has detuning $\Delta_j = \omega_j - \omega_i$. Neglecting exchange during this single-spin pulse, Eq. (4.7) gives

$$P_{j,\mathrm{flip}}(\tau_\pi) = \frac{\Omega_j^2}{\Delta_j^2 + \Omega_j^2} \sin^2\left(\frac{\tau_\pi}{2}\sqrt{\Delta_j^2 + \Omega_j^2}\right) \le \frac{\Omega_j^2}{\Delta_j^2 + \Omega_j^2}. \tag{8.1}$$

Thus $|\Delta_j| \gg |\Omega_j|$ suppresses an unwanted flip throughout the pulse. The resonance separation and the drive strength induced on the spectator must be measured to choose a drive frequency and amplitude that satisfy this condition (Appendix V.2.2). If both spins are resonant and appreciably driven, frequency selection alone does not isolate the target. For a fixed frequency separation, increasing the shared drive amplitude to shorten a gate can reduce this selectivity.

Spatial control acts on the other quantity in this condition: it reduces Ω_j. With electric-dipole spin resonance, a local gate can move the target electron more strongly than distant electrons. The transverse magnetic-field gradient converts that motion into a spin drive, as derived in Eq. (M.6). The longitudinal field variation that separates resonance frequencies and the transverse gradient that enables this drive have different roles, even when one micromagnet

provides both [19]. Electrical fields extend beyond one dot, so locality must be assessed through the resulting couplings rather than the electrode's name.

Suppressing spectator flips does not remove every action on its state. A far-detuned drive also shifts the spectator's precession frequency. Relative to an otherwise identical wait, it produces an additional rotation $R_{z,j}(-\phi_j)$ with

$$\phi_j \simeq \int_0^\tau \frac{\Omega_j(t)^2}{2\Delta_j}\, dt. \tag{8.2}$$

This expression assumes fixed detuning, $|\Omega_j/\Delta_j| \ll 1$, and switching slow enough to follow the dressed eigenstates. A spectator superposition can therefore acquire a phase even when few flips occur. Measure the additional spectator phase produced by the target pulse and include its reproducible part in the later control reference (Appendix V.2.2).

8.2 Selecting exchange links and parallel gates

To select an exchange gate on neighboring dots, change their tunnel barrier and, where needed, compensate the accompanying shifts of the dot energies with other gate voltages. This changes the virtual tunneling that produces J. In the four-dot chain of Fig. 8.1, a gate on spins 2 and 3 requires a useful J_{23} and sufficiently small unwanted actions from J_{12} and J_{34} during the same interval. A residual coupling of scale J_{off} acting for time τ has exchange area of scale $J_{\text{off}}\tau/\hbar$; the required suppression therefore depends on gate duration and error tolerance. Measure the selected and residual exchange couplings at the voltages used for the complete layer. Adjust the barrier and compensating plunger voltages so that the intended exchange areas reach their targets while unwanted areas remain within the error allowance (Appendix V.2.1). Barrier control and compensation have been used to select interactions in a silicon spin array [19].

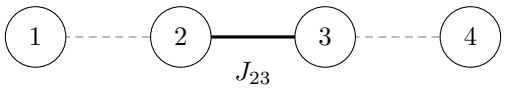


Figure 8.1: A four-dot exchange chain. The thick J_{23} link is selected for a gate on spins 2 and 3; the dashed links are available but intended to have small exchange during this pulse. Dots 1 and 3 have no direct exchange edge.

The same chain shows when two pair gates can be independent during one interval. Selecting $(1, 2)$ and $(3, 4)$ assigns each spin to only one pair. If the coupling between the pairs is negligible, their evolution factorizes:

$$U_{12,34} = U_{12} \otimes U_{34}. \tag{8.3}$$

Selecting $(1, 2)$ and $(2, 3)$ instead makes spin 2 participate in both interactions, generating a three-spin evolution that must be designed as such. Exchange gates on edges with no shared spin are therefore candidates for one parallel layer, provided unselected couplings and spectator responses are also controlled. Appendix V.2.1 develops the resulting scheduling conditions.

The physical exchange graph describes couplings between spins. Its edges have different logical meanings for different encodings: an internal edge of a three-spin block rotates one exchange-only qubit, whereas an edge between blocks participates in a two-qubit sequence. For two singlet–triplet blocks, Eq. (6.16) contributes a $Z_A Z_B$ interaction whenever κ is nonzero, including during local rotations. The same voltages can change both the local Hamiltonians and κ. Independent logical gates therefore require the unwanted interaction to be suppressed or refocused; otherwise the full coupled evolution must be used. Appendix O.2.4 works through this control constraint.

8.3 Acting on qubits without a direct link

The four-dot graph has no exchange edge between sites 1 and 3. One way to act on the states at those sites is to use a SWAP to bring them onto an available edge. If S_{23} exchanges the states at sites 2 and 3 and U_{12} is the desired operation available on edge $(1, 2)$, then

$$U_{13} = S_{23}U_{12}S_{23}. \tag{8.4}$$

The rightmost SWAP brings the original state of site 3 to site 2, and the leftmost restores the placement after the gate. This identity holds for unknown or entangled inputs, because SWAP permutes states without measuring them. Each dot remains occupied. For an encoded qubit, swapping a single constituent spin generally does not exchange the complete logical state; its logical permutation must be constructed separately. Appendix W.1 develops SWAP routing and the operations required along a longer path.

Shuttling provides another route by moving the electron itself between dots. The transfer must avoid unwanted charge excitation and spin flips, and the phase accumulated in the changing local field must remain known or reproducible. The measured interdot coupling and voltage-to-detuning response constrain the allowed plunger-ramp speed, and the measured spin phase sets the correction after transfer (Appendix W.2). A shared quantum mode can instead mediate an interaction while the electrons remain in place; the useful interaction must compete with spin decoherence and loss from the mode. The measured spin–mode couplings and loss rates determine the usable detuning and interaction time (Appendix W.3).

The controls and detectors must also distinguish the intended objects in an array. The measured voltage responses of the target and affected neighbours determine the compensating electrode changes needed to preserve their required energies and couplings (Appendix F.6). When several spins are read out, their converted charge records must be distinguishable through the available sensors or through measurements separated in time. Mean charge signals alone do not specify joint outcome probabilities; correlations require retaining the associated outcomes from each repetition. Appendix S.7 develops the conditions for separating those signals.

A Calculating the Confining Potential

Section 1 describes how the device confines electrons and uses V_{conf} in the orbital Hamiltonian. This appendix calculates that potential from the device geometry, material properties, gate voltages, and background charge. The gate and reservoir surfaces in Fig. 1.1 set electrostatic boundary conditions; solving for the electric potential and adding the material band edge gives the electron's potential energy.

Our target is the spatial function $V_{\text{conf}}(\mathbf{r})$ in the electron Hamiltonian. We first specify where the electric potential $\Phi_0(\mathbf{r})$ is to be found and the surfaces on which its boundary and interface conditions apply. Figure A.1 gives these regions and surfaces for the reservoir–barrier–dot arrangement. The x, y coordinates lie in the material plane, and z is perpendicular to the layers.

A.1 Define the regions and boundaries

We first use a fixed-background electrostatic model. The gates, electrical contact, and populated reservoir are treated as ideal conductors. For the reservoir, this approximates strong electrostatic screening. Each conductor has a constant potential fixed by the external circuit.

The unknown potential is calculated in three material regions: the silicon outside the idealized reservoir, $\mathcal{D}_{\text{Si}}$; the oxide, $\mathcal{D}_{\text{ox}}$; and the surrounding air, $\mathcal{D}_{\text{air}}$. These regions, joined at their material interfaces, form the solution region $\mathcal{D}$. The dot lies within $\mathcal{D}_{\text{Si}}$. Its few electrons will be included through their explicit interaction in Section E; here we calculate their external potential.

The boundary of $\mathcal{D}$ has two parts. The conductor surfaces form Γ_{D}: $\mathcal{S}_g$ is the surface of gate g, and $\mathcal{S}_{\text{R}}$ is the surface of the connected contact–reservoir region. The outer faces that limit the simulated volume form Γ_{N}. Thus $\partial\mathcal{D} = \Gamma_{\text{D}} \cup \Gamma_{\text{N}}$. Within $\mathcal{D}$, Σ denotes the silicon–oxide and oxide–air interfaces. Layer thicknesses and conductor sizes and positions specify the three-dimensional geometry; the figure shows an x–z cross-section.

A.2 Specify material data and boundary conditions

Electric fields displace bound charges in a material. Each material's permittivity ϵ_{d} describes the resulting polarization. Its value in each part of the device defines the function $\epsilon_{\text{d}}(\mathbf{r})$ over $\mathcal{D}$. The material conduction-band edge $E_{\text{c}}^0(\mathbf{r})$ is another input, incorporating band offsets and strain with energies measured from a common reference. Band offsets are energy steps at material boundaries. Strain is an elastic deformation of the crystal that changes its band energies; its effect is included in the supplied material profile.

The remaining source is a prescribed background charge density $\rho_{\text{bg}}(\mathbf{r})$, measured in charge per unit volume. A charge-free baseline sets $\rho_{\text{bg}} = 0$ throughout $\mathcal{D}$; a specified impurity-charge profile can be added here. The reservoir's screening enters through its conducting boundary. Its surface charge will adjust to maintain the prescribed potential.

On the gate surfaces $\mathcal{S}_g \subset \Gamma_{\text{D}}$, impose

$$\Phi_0|_{\mathcal{S}_g} = V_g + \Phi_g^{\text{off}}, \tag{A.1}$$

where V_g is the applied voltage and Φ_g^{off} is a specified built-in offset from the material and voltage-reference conventions. On $\mathcal{S}_{\text{R}}$, the contact and reservoir have the prescribed potential Φ_{R}. This is an electrostatic boundary value; the reservoir electrochemical potential μ_{R} sets the occupation of electron states.

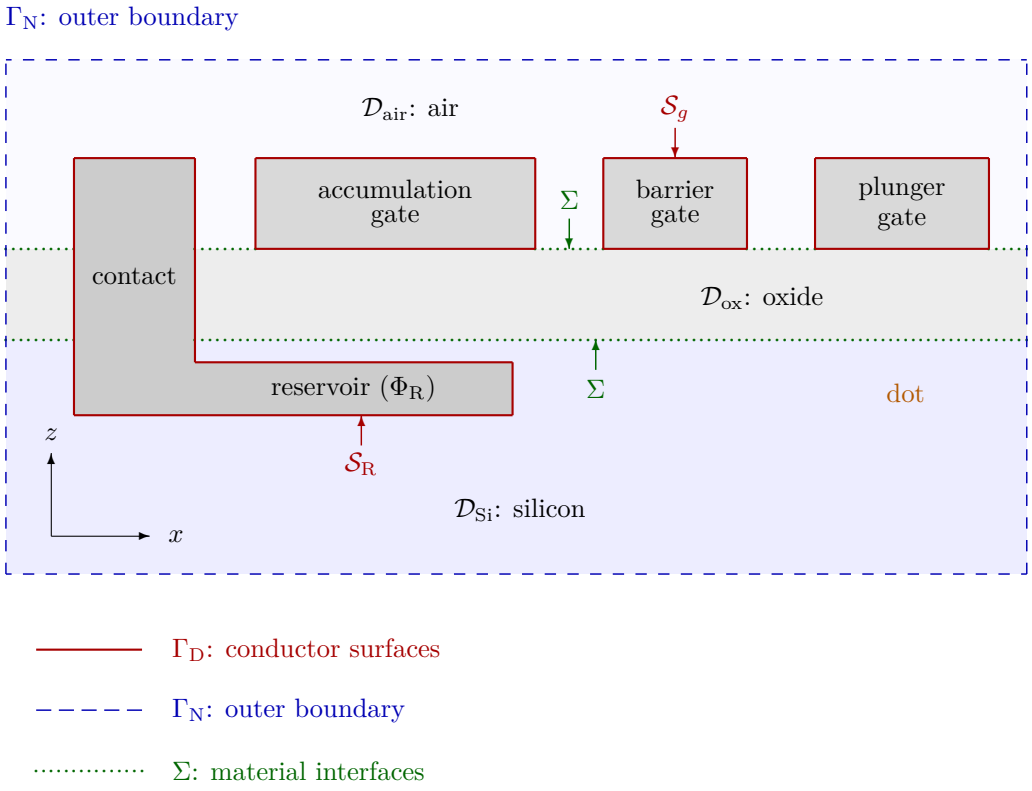


Figure A.1: Regions and boundaries of the electrostatic model. The labeled silicon, oxide, and air regions form $\mathcal{D}$. Solid red lines mark conductor surfaces Γ_D; dashed blue lines mark the outer boundary Γ_N; dotted green lines mark material interfaces Σ. The y direction is perpendicular to the drawing; the undrawn front and back outer faces also belong to Γ_N. The schematic is not to scale.

These prescribed conductor values define a function Φ_D on Γ_D. All potentials refer to the same reference conductor.

On the outer faces Γ_N, take zero normal electric displacement: $\mathbf{n} \cdot \epsilon_d \nabla \Phi_0 = 0$, where $\mathbf{n}$ is the outward normal. The outer faces are placed far enough away that moving them farther changes the potential near the dot negligibly.

Across the material interfaces Σ, the solutions in neighboring materials must join. We assume these interfaces carry no additional charge, such as trapped charge; material polarization is already included in ϵ_d. The potential and normal electric displacement are then continuous across each interface.

A.3 Solve the resulting boundary-value problem

The electric field is $\mathbf{E} = -\nabla \Phi_0$. Substitution into Gauss's law gives the problem for the unknown Φ_0:

$$\begin{aligned} -\nabla \cdot [\epsilon_d \nabla \Phi_0] &= \rho_{bg} && \text{in } \mathcal{D} \setminus \Sigma, \\ \Phi_0 &= \Phi_D && \text{on } \Gamma_D, \\ \mathbf{n} \cdot \epsilon_d \nabla \Phi_0 &= 0 && \text{on } \Gamma_N, \\ \Phi_{0,1} &= \Phi_{0,2} && \text{on } \Sigma, \\ \epsilon_{d,1}\, \partial_n \Phi_{0,1} &= \epsilon_{d,2}\, \partial_n \Phi_{0,2} && \text{on } \Sigma. \end{aligned} \tag{A.2}$$

The subscripts $1, 2$ identify the two sides of each material interface; ∂_n uses the same normal direction on both sides. The geometry fixes the domain and interfaces, material data fix the coefficients, and the preceding steps fix the source and every boundary value. With positive permittivity and a nonempty fixed-potential boundary on each connected component, this is a closed linear problem [20].

A.4 Convert the electric potential to electron energy

The resulting electron potential energy is

$$V_{\mathrm{conf}}(\mathbf{r}) = E_{\mathrm{c}}^{0}(\mathbf{r}) - e\Phi_0(\mathbf{r}). \tag{A.3}$$

Appendix B.8 inserts this function into the effective-mass Schrödinger equation to obtain dot orbitals and energies. For several retained valleys, indexed by ν, use their respective material profiles: $V_{\mathrm{conf},\nu} = E_{\mathrm{c},\nu}^{0} - e\Phi_0$.

This construction uses a fixed background charge and ideal reservoir screening. To resolve the reservoir's finite charge response, include its interior in $\mathcal{D}$ and supply a carrier-density model with its material parameters, temperature, and electrochemical potential. The charge then depends on Φ_0, and the source and potential are solved self-consistently [20]. The corresponding treatment of quantum-confined electrons is a many-electron mean-field approximation, developed in Section E.1.

B Orbital States in the Effective-Mass Approximation

Section 1 uses Eq. (1.2) to describe a localized orbital. This appendix derives that equation from the crystal problem and then shows how to obtain bound orbitals from a specified confining potential.

Our goal is to derive a Schrödinger equation for a confined conduction electron without resolving its wave function inside every crystal cell. The lattice will enter through an effective mass, while the dot potential will determine a slowly varying envelope. The main issue is how to derive that envelope equation when the envelope itself is unknown.

We first expand the microscopic state in a complete basis with ordinary numerical coefficients. Projection gives equations for those coefficients; a Fourier transform then gives coupled equations for the envelopes. Finally, we express the small other-band components in terms of the conduction-band envelope and substitute them back. This is a fixed-reference Luttinger–Kohn construction [21]. We work in one dimension to expose each step and give the three-dimensional result at the end.

B.1 The crystal reference and the confined state

Consider the spinless, one-electron Hamiltonian

$$H_{\mathrm{free}} = \frac{p^2}{2m_e} + V_{\mathrm{lat}}(x), \qquad p = -i\hbar\partial_x, \qquad V_{\mathrm{lat}}(x+a) = V_{\mathrm{lat}}(x), \tag{B.1}$$

where a is the lattice period, m_e is the vacuum electron mass, and V_{lat} is a real local potential energy. Here "free" means without dot confinement, not without the crystal lattice. We take this crystal model and its Bloch eigenstates as given:

$$H_{\mathrm{free}}\chi_{n,k} = E_n(k)\chi_{n,k}, \qquad \chi_{n,k}(x) = e^{ikx}u_{n,k}(x), \qquad u_{n,k}(x+a) = u_{n,k}(x). \tag{B.2}$$

Throughout this appendix, n and m label crystal bands, not the discrete orbitals of a quantum dot.

Choose a nondegenerate conduction-band minimum at k_0, with band index c. We abbreviate $E_n = E_n(k_0)$ and $\chi_n = \chi_{n,k_0}$. Adding a dot potential gives the eigenvalue problem

$$H = H_{\text{free}} + V_{\text{conf}}(x), \qquad H\psi = E\psi. \tag{B.3}$$

If the potential supports a bound state, ψ is localized, whereas the reference state χ_{c} is extended. The eventual approximation concerns small changes in band character near k_0; it does not require the localized state to be close to the extended state in norm.

Locally, wherever $\chi_{\text{c}} \neq 0$, one can write $\psi = \chi_{\text{c}} F$ exactly. Product differentiation gives

$$\left[-\frac{\hbar^2}{2m_e}\partial_x^2 - \frac{\hbar^2}{m_e}\frac{\chi_{\text{c}}'}{\chi_{\text{c}}}\partial_x + V_{\text{conf}}\right] F = (E - E_{\text{c}})F. \tag{B.4}$$

Here a prime denotes an x derivative. This is not yet an effective-mass equation: F can contain microscopic variations, and the quotient can be singular at nodes of χ_{c}. Dropping the coefficient involving χ_{c}' would leave m_e, not produce a material-dependent mass. We therefore resolve the microscopic components before making an approximation.

B.2 A complete basis at a fixed reference wave vector

Completeness within one cell. Substituting the Bloch form into Eq. (B.2) gives an eigenproblem for the periodic part:

$$H(k)u_{n,k} = E_n(k)u_{n,k}, \qquad H(k) = \frac{(p + \hbar k)^2}{2m_e} + V_{\text{lat}}. \tag{B.5}$$

At fixed k_0, this is one operator on $[0, a]$ with periodic boundary conditions on both the function and its derivative. Fixing k_0 fixes an operator parameter; it does not select one of its eigenstates. The band index enumerates all of them.

For a sufficiently regular real periodic potential, this self-adjoint cell operator has a discrete spectrum and a complete orthonormal eigenbasis. The mathematical reason is that it has compact resolvent: for a constant C chosen to make $H(k_0) + C$ positive, the inverse $(H(k_0) + C)^{-1}$ is compact. In the zero-potential Fourier basis its eigenvalues tend to zero as the Fourier wave vector grows; a bounded periodic potential preserves this property. The spectral theorem then supplies the complete basis, rather than self-adjointness alone. We use the cell-average normalization

$$\langle u_{n,k_0}, u_{m,k_0}\rangle_{\text{cell}} \equiv \frac{1}{a}\int_0^a u_{n,k_0}^*(x)u_{m,k_0}(x)\,dx = \delta_{nm}. \tag{B.6}$$

For example, when $V_{\text{lat}} = 0$, the periodic eigenfunctions can be chosen as e^{iGx}, with $G = 2\pi\ell/a$ and integer ℓ. These are precisely the Fourier basis of a cell. The periodic potential mixes these modes; its complete eigenbasis still spans the same function space.

The periodic parts at any other wave vector belong to this same space. Consequently,

$$u_{j,k_0+q}(x) = \sum_n b_{nj}(q)u_{n,k_0}(x), \qquad b_{nj}(q) = \langle u_{n,k_0}, u_{j,k_0+q}\rangle_{\text{cell}}. \tag{B.7}$$

Here j is another band index. With all bands retained, this is an exact change between two orthonormal cell bases. It does not claim that u_{j,k_0+q} is an eigenfunction of $H(k_0)$.

Completeness across the crystal. The fixed-k_0 states χ_n alone do not span arbitrary wave functions over the entire crystal. To include changes from cell to cell, introduce

$$B_{nq}(x) = \frac{1}{\sqrt{2\pi}} e^{iqx} \chi_n(x) = \frac{1}{\sqrt{2\pi}} e^{i(k_0+q)x} u_{n,k_0}(x), \qquad q \in \mathcal{Q} = [-\pi/a, \pi/a). \tag{B.8}$$

The interval $k_0 + \mathcal{Q}$ contains one representative of every crystal quasi-momentum modulo $2\pi/a$. Each B_{nq} has quasi-momentum $k_0 + q$, but is generally not an energy eigenstate there. Equation (B.7) expresses every true Bloch state as a combination of these functions at the same q, establishing completeness.

To verify orthogonality, expand the periodic product as

$$u^*_{n,k_0} u_{m,k_0} = \sum_G S_{nm}(G) e^{iGx}, \qquad S_{nm}(G) = \frac{1}{a} \int_0^a u^*_{n,k_0} u_{m,k_0} e^{-iGx}\, dx. \tag{B.9}$$

Integration over the whole line then gives

$$\int_{\mathbb{R}} B^*_{nq}(x) B_{mq'}(x)\, dx = \sum_G S_{nm}(G) \delta(q' - q + G) = \delta_{nm} \delta(q - q'). \tag{B.10}$$

The last equality follows because two points of the half-open interval $\mathcal{Q}$ cannot differ by a nonzero reciprocal lattice vector, and $S_{nm}(0) = \delta_{nm}$. The continuous basis is delta-normalized; a finite periodic crystal gives the corresponding discrete sums.

B.3 Unknown envelopes and an exact projection

We can now expand the unknown state without treating a position-dependent function as a numerical coefficient:

$$\psi(x) = \sum_n \int_{\mathcal{Q}} dq\, A_n(q) B_{nq}(x), \qquad A_n(q) = \int_{\mathbb{R}} B^*_{nq}(x) \psi(x)\, dx. \tag{B.11}$$

For fixed labels n, q, $A_n(q)$ is an unknown complex number independent of x. Define its Fourier transform by

$$f_n(x) = \frac{1}{\sqrt{2\pi}} \int_{\mathcal{Q}} dq\, A_n(q) e^{iqx}, \qquad \phi(x) \equiv f_{\mathrm{c}}(x). \tag{B.12}$$

Rearranging the exact expansion gives

$$\psi = \sum_n \chi_n f_n = \chi_{\mathrm{c}} \phi + \sum_{n \neq \mathrm{c}} \chi_n f_n. \tag{B.13}$$

Both ϕ and the other f_n vary with position and are unknown. Their Fourier restriction to the same interval $\mathcal{Q}$ is essential: arbitrary functions multiplying the χ_n would give a nonunique decomposition. Given ψ, the complete basis instead determines every A_n, and hence every f_n, uniquely. No assumption of slow variation has yet been made; that will require concentration near $q = 0$.

Projection must use Eq. (B.11) and whole-crystal inner products. At a fixed x, the values $\chi_n(x)$ are just complex numbers; one cannot compare their function-valued coefficients as independent coordinates. Nor can $f_m(x)$ be pulled out of the cell integral $\int_0^a \chi^*_n(x) \chi_m(x) f_m(x)\, dx$.

The crystal part of the projected equation. Define the momentum matrix elements at the reference wave vector:

$$p_{nm} = \frac{1}{a}\int_0^a u^*_{n,k_0}(p+\hbar k_0)u_{m,k_0}\,dx = \frac{1}{a}\int_0^a \chi^*_n p\chi_m\,dx, \qquad p_{mn} = p^*_{nm}. \tag{B.14}$$

These are numbers, with units of momentum. Acting on a basis function and differentiating the factor e^{iqx} gives

$$H_{\text{free}}B_{mq} = \left(E_m + \frac{\hbar^2 q^2}{2m_e}\right)B_{mq} + \frac{\hbar q}{m_e}\sum_n p_{nm}B_{nq}. \tag{B.15}$$

The sum arises by expanding $(p+\hbar k_0)u_{m,k_0}$ in the complete cell basis. There are no unknown envelopes in that cell expansion.

The confining potential and the full equation. Let

$$V_{nm}(q,q') = \int_{\mathbb{R}} B^*_{nq}(x)V_{\text{conf}}(x)B_{mq'}(x)\,dx. \tag{B.16}$$

Multiplying $H\psi = E\psi$ by B^*_{nq} and integrating now produces the exact coupled equations

$$\begin{aligned} 0 = &\left(E_n + \frac{\hbar^2 q^2}{2m_e} - E\right)A_n(q) + \frac{\hbar q}{m_e}\sum_m p_{nm}A_m(q) \\ &+ \sum_m \int_{\mathcal{Q}} dq'\,V_{nm}(q,q')A_m(q'). \end{aligned} \tag{B.17}$$

This is a representation of the microscopic problem, not yet a reduced model. In particular, both changes of quasi-momentum and mixing of the reference-band components are present.

B.4 The smooth-potential approximation

We now select low-energy states concentrated near $q = 0$ and a potential varying on lengths much greater than a. The reason a smooth potential becomes approximately diagonal in the reference-band index can be seen from its Fourier transform. With the convention

$$\widetilde{V}(s) = \frac{1}{2\pi}\int_{\mathbb{R}} dx\,e^{-isx}V_{\text{conf}}(x), \tag{B.18}$$

Eq. (B.9) gives the exact kernel

$$V_{nm}(q,q') = \sum_G S_{nm}(G)\widetilde{V}(q-q'-G). \tag{B.19}$$

For small q, q', every $G \neq 0$ samples a Fourier component at a lattice-scale wave vector. Smooth confinement has little weight there. Retaining $G = 0$ gives

$$V_{nm}(q,q') \simeq \delta_{nm}\widetilde{V}(q-q'). \tag{B.20}$$

Thus it is the potential's lack of large-wave-vector components, not the constancy of an unknown envelope, that justifies this step.

To obtain a local differential equation, we also neglect weight near the ends of $\mathcal{Q}$ and extend the low-wave-vector Fourier integrals to the real line. The convolution theorem then turns the potential term into $V_{\text{conf}}f_n$. At finite $\mathcal{Q}$, it would instead be the Fourier projection of that

product back into $\mathcal{Q}$; replacing it by the local product is valid when the discarded weight is negligible. This extension is an approximation, not a second exact basis expansion.

The remaining Fourier identities are

$$\frac{1}{\sqrt{2\pi}}\int dq\, qA_n(q)e^{iqx} = -if_n'(x), \qquad \frac{1}{\sqrt{2\pi}}\int dq\, q^2A_n(q)e^{iqx} = -f_n''(x).$$

Using them in Eq. (B.17) gives the leading long-wavelength envelope system

$$(\Delta_n + V_{\rm conf} - \epsilon)f_n - \frac{\hbar^2}{2m_e}f_n'' - \frac{i\hbar}{m_e}\sum_m p_{nm}f_m' = 0, \quad \Delta_n = E_n - E_{\rm c}, \quad \epsilon = E - E_{\rm c}. \tag{B.21}$$

Here $\Delta_{\rm c} = 0$. All envelopes are still unknown functions. Equation (B.21) was obtained by transforming projected equations for numerical amplitudes, not by pulling functions out of a cell integral.

B.5 Eliminating the other-band envelopes

The $n = {\rm c}$ component describes the desired low-energy state. The other components have nonzero reference energy separations Δ_n. When those separations dominate the envelope energy and derivative couplings, their amplitudes can be expressed perturbatively in terms of $\phi = f_{\rm c}$. They need not be known beforehand.

Let ℓ be a typical envelope length and $\Delta_{\rm min} = \min_{n\neq {\rm c}}|\Delta_n| > 0$. We require $\ell \gg a$, envelope kinetic and local energy scales small compared with $\Delta_{\rm min}$, and weak derivative coupling to the excluded bands. In particular,

$$\frac{\hbar^2}{2m_e\ell^2},\ |V_{\rm conf} - \epsilon| \ll \Delta_{\rm min}, \qquad \frac{\hbar|p_{n{\rm c}}|}{m_e\ell|\Delta_n|} \ll 1 \quad (n \neq {\rm c}), \tag{B.22}$$

with the analogous smallness of derivative couplings within the excluded subspace. Local energy conditions concern the region appreciably occupied by the state. A nearby degenerate band must be retained, not eliminated with a small denominator. The Δ_n can be positive or negative: "other bands" includes bands below the chosen conduction band. We retain terms through second order in slow derivatives and to leading order in the ratio of envelope energy to band separation.

At the minimum, the conduction-band diagonal momentum vanishes. Differentiating the cell eigenvalue equation and using normalization gives the Hellmann–Feynman identity

$$\left.\frac{dE_{\rm c}(k)}{dk}\right|_{k_0} = \left\langle u_{{\rm c},k_0}, \left.\frac{\partial H(k)}{\partial k}\right|_{k_0} u_{{\rm c},k_0}\right\rangle_{\rm cell} = \frac{\hbar}{m_e}p_{\rm cc} = 0. \tag{B.23}$$

Consequently, the conduction component of Eq. (B.21) is

$$-\frac{\hbar^2}{2m_e}\phi'' + V_{\rm conf}\phi - \frac{i\hbar}{m_e}\sum_{n\neq {\rm c}} p_{{\rm c}n}f_n' = \epsilon\phi. \tag{B.24}$$

For $n \neq {\rm c}$, the leading source in its equation is the derivative of ϕ. To first order in slow spatial variation, balance that source against the band separation:

$$\Delta_n f_n - \frac{i\hbar}{m_e}p_{n{\rm c}}\phi' \simeq 0, \qquad f_n \simeq \frac{i\hbar p_{n{\rm c}}}{m_e\Delta_n}\phi'. \tag{B.25}$$

Derivatives of these already small components give higher orders in their own equations; their local energy corrections are suppressed by Eq. (B.22). The displayed term is enough to

determine the leading second-derivative correction in the conduction equation. Although f_n is small, setting it to zero before this substitution would discard the mass correction.

The crystal parameters are position independent here. Substituting f_n' from Eq. (B.25) into Eq. (B.24), with $p_{cn} = p_{nc}^*$, yields

$$\left[-\frac{\hbar^2}{2m_e} + \frac{\hbar^2}{m_e^2}\sum_{n\neq \mathrm{c}} \frac{|p_{\mathrm{nc}}|^2}{\Delta_n}\right]\phi'' + V_{\mathrm{conf}}\phi = \epsilon\phi. \tag{B.26}$$

Define the effective mass by collecting the coefficient of $-\hbar^2\phi''/2$:

$$\boxed{\frac{1}{m^*} = \frac{1}{m_e} + \frac{2}{m_e^2}\sum_{n\neq \mathrm{c}} \frac{|p_{\mathrm{nc}}|^2}{E_{\mathrm{c}} - E_n}}. \tag{B.27}$$

The desired single-envelope equation is therefore

$$\boxed{\left[-\frac{\hbar^2}{2m^*}\partial_x^2 + V_{\mathrm{conf}}(x)\right]\phi(x) = \epsilon\phi(x)}. \tag{B.28}$$

The other-band amplitudes describe components of the same stationary state in the reference basis, not real transitions that populate separate excited states in time. Their coupling back to the conduction component changes its kinetic energy.

B.6 Band curvature and reconstruction of the microscopic state

Why the mass equals the band curvature. Set $V_{\mathrm{conf}} = 0$ and examine one small wave-vector offset q. In the fixed cell basis,

$$H(k_0 + q) = H(k_0) + \frac{\hbar q}{m_e}(p + \hbar k_0) + \frac{\hbar^2 q^2}{2m_e}. \tag{B.29}$$

The first-order admixture of band $n \neq \mathrm{c}$ is $\hbar q p_{nc}/[m_e(E_{\mathrm{c}} - E_n)]$. Feeding it back into the conduction component gives

$$E_{\mathrm{c}}(k_0 + q) - E_{\mathrm{c}} = \frac{\hbar^2 q^2}{2m_e} + \frac{\hbar^2 q^2}{m_e^2}\sum_{n\neq \mathrm{c}} \frac{|p_{\mathrm{nc}}|^2}{E_{\mathrm{c}} - E_n} + O(q^3). \tag{B.30}$$

There is no linear term because $p_{\mathrm{cc}} = 0$. Comparison with Eq. (B.27) gives

$$\frac{1}{m^*} = \frac{1}{\hbar^2}\left.\frac{d^2E_{\mathrm{c}}(k)}{dk^2}\right|_{k_0}. \tag{B.31}$$

Thus band-curvature matching and elimination of the reference-band components give the same mass. Matching identifies the answer from known dispersion; the projection derivation explains which microscopic amplitudes generate it.

What the envelope approximates. After solving Eq. (B.28) with the appropriate bound-state boundary conditions, reconstruct

$$\psi(x) \simeq \chi_{\mathrm{c}}(x)\phi(x) + \sum_{n\neq \mathrm{c}} \chi_n(x)\frac{i\hbar p_{\mathrm{nc}}}{m_e\Delta_n}\phi'(x). \tag{B.32}$$

The leading approximation is $\psi \simeq \chi_c \phi$; the displayed derivative term is its first microscopic correction. Where division by χ_c is permitted,

$$F = \frac{\psi}{\chi_c} \simeq \phi + \sum_{n \neq c} \frac{\chi_n}{\chi_c} \frac{i\hbar p_{nc}}{m_e \Delta_n} \phi'. \tag{B.33}$$

This gives the relation between the exact multiplicative correction F and the leading smooth envelope ϕ. They are not exactly the same unknown. Equation (B.32) also remains meaningful at nodes where the quotient does not.

The complete-basis normalization satisfies $\|\psi\|^2 = \sum_n \int_{\mathbb{R}} |f_n(x)|^2 \, dx$. For weak admixture, $\int |\phi|^2 dx = 1$ is the leading normalization; retaining the other components changes it at second order in their relative amplitudes.

B.7 Three dimensions and the range of validity

In three dimensions, the linear coupling is $\hbar \mathbf{q} \cdot \mathbf{p}_{nm}/m_e$. Define its Cartesian components by

$$p^{\alpha}_{nm} = \langle u_{n,\mathbf{k}_0}, (-i\hbar \partial_{r_\alpha} + \hbar k_{0,\alpha}) u_{m,\mathbf{k}_0} \rangle_{\text{cell}}, \qquad \alpha = x, y, z,$$

where the cell inner product is now the integral divided by the cell volume. The same elimination gives a real symmetric inverse-mass tensor:

$$(\mathbf{M}^{-1})_{\alpha\beta} = \frac{\delta_{\alpha\beta}}{m_e} + \frac{2}{m_e^2} \operatorname{Re} \sum_{n \neq c} \frac{p^{\alpha}_{cn} p^{\beta}_{nc}}{E_c - E_n} = \frac{1}{\hbar^2} \left. \frac{\partial^2 E_c}{\partial k_\alpha \partial k_\beta} \right|_{\mathbf{k}_0}. \tag{B.34}$$

Accordingly,

$$\left[-\frac{\hbar^2}{2} \sum_{\alpha,\beta} (\mathbf{M}^{-1})_{\alpha\beta} \partial_{r_\alpha} \partial_{r_\beta} + V_{\text{conf}}(\mathbf{r}) \right] \phi = \epsilon \phi. \tag{B.35}$$

At a nondegenerate minimum with positive curvature, the tensor is positive definite. This is the constant inverse-mass tensor used in Eq. (1.2). In rotated coordinates s_j along its principal axes, with eigenvalues $1/m_j$ for $j = 1, 2, 3$, the kinetic operator reduces to $-\sum_{j=1}^{3} \hbar^2 \partial^2_{s_j}/(2m_j)$. These axes need not coincide with the device coordinates x, y, z.

The complete expansion and its projection were exact within the stated local, spinless crystal model. The reduced equation additionally assumes low wave-vector offsets, smooth potential coupling, and weak admixture of separated bands, retaining the leading quadratic dispersion. Strong confinement can require higher dispersion terms; near-degenerate bands require coupled envelopes rather than the scalar mass formula.

For silicon, the restriction to one minimum is a separate single-valley assumption. Other valleys are not removed by an energy denominator at that minimum. Atomic-scale interfaces can couple valleys and require a multivalley description; a bulk effective-mass tensor alone does not determine valley splitting [2]. Spatially changing materials, nonlocal crystal operators, and spin–orbit terms also require extensions of the model used here.

B.8 From a three-dimensional well to bound orbitals

We first consider one valley ν and neglect coupling to other valleys. The potential $V_{\text{conf},\nu}$ is supplied by the electrostatic calculation in Appendix A; we now find its bound states. Assuming smooth confinement on the lattice scale, energies near this conduction-band minimum, and weak mixing with other bands, the envelope Hamiltonian is

$$h_\nu = -\frac{\hbar^2}{2} \nabla \cdot [\mathbf{M}_\nu^{-1}(\mathbf{r}) \nabla] + V_{\text{conf},\nu}(\mathbf{r}), \qquad h_\nu F_{n\nu} = \epsilon_{n\nu} F_{n\nu}. \tag{B.36}$$

$\mathbf{M}_\nu$ is the effective-mass tensor, and $F_{n\nu}$ is the slowly varying spatial envelope of orbital n. For a constant tensor this is the Hamiltonian of Eq. (1.2), whose relation to the microscopic crystal problem has been derived above. A spatially varying mass is written in divergence form so that the kinetic operator respects the appropriate interface matching.

Strong vertical confinement can simplify this three-dimensional problem. Suppose, for illustration, that one valley has a separable potential $V_z(z) + V_\parallel(x, y)$. Assume additionally that the mass tensor is constant and its principal axes align with the device coordinates x, y, z, with corresponding masses m_x, m_y, m_z. First solve

$$\left[-\frac{\hbar^2}{2m_z}\frac{d^2}{dz^2} + V_z(z)\right]\zeta_0(z) = \epsilon_z\zeta_0(z). \tag{B.37}$$

If the vertical excitation gap is large compared with the energies being used, the electron stays in this state. Writing $F_n(x, y, z) = f_n(x, y)\zeta_0(z)$ then gives

$$\left[-\frac{\hbar^2}{2m_x}\partial_x^2 - \frac{\hbar^2}{2m_y}\partial_y^2 + V_\parallel(x, y)\right] f_n = (\epsilon_n - \epsilon_z)f_n. \tag{B.38}$$

The lateral potential governs motion in the material plane within the chosen vertical state. A well in this potential supports localized dot orbitals.

In a general device, a gate voltage changes both the vertical state and lateral confinement. The full three-dimensional equation includes this coupled response. A reduced-dimensional model is useful when the vertical excitation gap is sufficiently large and its dependence on the lateral coordinates is treated consistently.

The resulting orbital energies and tails have different uses. The energy gap to an excited orbital determines whether that orbital can be omitted from a low-energy model. The tails extending into a barrier determine coupling to another dot. Both must be resolved numerically. Artificial hard walls are acceptable only where moving them farther away no longer changes the states or couplings of interest.

C Valley States and Interface Coupling

Interface coupling can mix the valleys retained in a dot and correlate their spatial envelopes. This appendix develops the multivalley block Hamiltonian of Eq. (1.3), then projects it onto one common orbital of two equivalent valleys. This projection gives the two-level Hamiltonian and valley eigenstates in Eqs. (1.4) and (1.5), and states the conditions under which their splitting is a useful description of the valley gap.

C.1 Coupled valley envelopes

The silicon conduction band has several energy minima at different wave vectors $\mathbf{k}_\nu$. The Bloch states near each minimum span a low-energy space $\mathcal{H}_\nu$. Each space contains many wave vectors and therefore many possible spatial wave packets. The retained wave-vector regions are disjoint, so these valley spaces are mutually orthogonal.

Within the effective-mass description, the expansion coefficients for wave vectors near one minimum are represented in position space by a slowly varying envelope $F_\nu(\mathbf{r})$. Retaining N_v valleys gives the direct-sum space and its envelope representation:

$$\mathcal{H}_{\mathrm{mv}} = \bigoplus_{\nu=1}^{N_v} \mathcal{H}_\nu, \qquad \mathbf{F}(\mathbf{r}) = \begin{pmatrix} F_1(\mathbf{r}) \\ \vdots \\ F_{N_v}(\mathbf{r}) \end{pmatrix}. \tag{C.1}$$

Each component is a function describing the spatial state within one valley. The state is normalized by $\sum_\nu \int |F_\nu|^2 \, d^3r = 1$.

Let $h_{0,\nu}$ denote the single-valley orbital Hamiltonian h_ν of Eq. (B.36). It acts within the νth component of the direct-sum space. An abrupt interface can transfer amplitude between valleys and mix their spatial states [6]. We describe transfer from valley ν' to valley ν by an operator $\hat{v}_{\nu\nu'}$. The full Hamiltonian is

$$h_{\mathrm{mv}} = \begin{pmatrix} h_{0,1} & \hat{v}_{12} & \cdots & \hat{v}_{1,N_v} \\ \hat{v}_{21} & h_{0,2} & \cdots & \hat{v}_{2,N_v} \\ \vdots & \vdots & \ddots & \vdots \\ \hat{v}_{N_v,1} & \hat{v}_{N_v,2} & \cdots & h_{0,N_v} \end{pmatrix}. \tag{C.2}$$

Hermiticity requires $\hat{v}_{\nu'\nu} = \hat{v}^\dagger_{\nu\nu'}$. All diagonal terms are included in $h_{0,\nu}$; the coupling operators $\hat{v}_{\nu\nu'}$ connect distinct valleys.

We use the local multivalley effective-mass model, in which each off-diagonal operator multiplies an envelope by a position-dependent coupling function [22]:

$$(\hat{v}_{\nu\nu'} F_{\nu'})(\mathbf{r}) = v_{\nu\nu'}(\mathbf{r}) F_{\nu'}(\mathbf{r}), \qquad v_{\nu'\nu}(\mathbf{r}) = v^*_{\nu\nu'}(\mathbf{r}).$$

Thus components at the same position are coupled. The functions $v_{\nu\nu'}$ encode the microscopic interface structure and are additional inputs to this effective model.

C.2 A two-valley doublet within one orbital

Consider the two lowest, equivalent valleys with orthogonal labels $|\nu_1\rangle$ and $|\nu_2\rangle$. Suppose that their uncoupled orbital blocks h_{0,ν_1} and h_{0,ν_2} share a normalized ground orbital ϕ_0 with energy ϵ_0. Retain the states $|\phi_0\rangle \otimes |\nu_1\rangle$ and $|\phi_0\rangle \otimes |\nu_2\rangle$ as the lowest orbital–valley pair, with all other states higher in energy. The approximation requires coupling matrix elements to other orbitals and valleys to be small compared with their energy separations from this pair. It need not be small compared with the separation within the pair, which is zero before the valleys are coupled.

Write $v_{12}(\mathbf{r}) = v_{\nu_1\nu_2}(\mathbf{r})$. Projection of the local interface coupling onto the common orbital gives

$$v_0 = \langle \phi_0 | v_{12} | \phi_0 \rangle = \int d^3r \, \phi_0^*(\mathbf{r}) v_{12}(\mathbf{r}) \phi_0(\mathbf{r}). \tag{C.3}$$

Hermiticity supplies the conjugate matrix element in the reverse direction. Define $P_0 = |\phi_0\rangle\langle\phi_0| \otimes (|\nu_1\rangle\langle\nu_1| + |\nu_2\rangle\langle\nu_2|)$. In the ordered valley basis $(|\nu_1\rangle, |\nu_2\rangle)$, with the common orbital suppressed, the restriction of $P_0 h_{\mathrm{mv}} P_0$ is

$$H_{\mathrm{v}} = \epsilon_0 I + \begin{pmatrix} 0 & v_0 \\ v_0^* & 0 \end{pmatrix}. \tag{C.4}$$

The characteristic equation is $(\epsilon - \epsilon_0)^2 - |v_0|^2 = 0$. For $v_0 \neq 0$, write $v_0 = |v_0| e^{i\theta_{\mathrm{v}}}$. The energies, normalized valley eigenstates, and their gap are

$$\begin{aligned} \epsilon_\pm &= \epsilon_0 \pm |v_0|, \\ |v_\pm\rangle &= \frac{|\nu_1\rangle \pm e^{-i\theta_{\mathrm{v}}} |\nu_2\rangle}{\sqrt{2}}, \\ \Delta_{\mathrm{val}} &= \epsilon_+ - \epsilon_- = 2|v_0|. \end{aligned} \tag{C.5}$$

Thus the interface selects two orthogonal valley superpositions and separates their energies. If $v_0 = 0$, the pair remains degenerate and θ_{v} is undefined; any orthonormal pair within this valley space is then an eigenbasis.

Within this projection, the lowest orbital–valley state is $|\chi_0\rangle \simeq |\phi_0\rangle \otimes |v_-\rangle$ for $v_0 \neq 0$. The common orbital and equivalent diagonal energies are assumptions of this two-valley limit. If interface coupling appreciably mixes other orbitals or valleys, the full block Hamiltonian in Eq. (C.2) must be used instead. Its lowest state can have different spatial envelopes in its valley components, so $|\chi_0\rangle$ need not factor into one orbital and one valley state.

D Screening of the Electron–Electron Interaction

The many-electron Hamiltonian in Eq. (1.7) uses a pair interaction $W(\mathbf{r}, \mathbf{r}')$. An electron's field polarizes the surrounding dielectric and redistributes charge on nearby conductors. The resulting field determines the force on a second electron. This appendix calculates the corresponding pair interaction for fixed electrode voltages. The applied voltages set the background confining potential described in Appendix A; the response to an added charge sets W.

In an infinite uniform dielectric of permittivity ϵ_d, the pair interaction is

$$W(\mathbf{r}, \mathbf{r}') = \frac{e^2}{4\pi\epsilon_\mathrm{d}|\mathbf{r} - \mathbf{r}'|}. \tag{D.1}$$

Here $e > 0$ is the magnitude of the electron charge. The permittivity already accounts for the material's dielectric polarization. A real device also has nearby metal electrodes. An electron redistributes charge on those conductors, and the redistributed charge changes the field felt by another electron. This modification of the interaction is called *screening*.

For example, consider two electrons at the same distance d from an infinite conducting plane, with separation R parallel to the plane. At fixed conductor potential, the induced field can be calculated by replacing the conductor with an opposite-sign image charge behind it. The pair interaction becomes

$$W(R) = \frac{e^2}{4\pi\epsilon_\mathrm{d}} \left(\frac{1}{R} - \frac{1}{\sqrt{R^2 + 4d^2}} \right). \tag{D.2}$$

The second term reduces the repulsion. Thus the same two orbital shapes can have different interaction energies when a metal surface is moved closer to them. Actual patterned gates require their own geometry-dependent calculation [23].

The general calculation is still electrostatics. Let $G(\mathbf{r}, \mathbf{r}')$ be the change in potential at $\mathbf{r}$ per unit test charge placed at $\mathbf{r}'$, with electrode voltages held fixed. For a linear dielectric response it obeys

$$-\nabla_\mathbf{r} \cdot [\epsilon_\mathrm{d}(\mathbf{r}) \nabla_\mathbf{r} G(\mathbf{r}, \mathbf{r}')] = \delta(\mathbf{r} - \mathbf{r}'), \qquad W(\mathbf{r}, \mathbf{r}') = e^2 G(\mathbf{r}, \mathbf{r}'). \tag{D.3}$$

Here δ is the three-dimensional Dirac delta function. Since the conductor voltages are held fixed, the test charge induces zero potential change on their surfaces, giving $G = 0$ there. The background potential Φ_0 carries the applied voltages separately. If a reservoir responds appreciably to the test charge, its screening response must also be included. We assume that this environment responds linearly and rapidly enough to use a static pair interaction.[3]

E Calculating Many-Electron Energies

Section 1 defines H_N and its ground energy E_N; Section 2 compares these energies to determine the electron number. This appendix calculates the many-electron energies in an orbital basis,

[3] An electron can also interact with the polarization that it itself induces in a boundary. After removal of the divergent bare self-energy, this gives a one-electron polarization energy, to be included in the single-electron Hamiltonian h_1 in Appendix E if retained.

using the screened pair interaction W from Appendix D. The localized basis and its matrix elements also give the several-dot description in Section 1.4, including the tunnel matrix in Eq. (1.9). The orbitals are those described in Appendix B, with valley components from Appendix C when they need to be retained.

With two electrons in the device, their mutual repulsion adds an energy that depends on both positions. For N electrons, we therefore use

$$H_N = \sum_{a=1}^{N} h_1(a) + \sum_{a<b} W(\mathbf{r}_a, \mathbf{r}_b). \tag{E.1}$$

Here $h_1(a)$ acts on electron a and includes kinetic energy, external confinement, and any retained valley and spin terms. W is the interaction energy of a pair. The restriction $a < b$ counts each pair once and excludes interaction of an electron with itself. H_N acts on antisymmetric N-electron states.

E.1 Self-consistent mean-field approximation: Schrödinger–Poisson

One way to approximate the interaction term in Eq. (E.1) is to replace the electrons' position-dependent repulsion by the electrostatic potential of their average charge density. If $n_q(\mathbf{r})$ is the electron number density, this gives the Hartree potential energy and the mean-field one-particle operator

$$V_{\mathrm{H}}[n_q](\mathbf{r}) = \int W(\mathbf{r}, \mathbf{r}') n_q(\mathbf{r}')\, d^3r', \qquad h_{\mathrm{MF}}[n_q] = h_1 + V_{\mathrm{H}}[n_q]. \tag{E.2}$$

The density determines the potential, while the occupied states in that potential determine the density. A self-consistent solution must satisfy both requirements simultaneously.

Assume spin-degenerate levels. Let $\mathbf{F}_a$ be a normalized multivalley orbital with energy ϵ_a^{MF}, and let p_a be its mean occupation, counting both spins. At temperature T, with chemical potential μ and Boltzmann constant k_B, the *Schrödinger–Poisson equations* are

$$\begin{cases} [h_1 + V_{\mathrm{H}}[n_q]]\mathbf{F}_a = \epsilon_a^{\mathrm{MF}} \mathbf{F}_a, \\ n_q(\mathbf{r}) = \displaystyle\sum_a p_a \sum_\nu |F_{a\nu}(\mathbf{r})|^2, \\ p_a = \dfrac{2}{1 + \exp[(\epsilon_a^{\mathrm{MF}} - \mu)/(k_B T)]}. \end{cases} \tag{E.3}$$

The unknowns are the orbitals, orbital energies, density, and occupations. Choose the orbitals orthonormal, with $\sum_\nu \int |F_{a\nu}|^2\, d^3r = 1$, and use the envelope boundary and interface conditions of h_1. Set $n_q = 0$ outside the region used for the envelope calculation.

The occupation condition closes the problem in one of two ways. For a prescribed mean electron number N, determine μ along with the other unknowns by imposing

$$\sum_a p_a = \int n_q(\mathbf{r})\, d^3r = N. \tag{E.4}$$

At zero temperature, fill the lowest self-consistent levels subject to this number constraint. For equilibrium with an electron reservoir, instead set $\mu = \mu_{\mathrm{R}}$; then $\sum_a p_a$ is an output. Spin-resolved levels use individual occupations with numerator one in the last line of Eq. (E.3).

To solve these equations, start with a trial density, calculate $V_{\mathrm{H}}[n_q]$ by integration using Eq. (E.2), find the orbitals of $h_1 + V_{\mathrm{H}}[n_q]$, and update their occupations and density. At

convergence, the density reconstructed from the orbitals is the same density used to calculate the Hartree potential [24].

This electrostatic mean-field treatment retains the Hartree potential and omits exchange and correlation.[4]

After convergence, a finite set of occupied and unoccupied orbitals can also be used to construct an antisymmetric basis for a subsequent many-electron calculation. Denote the electron number used to generate these orbitals by N_{ref}. For a target N-electron problem, one may choose $N_{\mathrm{ref}} = N$; to compare several electron numbers, a common reference basis can be used for all of them. The reference number selects the orbital shapes, not the electron number of the later calculation. Finite-basis results can depend on this choice, so check convergence as more orbitals are retained.

E.2 Basis choice and second quantization

To write H_N in second quantization, choose a finite set of orthonormal one-electron spatial states $w_i(\mathbf{r})$, $i = 1, \ldots, M$. Several states can be retained on each dot. We neglect spin mixing. A spatial state w_i may already include local valley coupling, as in Section 1.4; its matrix elements then use the full orbital–valley inner product. In the single-valley limit, w_i is a normalized scalar envelope with the common crystal Bloch factor suppressed. The scalar integral expressions below refer to that limit. The states w_i may be obtained directly from h_1 or from the self-consistent calculation above. A dot model is easier to interpret if each state is concentrated near a particular dot. When using states from separate isolated-well calculations, orthonormalize them first.

Consider two neighboring wells in a fixed spin sector. Their exact one-electron eigenstates generally spread over both wells. Linear combinations within their low-energy subspace can instead give states w_1, w_2 localized mainly on the left and right. In that basis,

$$h_{ij} = \langle w_i | h_1 | w_j \rangle, \qquad h_{\mathrm{pair}} = \begin{pmatrix} \epsilon_1 & -t_{12} \\ -t_{12}^* & \epsilon_2 \end{pmatrix}, \quad \epsilon_i = h_{ii}, \quad t_{12} = -h_{12}. \tag{E.5}$$

For scalar envelopes, $h_{ij} = \int d^3r\, w_i^*(\mathbf{r}) h_1 w_j(\mathbf{r})$; if valley components are retained, the inner product also sums over those components. The diagonal entries are the energies of the chosen localized states. The off-diagonal entry coherently transfers amplitude between them: this is the tunnel coupling.

Diagonalizing the matrix gives the level separation

$$\Delta E = \sqrt{(\epsilon_1 - \epsilon_2)^2 + 4|t_{12}|^2}. \tag{E.6}$$

At equal dot energies, $\Delta E = 2|t_{12}|$. Away from this point, both the energy offset and tunnel coupling contribute to the splitting.

For self-consistent orbitals, compute the same matrix elements from h_1. The interaction between the dot electrons will be included explicitly through W, so their Hartree potential is not part of the one-body matrix.

In the same basis, the pair matrix element is $\mathcal{V}_{ijkl} = \langle w_i w_j | W | w_k w_l \rangle$, with products referring to the two electron coordinates. For scalar envelopes,

$$\mathcal{V}_{ijkl} = \int d^3r\, d^3r'\, w_i^*(\mathbf{r}) w_j^*(\mathbf{r}') W(\mathbf{r}, \mathbf{r}') w_k(\mathbf{r}) w_l(\mathbf{r}'). \tag{E.7}$$

[4] Using the total density in V_{H} also introduces a spurious self-interaction unless it is explicitly corrected. In particular, the pair-interaction term of H_N vanishes for $N = 1$.

This coefficient connects a pair initially in spatial states k, l to a pair in states i, j. Different choices of these indices describe charge repulsion or transitions between orbital configurations.

With these one-body and two-body matrix elements, we can write the many-electron Hamiltonian using $c^\dagger_{i\sigma}$ and $c_{i\sigma}$, which add or remove an electron in spatial state i with spin $\sigma \in \{\uparrow, \downarrow\}$. Their fermionic anticommutation rules enforce the required antisymmetry. Retaining all matrix elements gives

$$H^{(M)} = \sum_{ij,\sigma} h_{ij} c^\dagger_{i\sigma} c_{j\sigma} + \frac{1}{2} \sum_{ijkl,\sigma\sigma'} \mathcal{V}_{ijkl} c^\dagger_{i\sigma} c^\dagger_{j\sigma'} c_{l\sigma'} c_{k\sigma}. \tag{E.8}$$

The factor $1/2$ avoids counting the same electron pair twice. This is the full one-body-plus-pair Hamiltonian within the selected orbital space [25]. If spin mixing is included, the one-body term becomes $\sum_{ij,\sigma\sigma'} h^{\sigma\sigma'}_{ij} c^\dagger_{i\sigma} c_{j\sigma'}$. Additional valley states can likewise be included in the state index.

Within the retained orbital space, a unitary basis change transforms both h_{ij} and $\mathcal{V}_{ijkl}$ while preserving the spectrum of $H^{(M)}$. Individual matrix elements refer to a specified basis.

Grouping states by dot occupation. In a localized basis, assign each spatial state to a dot. If $\mathcal{I}_d$ is the set of orbital indices assigned to dot d, its number operator is

$$\hat{N}_d = \sum_{i \in \mathcal{I}_d, \sigma} c^\dagger_{i\sigma} c_{i\sigma}.$$

At fixed total N, let $P_{\mathbf{N}}$ project onto the common eigenspace of these operators with occupations $\mathbf{N} = (N_1, N_2, \ldots)$, where N_d is the eigenvalue of $\hat{N}_d$ and $\sum_d N_d = N$. These projectors satisfy

$$P_{\mathbf{N}} P_{\mathbf{N}'} = \delta_{\mathbf{N}\mathbf{N}'} P_{\mathbf{N}}, \qquad \sum_{\mathbf{N}} P_{\mathbf{N}} = I_N, \tag{E.9}$$

where the sum includes all allowed occupation tuples at the fixed total N, and I_N is the identity in the retained N-electron space. Inserting this resolution of the identity on both sides of $H^{(M)}$ gives the block decomposition used in Eq. (1.8):

$$\begin{aligned} H^{(M)} &= H^{(0)} + \mathcal{T}, \\ H^{(0)} &= \sum_{\mathbf{N}} P_{\mathbf{N}} H^{(M)} P_{\mathbf{N}}, \quad \mathcal{T} = \sum_{\mathbf{N} \neq \mathbf{N}'} P_{\mathbf{N}} H^{(M)} P_{\mathbf{N}'}. \end{aligned} \tag{E.10}$$

The block $H_{\mathbf{N}}$, written explicitly as $H_{(N_1,N_2,\ldots)}$ in Eq. (1.8), is the restriction of $P_{\mathbf{N}} H^{(M)} P_{\mathbf{N}}$ to its charge subspace. Diagonalizing each block separately gives its ground and excited energies $E_{\mathbf{N},a}$. All terms that preserve every dot occupation remain in these blocks, including interdot repulsion and direct Coulomb exchange. The off-diagonal operator $\mathcal{T}$ includes single-electron hopping and any interaction matrix elements that change dot occupations, such as pair hopping. Dropping $\mathcal{T}$ therefore defines a reference with no charge transfer; it still retains interactions between different dots.

If $|\mathbf{N}, a\rangle$ and $|\mathbf{N}', b\rangle$ are nondegenerate block eigenstates, first-order perturbation theory gives the admixture of the latter into the former as

$$\frac{\langle \mathbf{N}', b | \mathcal{T} | \mathbf{N}, a \rangle}{E_{\mathbf{N},a} - E_{\mathbf{N}',b}}.$$

This expression explains why the relevant comparison is with the energy separation of the connected states. When that separation is comparable to their coupling, retain both states and diagonalize the coupled block. For the one-electron two-dot matrix in Eq. (E.5), write

$t_{12} = |t_{12}|e^{i\varphi}$. At $\epsilon_1 = \epsilon_2 = \epsilon_0$ and $|t_{12}| > 0$, its eigenstates are $(|w_1\rangle \pm e^{-i\varphi}|w_2\rangle)/\sqrt{2}$, with energies $\epsilon_0 \mp |t_{12}|$. The equal weights give strong charge mixing even when $|t_{12}|$ is small compared with gaps to other states.

This grouping preserves all matrix elements in the selected orbital space. The Hubbard model in Appendix J.1 makes additional choices about both the spatial states and the interaction terms to retain.

E.3 Solving the many-electron matrix: CI and FCI

We now solve the eigenvalue problem for $H^{(M)}$ in Eq. (E.8), retaining all its one-body and two-body matrix elements. At fixed electron number N, choose a basis of Slater determinants $|D_I\rangle$, each specifying which N spin orbitals are occupied. Expand the interacting state and solve the matrix eigenvalue problem

$$|\Psi_{N\alpha}\rangle = \sum_I C_I^{(N\alpha)}|D_I\rangle, \qquad \sum_J \langle D_I|H^{(M)}|D_J\rangle C_J^{(N\alpha)} = E_{N\alpha}C_I^{(N\alpha)}. \tag{E.11}$$

Here α labels many-electron eigenstates. This is *configuration interaction* (CI).

With M spatial orbitals and two spin states for each, there are

$$\dim \mathcal{H}_N^{(M)} = \binom{2M}{N} \tag{E.12}$$

Slater determinants at fixed N. *Full configuration interaction* (FCI) includes them all, or all states in an exactly invariant symmetry sector containing the state of interest. Truncated CI uses a selected subset. FCI solves the complete matrix in the retained orbital space; the remaining errors come from the orbital truncation, numerical precision, and the underlying device model [25].

The lowest eigenvalue gives E_N. Repeating for different electron numbers supplies the quantities needed in Section 2:

$$N_{\mathrm{eq}}(\mu_{\mathrm{R}}) \in \arg\min_N [E_N - \mu_{\mathrm{R}}N], \qquad \mu_N = E_N - E_{N-1}. \tag{E.13}$$

Here N_{eq} is the zero-temperature equilibrium charge, and μ_N is the energy for adding the Nth electron. If successive charge sectors are stable, μ_N is their transition threshold; otherwise the minimization may skip a sector. The set notation allows degeneracy at a transition. For a localized basis, summing $\langle c_{i\sigma}^\dagger c_{i\sigma}\rangle$ over the spins and orbitals assigned to a dot gives its mean electron number. These mean occupations can be fractional at fixed total N.

For a fixed physical Hamiltonian, enlarging a nested orbital basis can only lower or leave unchanged the FCI ground energy by the variational principle. Check the convergence of energy differences such as J and μ_N separately: monotonic convergence of each total energy permits either direction of change in their difference.

F How Voltages Set and Preserve Model Parameters

A gate electrode changes the potential over a finite part of the device. Its voltage can therefore shift a dot level, reshape a localized orbital and change tunneling through a nearby barrier at the same time. Moving an addition threshold in Section 2 requires knowing its voltage response. Tuning the exchange barrier at fixed thresholds in Section 4.3 requires coordinating several electrodes. This appendix derives the parameter responses and then solves for the voltage combinations that produce a specified change. The same construction is extended to an array, where the selection problem of Section 8 requires constraints on affected neighbours as well as on the target.

F.1 How much does a gate voltage move an energy level?

Consider a concrete operation: increase the voltage on the metal electrode above a dot. The resulting electric potential change lowers the electron potential energy. The oxide, surrounding conductors, and electron position determine how strongly the dot responds.

Let $\mathbf{V}$ denote the collection of gate voltages. For a small change around a specified operating point, define the dimensionless response $\eta_g(\mathbf{r})$ by

$$\delta\Phi_0(\mathbf{r}) = \sum_g \eta_g(\mathbf{r})\,\delta V_g, \qquad \eta_g(\mathbf{r}) = \frac{\partial\Phi_0(\mathbf{r})}{\partial V_g}. \tag{F.1}$$

It is obtained by changing a gate boundary voltage in the electrostatic problem. It can be calculated by differencing nearby solutions; if the charge response is nonlinear, it depends on the chosen operating point.

For a normalized, nondegenerate eigenstate w_a of an isolated dot, assume initially that voltage dependence enters only through $-e\Phi_0$. Differentiating its eigenvalue equation gives the Hellmann–Feynman relation

$$\frac{\partial\epsilon_a}{\partial V_g} = \langle w_a|\frac{\partial h_1}{\partial V_g}|w_a\rangle = -e\int d^3r\,|w_a(\mathbf{r})|^2\eta_g(\mathbf{r}). \tag{F.2}$$

Terms containing derivatives of w_a cancel by normalization and the eigenvalue equation. Thus the electron samples the gate's potential response over its probability distribution.

The resulting voltage-to-energy conversion is called a *lever arm*. Using a dimensionless convention,

$$\alpha_{ag} \equiv -\frac{1}{e}\frac{\partial\epsilon_a}{\partial V_g}, \qquad \delta\epsilon_a = -e\sum_g \alpha_{ag}\delta V_g. \tag{F.3}$$

Some literature instead quotes $e\alpha_{ag}$, an energy per voltage. The convention must be stated. If voltage also changes interface couplings or other terms in h_1, their derivatives must be included in Eq. (F.2).

An addition threshold depends on the energies of two many-electron states. Its voltage response is

$$\frac{\partial\mu_N}{\partial V_g} = \frac{\partial E_N}{\partial V_g} - \frac{\partial E_{N-1}}{\partial V_g}. \tag{F.4}$$

For example, a second electron in a frozen single orbital has addition energy $\epsilon + U$, so its shift contains both $\partial\epsilon/\partial V_g$ and $\partial U/\partial V_g$. This is why converting a charge transition into a model parameter requires identifying which energy difference is being measured.

F.2 Changing tunneling and repulsion

Changing the electrode between two dots changes the barrier through which their wave functions extend. Its most direct effect is therefore on t_{12}.

A one-dimensional barrier gives a useful estimate. Let s measure distance along a tunneling path, $V_b(s)$ be its potential energy, and E_0 the approximate bound-state energy. In the forbidden interval $s_1 < s < s_2$, where $V_b > E_0$, an effective mass m_b gives the leading exponential dependence

$$|t_{12}| \;\propto\; \exp\left[-\int_{s_1}^{s_2}\frac{\sqrt{2m_b[V_b(s)-E_0]}}{\hbar}\,ds\right]. \tag{F.5}$$

This WKB estimate explains why a modest change in barrier height or width can strongly change tunneling. The prefactor and the actual multidimensional coupling should be calculated

from the wave functions. Here t_{12} is a coherent Hamiltonian matrix element with units of energy.

Gate voltages also change the orbital sizes and positions, so they change the interaction integrals. For a family of similarly shaped orbitals of radius ℓ in a uniform dielectric, U scales roughly as $e^2/(4\pi\epsilon_{\mathrm{d}}\ell)$. Spreading an orbital out reduces the cost of placing two electrons in it. For well-separated dots at distance R, the unscreened direct interaction scales as $V_{12} \sim e^2/(4\pi\epsilon_{\mathrm{d}}R)$. The scaling estimates apply to unscreened interactions in a uniform dielectric. Nearby metals modify them through the screening described in Eq. (D.2).

Consequently, geometry affects more than the shape of V_{conf}. Oxide thickness affects gate control and screening; dot separation affects tunneling and interdot repulsion; interface structure affects the valley states. The integrals must be recalculated, or their changes justified as negligible, when these inputs change appreciably.

F.3 Why one physical control changes several model parameters

Consider a two-dot device with a plunger above each dot and an electrode between them. Changing the middle electrode may increase tunneling while also shifting both dot energies. This simultaneous response to one gate voltage is called *gate cross-coupling*. To keep the energies fixed, one can simultaneously adjust the two plungers. Near a reference voltage setting, the required compensation obeys

$$\delta\epsilon_i = \sum_g \frac{\partial\epsilon_i}{\partial V_g}\delta V_g = 0, \qquad i = 1, 2. \tag{F.6}$$

For the Hubbard on-site energies, evaluate the derivatives including the voltage dependence of the chosen localized basis. Independent plunger responses allow these two linear equations to determine compensating voltages locally. The interaction parameters U_i, V_{12} and orbital shapes can still vary along this compensated voltage change.

The exchange coupling inherits these simultaneous changes. For two dots with equal U, define $D = U - V_{12}$ and $\varepsilon = \epsilon_1 - \epsilon_2$. Writing Eq. (4.12) with $t_c = |t_{12}|$ and $V = V_{12}$ gives

$$J_{\mathrm{se}}(\varepsilon) = \frac{4|t_{12}|^2 D}{D^2 - \varepsilon^2}, \qquad D > |\varepsilon|, \quad |t_{12}| \ll D - |\varepsilon|. \tag{F.7}$$

Holding t_{12} and D fixed, the first derivative with respect to detuning vanishes at $\varepsilon = 0$. This explains why operation near equal dot energies can reduce first-order sensitivity to detuning fluctuations [26]. The sensitivity to changes in the tunnel amplitude and interaction energy follows, at this point, from the first-order relation

$$\frac{\delta J_{\mathrm{se}}}{J_{\mathrm{se}}} = 2\frac{\delta|t_{12}|}{|t_{12}|} - \frac{\delta D}{D}. \tag{F.8}$$

A voltage that changes the barrier can therefore tune exchange even when the dot energies remain aligned.

The spin splitting can change as well. If a dot moves through the field of a micromagnet, its mean field changes. For a scalar, position-independent electron g factor g_{e} and a retained normalized orbital w_i, the Zeeman vector is

$$\mathbf{b}_i = g_{\mathrm{e}}\mu_B \int d^3r\, |w_i(\mathbf{r})|^2 \mathbf{B}(\mathbf{r}). \tag{F.9}$$

Here μ_B is the Bohr magneton and $\mathbf{B}$ is the applied plus micromagnet field. Changing w_i changes $\mathbf{b}_i$ even when the magnet is unchanged. This connects the device calculation back to electrically driven spin control in Section 4.2.

F.4 Choosing voltages for specified addition-energy changes

Loading requires moving an addition energy below a reservoir chemical potential; a compensated exchange pulse instead holds the addition thresholds fixed while changing tunneling. Because one electrode can produce both effects, we use the combined response to all varied voltages to select the desired change. For this local calculation, the response can be measured without recalculating the full device potential for every pulse.

The many-electron Hamiltonian identifies what this measured response represents. For a small voltage change, write

$$\delta H = \sum_j \widehat{W}_j \, \delta V_j, \qquad \widehat{W}_j = \left.\frac{\partial H}{\partial V_j}\right|_{\mathbf{V}^0}.$$

For an isolated, nondegenerate state on a specified charge branch, the Hellmann–Feynman relation gives

$$\frac{\partial (E_N - E_{N-1})}{\partial V_j} = \langle \widehat{W}_j \rangle_N - \langle \widehat{W}_j \rangle_{N-1}.$$

This is the addition-energy derivative in Eq. (F.4), expressed through the voltage-dependent Hamiltonian.

Let $\mu_i(\mathbf{V})$ be the addition energy of dot i for a specified charge transition. Near reference voltages $\mathbf{V}^0$, the response to the N_g varied gate electrodes is

$$d\mu_i = \sum_{j=1}^{N_g} G_{ij} \, dV_j, \qquad G_{ij} = \left.\frac{\partial \mu_i}{\partial V_j}\right|_{\mathbf{V}^0}, \qquad d\boldsymbol{\mu} = G \, d\mathbf{V}. \tag{F.10}$$

The index j includes plunger and barrier electrodes. The reservoir contact voltages and magnetic field are held fixed in the examples below. Together with the chosen charge configuration, these reference settings define the *operating point*. Each matrix entry has units of energy per voltage. For a finite small step, the differential relation gives $\Delta\boldsymbol{\mu} \simeq G\Delta\mathbf{V}$.

Suppose we want to increase $\mu_1 - \mu_2$ without changing $(\mu_1 + \mu_2)/2$. The required energy changes have opposite signs, whereas a pulse on one plunger may shift both energies in the same direction. We use the measured response matrix to find voltages whose combined effects give the desired pair of changes.

Consider two dots controlled by two plunger voltages V_1, V_2 and one barrier voltage V_B. The full relation is

$$\begin{pmatrix} d\mu_1 \\ d\mu_2 \end{pmatrix} = \underbrace{\begin{pmatrix} G_{11} & G_{12} & G_{1\mathrm{B}} \\ G_{21} & G_{22} & G_{2\mathrm{B}} \end{pmatrix}}_{G} \begin{pmatrix} dV_1 \\ dV_2 \\ dV_\mathrm{B} \end{pmatrix}. \tag{F.11}$$

F.4.1 Obtaining the response matrix

The control calculation needs all six entries, including those of the barrier electrode. The charge-transition slopes and finite-bias measurements provide these entries; we now assemble them into G. On a dot–reservoir transition, $\mu_i = \mu_\mathrm{R}$. Vary electrode j and adjust the corresponding plunger V_i to stay on that transition, with the remaining electrode fixed. The measured slope is

$$c_{ij} = \left.\frac{dV_i}{dV_j}\right|_{\mu_i = \mu_\mathrm{R}}, \qquad j \neq i.$$

Since $0 = G_{ii}dV_i + G_{ij}dV_j$, it gives $G_{ij} = -G_{ii}c_{ij}$. The four slopes $c_{12}, c_{1\mathrm{B}}, c_{21}, c_{2\mathrm{B}}$ determine the cross-responses relative to the two plunger responses. Measure the latter, $g_1 = G_{11}$ and $g_2 = G_{22}$, using the finite-bias boundaries of Section G.1.4. The complete matrix is then

$$G = \begin{pmatrix} g_1 & -g_1c_{12} & -g_1c_{1\mathrm{B}} \\ -g_2c_{21} & g_2 & -g_2c_{2\mathrm{B}} \end{pmatrix}. \tag{F.12}$$

The transition slopes give ratios; the finite-bias measurement supplies the energy scale. Use nearby smooth segments for the specified charge sectors, since a change in a neighbouring occupation can shift a transition discontinuously.

F.4.2 Solving for the voltage change

With G known, the requested change $d\boldsymbol{\mu}^\star$ is the input and $d\mathbf{V}$ is the unknown. The control problem is $G\,d\mathbf{V} = d\boldsymbol{\mu}^\star$. A rank-two 2×3 matrix leaves one free voltage direction: two energy conditions do not uniquely determine three voltages. To write the solutions explicitly, partition the already measured matrix as

$$G = \begin{pmatrix} G_{\mathrm{P}} & \mathbf{g}_{\mathrm{B}} \end{pmatrix}, \qquad d\mathbf{V}_{\mathrm{P}} = \begin{pmatrix} dV_1 \\ dV_2 \end{pmatrix},$$

where G_{P} contains the first two columns and $\mathbf{g}_{\mathrm{B}}$ is the third. If G_{P} is invertible, specifying dV_{B} gives

$$d\mathbf{V}_{\mathrm{P}} = G_{\mathrm{P}}^{-1}\left(d\boldsymbol{\mu}^\star - \mathbf{g}_{\mathrm{B}}dV_{\mathrm{B}}\right). \tag{F.13}$$

Thus keeping the barrier voltage fixed is one possible choice; the full response matrix also permits solutions that change it. Coordinated voltage directions chosen to control specified physical quantities are called *virtual gates*.

F.4.3 An opposite shift of two addition energies

For a numerical illustration, take

$$G = \begin{pmatrix} -100 & -25 & -10 \\ -20 & -80 & -15 \end{pmatrix} \frac{\mu\mathrm{eV}}{\mathrm{mV}}.$$

These are illustrative response values, not experimental data. Let $\varepsilon_\mu = \mu_1 - \mu_2$ and $\bar{\mu} = (\mu_1 + \mu_2)/2$. Increasing ε_μ by $80\,\mu\mathrm{eV}$ at fixed $\bar{\mu}$ requires $(\Delta\mu_1, \Delta\mu_2) = (40, -40)\,\mu\mathrm{eV}$. Choosing $\Delta V_{\mathrm{B}} = 0$ gives

$$\begin{pmatrix} \Delta V_1 \\ \Delta V_2 \\ \Delta V_{\mathrm{B}} \end{pmatrix} \simeq \begin{pmatrix} -0.56 \\ 0.64 \\ 0 \end{pmatrix} \mathrm{mV}. \tag{F.14}$$

Substitution gives $56 - 16 = 40$ and $11.2 - 51.2 = -40$, in μeV, for the two energy changes. More generally, with $dV_{\mathrm{B}} = 0$,

$$d\mathbf{V}_{\mathrm{P}} = G_{\mathrm{P}}^{-1} \begin{pmatrix} 1 & 1/2 \\ 1 & -1/2 \end{pmatrix} \begin{pmatrix} d\bar{\mu} \\ d\varepsilon_\mu \end{pmatrix}. \tag{F.15}$$

The difference ε_μ concerns the chosen addition transitions. Its relation to the orbital detuning $\epsilon_1 - \epsilon_2$ also depends on interaction energies.

The calculation has now supplied a voltage vector for the requested energy shift. Apply that small step, then locate the transitions again. Their displacements, converted to energy with the measured responses, test whether the requested energy changes occurred. If the chosen G_{P} is nearly singular, its inverse requires large compensating voltages. Another independent pair of columns of the full G may provide a better set of controls.

F.5 Changing tunneling at fixed addition energies

An exchange pulse should strengthen the interdot coupling without unintentionally moving the dot–reservoir loading thresholds. We therefore seek a voltage direction that keeps both addition energies fixed but changes t_c. To first order, the first requirement is simply

$$G\, d\mathbf{V} = 0. \tag{F.16}$$

For the rank-two matrix above, its nullspace is one-dimensional. If G_{P} is invertible, choose a vector in that nullspace with barrier component equal to one:

$$\mathbf{n} = \begin{pmatrix} -G_{\mathrm{P}}^{-1}\mathbf{g}_{\mathrm{B}} \\ 1 \end{pmatrix}, \qquad d\mathbf{V} = \mathbf{n}\, dv, \qquad G\mathbf{n} = 0. \tag{F.17}$$

Here $dv = dV_{\mathrm{B}}$ parametrizes the coordinated voltage change. Equivalently, after choosing the barrier step, the plunger compensation is

$$d\mathbf{V}_{\mathrm{P}} = -G_{\mathrm{P}}^{-1}\mathbf{g}_{\mathrm{B}}\, dV_{\mathrm{B}}. \tag{F.18}$$

The barrier pulse and its two compensating plunger pulses are applied together. Their energy shifts cancel even though the potential between the dots, and hence the wave-function overlap, can change.

For the illustrative matrix above,

$$\mathbf{n} = \begin{pmatrix} -17/300 \\ -13/75 \\ 1 \end{pmatrix}.$$

A 0.30 mV barrier step therefore requires plunger steps -0.017 mV and -0.052 mV. Their combined first-order shifts of both μ_1 and μ_2 vanish. Whether this direction changes tunneling is a separate, measurable property. Define $w_j = \partial t_c/\partial V_j$ at the operating point. Along the coordinated direction,

$$dt_c = \sum_j w_j\, dV_j = \left(\sum_j w_j n_j\right) dv. \tag{F.19}$$

A nonzero coefficient permits tunneling control at fixed addition energies. If it vanishes, these three controls cannot achieve that change to first order at this operating point.

To measure the coefficient, apply several small values of v in $\mathbf{V}(v) = \mathbf{V}^0 + \mathbf{n}v$. At each setting use the charge-distribution measurement of Section K.1 to determine $t_c(v)$, and recheck the addition thresholds. The response must be measured along this coordinated path: changing the plungers also affects tunneling.

The scan gives $t_c(v)$ along a path that keeps the addition energies fixed. For spin gates, the same voltage path can be characterized by its exchange coupling $J(v)$ using Section N.2.

For several couplings, append response rows such as $d\ln(t_{ij}/t_{\mathrm{ref}}) = \sum_g B_{ij,g}\, dV_g$, where $t_{\mathrm{ref}} > 0$ is a fixed reference energy. The combined matrix can prescribe a change of t_{12} with t_{23} and the addition energies held fixed, provided its rank permits those independent constraints. Such control of energies and tunnel couplings has been demonstrated using virtual gates [27].

F.5.1 Turning voltage settings into a pulse

The energy-shifting and compensated barrier directions solve different problems: the first loads or unloads electrons, whereas the second changes their coupling without moving the loading

thresholds. A change between two settings means a change of the physical voltage vector, not an instantaneous change of a model parameter by itself. For a locally linear path, a pulse can be programmed as

$$\mathbf{V}(t) = \mathbf{V}^{\mathrm{A}} + f(t)(\mathbf{V}^{\mathrm{B}} - \mathbf{V}^{\mathrm{A}}), \qquad 0 \le f(t) \le 1.$$

The function f rises from zero to one, holds, and eventually returns or proceeds to another setting. For an exchange pulse, the displacement can follow the compensated direction $\mathbf{n}$; for loading, it must instead move an addition energy across a reservoir chemical potential. The endpoint conditions depend on whether the operation loads an electron, retains a fixed charge sector, or converts a spin state into charge motion.

The compensation is local. For a larger excursion, measure or update the response along the path. The static matrix sets the voltage combinations; the measured time-dependent response sets the pulse durations and tests the effect of the edges.

F.6 Compensating a voltage change across an array

Changing the detuning of a target pair may shift the spin gaps and exchange couplings of neighbouring dots. Near an operating point, collect the relevant energies and couplings in a vector q, and the electrode voltages in V. Their measured local response is

$$\delta q_a = \sum_g R_{ag} \delta V_g.$$

To change one target quantity while holding others fixed, the requested δq must specify both the target and the spectator constraints. The requested change must lie in the range of the measured response matrix. This extends the virtual-gate construction of Eq. (F.10) from two addition energies to all parameters that the operation must change or preserve. The response may be short-ranged, but its spatial extent is an experimental property rather than a consequence of drawing nearby electrodes.

Consider a square plunger-to-addition-energy response $\delta\boldsymbol{\mu} = G\delta\mathbf{V}$, with $G = D + K$. The diagonal matrix D describes each plunger's response on its own dot; K contains cross responses. If we use only D^{-1}, the voltage $D^{-1}\delta\boldsymbol{\mu}$ produces an extra energy shift $KD^{-1}\delta\boldsymbol{\mu}$. For $\|D^{-1}K\| < 1$, a Neumann expansion cancels these successive cross effects:

$$G^{-1} = [I - D^{-1}K + (D^{-1}K)^2 - \cdots]D^{-1}. \tag{F.20}$$

The equivalent iteration is

$$\delta\mathbf{V}^{(k+1)} = D^{-1}\big(\delta\boldsymbol{\mu} - K\delta\mathbf{V}^{(k)}\big).$$

If K connects only neighbouring dots, each additional order reaches farther through the array. When the response is strong or long-ranged, the truncated local inverse may be inadequate and the wider matrix must be used.

This inverse is only as current as its measured response. Changing a barrier can alter the two neighbouring spin gaps, their exchange, and the reference distribution of a shared sensor. Before reusing a gate pulse, recheck the observables that depend on that barrier; a changed sensor distribution must be separated from a changed gate outcome. A feedback procedure for updating calibrated controls is developed in Eq. (Y.2). Its use in an array also requires tracking which other calibrated operations depend on the control being updated.

G Electrical Detection of Charge

Sections 2 and 5 use electrical signals to identify changes in electron number and resolve the target charge. This appendix derives the current through the target and the response of a separate charge sensor.

An electron entering or leaving a target dot can be observed in two ways. If the target lies between a source and drain, repeated transfer through it produces a current. If continuous transport is unavailable or undesirable, a nearby sensor responds to the target occupation. Both methods locate the same charge boundaries, but they record different electrical signals.

G.1 Current through a dot

G.1.1 Physical picture

We seek the gate voltage V_g at which the dot's equilibrium electron number changes from $N-1$ to N. With all other electrode voltages fixed, the addition energy is $\mu_N(V_g) = E_N(V_g) - E_{N-1}(V_g)$, where E_N is the N-electron ground-state energy.

At zero temperature, a dot in equilibrium with a reservoir at chemical potential μ_{R} changes between $N-1$ and N electrons at

$$\mu_N(V_g) = \mu_{\mathrm{R}}. \tag{G.1}$$

To locate this transition through a current, we couple the dot to a source that supplies electrons and a drain that receives them, with chemical potentials $\mu_{\mathrm{S}} > \mu_{\mathrm{D}}$ held fixed during each gate scan. We treat tunneling as successive single-electron events and neglect the energy width of the dot states caused by coupling to the reservoirs. We choose the interval between the reservoir chemical potentials narrow enough that only the $N-1 \leftrightarrow N$ ground-state transition can enter it during this scan. For current to flow, the source must have an occupied state at μ_N, and the drain an empty one. At zero temperature, the three regimes are

$$\begin{aligned} \mu_{N-1} < \mu_{\mathrm{D}} < \mu_{\mathrm{S}} < \mu_N: &\quad \text{blocked at } N-1, \\ \mu_{\mathrm{D}} < \mu_N < \mu_{\mathrm{S}}: &\quad \text{transport between } N-1 \text{ and } N, \\ \mu_N < \mu_{\mathrm{D}} < \mu_{\mathrm{S}} < \mu_{N+1}: &\quad \text{blocked at } N. \end{aligned} \tag{G.2}$$

The two fixed-charge regimes have zero sequential current and are called *Coulomb-blockade regions*.

The energy interval between μ_{D} and μ_{S} is called the *bias window*. We control its width with the source–drain bias V_b, defined by

$$\mu_{\mathrm{S}} - \mu_{\mathrm{D}} = eV_b, \qquad e > 0. \tag{G.3}$$

The window has width $e|V_b|$; the sign of V_b determines which reservoir has the higher chemical potential. For negative bias, interchange source and drain in Eq. (G.2). At positive bias with μ_N inside the window, an electron tunnels from the source into the dot, changing its electron number from $N-1$ to N. An electron then tunnels from the dot into the drain, returning the dot to $N-1$ electrons. Each cycle transfers one electron from source to drain, and repeated cycles produce a steady current.

Our main scan varies V_g to change the dot potential and hence μ_N, while the reservoir voltages keep the window fixed. Between scans, we also adjust V_b to select the window width. In Figure G.1, increasing V_g lowers μ_N through the window. The current turns on at $\mu_N = \mu_{\mathrm{S}}$, when electron entry becomes possible, and turns off at $\mu_N = \mu_{\mathrm{D}}$, when electron exit is blocked. The measured current edges thus identify the gate voltages that align μ_N with each reservoir.

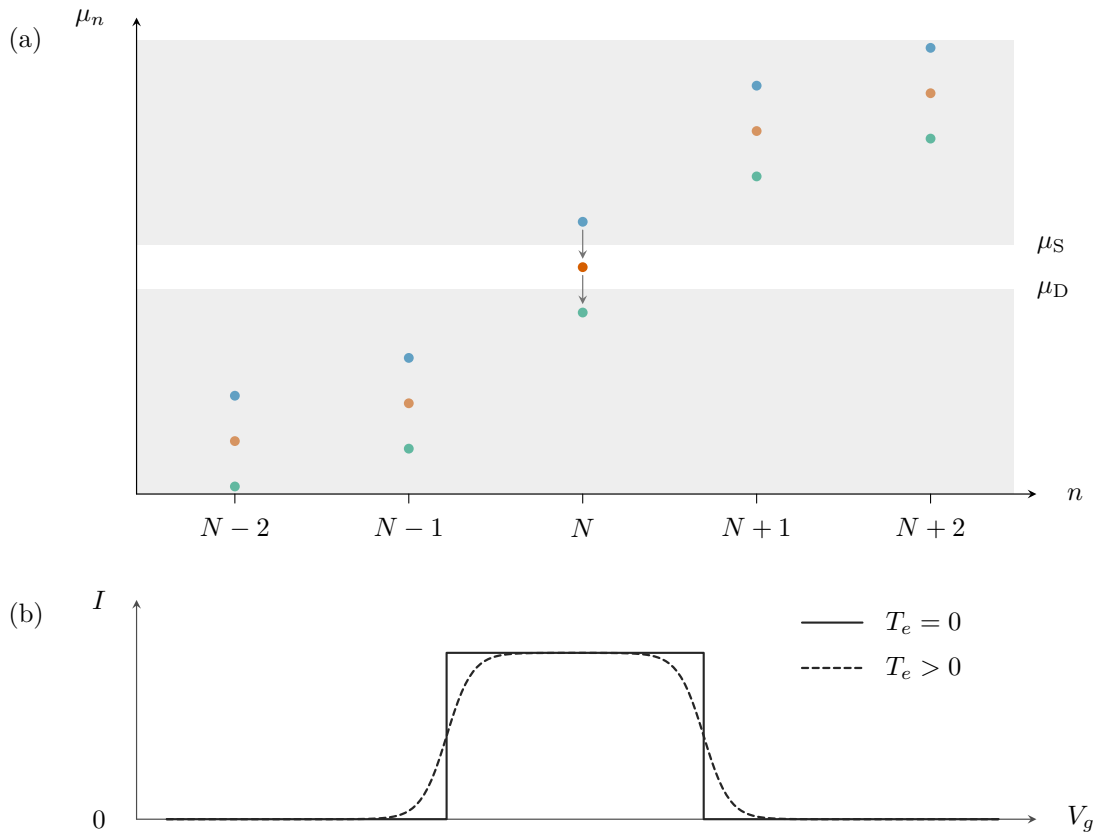


Figure G.1: Gate-controlled onset and suppression of transport. (a) Three colored sets of addition thresholds $\mu_n = E_n - E_{n-1}$ at increasing gate voltage; the arrows show their downward shift. The reservoir chemical potentials remain fixed. A translucent gray mask covers thresholds outside the narrow white zero-temperature bias window; only μ_N at the middle setting lies inside. (b) Current for the same bias and constant tunnel rates: zero temperature (solid) and finite temperature (dashed). The dashed curve follows Eq. (G.9), with $k_B T_e/(\mu_S - \mu_D) = 0.05$.

As the two reservoir chemical potentials approach a common value μ_{R}, these edges approach the equilibrium transition voltage in Eq. (G.1).

Repeating the gate scan at different positive and negative biases gives a two-dimensional current map $I(V_g, V_b)$. The Coulomb-blockade regions in Eq. (G.2) form *Coulomb diamonds* in this map (Figure G.2) [7].

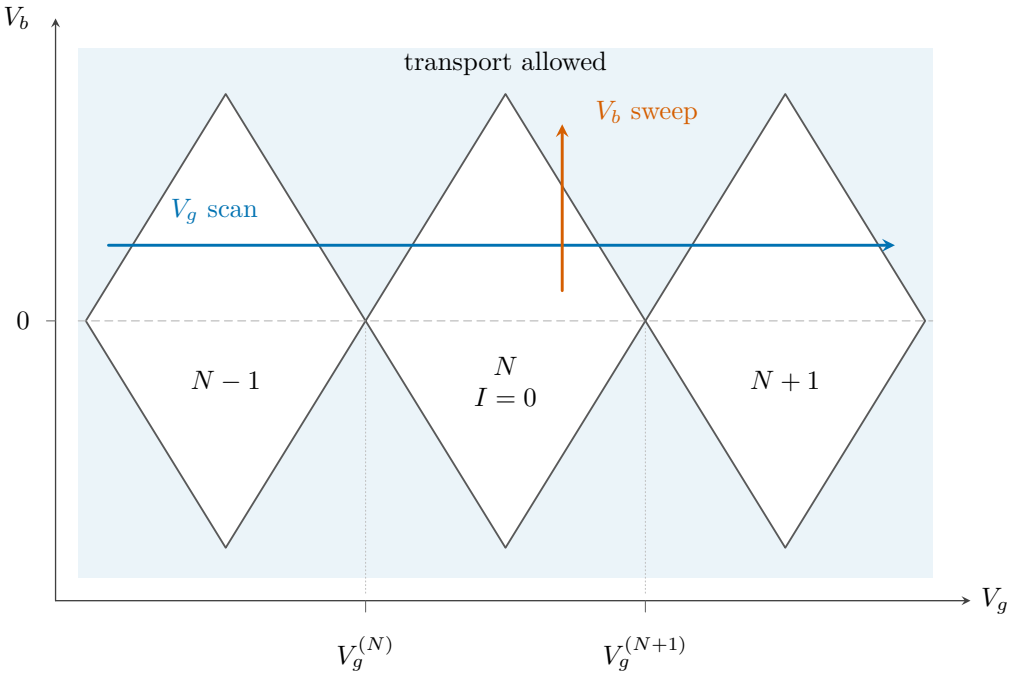


Figure G.2: Gate and bias control of single-dot transport. White diamonds are fixed-charge regions with zero sequential current in the zero-temperature, negligible-broadening model. The surrounding tint marks energetically allowed transport, not the current magnitude. The blue horizontal arrow shows the main gate scan at fixed bias; the orange vertical arrow shows increasing positive bias at fixed gate voltage, widening the reservoir window. The voltages $V_g^{(N)}$ and $V_g^{(N+1)}$ are the zero-bias entry points of the Nth and $(N+1)$th electrons. Equal diamond sizes and symmetric slopes are chosen for illustration; this is a schematic, not experimental data.

G.1.2 Deriving the current

We treat tunneling as successive, incoherent events between the two charge states, keeping one nondegenerate ground state for each. At zero temperature with μ_N inside the window, let Γ_{S} and Γ_{D} be the rates for entry from the source and exit into the drain, in inverse seconds. Each rate is conditional on the dot occupying the appropriate initial charge state; Appendix Q.1 derives their probability interpretation and the rate equation from quantum transition amplitudes. If p is the probability that the dot contains N electrons, then

$$\dot{p} = \Gamma_{\mathrm{S}}(1-p) - \Gamma_{\mathrm{D}} p. \tag{G.4}$$

The first term counts entries when the dot has $N-1$ electrons; the second counts exits when it has N. This is the *sequential tunneling* description. Define I as $e > 0$ times the net electron-transfer rate from source to drain; conventional electric current has the opposite

direction. In the stationary state, that transfer rate is $\Gamma_{\mathrm{D}} p_{\mathrm{st}}$. Setting $\dot{p} = 0$ therefore gives

$$p_{\mathrm{st}} = \frac{\Gamma_{\mathrm{S}}}{\Gamma_{\mathrm{S}} + \Gamma_{\mathrm{D}}}, \qquad I = e\Gamma_{\mathrm{D}} p_{\mathrm{st}} = e\frac{\Gamma_{\mathrm{S}}\Gamma_{\mathrm{D}}}{\Gamma_{\mathrm{S}} + \Gamma_{\mathrm{D}}}. \tag{G.5}$$

With constant tunnel rates, this current is constant inside the window and zero outside, giving the solid curve in Figure G.1(b).

At finite temperature, reservoir states near the chemical potential are only partially occupied. Their occupation probability is the Fermi function

$$f_r(E) = \frac{1}{1 + \exp[(E - \mu_r)/(k_B T_e)]}, \qquad r = \mathrm{S}, \mathrm{D}, \tag{G.6}$$

where T_e is the reservoir electron temperature. To retain the two-charge-state description, the excluded transitions must remain outside the window by energy margins large compared with $k_B T_e$. An entry event requires an occupied reservoir state, whereas an exit event requires an empty one. The rates become

$$\Gamma_r^{\mathrm{in}} = \Gamma_r f_r(\mu_N), \qquad \Gamma_r^{\mathrm{out}} = \Gamma_r[1 - f_r(\mu_N)]. \tag{G.7}$$

Here Γ_r is the rate before the occupation factor is applied. For a tunneling matrix element v_r and a reservoir density of states ρ_r per unit energy, Fermi's golden rule gives $\Gamma_r = 2\pi|v_r|^2\rho_r/\hbar$. We treat these rates as nearly constant across the transition. For the following finite-temperature line shape, assume $\hbar(\Gamma_{\mathrm{S}} + \Gamma_{\mathrm{D}}) \ll k_B T_e$, so that thermal broadening dominates the level width caused by tunneling, called *lifetime broadening*. Spin or orbital degeneracy is handled by including the corresponding states and rates separately [7].

Including entry and exit through both barriers generalizes the zero-temperature rate equation to

$$\dot{p} = (1 - p)\sum_r \Gamma_r f_r(\mu_N) - p\sum_r \Gamma_r[1 - f_r(\mu_N)]. \tag{G.8}$$

Its stationary solution gives

$$\begin{aligned} p_{\mathrm{st}} &= \frac{\Gamma_{\mathrm{S}} f_{\mathrm{S}}(\mu_N) + \Gamma_{\mathrm{D}} f_{\mathrm{D}}(\mu_N)}{\Gamma_{\mathrm{S}} + \Gamma_{\mathrm{D}}}, \\ I &= e\Gamma_{\mathrm{S}}[f_{\mathrm{S}}(\mu_N) - p_{\mathrm{st}}] = e\frac{\Gamma_{\mathrm{S}}\Gamma_{\mathrm{D}}}{\Gamma_{\mathrm{S}} + \Gamma_{\mathrm{D}}}[f_{\mathrm{S}}(\mu_N) - f_{\mathrm{D}}(\mu_N)]. \end{aligned} \tag{G.9}$$

Changing the bias changes the window and hence the current through the factor $f_{\mathrm{S}}(\mu_N) - f_{\mathrm{D}}(\mu_N)$. Well inside the bias window, $f_{\mathrm{S}}(\mu_N) \simeq 1$ and $f_{\mathrm{D}}(\mu_N) \simeq 0$, recovering Eq. (G.5). Near either edge, the gradual change in reservoir occupation rounds the zero-temperature current step over an energy range of order $k_B T_e$, as illustrated by the dashed curve in Figure G.1(b).

At zero bias, the two occupation factors are equal, so the net current vanishes even at a charge transition. At finite temperature and for $e|V_b| \ll k_B T_e$, the current is approximately linear in bias: $I \simeq GV_b$, where $G = \partial I/\partial V_b|_{V_b=0}$ is the linear conductance. With a common equilibrium chemical potential μ_{R}, expand the occupation difference as $f_{\mathrm{S}}(\mu_N) - f_{\mathrm{D}}(\mu_N) \simeq -eV_b f'(\mu_N)$, where f is the common zero-bias Fermi function. Substitution gives

$$G = \frac{e^2}{4k_B T_e}\frac{\Gamma_{\mathrm{S}}\Gamma_{\mathrm{D}}}{\Gamma_{\mathrm{S}} + \Gamma_{\mathrm{D}}}\operatorname{sech}^2\left(\frac{\mu_N - \mu_{\mathrm{R}}}{2k_B T_e}\right). \tag{G.10}$$

In this constant-rate model, the conductance peak is centered at the gate voltage satisfying $\mu_N(V_g) = \mu_{\mathrm{R}}$.

G.1.3 Experimental procedure

To perform the measurement, connect each reservoir through its conducting contact to an external voltage output. Two outputs sharing a reference conductor, called ground, can set the source and drain voltages V_S and V_D independently. The sign convention in Eq. (G.3) corresponds to $V_b = V_D - V_S$. One choice is $V_D = 0$ and $V_S = -V_b$. The electron path through the device is source–dot–drain, with a tunnel barrier on either side of the dot (Figure G.3). A separate voltage output sets the gate voltage V_g that changes the dot potential.

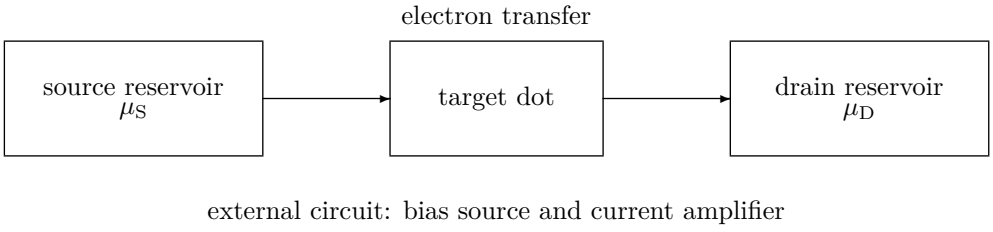


Figure G.3: Current through the target dot. Arrows indicate electron transfer from source to dot and from dot to drain. Reservoir contacts connect to an external circuit that sets their voltages and measures the current.

A current amplifier in one external branch converts the current into a voltage, $V_{out} = R_{amp}I$, where R_{amp} is the measured gain in volts per ampere. A digitizer samples this voltage, from which the current is $I = V_{out}/R_{amp}$. A transfer rate of 10^6 electrons per second gives only about 0.16 pA, which sets the scale of the required amplification.

Hold the reservoir voltages and barrier-gate settings fixed, and step V_g through the range of interest. After the current settles at each point, store it together with V_g. At a bias for which the current has a resolved plateau, its two edges give the gate voltages satisfying $\mu_N = \mu_S$ and $\mu_N = \mu_D$. In the linear regime $e|V_b| \ll k_B T_e$, use I/V_b to obtain the conductance and locate its peak. The peak position gives the equilibrium transition voltage in the small-bias limit.

To obtain the current map $I(V_g, V_b)$, repeat these gate scans for a sequence of positive and negative biases. Keep the drain voltage fixed and change the source voltage between scans to adjust the window width and reverse the transport direction.

Keep the temperature low enough to resolve neighboring transitions, and the tunnel rates sufficiently constant across the measured feature for the derived current formulas to apply. If the bias window admits another transition, reduce $|V_b|$ and repeat the gate scan to recover an isolated transition before using the two-charge-state current formula.

G.1.4 Measuring the voltage response of an addition energy

A gate scan with a very small bias locates the conductance peak. In the zero-bias limit, its position V_g^0 satisfies $\mu_N(V_g^0, 0) = \mu_R$, as in Eq. (G.10). This determines the addition energy at one voltage setting. We now want to determine how μ_N changes with gate voltage V_g and bias V_b.

Keep the drain voltage fixed and vary the source voltage, so that $\mu_D = \mu_R$ and $\mu_S = \mu_R + eV_b$. Choosing the fixed μ_R as the energy zero, a first-order expansion about $(V_g^0, 0)$ gives

$$\mu_N(V_g, V_b) \simeq -e\alpha(V_g - V_g^0) + e\beta V_b. \tag{G.11}$$

Here $\alpha = -(1/e)\partial\mu_N/\partial V_g$ and $\beta = (1/e)\partial\mu_N/\partial V_b$ are evaluated at $(V_g^0, 0)$, with the other electrode settings fixed.

To measure these coefficients, widen the window enough to resolve its two current edges, while keeping the transition isolated. Repeat the gate scan at several nearby positive biases and record the gate voltages at both edges. The two edges satisfy $\mu_N = \mu_D = 0$ and $\mu_N = \mu_S = eV_b$. Substituting these two conditions into Eq. (G.11) gives the local edge lines

$$V_b = \frac{\alpha}{\beta}(V_g - V_g^0), \qquad V_b = -\frac{\alpha}{1-\beta}(V_g - V_g^0). \tag{G.12}$$

Fit these edges in the (V_g, V_b) map, with bias on the vertical axis. For $\alpha > 0$ and $0 < \beta < 1$, their slopes are $m_+ > 0$ and $m_- < 0$. Solving for the two coefficients gives

$$\alpha = \frac{m_+|m_-|}{m_+ + |m_-|}, \qquad \beta = \frac{|m_-|}{m_+ + |m_-|}. \tag{G.13}$$

The scan thus determines the desired local responses $\partial\mu_N/\partial V_g = -e\alpha$ and $\partial\mu_N/\partial V_b = e\beta$.

As an additional use, α converts the separation of adjacent conductance peaks into an estimate of the addition-energy spacing. Let their zero-bias positions be $V_g^{(N)}$ and $V_g^{(N+1)}$, with $\Delta V_g = V_g^{(N+1)} - V_g^{(N)}$. If both thresholds shift approximately with a common response $-e\alpha$ over this interval, then their energy difference at the same gate voltage is

$$E_{\mathrm{add}}(N) = \mu_{N+1}(V_g, 0) - \mu_N(V_g, 0) \simeq e\alpha\Delta V_g.$$

For this estimate, use a representative α measured near the two peaks.

G.2 A separate charge sensor

We want to determine the target dot's electron number even when electron transfer through it is too slow to give a measurable steady current. A nearby detector can instead sense the electric field of the target charge. The measured current passes through the detector, so a charge change in the target can be detected without repeated electron transfer through the target. The sensor can also distinguish target charge configurations after the target's reservoir barrier has been closed, because their electric interaction with the sensor remains.

For a concrete setup, let the target exchange electrons with one reservoir at chemical potential μ_R. This reservoir can supply or accept an electron as the target gate voltage V_g is varied. At low temperature, the equilibrium electron number minimizes $E_N(V_g) - \mu_R N$, with all other electrode voltages fixed. A boundary between neighboring charge states therefore occurs at $\mu_N(V_g) = E_N(V_g) - E_{N-1}(V_g) = \mu_R$. Holding the reservoir contact voltage fixed keeps μ_R fixed; scanning V_g changes the target potential and hence μ_N. We seek the gate voltage at which the electron number changes, using the detector signal to identify that change.

We model the detector as a second dot connected to its own source and drain, using the sequential-tunneling description of Section G.1. Its steady current $I_s(\mu_s)$ follows Eq. (G.9), with μ_N replaced by the sensor's addition energy μ_s and with the Fermi functions and tunnel rates taken for the sensor's own reservoirs. With these reservoir voltages and tunnel rates held fixed, a change in the target charge shifts μ_s through the electric interaction between the dots and thereby changes I_s. This is *charge sensing*.

For a simple density interaction $H_{\mathrm{int}} = U_{\mathrm{sd}}\hat{N}\hat{n}_s$, $\hat{N}$ counts target electrons, $\hat{n}_s$ counts sensor electrons, and U_{sd} is an interaction energy. Adding one sensor electron increases this interaction energy by $U_{\mathrm{sd}}N$, so the same quantity contributes to the sensor's addition energy. A change δN in the target electron number shifts the sensor addition energy by $\delta\mu_s = U_{\mathrm{sd}}\,\delta N$. This shift moves the sensor threshold relative to the fixed source and drain chemical potentials, changing the availability of tunneling through the sensor. For a small shift about the chosen

sensor setting, the current change is

$$\delta I_{\mathrm{s}} \simeq \frac{\partial I_{\mathrm{s}}}{\partial \mu_{\mathrm{s}}} U_{\mathrm{sd}}\, \delta N. \tag{G.14}$$

We therefore choose a sensor setting on a sloping part of its independently measured current curve, where $\partial I_{\mathrm{s}}/\partial \mu_{\mathrm{s}}$ is appreciable. For a larger shift, the two current levels are calibrated directly. A narrow conducting constriction, called a *quantum point contact*, can play the same role: target charge changes its barrier and hence its transmission.

For two target dots, let $U_{\mathrm{s}1}$ and $U_{\mathrm{s}2}$ be their respective interaction energies with the sensor. The sensor addition energy then changes by

$$\delta \mu_{\mathrm{s}} = U_{\mathrm{s}1} \delta N_1 + U_{\mathrm{s}2} \delta N_2. \tag{G.15}$$

A transfer $(1,1) \to (0,2)$ has $\delta N_1 = -1$ and $\delta N_2 = +1$. It therefore shifts the sensor threshold by $U_{\mathrm{s}2} - U_{\mathrm{s}1}$ even though the total target electron number is unchanged. Distinguishing these configurations requires different sensitivity to charge on the two dots; equal couplings would give no signal for the transfer in this model. This spatial sensitivity is needed for the singlet–triplet readout in Section 5.2.

For several target dots, a local model of the recorded sensor signal is

$$s(t) = s_{\mathrm{bg}}(\mathbf{V}) + \sum_i \kappa_i N_i(t) + \xi(t). \tag{G.16}$$

Here s may be a sensor current or an amplified voltage, s_{bg} is the response at fixed target charge to the applied gate voltages $\mathbf{V}$, κ_i is the signal change per electron on dot i, and ξ is detector noise. This expression assumes approximately constant sensitivity over the small range being measured. It is a model for the signal conditioned on a charge record; averaging many records replaces $N_i(t)$ by $\langle \widehat{N}_i(t) \rangle$.

A tunnel event produces a step between sensor levels. Monitoring these steps reveals loading and unloading even when the target has only one reservoir, so no source–drain path exists through it. If the detector resolves individual events, this is a time-resolved charge measurement [28]. In particular, an electron leaving a dot and a subsequent loading event produce two steps that delimit a temporarily empty interval. Sensor noise limits the distinction between current levels, and a finite response time can hide an empty interval that is too short.

In the experiment, first scan the sensor gate to find a conductance peak, then set the sensor on its side, where $\partial I_{\mathrm{s}}/\partial \mu_{\mathrm{s}}$ is large. Establish the two sensor signal levels corresponding to neighboring target charge states. Hold the target reservoir voltage fixed and scan the target plunger gate, allowing time for electron exchange with the reservoir at each setting. A step between the calibrated levels marks loading or unloading. If the scan also shifts the sensor directly, compensate with its gate or measure and subtract the smooth background s_{bg}.

For a time-resolved record, keep the target settings fixed near a loading or unloading condition and record the sensor signal as a function of time. Individual steps then give event times. Figure G.4 shows the connections for a sensor coupled to two target dots.

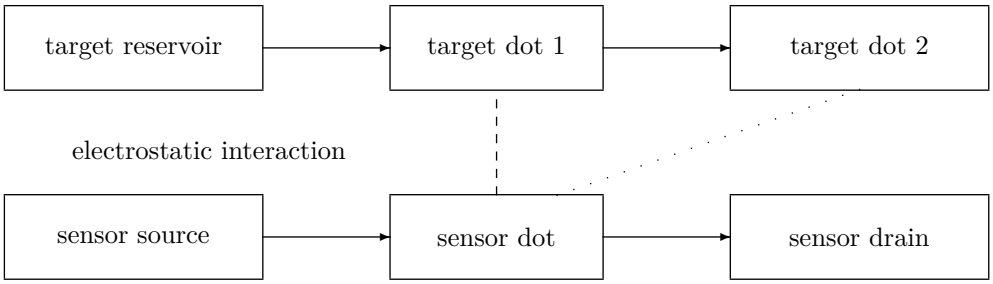


Figure G.4: Charge detection with a separate sensor. Current passes through the sensor dot. The target charges change its addition threshold through the electric interaction between the dots. Horizontal arrows indicate tunnel connections; broken lines indicate electric interactions.

H Double-Dot Charge Stability Diagrams

Section 2.2 uses the charge regions in Fig. 2.2 to choose voltages for preparing one electron on each dot. This appendix derives their boundaries from the electron-number condition and relates them to measured charge-transition lines.

H.1 Charge regions and their boundaries

For two dots, the desired result is a map of the charge configuration (N_1, N_2) as a function of the two plunger voltages V_1, V_2. This locates settings for loading each dot and for transferring an electron between dots. Keep both reservoirs at a common μ_R and first consider weak interdot tunneling, so that the occupations label approximately distinct configurations. Let $E_{N_1,N_2}(\mathbf{V})$ be the lowest energy of each configuration before tunneling mixes it with the others. The equilibrium pair minimizes

$$\Omega_{N_1,N_2}(\mathbf{V}) = E_{N_1,N_2}(\mathbf{V}) - \mu_R(N_1 + N_2). \tag{H.1}$$

This extends the electron-number construction in Section 2 to two separately resolved dots.

A boundary occurs when two minimizing configurations have equal Ω. Adding an electron from a reservoir to dot 1 gives

$$\mu_1(N_1 + 1, N_2) \equiv E_{N_1+1,N_2} - E_{N_1,N_2} = \mu_R. \tag{H.2}$$

There is an analogous condition for dot 2. An interdot transfer conserves total electron number, so the reservoir term cancels:

$$E_{N_1+1,N_2-1} = E_{N_1,N_2}. \tag{H.3}$$

These are different physical events: the first changes the total charge, while the second redistributes it.

Repulsion between dots makes the addition threshold of one depend on the occupation of the other. A line for adding to dot 1 therefore shifts when dot 2 gains an electron. Together with interdot-transfer segments, the shifted lines commonly form hexagonal cells, called the *honeycomb diagram* [29]. Figure 2.2 shows one cell and its neighbors. Its polygon is a region in voltage space, not the spatial shape of a dot. Varying orbital spacings, interactions, or gate responses changes the cell sizes and slopes.

The slopes follow directly from the boundary equations. Locally write the change in the relevant addition threshold as

$$\delta\mu_i = -e(\alpha_{i1}^{(\mu)}\delta V_1 + \alpha_{i2}^{(\mu)}\delta V_2), \qquad \left.\frac{dV_2}{dV_1}\right|_{\mu_i=\mu_R} = -\frac{\alpha_{i1}^{(\mu)}}{\alpha_{i2}^{(\mu)}}. \tag{H.4}$$

Here $\alpha_{ig}^{(\mu)} = -(1/e)\partial\mu_i/\partial V_g$ is a dimensionless addition-threshold lever arm. The superscript distinguishes it from a single-orbital lever arm in Eq. (F.3). A slope gives their ratio, but not an absolute energy scale.

Charge sensing can reveal the boundary lines throughout the map. Near-zero-bias sequential transport through the double dot is concentrated near the triple points, where three charge regions meet; Section H.3 explains this transport condition. To obtain the map by charge sensing, fix the reservoir and barrier-gate voltages. Sweep V_1 at a sequence of fixed V_2 values and record the charge-sensor signal at each pair. The resulting two-dimensional record is a *charge stability diagram* (CSD). Its changes can be compared with the boundaries just derived. Establish an empty reference region, then count loading transitions to assign absolute electron numbers. If tunneling becomes too slow, an occupied region can appear empty on the measurement timescale; the waiting-time test below is therefore part of this assignment.

H.2 Extracting transition lines from a sensor record

We want to make the boundaries between charge regions easy to identify in the measured voltage map. Recall the local sensor model:

$$s(t) = s_{\mathrm{bg}}(\mathbf{V}) + \sum_i \kappa_i N_i(t) + \xi(t). \tag{G.16}$$

Within each region, the fixed occupations give a fixed charge contribution $\sum_i \kappa_i N_i$, while s_{bg} varies smoothly with gate voltage. When the scan crosses a boundary where dot i gains an electron, the extra charge shifts the sensor's addition energy and hence its current. The change $N_i \to N_i + 1$ changes the charge contribution by κ_i, producing a step on top of the smooth background. At finite temperature, the mean signal changes smoothly between the two levels.

Differentiating with respect to the scanned gate voltage turns this step into a peak or dip, highlighting the boundary as a line in the two-dimensional map; within each region, the fixed charge contributes no derivative. For zero-mean detector noise, averaging the signal and differentiating with respect to V_g, with the other voltages fixed, gives

$$\frac{\partial \bar{s}}{\partial V_g} = \frac{\partial s_{\mathrm{bg}}}{\partial V_g} + \sum_i \kappa_i \frac{\partial \langle \widehat{N}_i \rangle}{\partial V_g}. \tag{H.5}$$

Clear steps can also be located directly in the original signal; differentiation is optional, and numerical differentiation can amplify measurement noise.

An interdot transfer from dot 2 to dot 1 changes the signal by approximately $\kappa_1 - \kappa_2$. Its line can therefore be faint when the two sensor couplings are similar.

The voltage derivative can also be measured by applying a small gate modulation. Let $S(V_g)$ be the mean sensor signal at a fixed gate voltage V_g, with all other voltages fixed. Apply $V_g(t) = V_g^0 + v_{\mathrm{ac}} \cos \omega_m t$ slowly enough that the charge follows the modulation. Omitting detector noise for the derivation, $s(t) = S[V_g(t)]$. A Taylor expansion about V_g^0 gives

$$s(t) = S(V_g^0) + v_{\mathrm{ac}} S'(V_g^0) \cos \omega_m t + O(v_{\mathrm{ac}}^2).$$

Multiply both sides by $\cos \omega_m t$ and integrate over a time τ_{av} containing an integer number of modulation periods. Keeping the first-order terms gives

$$\begin{aligned} \frac{1}{\tau_{\mathrm{av}}} \int_0^{\tau_{\mathrm{av}}} s(t) \cos \omega_m t \, dt &\simeq \frac{S(V_g^0)}{\tau_{\mathrm{av}}} \int_0^{\tau_{\mathrm{av}}} \cos \omega_m t \, dt \\ &\quad + \frac{v_{\mathrm{ac}} S'(V_g^0)}{\tau_{\mathrm{av}}} \int_0^{\tau_{\mathrm{av}}} \cos^2 \omega_m t \, dt \\ &= \frac{v_{\mathrm{ac}}}{2} S'(V_g^0). \end{aligned}$$

The two integrals on the right are 0 and $\tau_{\mathrm{av}}/2$, respectively. Writing the time average as $\overline{f} = \tau_{\mathrm{av}}^{-1} \int_0^{\tau_{\mathrm{av}}} f(t)\, dt$, we obtain

$$\frac{2}{v_{\mathrm{ac}}} \overline{s(t) \cos \omega_m t} \simeq \left. \frac{dS}{dV_g} \right|_{V_g^0}. \tag{H.6}$$

An instrument that performs this reference multiplication and averaging is a *lock-in amplifier*. Detector noise adds uncertainty to this estimate. The Taylor approximation requires a modulation amplitude small compared with the voltage scale over which the response varies. If the charge cannot follow the modulation, the measured amplitude and phase also contain dynamics.

To locate the charge-transition boundaries experimentally, we construct a map of $\partial\overline{s}/\partial V_1$ or $\partial\overline{s}/\partial V_2$ over the (V_1, V_2) plane. There are two ways to obtain this map.

For numerical differentiation, scan V_1 at a sequence of fixed V_2 values and record the mean sensor signal at each voltage pair. Divide the signal difference between neighboring samples by their voltage separation to estimate the corresponding derivative. Plotting these estimates over (V_1, V_2) highlights the charge-transition boundaries.

For lock-in measurement, scan the same voltage pairs, but apply a small modulation to V_1 or V_2 at each pair. Record the lock-in response and use Eq. (H.6) to obtain the corresponding derivative directly. Choose a modulation small compared with the feature width and slow enough for charge to follow it.

To map equilibrium charge states, allow the electrons to equilibrate at each gate-voltage setting before recording. Otherwise a charge transition can be missed because an electron has not yet entered. For an initially empty accessible state with constant loading rate Γ_{in} and negligible unloading, the probability of loading during a waiting time τ is

$$P_{\mathrm{load}}(\tau) = 1 - e^{-\Gamma_{\mathrm{in}}\tau}. \tag{H.7}$$

The wait must therefore be long compared with $\Gamma_{\mathrm{in}}^{-1}$, as well as any relevant relaxation times. To check whether insufficient waiting caused a missing line, repeat that region with a longer wait at each voltage setting. A line restored by allowing more time to load reveals a failure to equilibrate in the faster scan, an important test in the last-electron regime [30].

H.3 Detecting transport cycles: triangles and triple points

We want to identify the gate-voltage pairs that support a current through two dots in series. First consider positive finite bias, $\mu_{\mathrm{S}} - \mu_{\mathrm{D}} = eV_b > 0$. A forward cycle requires an electron to enter dot 1 from the source, transfer to dot 2, and leave into the drain. Denote the configurations by $A = (N_1, N_2)$, $B = (N_1 + 1, N_2)$, and $C = (N_1, N_2 + 1)$. The cycle is $A \to B \to C \to A$. Define the addition energies $\widetilde{\mu}_1 = E_B - E_A$ and $\widetilde{\mu}_2 = E_C - E_A$, with all energies evaluated at the chosen bias. At zero temperature and negligible tunnel broadening, entry, downhill transfer, and exit require

$$\mu_{\mathrm{S}} \geq \widetilde{\mu}_1 \geq \widetilde{\mu}_2 \geq \mu_{\mathrm{D}}. \tag{H.8}$$

The first inequality permits loading, the middle one permits interdot transfer from B to C without absorbing energy, since $E_C - E_B = \widetilde{\mu}_2 - \widetilde{\mu}_1$, and the last permits unloading. In the $(\widetilde{\mu}_1, \widetilde{\mu}_2)$ plane they enclose a triangle. Approximately linear gate responses map it to a triangle in (V_1, V_2), whose extent grows with eV_b.

The interior requires a way to release the interdot energy difference, for example by emitting a lattice vibration, or phonon. Purely elastic transfer between two sharp levels is resonant

along $\widetilde{\mu}_1 = \widetilde{\mu}_2$ instead. Excited states create additional resonances, and spin selection can block an otherwise allowed cycle. A bias triangle is therefore an allowed transport region, not necessarily a uniformly bright patch [29].

Now reduce the bias toward zero. As both reservoir chemical potentials approach a common μ_{R}, Eq. (H.8) forces

$$\widetilde{\mu}_1 = \widetilde{\mu}_2 = \mu_{\mathrm{R}}.$$

Because B and C each contain one more electron than A, their grand-energy differences are $\Omega_B - \Omega_A = \widetilde{\mu}_1 - \mu_{\mathrm{R}}$ and $\Omega_C - \Omega_A = \widetilde{\mu}_2 - \mu_{\mathrm{R}}$. Hence

$$\Omega_A = \Omega_B = \Omega_C. \tag{H.9}$$

These two independent equalities select isolated points in the two-dimensional voltage plane: the finite-bias triangles shrink to *triple points*. The nearby complementary cycle through $(N_1 + 1, N_2 + 1)$ gives the other member of a triple-point pair. Near zero bias, the current is confined to small neighborhoods of these points, with finite temperature broadening the features. At exactly zero DC bias, the net current vanishes; it is the linear conductance that has peaks near the triple points.

To locate these transport regions experimentally, connect the two dots in series between source and drain, choose a positive finite bias, and record the current while scanning the two plunger voltages. Keep the reservoir voltages fixed during each map. Repeat at successively smaller biases to follow the triangles as they shrink toward the triple points. Reversing the bias probes transport in the opposite direction. Compare these maps with charge-sensor boundaries acquired at the corresponding settings. A sensor line can identify a charge transition even where no complete transport cycle is resolved. The triangle edges, rather than their brightness alone, are the features to compare with Eq. (H.8).

I Spectroscopy of Dot Excitations

The two-level restriction in Section 4.1 depends on the gaps to excluded orbital and valley states. This appendix explains how those gaps can be inferred from the opening of additional transport or loading channels, including the effect of spin–valley mixing near an avoided crossing.

Let $|N, a\rangle$ denote an N-electron state, with $a = 0$ for the ground state and $a = 1$ for the first excited state. A narrow source–drain window allows the transport cycle

$$|N-1, 0\rangle \xrightarrow{\text{source}\to\text{dot}} |N, 0\rangle \xrightarrow{\text{dot}\to\text{drain}} |N-1, 0\rangle.$$

Increasing the source–drain bias widens the window and can also allow the cycle

$$|N-1, 0\rangle \xrightarrow{\text{source}\to\text{dot}} |N, 1\rangle \xrightarrow{\text{dot}\to\text{drain}} |N-1, 0\rangle.$$

The additional entry process can change the current.

Write $E_{N,a}$ for the energy of $|N, a\rangle$. Starting from an $(N-1)$-electron state b, the energy needed to add an electron and leave the dot in state a is

$$\mu_N^{ab} = E_{N,a} - E_{N-1,b}. \tag{I.1}$$

For both entry processes starting from $b = 0$, subtraction cancels the initial-state energy:

$$\mu_N^{a0} - \mu_N^{00} = E_{N,a} - E_{N,0} \equiv \Delta_a. \tag{I.2}$$

All energies in this equality refer to the same electrode settings. This is why we compare transitions with a common initial state and the same final electron number.

I.1 Excitation energies from current

At zero temperature and positive bias, use the sequential-tunneling model with sharp dot levels from Section G.1. The narrow-window situation described above satisfies

$$\mu_{\mathrm{D}} < \mu_N^{00} < \mu_{\mathrm{S}} < \mu_N^{10},$$

where $a = 1$ labels the first excited state. The source has occupied electron states at the energy μ_N^{00} needed for ground-state entry, but none at μ_N^{10}. Hold the drain voltage fixed and increase μ_{S} by increasing positive V_b. Entry into state a becomes energetically allowed when $\mu_N^{a0} < \mu_{\mathrm{S}}$; the equality $\mu_N^{a0} = \mu_{\mathrm{S}}$ defines its source-loading boundary. These addition energies are evaluated at the applied voltages, since changing the bias can also shift the dot energies. Figure I.1 shows this opening of a second entry process. An electron entering the excited state may leave directly, or first relax to the ground state and then leave. Either route can contribute to the current.

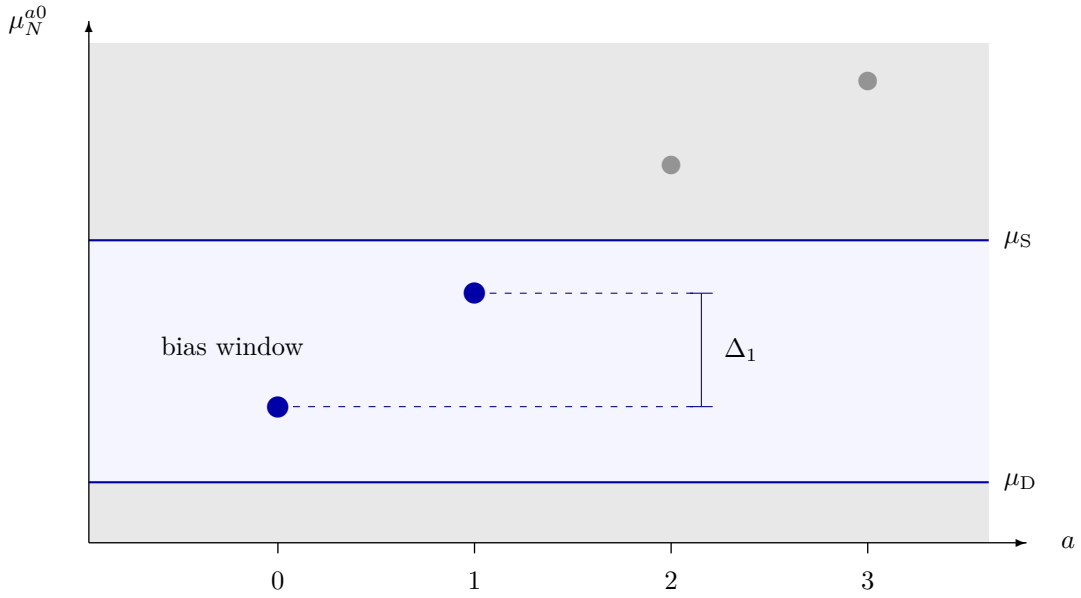


Figure I.1: Bias window for transitions from the $(N-1)$-electron ground state to N-electron states labeled by a. Each point is an addition energy $\mu_N^{a0} = E_{N,a} - E_{N-1,0}$. The wider window includes the addition energies for the ground state ($a = 0$) and first excited state ($a = 1$). Their difference gives $\Delta_1 = \mu_N^{10} - \mu_N^{00} = E_{N,1} - E_{N,0}$. Gray points mark higher excited-state transitions outside the window. Transitions to other charge sectors are excluded.

To see explicitly why the current changes, consider fast relaxation within the N-electron sector, followed by escape from its ground state at rate Γ_{D}. Let every escape return the dot to its $(N-1)$-electron ground state. While the ground-state transition remains inside the window, the total source entry rate is

$$\Gamma_{\mathrm{in}} = \begin{cases} \Gamma_{\mathrm{S0}}, & \mu_N^{00} < \mu_{\mathrm{S}} < \mu_N^{10}, \\ \Gamma_{\mathrm{S0}} + \Gamma_{\mathrm{S1}}, & \mu_N^{10} < \mu_{\mathrm{S}} < \mu_N^{20}. \end{cases}$$

Here $\Gamma_{\mathrm{S}a}$ is the entry rate into state a when that process is allowed. Rates into orthogonal final states add, as explained in Appendix Q.1. With negligible time spent in the excited state

before relaxation, the rate equation of Section G.1.2 becomes

$$\dot{p} = \Gamma_{\mathrm{in}}(1-p) - \Gamma_{\mathrm{D}}p, \qquad I = e\frac{\Gamma_{\mathrm{in}}\Gamma_{\mathrm{D}}}{\Gamma_{\mathrm{in}}+\Gamma_{\mathrm{D}}}, \tag{I.3}$$

where p is the probability of N electrons. The new entry rate therefore produces a current step at $\mu_N^{10} = \mu_{\mathrm{S}}$. Finite temperature rounds the step. If relaxation is slow, the excited-state occupation needs its own rate equation; the current change can then have a different size or sign. A state with negligible entry matrix element may produce no resolved feature.

In the experiment, first locate the ground-state diamond edge and neighboring charge transitions. Then increase positive V_b in steps, keeping the drain voltage fixed, and scan V_g at each bias. Store the current $I(V_g, V_b)$. Taking its numerical bias derivative $\partial I/\partial V_b$ emphasizes the current steps. Their positions trace curves in this *differential-conductance map*: one follows $\mu_N^{00} = \mu_{\mathrm{S}}$ and another follows $\mu_N^{a0} = \mu_{\mathrm{S}}$. We compare this pair of source-alignment curves, rather than the brightness of arbitrary features in the map.

At a fixed bias, let these two curves occur at V_g^0 and V_g^a. If the addition energies shift with the same approximately constant response $\partial\mu/\partial V_g = -e\alpha$ across their separation, then

$$\begin{aligned} 0 &= \mu_N^{a0}(V_g^a) - \mu_N^{00}(V_g^0) \simeq \Delta_a - e\alpha(V_g^a - V_g^0), \\ \Delta_a &\simeq e\alpha(V_g^a - V_g^0). \end{aligned} \tag{I.4}$$

Here Δ_a is evaluated at V_g^0, and $\alpha > 0$ is determined from the local voltage response as in Section G.1.4.

Increasing the bias can also permit another electron to enter, giving $N \to N+1$ and eventually $N+2$. These are different transitions, not excitations at fixed N. For the interpretation above, other charge sectors must remain inaccessible. In particular, to forbid further loading from every occupied N-electron state a, require

$$E_{N+1,0} - E_{N,a} = (E_{N+1,0} - E_{N,0}) - \Delta_a > \mu_{\mathrm{S}}. \tag{I.5}$$

An excited state reduces the cost of adding the next electron, so checking only the ground-state cost is insufficient. At finite temperature the margin should be large compared with $k_B T_e$. Use the neighboring charge boundaries to delimit the scan and check this condition as excitations become accessible. If other charge states or initial excited states participate, include their transitions explicitly before applying Eq. (I.2).

I.2 Excitation energies from loading probabilities

For a dot coupled to one reservoir, we can instead prepare $N-1$ electrons and ask whether an electron enters during a fixed time. As the gate lowers successive entry thresholds below the reservoir chemical potential, more final states become available and loading becomes faster. No steady current through the target is needed.

Prepare the $(N-1)$-electron ground state before every trial. Let V_L be the gate voltage during loading and Γ_a the bare entry rate into state a. The occupied reservoir states give

$$\Gamma_{\mathrm{load}}(V_L) = \sum_a \Gamma_a f_{\mathrm{R}}[\mu_N^{a0}(V_L)]. \tag{I.6}$$

At zero temperature this rate is zero before the first threshold, Γ_0 after the ground-state threshold, and $\Gamma_0 + \Gamma_1$ after the first excited-state threshold. Assume unloading during the loading interval is negligible, either because the occupied states lie sufficiently below μ_{R} or because rapid internal relaxation traps the loaded electron. Also exclude further loading using Eq. (I.5) with μ_{S} replaced by μ_{R}.

If $q(t)$ is the probability that no electron has yet entered, then $\dot{q} = -\Gamma_{\text{load}} q$ with $q(0) = 1$. The probability of finding N electrons after a loading time τ_L is therefore

$$P_N(\tau_L) = 1 - e^{-\Gamma_{\text{load}}\tau_L}, \qquad \Gamma_{\text{load}} = -\frac{\ln[1 - P_N(\tau_L)]}{\tau_L}. \tag{I.7}$$

To measure P_N, pulse the gate to V_L, wait τ_L, then close the reservoir barrier to retain the charge during sensor readout. The fraction of repeated trials reporting N electrons estimates P_N. Repeat at different V_L and loading times; choose times for which the probability is not saturated at one. Equation (I.7) then converts the counts into a loading-rate curve. Its successive onsets give the threshold voltages [28].

If unloading is appreciable, extract the rate from the initial growth instead. For times short compared with the tunneling waiting times, the rate description gives

$$P_N(V_L, \tau_L) \simeq \Gamma_{\text{load}}(V_L)\tau_L. \tag{I.8}$$

Entry followed by exit requires two events and contributes only at higher order in τ_L. Repeated measurements at several short loading times therefore give Γ_{load} from the initial slope.

Two onsets aligned with the same μ_{R} give a gap by the same subtraction as Eq. (I.4), using the voltage response for this loading configuration. Without an independent voltage-to-energy conversion, this experiment determines threshold separations in volts, not absolute gaps in energy units.

I.3 Spin–valley mixing

Consider loading one electron into a dot whose orbital excitations lie outside the energy range being probed. The valley eigenstates $|v_-\rangle$ and $|v_+\rangle$ include the interface-induced valley coupling described in Section C. Their separation is $\Delta_v > 0$. At zero field, each valley has two degenerate spin states. As the magnetic field B increases, $|v_-, \uparrow\rangle$ rises toward $|v_+, \downarrow\rangle$. A spin-dependent coupling mixes this pair and prevents their energies from crossing. We measure the minimum separation to determine that coupling.

Take the field along z, with $B \geq 0$ and the same $g > 0$ in both valleys. Neglect changes of the orbital and valley energies with field. Before including spin–valley mixing, the four energies, measured relative to $|v_-, \downarrow\rangle$, are[5]

$$\begin{array}{c|cc} & \downarrow & \uparrow \\ \hline v_- & 0 & E_Z \\ v_+ & \Delta_v & \Delta_v + E_Z \end{array}, \qquad E_Z = g\mu_B B. \tag{I.9}$$

The zero-field valley gap is obtained by applying the loading spectroscopy of Section I.2 to an empty dot. If μ_- and μ_+ are the addition energies into the two valleys, their difference is

$$\Delta_v = \mu_+ - \mu_-. \tag{I.10}$$

The two degenerate spins in each valley contribute to the loading rate at the same energy. Under the assumed exclusion of orbital excitations, these two distinct loading energies correspond to the two valleys.

[5]This four-state description is specific to one electron. For several electrons, exchange can separate different total-spin multiplets already at $B = 0$, and one energy can contain more than one multiplet.

I.4 The spin–valley avoided crossing

As B increases, the spin excitation energy E_Z approaches the valley gap Δ_v, bringing $|v_-,\uparrow\rangle$ and $|v_+,\downarrow\rangle$ close in energy. Without coupling, the two energies would cross at $g\mu_B B = \Delta_v$. A spin-dependent interaction, such as spin–orbit interaction, can couple these states. Denote this interaction by V_{sv} and its matrix element by

$$\lambda = \langle v_-,\uparrow|V_{\mathrm{sv}}|v_+,\downarrow\rangle.$$

To describe the crossing, retain this pair of nearly degenerate states. The two remaining states lie about Δ_v below and above the pair. Couplings to them therefore give small perturbative corrections when their matrix elements are much smaller than Δ_v. The other opposite-spin, opposite-valley pair, $|v_-,\downarrow\rangle$ and $|v_+,\uparrow\rangle$, can also couple, but its separation is $\Delta_v + E_Z$, rather than the small difference $\Delta_v - E_Z$. Neglecting the off-resonant corrections and diagonal shifts from V_{sv}, we obtain, in the basis $(|v_-,\uparrow\rangle, |v_+,\downarrow\rangle)$,

$$H_{\mathrm{cross}} = \begin{pmatrix} E_Z & \lambda \\ \lambda^* & \Delta_v \end{pmatrix}, \qquad \Delta_\pm = \frac{E_Z + \Delta_v}{2} \pm \sqrt{\frac{(E_Z - \Delta_v)^2}{4} + |\lambda|^2}. \tag{I.11}$$

Here Δ_- and Δ_+ are the excitation energies of the two mixed states, relative to the unperturbed $|v_-,\downarrow\rangle$ ground state [31]. Their separation is

$$d(B) = \Delta_+ - \Delta_- = \sqrt{(g\mu_B B - \Delta_v)^2 + 4|\lambda|^2}. \tag{I.12}$$

For nonzero λ the levels remain separated even at the field where the uncoupled energies coincide: this is the *avoided crossing*. If g, Δ_v, and λ are approximately constant over the field range,

$$B_* = \frac{\Delta_v}{g\mu_B}, \qquad d(B_*) = \min_B d(B) = 2|\lambda|. \tag{I.13}$$

Measuring this minimum separation therefore determines the magnitude of the coupling.

I.5 Extracting the coupling from loading spectra

In loading spectroscopy, an electron can enter the empty dot and occupy either mixed state. The two processes have the same initial state, so their addition-energy difference equals $\Delta_+ - \Delta_-$, as in Section I.2. We use this difference to measure $d(B)$.

At each fixed B, use that section's $N = 1$ loading protocol, keeping entry of a second electron forbidden. Scan the loading gate voltage V_L and obtain the loading-rate curve from Eq. (I.8). After the ground-state loading step, entry into each mixed state adds a further step. Fit these steps with the Fermi functions in Eq. (I.6) to locate the two alignment voltages: $V_{L,1}(B)$ for the state at Δ_- and $V_{L,2}(B)$ for the state at Δ_+.

Suppose both addition energies have the same approximately constant response $\partial\mu/\partial V_L = -e\alpha_L$, with $\alpha_L > 0$, across this interval. Then Eq. (I.4) gives

$$d(B) \simeq e\alpha_L\big[V_{L,2}(B) - V_{L,1}(B)\big]. \tag{I.14}$$

If α_L is not already known, an independently measured spin gap can provide an energy reference. At a field where $E_Z \ll \Delta_v$ is resolved, the spin-resonance measurement of Section N.1.1 gives a frequency f_{res} with $hf_{\mathrm{res}} = E_Z$. Use the loading measurement at that field to find the voltage separation δV_L^s between $|v_-,\downarrow\rangle$ and $|v_-,\uparrow\rangle$. Their known energy separation gives

$$e\alpha_L \simeq \frac{E_Z}{\delta V_L^s} = \frac{hf_{\mathrm{res}}}{\delta V_L^s}. \tag{I.15}$$

Using this response for the crossing measurement requires it to remain approximately unchanged over the relevant gate and field range. The common-response approximation also requires the gap to vary negligibly between the two loading voltages.

Repeat the loading scan at fields on both sides of B_*, with the same gate settings apart from the scan. Convert each voltage separation to $d(B)$ and compare the resulting curve with Eq. (I.12). Its minimum gives $2|\lambda|$; the points on either side check the predicted field dependence. Both loading steps must be detectable, and their minimum separation must exceed the experimental energy resolution for this direct extraction.

The measured coupling quantifies the departure from a pure lower-valley spin state. Away from the crossing, where $|\lambda| \ll |\Delta_v - E_Z|$, first-order perturbation theory gives

$$|\widetilde{\uparrow}\rangle \simeq |v_-,\uparrow\rangle - \frac{\lambda^*}{\Delta_v - E_Z}|v_+,\downarrow\rangle, \qquad w_v \simeq \frac{|\lambda|^2}{(\Delta_v - E_Z)^2}. \tag{I.16}$$

Here w_v is the leading higher-valley weight due to this coupling, not a gate-error probability. It estimates how far from the crossing the qubit should operate to keep that admixture small, using λ at the operating field or assuming that its variation from the measured crossing is negligible.

J From Orbital Hamiltonians to Spin Exchange

The general many-electron matrix in Appendix E.2 retains all interactions within a chosen orbital space. This appendix develops the more restrictive model used to derive the exchange interaction in Section 4.3. We first specify the conditions for a single-spatial-state Hubbard model. We then eliminate its high-energy charge states to obtain spin exchange, check the splitting against exact diagonalization, and calculate the singlet charge admixture that enters interactions between encoded qubits.

J.1 One spatial state per dot: assumptions of the Hubbard model

For two neighboring spin qubits, the relevant charge states include one electron on each dot and the doubly occupied states reached by moving an electron between them. The one-electron spatial states are concentrated near individual dots, but their tails extend into neighboring wells and permit tunneling. Repulsion changes the energy cost of this transfer. We need a model that retains these charge states and the tunneling between them before deriving their effect on the spins. Starting from Appendix E.2, we construct such a model for dots with a small number of electrons.

Restricting the spatial states. Choose one orthonormal localized spatial state w_i per dot, including its local valley composition, and retain its two spin states. We neglect spin-flip tunneling. The retained spatial state has energy ϵ_i and can hold zero, one, or two electrons; the doubly occupied state is a spin singlet. We use this same spatial basis for the charge configurations being compared. The model therefore retains changes in local occupation, including double occupation during tunneling.

This restriction assumes that other orbital–valley states have negligible thermal occupation and that their mixing into the retained states gives corrections below the required accuracy. The condition must hold in every charge sector used in the calculation, including the doubly occupied states that enter a virtual tunneling process. A large single-electron excitation gap alone does not establish this: electron repulsion can change the ordering and composition of

the two-electron states. If excited orbitals or valleys contribute appreciably, the local spatial basis must be enlarged.

Putting two opposite-spin electrons on the same dot costs an additional repulsion energy U_i. Occupying different dots i, j gives a direct repulsion V_{ij}. Write $|w_i(\mathbf{r})|^2$ for the spatial probability density, summed over retained valley components when needed. From the pair matrix elements,

$$\begin{aligned} U_i &= \mathcal{V}_{iiii} = \int d^3r\, d^3r'\, |w_i(\mathbf{r})|^2 W(\mathbf{r},\mathbf{r}') |w_i(\mathbf{r}')|^2, \\ V_{ij} &= \mathcal{V}_{ijij} = \int d^3r\, d^3r'\, |w_i(\mathbf{r})|^2 W(\mathbf{r},\mathbf{r}') |w_j(\mathbf{r}')|^2. \end{aligned} \tag{J.1}$$

These integrals calculate the repulsion energies from the orbital charge distributions and their screened interaction.

Restricting the interaction terms. Within this spatial basis, a further approximation retains only U_i and V_{ij} from the pair interaction. This requires the remaining pair matrix elements to give negligible corrections to the energies or dynamics being calculated. The spatial-state truncation alone does not imply this condition. Keeping these density interactions together with the one-electron tunneling from Eq. (E.5) gives the *extended Hubbard Hamiltonian*

$$\begin{aligned} H_{\mathrm{Hub}} &= \sum_i \epsilon_i n_i - \sum_{i<j,\sigma} \left(t_{ij} c_{i\sigma}^\dagger c_{j\sigma} + t_{ij}^* c_{j\sigma}^\dagger c_{i\sigma} \right) \\ &\quad + \sum_i U_i n_{i\uparrow} n_{i\downarrow} + \sum_{i<j} V_{ij} n_i n_j, \\ n_{i\sigma} &= c_{i\sigma}^\dagger c_{i\sigma}, \qquad n_i = n_{i\uparrow} + n_{i\downarrow}. \end{aligned} \tag{J.2}$$

The operator n_i counts electrons on dot i. Each displayed term now has a specific origin: localized-state energy, coherent transfer, same-dot repulsion, and interdot repulsion. Local Zeeman terms can be added when resolving spins.

On a charge configuration with $N_i = 0, 1, 2$, the operators n_i and $n_{i\uparrow} n_{i\downarrow}$ have eigenvalues N_i and $N_i(N_i - 1)/2$, respectively. In the occupation-number basis of this Hubbard model, the diagonal matrix element is therefore

$$E_{\mathrm{ch}}(\{N_i\}) = \sum_i \epsilon_i N_i + \sum_i U_i \frac{N_i(N_i-1)}{2} + \sum_{i<j} V_{ij} N_i N_j. \tag{J.3}$$

Tunneling supplies off-diagonal matrix elements between these basis states, so the many-electron eigenenergies require diagonalizing H_{Hub}. The expression above relies on the spatial-state and interaction restrictions already specified; spin-dependent energies are omitted here.

This model omits other allowed pair matrix elements. For example, the Coulomb interaction connects the spatial product states $w_i(\mathbf{r}_1) w_j(\mathbf{r}_2)$ and $w_j(\mathbf{r}_1) w_i(\mathbf{r}_2)$, with matrix element $K_{ij} = \mathcal{V}_{ijji}$. In the subspace with one electron in each orbital and no tunneling, their symmetric spatial combination has energy $V_{ij} + K_{ij}$, and their antisymmetric combination has energy $V_{ij} - K_{ij}$. Fermionic antisymmetry pairs these with a spin singlet and triplet, respectively. This *direct Coulomb exchange* therefore contributes $E_T - E_S = -2K_{ij}$ through the exchange symmetry of the spatial state at zero tunneling. When calculating a small spin-exchange splitting, the size of $2K_{ij}$ must be judged against that splitting and the accuracy sought; its being small compared with U_i is insufficient.

The full interaction can also transfer two electrons together between orbitals, or transfer one with an amplitude depending on the other electron's orbital. These are called pair hopping and occupation-dependent hopping [32]. They can be restored from $\mathcal{V}_{ijkl}$ if their effects matter.

The Hubbard model obtained under these conditions still contains empty and doubly occupied dots, and allows tunneling to mix charge configurations. A Hamiltonian acting only on spins requires the further charge-state elimination in Appendix J.2: there we restrict to one electron per dot and require the positive double-occupation costs to be much larger than the tunnel coupling.

J.2 Keeping spin states and eliminating costly excitations

Starting from the Hubbard Hamiltonian in Eq. (J.2), we seek a description in which the electron number and lowest orbital–valley state on each dot are fixed, while spins remain free to evolve. Virtual tunneling through excited charge states shifts the energies within this low-energy set. We retain these shifts when deriving its effective Hamiltonian.

To derive their effect, let P project onto the states we want to keep, and $Q = I - P$ onto the remaining states, with I the identity. Write an eigenstate as $|\Psi\rangle = |p\rangle + |q\rangle$, where $|p\rangle = P|\Psi\rangle$ and $|q\rangle = Q|\Psi\rangle$. Its two coupled equations are

$$\begin{aligned} PHP|p\rangle + PHQ|q\rangle &= E|p\rangle, \\ QHP|p\rangle + QHQ|q\rangle &= E|q\rangle. \end{aligned} \tag{J.4}$$

Solving the second for $|q\rangle$ and substituting into the first gives

$$\begin{aligned} |q\rangle &= (E - QHQ)^{-1} QHP|p\rangle, \\ H_{\text{eff}}(E) &= PHP + PHQ(E - QHQ)^{-1}QHP. \end{aligned} \tag{J.5}$$

The inverse is taken only in the excluded space, away from its eigenvalues. Reading the second term from right to left, QHP takes a retained state out, the energy denominator weights the intermediate amplitude, and PHQ brings it back. This is the origin of a virtual correction.

If the coupling between the two subspaces is small compared with their energy separation, this expression can be expanded. An equivalent construction, the Schrieffer–Wolff transformation, produces an energy-independent low-energy Hamiltonian order by order [33]. With one electron per dot and approximately spin-rotation-invariant exchange, its leading form is

$$H_{\text{spin}} = E_{\text{off}} + \sum_i \mathbf{b}_i \cdot \mathbf{s}_i + \sum_{i<j} J_{ij} \mathbf{s}_i \cdot \mathbf{s}_j + \delta H_{\text{spin}}, \qquad \mathbf{s}_i = \boldsymbol{\sigma}_i/2. \tag{J.6}$$

E_{off} is a common energy shift, $\mathbf{b}_i$ is the local Zeeman energy vector, and $\boldsymbol{\sigma}_i$ is the vector of Pauli operators for spin i. The remainder δH_{spin} contains additional terms, such as anisotropic exchange and multispin interactions.

For two dots, a hop from the $(1,1)$ configuration produces $(2,0)$ or $(0,2)$. These states are intermediate because a second hop returns to $(1,1)$. With detuning $\varepsilon = \epsilon_1 - \epsilon_2$, their costs and leading tunneling-induced exchange are

$$\begin{aligned} \Delta_{20} &= U_1 - V_{12} + \varepsilon, & \Delta_{02} &= U_2 - V_{12} - \varepsilon, \\ J_{\text{se}} &= 2|t_{12}|^2 \left(\frac{1}{\Delta_{20}} + \frac{1}{\Delta_{02}} \right). \end{aligned} \tag{J.7}$$

Both costs must be positive and much larger than $|t_{12}|$. A small direct-exchange correction contributes $-2K_{12}$ at leading order. If a charge, orbital, or valley excitation becomes comparable to its coupling to the retained states, it should be kept explicitly in the retained space.

Fermionic matrix elements and charge elimination. The exchange result used in Eq. (4.10) follows by retaining the singlet's two virtual charge channels. Specialize Eq. (J.2) to two dots with $U_1 = U_2 = U$, $V_{12} = V$, and real $t_{12} = t_c$. Write H_t for its tunneling term. The charge costs Δ_{20}, Δ_{02} are given in Eq. (4.9). We evaluate the tunneling matrix elements in the $(1,1)$ singlet–triplet basis

$$\begin{aligned} |S_{11}\rangle &= \frac{|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle}{\sqrt{2}}, \quad |T_{0,11}\rangle = \frac{|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle}{\sqrt{2}}, \\ |T_{+,11}\rangle &= |\uparrow\uparrow\rangle, \qquad\qquad |T_{-,11}\rangle = |\downarrow\downarrow\rangle. \end{aligned} \tag{J.8}$$

The subscript 11 labels charge occupation; $|T_{11}\rangle$ below denotes any triplet. Fix the doubly occupied singlet phases by $|D_{20}\rangle = c^\dagger_{1\uparrow}c^\dagger_{1\downarrow}|\text{vac}\rangle$ and $|D_{02}\rangle = c^\dagger_{2\uparrow}c^\dagger_{2\downarrow}|\text{vac}\rangle$, where $|\text{vac}\rangle$ is the empty-dot state. For product states, create the dot-1 electron before the dot-2 electron, so $|\uparrow\downarrow\rangle = c^\dagger_{1\uparrow}c^\dagger_{2\downarrow}|\text{vac}\rangle$. Applying H_t gives

$$\begin{aligned} H_t|\uparrow\downarrow\rangle &= -t_c(|D_{20}\rangle + |D_{02}\rangle), \\ H_t|\downarrow\uparrow\rangle &= +t_c(|D_{20}\rangle + |D_{02}\rangle). \end{aligned} \tag{J.9}$$

The two amplitudes add for $|S_{11}\rangle$ and cancel for $|T_{0,11}\rangle$. Pauli exclusion blocks the polarized triplets. Thus the singlet couples to both doubly occupied states, while every triplet is uncoupled:

$$|\langle D_{20}|H_t|S_{11}\rangle| = |\langle D_{02}|H_t|S_{11}\rangle| = \sqrt{2}|t_c|, \qquad \langle D_{20}|H_t|T_{11}\rangle = \langle D_{02}|H_t|T_{11}\rangle = 0. \tag{J.10}$$

Eliminating charge excitations. We now calculate the singlet energy shift caused by its two tunneling channels. In the basis $(|S_{11}\rangle, |D_{20}\rangle, |D_{02}\rangle)$, subtracting the reference energy $E_{11} = \epsilon_1 + \epsilon_2 + V$ gives the matrix

$$M_S = \begin{pmatrix} 0 & -\sqrt{2}t_c & -\sqrt{2}t_c \\ -\sqrt{2}t_c & \Delta_{20} & 0 \\ -\sqrt{2}t_c & 0 & \Delta_{02} \end{pmatrix}. \tag{J.11}$$

Let $|\widetilde{S}\rangle = a_S|S_{11}\rangle + a_{20}|D_{20}\rangle + a_{02}|D_{02}\rangle$ be its normalized low-energy eigenstate, with energy shift δE_S. To eliminate the excited charge states, solve for their amplitudes a_{20}, a_{02} in terms of a_S. The eigenvalue equation reads

$$\begin{aligned} \delta E_S a_S &= -\sqrt{2}t_c(a_{20} + a_{02}), \\ (\Delta_{20} - \delta E_S)a_{20} &= \sqrt{2}t_c a_S, \\ (\Delta_{02} - \delta E_S)a_{02} &= \sqrt{2}t_c a_S. \end{aligned} \tag{J.12}$$

For $|\delta E_S| \ll \Delta_{20}, \Delta_{02}$, the last two equations give

$$a_{20} \simeq \frac{\sqrt{2}t_c}{\Delta_{20}}a_S, \qquad a_{02} \simeq \frac{\sqrt{2}t_c}{\Delta_{02}}a_S. \tag{J.13}$$

Substitution into the first line of Eq. (J.12) gives the singlet shift; the triplet has no corresponding channel:

$$\delta E_S = -\frac{2t_c^2}{\Delta_{20}} - \frac{2t_c^2}{\Delta_{02}}, \qquad \delta E_T = 0. \tag{J.14}$$

Their difference is the exchange energy $J = E_T - E_S$ in Eq. (4.10). The correction within $(1,1)$ is $-JP_S$, where $P_S = |S_{11}\rangle\langle S_{11}|$. Since $\mathbf{s}_1 \cdot \mathbf{s}_2$ has singlet and triplet eigenvalues $-3/4$ and $1/4$, $P_S = I/4 - \mathbf{s}_1 \cdot \mathbf{s}_2$. Dropping the common energy shift $-J/4$ gives $H_{\text{ex}} = J\mathbf{s}_1 \cdot \mathbf{s}_2$. Squaring the small charge amplitudes, with $|a_S|^2 \simeq 1$, gives the total double-occupation weight in the low-energy singlet:

$$P_D = |a_{20}|^2 + |a_{02}|^2 \simeq 2t_c^2\left(\frac{1}{\Delta_{20}^2} + \frac{1}{\Delta_{02}^2}\right) \ll 1. \tag{J.15}$$

Comparison with exact diagonalization. We can check this low-energy expansion against exact diagonalization of the two-site Hubbard Hamiltonian. Take $U_1 = U_2 = U$, $V_{12} = V$, zero detuning, and real hopping $t_{12} = t$, with Zeeman and direct-exchange terms omitted. Set $D = U - V > 0$. With two electrons, tunneling couples the $(1,1)$ singlet $|S_{11}\rangle$ to the symmetric combination of the doubly occupied states, $|D_+\rangle = (|\uparrow\downarrow, 0\rangle + |0, \uparrow\downarrow\rangle)/\sqrt{2}$. Relative to the triplet energy, the matrix in the basis $(|S_{11}\rangle, |D_+\rangle)$ and its exact singlet–triplet splitting are

$$M_S = \begin{pmatrix} 0 & -2t \\ -2t & D \end{pmatrix}, \qquad J_{\text{exact}} \equiv E_T - E_S = \frac{\sqrt{D^2 + 16t^2} - D}{2}. \tag{J.16}$$

Diagonalization retains double occupation explicitly. Expanding for $|t| \ll D$ gives $J_{\text{exact}} \simeq 4t^2/D$, agreeing with Eq. (J.7) when $\Delta_{20} = \Delta_{02} = D$.

Virtual processes can also involve more than two dots. On a four-dot loop, four successive hops can restore one electron per dot while cyclically permuting the spins. For comparable hopping amplitudes t and charge gaps Δ, the resulting cyclic exchange is of order t^4/Δ^3. The effective multispin term thus arises from successive virtual transfers generated by one-electron tunneling and pairwise repulsion.

Finally, a small model can be constructed either by evaluating matrix elements in a chosen basis or by adjusting its parameters to reproduce selected energies from a larger calculation. The second procedure can incorporate excluded-state effects into effective parameters. Matching theoretical energies in this way is a theoretical model reduction. Its accuracy must be tested in the states and voltage range where the small model will be used.

K Measuring Interdot Tunneling

The exchange result in Section 4.3 depends on the coherent interdot tunnel coupling. This appendix shows how to determine that parameter from the charge distribution of one electron shared between two dots.

Consider one electron confined to two neighboring dots. Let ϵ_1 and ϵ_2 be the energies of the states localized in the left and right dots. Changing the gate voltages varies their energy difference $\varepsilon = \epsilon_1 - \epsilon_2$, called detuning. We follow n_L, the probability of finding the electron in the left dot, as a function of ε. First consider the ground state at zero temperature. Without tunneling, the electron occupies whichever dot has lower energy:

$$n_L(\varepsilon) = \begin{cases} 1, & \varepsilon < 0, \\ 0, & \varepsilon > 0. \end{cases}$$

Tunneling couples these two localized states. Let $t_c = |t_{12}|$ be the magnitude of its matrix element, in energy units. A large energy difference favors localization in the lower-energy dot, while for $|\varepsilon|$ comparable to t_c the ground state contains substantial contributions from both dots. At a fixed nonzero t_c, the curve of n_L against ε therefore falls continuously from near 1 to near 0, instead of making a step at zero. A larger t_c sustains this mixing at a larger energy difference, so the fall spans a wider interval of ε. We determine t_c from the shape of this $n_L(\varepsilon)$ curve. A charge sensor with different responses to occupation of the two dots lets us infer n_L from its mean signal.

K.1 Equilibrium charge distribution

Keep one orbital–valley state per dot and a fixed spin state. The localized charge states are $|L\rangle = |1,0\rangle$ and $|R\rangle = |0,1\rangle$. Removing their common mean energy and choosing the relative

basis phase gives

$$H_c = \begin{pmatrix} \varepsilon/2 & -t_c \\ -t_c & -\varepsilon/2 \end{pmatrix}, \qquad E_\pm = \pm\frac{\Delta_c}{2}, \quad \Delta_c = \sqrt{\varepsilon^2 + 4t_c^2}. \tag{K.1}$$

For $t_c > 0$, define $0 < \theta < \pi$ by $\cos\theta = \varepsilon/\Delta_c$ and $\sin\theta = 2t_c/\Delta_c$. The normalized eigenstates are

$$|g\rangle = \sin\frac{\theta}{2}|L\rangle + \cos\frac{\theta}{2}|R\rangle, \qquad |e\rangle = \cos\frac{\theta}{2}|L\rangle - \sin\frac{\theta}{2}|R\rangle, \tag{K.2}$$

with energies E_- and E_+, respectively. At zero temperature the electron occupies $|g\rangle$, so the mean left-dot occupation is

$$n_L^{(g)} = |\langle L|g\rangle|^2 = \frac{1}{2}\left(1 - \frac{\varepsilon}{\sqrt{\varepsilon^2 + 4t_c^2}}\right). \tag{K.3}$$

It approaches one for $\varepsilon \ll -2t_c$, one half at zero, and zero for $\varepsilon \gg 2t_c$. Its central slope is $dn_L^{(g)}/d\varepsilon|_0 = -1/(4t_c)$: increasing t_c makes the fall in $n_L^{(g)}$ less steep as a function of ε, quantifying the broader transition described above.

Finite temperature also broadens this curve by populating $|e\rangle$, whose left-dot occupation is $n_L^{(e)} = (1 + \varepsilon/\Delta_c)/2$. If the two levels reach thermal equilibrium at temperature T_e, then $p_e/p_g = e^{-\Delta_c/(k_B T_e)}$ and $p_e + p_g = 1$. Taking $n_L = p_g n_L^{(g)} + p_e n_L^{(e)}$ gives

$$n_L \equiv \langle \widehat{N}_1 \rangle = \frac{1}{2}\left[1 - \frac{\varepsilon}{\Delta_c}\tanh\!\left(\frac{\Delta_c}{2k_B T_e}\right)\right]. \tag{K.4}$$

For $t_c \ll k_B T_e$, this approaches $n_L = 1/[1 + e^{\varepsilon/(k_B T_e)}]$, which hardly depends on t_c. A thermally dominated curve cannot accurately resolve a much smaller coupling even when the temperature is known.

K.2 From the sensor record to the coupling

Because the total charge remains one, $\langle \widehat{N}_2 \rangle = 1 - n_L$. Substitution into the sensor model of Eq. (G.16) gives

$$\overline{s} = s_{\mathrm{bg}} + \kappa_2 + (\kappa_1 - \kappa_2) n_L. \tag{K.5}$$

The bar denotes an average of repeated or stationary records. The sensor must have $\kappa_1 \neq \kappa_2$ to detect redistribution at fixed total charge. After subtracting the smooth gate-dependent background, call the signal s'. Its values far on the left- and right-occupied sides are $s_L = \kappa_1$ and $s_R = \kappa_2$. The normalized measured curve is then

$$n_L = \frac{s' - s_R}{s_L - s_R}. \tag{K.6}$$

Load one electron and close its barriers to the reservoirs. Keep the interdot barrier setting fixed and scan a chosen plunger voltage v across the redistribution region. Allow the two-level system to relax at each setting before averaging the sensor signal. Use a scan range in which t_c is approximately constant and the detuning is approximately linear, $\varepsilon = \eta(v - v_0)$; here v_0 is the equal-energy setting and $\eta = \partial\varepsilon/\partial v$ has units of energy per voltage. The common shift of the two dot energies does not affect this fixed-charge model.

Record both nearly localized ends of the scan. There the dot occupation is nearly constant, so the remaining signal slope comes from the direct gate response. For a narrow scan with an approximately linear background, fit the two ends with a common slope and separate offsets,

then subtract that common slope. The two corrected plateaus supply s_L and s_R; any common constant offset cancels in the normalization. Normalize using Eq. (K.6), then compare the measured curve with Eq. (K.4). For independently known η and T_e, the remaining shape parameter is t_c; v_0 sets the center. In the low-temperature limit the central slope gives directly

$$t_c = \frac{|\eta|}{4|dn_L/dv|_{v_0}},$$

while fitting the full curve uses more of the data. This is a measurement of coupling through the equilibrium charge distribution, without transport through the double dot [34].

K.3 Spectroscopy with an oscillating gate

There is also a frequency-based measurement. An oscillating gate voltage changes the relative energies of the two dots. Because $|g\rangle$ and $|e\rangle$ are superpositions of the localized states, this perturbation can drive transitions between them. Populating $|e\rangle$ changes the mean charge distribution, which the same sensor can detect [35].

Suppose the drive changes detuning by $A_\varepsilon \cos(2\pi f t)$ while leaving t_c approximately fixed. With $\tau_z = |L\rangle\langle L| - |R\rangle\langle R|$,

$$H_d(t) = \frac{A_\varepsilon}{2}\cos(2\pi f t)\tau_z, \qquad \langle e|\tau_z|g\rangle = \sin\theta = \frac{2t_c}{\Delta_c}. \tag{K.7}$$

Thus the transition matrix element is nonzero. The weak-drive resonance condition is the same two-level condition derived in Section 4.2, now with gap Δ_c:

$$hf = \Delta_c(\varepsilon), \qquad \varepsilon_{\text{res}} = \pm\sqrt{(hf)^2 - 4t_c^2}, \quad hf \geq 2t_c. \tag{K.8}$$

Take $k_B T_e \ll \Delta_c$, so the drive-off state is approximately $|g\rangle$. Under a weak continuous drive, excitation competes with relaxation and produces a mean excited-state population p_e. A sensor averaging over many drive periods measures the population contribution; the rapidly oscillating coherence averages away. Since $n_L^{(e)} - n_L^{(g)} = \varepsilon/\Delta_c$, the change is

$$\delta\overline{s} = (\kappa_1 - \kappa_2)p_e\frac{\varepsilon}{\Delta_c}. \tag{K.9}$$

The resonantly increased population therefore gives a sensor feature away from zero detuning. At zero detuning both eigenstates have equal mean dot charge, so this particular signal vanishes even though the drive matrix element is nonzero.

At a fixed weak drive amplitude and frequency f, scan v and subtract the drive-off sensor record from the drive-on record. Locate the narrow features on the two sides of v_0, and repeat for several frequencies. Their resonance positions obey

$$(hf)^2 = \eta^2(v_{\text{res}} - v_0)^2 + 4t_c^2. \tag{K.10}$$

For a symmetric local response, the midpoint of each pair gives v_0. Plotting $(hf)^2$ against $(v_{\text{res}} - v_0)^2$ gives slope η^2 and intercept $4t_c^2$. This determines both $|\eta|$ and t_c, even when the feature at the minimum gap is invisible. Use resolved, weak-drive resonances for which the charge-contrast factor varies little across each linewidth; otherwise a signal extremum need not coincide with the resonance.

L Rotating Frame

Section 4.2 uses an effective rotating-frame Hamiltonian to describe transverse spin driving. Keeping the rapidly oscillating term during the transformation makes explicit the approximation behind Eq. (4.6) and fixes the phase convention used for the rotation axes.

Use the spin doublet of Section 4.1, with $|0\rangle = |\downarrow\rangle$ and $|1\rangle = |\uparrow\rangle$. This ordering exchanges the two physical-spin basis states, reversing the signs of the z and y Pauli operators while leaving the x operator unchanged:

$$\sigma_x = X, \qquad \sigma_y = -Y, \qquad \sigma_z = -Z.$$

Allow the real drive coefficient in Eq. (4.5) to have a slowly varying envelope $\Omega(t)$, keeping ϕ fixed. In the logical basis, its Hamiltonian is

$$H_{\mathrm{lab,q}}(t) = -\frac{E_Z}{2} Z + \hbar\Omega(t)\cos(\omega t + \phi) X. \tag{L.1}$$

Here E_Z is the static splitting, ω the carrier angular frequency, and $\Omega(t)$ the transverse drive coefficient in angular-frequency units.

The drive in Eq. (L.1) oscillates at ω, whereas near resonance the spin's motion relative to that carrier is slow. To describe the latter motion, write the laboratory state as

$$|\psi_{\mathrm{lab}}(t)\rangle = U_{\mathrm{rf}}(t)|\psi_{\mathrm{rot}}(t)\rangle, \qquad U_{\mathrm{rf}}(t) = e^{i\omega t Z/2}.$$

The rotating state evolves with

$$H_{\mathrm{rot}}(t) = U_{\mathrm{rf}}^{\dagger} H_{\mathrm{lab,q}}(t) U_{\mathrm{rf}} - i\hbar U_{\mathrm{rf}}^{\dagger}\dot{U}_{\mathrm{rf}}. \tag{L.2}$$

The second term contributes $+\hbar\omega Z/2$, reducing the longitudinal splitting to the difference between the Zeeman and carrier frequencies. The transverse operator becomes

$$U_{\mathrm{rf}}^{\dagger} X U_{\mathrm{rf}} = X\cos\omega t + Y\sin\omega t.$$

Multiplying this expression by the carrier separates the drive into a stationary part and a part oscillating at twice the carrier frequency:

$$\begin{aligned}&\cos(\omega t + \phi)(X\cos\omega t + Y\sin\omega t)\\ &\quad = \frac{1}{2}(X\cos\phi - Y\sin\phi) + \frac{1}{2}\big[X\cos(2\omega t + \phi) + Y\sin(2\omega t + \phi)\big].\end{aligned}$$

When $|\delta|, |\Omega(t)| \ll \omega$ and the envelope changes little during one carrier period, the 2ω term is fast compared with the rotation of interest. Dropping it is the rotating-wave approximation (RWA). The remaining Hamiltonian is

$$\begin{aligned}H_{\mathrm{rot,q}}(t) &= \frac{\hbar}{2}\left[-\delta Z + \Omega(t)(\cos\phi X - \sin\phi Y)\right],\\ \delta &= E_Z/\hbar - \omega.\end{aligned} \tag{L.3}$$

On resonance, a constant envelope therefore produces rotations at angular frequency $|\Omega|$. This is the Rabi frequency associated with the laboratory-frame coefficient defined above.

Equation (L.3) also identifies the role of each control: the frequency fixes the detuning, the phase fixes the transverse rotation axis, and the envelope fixes the rotation strength. For positive Ω, $\phi = 0, -\pi/2, \pi, +\pi/2$ select $+X, +Y, -X, -Y$, respectively. Turning the envelope off leaves the waiting Hamiltonian

$$H_{\mathrm{wait}} = -\frac{\hbar\delta}{2} Z.$$

These expressions use a constant reference frequency ω and neglect noise in its phase. With that reference held fixed, a time-dependent detuning comes from a changing spin splitting $E_Z(t)$. Population relaxation requires the open-system transition rates described in Section U.1.2; it is not generated by these coherent two-state Hamiltonians.

To recover the constant-envelope result of the main text, set $\Omega(t) = \Omega$ and substitute $\sigma_z = -Z$, $\sigma_x = X$ and $\sigma_y = -Y$. The transformation becomes $R(t) = e^{-i\omega t\sigma_z/2}$, with $|\psi(t)\rangle = R(t)|\psi_{\mathrm{r}}(t)\rangle$. For the laboratory Hamiltonian in Eq. (4.5), keeping the 2ω term gives the full expression

$$\begin{aligned} H_{\mathrm{r}} &= R^\dagger H R - i\hbar R^\dagger \dot{R} \\ &= \frac{\hbar\delta}{2}\sigma_z + \frac{\hbar\Omega}{2}(\cos\phi\,\sigma_x + \sin\phi\,\sigma_y) \\ &\quad + \frac{\hbar\Omega}{2}\big[\cos(2\omega t+\phi)\sigma_x - \sin(2\omega t+\phi)\sigma_y\big], \qquad \delta = \omega_Z - \omega. \end{aligned} \tag{L.4}$$

Here $\omega_Z = E_Z/\hbar$. Omitting the last term gives precisely the constant Hamiltonian used for the propagator in Eq. (4.6).

M Physical Origins of Transverse Spin Driving

Section 4.2 specifies a transverse term in the effective spin Hamiltonian. Its physical origin is contained in the off-diagonal matrix element of the applied perturbation, after any required orbital or valley reduction. In the static spin basis, write this contribution as

$$H_\perp(t) = d_x(t)\sigma_x + d_y(t)\sigma_y, \qquad d_x(t) - id_y(t) = \langle\uparrow|\delta H_{\mathrm{eff}}(t)|\downarrow\rangle. \tag{M.1}$$

Here d_x and d_y have energy units. The drive used in the main text corresponds to $d_x(t) = \hbar\Omega\cos(\omega t+\phi)$ and $d_y(t) = 0$. The microscopic control determines these coefficients; the rotating-frame calculation in Appendix L then determines the resulting spin dynamics.

M.1 Direct magnetic-field driving

An oscillating magnetic field couples directly to the electron's magnetic moment. For a positive isotropic effective g factor g_{e}, equal to g in the main text, take

$$\mathbf{B}(t) = B_0\hat{\mathbf{z}} + B_1\cos(\omega t+\phi)\hat{\mathbf{x}}.$$

The Zeeman interaction gives

$$\begin{aligned} H(t) &= \frac{E_Z}{2}\sigma_z + \frac{g_{\mathrm{e}}\mu_B B_1}{2}\cos(\omega t+\phi)\sigma_x, \\ E_Z &= g_{\mathrm{e}}\mu_B B_0, \qquad \Omega_{\mathrm{mag}} = \frac{g_{\mathrm{e}}\mu_B|B_1|}{2\hbar}. \end{aligned} \tag{M.2}$$

The transverse part has a nonzero matrix element between the Zeeman states. Comparing its coefficient with Eq. (4.5) gives the resonant Rabi frequency Ω_{mag} within the RWA. A field component parallel to the static quantization axis instead changes the splitting. The spin-only description also requires the isolation conditions of Section 4.1.

M.2 Electric driving in a magnetic-field gradient

An electric field acts directly on the electron's position. For charge $-e$, with $e > 0$, its interaction is $H_{\mathrm{el}} = e\mathcal{E}(t)x \otimes I_{\mathrm{spin}}$. If spin–orbit coupling is neglected, confinement is spin

independent and the Zeeman field is uniform, the two spin states share an orbital–valley state $|\chi_0\rangle$. Their electric-field matrix element factorizes as

$$\langle\chi_0,\uparrow|H_{\rm el}|\chi_0,\downarrow\rangle = e\mathcal{E}(t)\langle\chi_0|x|\chi_0\rangle\langle\uparrow|\downarrow\rangle = 0. \tag{M.3}$$

Thus orbital motion alone does not couple these two spin states. In a transverse magnetic-field gradient, however, a displaced electron samples a different transverse field. The same electric motion then supplies the time-dependent spin coupling. A micromagnet can produce this gradient [36, 37]. Spin transitions driven through electrically induced motion are called electric-dipole spin resonance (EDSR).

Take the undriven dot centre as $x = 0$, and approximate the local field by $\mathbf{B}(x) \simeq B_0\hat{\mathbf{z}} + b_\perp x\hat{\mathbf{x}}$, where $b_\perp = (\partial B_x/\partial x)_0$. For motion along x, use harmonic confinement with effective mass m^*, orbital frequency ω_0 and momentum p_x. An electric field $\mathcal{E}(t) = \mathcal{E}_{\rm ac}\cos\omega t$ then gives the combined orbital and spin Hamiltonian

$$H = \frac{p_x^2}{2m^*} + \frac{m^*\omega_0^2 x^2}{2} + e\mathcal{E}_{\rm ac}x\cos(\omega t) + \frac{E_Z}{2}\sigma_z + \frac{g_{\rm e}\mu_B b_\perp x}{2}\sigma_x. \tag{M.4}$$

The last term makes the direction of the spin splitting depend on position. We treat its mixing of orbital states as weak: $g_{\rm e}\mu_B|b_\perp|\ell_0 \ll \hbar\omega_0$, with ground-state position width $\ell_0 = \sqrt{\hbar/(2m^*\omega_0)}$. The driven displacement must also stay in the region where both the harmonic potential and the constant gradient are valid.

To leading order in this weak mixing, the orbital displacement can be calculated without spin backaction. Ehrenfest's equation for $x_c(t) = \langle x\rangle_t$ is a forced-oscillator equation:

$$\begin{aligned} m^*\ddot{x}_c + m^*\omega_0^2 x_c &= -e\mathcal{E}_{\rm ac}\cos(\omega t), \\ x_c(t) &= x_{\rm ac}\cos(\omega t), \qquad x_{\rm ac} = -\frac{e\mathcal{E}_{\rm ac}}{m^*(\omega_0^2-\omega^2)}. \end{aligned} \tag{M.5}$$

The solution shown omits free orbital oscillations, as appropriate to a drive switched on smoothly and far from orbital resonance. For $\omega \ll \omega_0$, it reduces to $x_{\rm ac} \simeq -e\mathcal{E}_{\rm ac}/(m^*\omega_0^2)$: the orbital approximately follows the moving potential minimum.

The moving orbital samples the transverse field $b_\perp x_c(t)$. Projecting the spin-dependent terms of Eq. (M.4) onto that orbital therefore replaces x by $x_c(t)$. The resulting spin Hamiltonian and the Rabi frequency obtained by comparison with Eq. (L.1) are

$$H_{\rm spin}(t) \simeq \frac{E_Z}{2}\sigma_z + \frac{g_{\rm e}\mu_B b_\perp x_{\rm ac}}{2}\cos(\omega t)\sigma_x, \qquad \Omega_{\rm mm} = \frac{g_{\rm e}\mu_B|b_\perp x_{\rm ac}|}{2\hbar}. \tag{M.6}$$

The effective drive thus has the same form as a directly applied oscillating magnetic field. Its transverse direction matters: a purely longitudinal gradient would modulate the Zeeman splitting without producing this off-diagonal term.

M.3 Electric driving through spin–orbit coupling

Spin–orbit coupling connects orbital motion to spin even when the applied Zeeman field is uniform. Consider a one-dimensional model in which momentum couples to a transverse spin component through $\alpha p_x\sigma_y$. The coefficient α has units of velocity, and the driven Hamiltonian is

$$H = H_{\rm orb} + e\mathcal{E}(t)x + \frac{E_Z}{2}\sigma_z + \alpha p_x\sigma_y, \qquad H_{\rm orb} = \frac{p_x^2}{2m^*} + \frac{m^*\omega_0^2 x^2}{2}. \tag{M.7}$$

The electric field produces the orbital displacement already calculated in Eq. (M.5). To extract its spin-dependent effect, it is useful to remove the momentum coupling to first order by a change of basis [38].

For nonzero α, introduce the signed spin–orbit length $\ell_{\rm so} = \hbar/(m^*\alpha)$ and the anti-Hermitian generator $A = ix\sigma_y/\ell_{\rm so}$. Expanding the unitary transformation gives $\widetilde{H} = e^A H e^{-A} = H + [A, H] + \cdots$. The required commutators are

$$\begin{aligned} [A, H_{\rm orb}] &= -\alpha p_x \sigma_y, \qquad [A, e\mathcal{E}(t)x] = 0, \\ \left[A, \frac{E_Z}{2}\sigma_z\right] &= -\frac{E_Z x}{\ell_{\rm so}}\sigma_x. \end{aligned} \tag{M.8}$$

The first cancels the original momentum coupling. The electric potential is unchanged, while the Zeeman term acquires a position-dependent transverse component. To first order, the transformed Hamiltonian is consequently

$$\widetilde{H} = H_{\rm orb} + e\mathcal{E}(t)x + \frac{E_Z}{2}\sigma_z - \frac{E_Z x}{\ell_{\rm so}}\sigma_x + O(\alpha^2). \tag{M.9}$$

This form makes the effect of electric motion explicit: the transverse coupling now depends on the electron's displacement.

For $E_Z, \hbar\omega \ll \hbar\omega_0$ and $\ell_0, |x_{\rm ac}| \ll |\ell_{\rm so}|$, we may project onto the slowly moving orbital and set $x = x_c(t)$ in the spin interaction. In the transformed spin basis, this yields

$$\widetilde{H}_{\rm spin}(t) \simeq \frac{E_Z}{2}\sigma_z - \frac{E_Z x_{\rm ac}}{\ell_{\rm so}}\cos(\omega t)\sigma_x, \qquad \Omega_{\rm so} = \frac{E_Z|x_{\rm ac}|}{\hbar|\ell_{\rm so}|}. \tag{M.10}$$

The retained spin states include the small orbital admixtures implicit in the basis transformation. In this model the drive vanishes when $\alpha = 0$ or $E_Z = 0$; interface, valley and anisotropic spin–orbit effects can modify the model in silicon.

Equations (M.6) and (M.10) identify the transverse coefficients produced by the two electric mechanisms. Once the orbital reduction is valid, both can be inserted into the same spin dynamics derived in Appendix L. Within the RWA, their constant-envelope spin-flip probabilities are given by Eq. (4.7), with Ω replaced by $\Omega_{\rm mm}$ or $\Omega_{\rm so}$.

Gate-dependent Zeeman coupling. The one-dimensional spin–orbit model is one approximation to the electrical response. More generally, orbital, valley and spin–orbit effects can be incorporated in a gate-dependent effective g matrix [39]. In a fixed reference basis for the retained doublet, write the Zeeman interaction to first order in the static applied field $\mathbf{B}_0$ as

$$H_Z(V) = \mathbf{d}(V)\cdot\boldsymbol{\sigma}, \qquad \mathbf{d}(V) = \frac{\mu_B}{2}\mathbf{g}(V)\mathbf{B}_0, \qquad \mathbf{n} = \frac{\mathbf{d}(V_0)}{|\mathbf{d}(V_0)|}. \tag{M.11}$$

The static splitting is $E_Z = 2|\mathbf{d}(V_0)|$. For a small modulation $V(t) = V_0 + v_{\rm ac}\cos\omega t$, linearization gives an oscillating term $\delta\mathbf{d}\cdot\boldsymbol{\sigma}\cos\omega t$. Its transverse component and resonant Rabi frequency are

$$\begin{aligned} \delta\mathbf{d} &= v_{\rm ac}\left.\frac{\partial\mathbf{d}}{\partial V}\right|_{V_0}, \quad \delta\mathbf{d}_\perp = \delta\mathbf{d} - \mathbf{n}(\mathbf{n}\cdot\delta\mathbf{d}), \\ \Omega_g &= \frac{|\delta\mathbf{d}_\perp|}{\hbar}. \end{aligned} \tag{M.12}$$

This result assumes the RWA and a drive sufficiently weak and far below transitions to excluded orbital and valley states for the doublet reduction to remain valid. A scalar $g(V)$ changes only the longitudinal splitting; transverse driving requires a change in the direction of $\mathbf{d}$. The g matrix describes the projected microscopic response. A spin–orbit contribution already included in it must therefore be counted only once.

M.4 Programming the drive amplitude and phase

For the electric-motion mechanisms, the displacement entering the spin Hamiltonian must be related to the voltage waveform at the dot. Separate the voltage that sets the operating point from the oscillating component used for spin rotation:

$$\begin{aligned} V_g(t) &= V_g^{\mathrm{DC}} + V_g^{\mathrm{p}}(t) + v_d(t), \\ v_d(t) &= a(t)\cos(\omega t + \phi). \end{aligned} \tag{M.13}$$

Here V_g^{DC} is the constant bias and $V_g^{\mathrm{p}}(t)$ is a slowly varying pulse used for loading, holding and readout. During a rotation, their sum fixes the holding point. The drive envelope $a(t)$ denotes the voltage that actually reaches the dot, including the effects of cables and filters on the programmed waveform.

At that operating point, let F_v be the force per unit drive voltage, so that $F_v v_d(t) = -e\mathcal{E}(t)$ in linear response. Substituting this force into Eq. (M.5), for a carrier slow compared with the orbital frequency, gives

$$x_c(t) \simeq \chi_x a(t)\cos(\omega t + \phi), \qquad \chi_x = \frac{F_v}{m^*\omega_0^2}.$$

The envelope must also vary slowly enough for the orbital to follow. For the magnetic-gradient mechanism, this displacement produces

$$B_{x,\mathrm{ac}}(t) = B_1(t)\cos(\omega t + \phi), \qquad B_1(t) = b_\perp \chi_x a(t), \tag{M.14}$$

and hence the signed drive coefficient

$$\Omega(t) = \frac{g_{\mathrm{e}}\mu_B b_\perp \chi_x a(t)}{2\hbar}.$$

This relation places the device voltage directly in Eq. (L.1). The sign of $b_\perp\chi_x$ can be absorbed into the phase convention. For the spin–orbit mechanism, the same displacement is inserted into Eq. (M.10).

The phase controls a different part of the rotation: pulses with the same frequency and envelope can act about different transverse axes. To set it, a controller combines two versions of the carrier separated by a quarter period. Their independently chosen, slowly varying amplitudes give

$$v_d(t) = v_I(t)\cos\omega t - v_Q(t)\sin\omega t, \qquad v_I(t) = a(t)\cos\phi, \quad v_Q(t) = a(t)\sin\phi. \tag{M.15}$$

Substituting v_I and v_Q yields $v_d(t) = a(t)\cos(\omega t + \phi)$, the waveform in Eq. (M.13). Changing their ratio and signs therefore sets ϕ at fixed carrier frequency and, through Eq. (L.3), sets the rotation axis. An IQ mixer driven by an arbitrary-waveform generator implements this combination. A direct digital synthesizer can generate the same waveform from a programmed phase word. No separate static gate voltage is needed to set the phase.

The phase of each pulse is defined relative to a common carrier clock. For pulse k, the controller turns on an envelope $a_k(t)$ with offset ϕ_k; turning that envelope off does not reset the clock. The difference $\phi_2 - \phi_1$ therefore remains set even if the drive is zero between the pulses. A fixed phase shift from the cables, filters and mixer adds the same offset to the device phases. It can be absorbed into the definition of X, leaving the relative rotation axes unchanged.

The relation between programmed relative phase and rotation axis can be checked with two calibrated resonant $\pi/2$ pulses. Prepare $|0\rangle$, give the first pulse phase zero and the second phase φ, and neglect detuning and decoherence during the sequence. Equation (L.3) then

gives the rotating-frame operators $U_0 = e^{-i\pi X/4}$ and $U_\varphi = e^{-i\pi(\cos\varphi X - \sin\varphi Y)/4}$. Their product has excited-state amplitude

$$\langle 1|U_\varphi U_0|0\rangle = -\frac{i}{2}(1 + e^{-i\varphi}),$$

so the ideal measured probability is

$$P_1(\varphi) = \frac{1 + \cos\varphi}{2}. \tag{M.16}$$

The maximum at $\varphi = 0$ corresponds to the same axis for both pulses, and the minimum at $\varphi = \pi$ to opposite axes; $\varphi = \pm\pi/2$ selects orthogonal axes. With pulse angles and detuning controlled, a shifted or distorted fringe indicates an error in the relation between programmed and device phase, such as an IQ quadrature error. Correcting that relation makes the programmed phase offsets produce the intended gate axes.

N Single-Spin Control and Exchange Gates

The drive Hamiltonian in Appendix L and the exchange interaction in Section 4.3 provide the physical controls for single-spin qubits. To implement a specified gate, their strengths must be related to the applied pulse amplitudes and durations. This appendix obtains those relations from spin oscillations, uses them to construct the gates in Section 6, and corrects systematic pulse errors.

The preparation and readout needed for those measurements are developed in Appendices R and S. Voltage compensation is treated in Appendix F.4.

N.1 Single-qubit gates for single-spin qubits

One electron in one dot stores the logical states $|0\rangle = |\downarrow\rangle$ and $|1\rangle = |\uparrow\rangle$. A static longitudinal field defines their splitting, while a driven transverse field supplies logical x and y rotations. Here resonance and rotation-rate measurements set the pulse parameters for one-qubit gates; repeated pulse sequences then reveal reproducible angle and axis errors.

N.1.1 Measuring the resonance and rotation rate

We use the rotating-frame Hamiltonian in Eq. (6.1). During each rectangular pulse of duration τ, the drive has a constant amplitude $\Omega(t) = \Omega_0 > 0$ and phase $\phi = 0$, giving

$$H_{\mathrm{rot,q}} = \frac{\hbar}{2}(\Omega_0 X - \delta Z), \qquad \delta = \frac{E_Z}{\hbar} - \omega. \tag{N.1}$$

Here ω is the drive angular frequency.

At fixed amplitude, scanning the drive frequency determines the resonance; varying the pulse duration at that resonance then determines the rotation rate. In both scans, prepare $|0\rangle$, apply the drive while the electron remains isolated, and use the single-spin readout of Section S.1. The fraction of temporary empty-dot records estimates $P_1 = P_\uparrow$, with the assignment correction of Eq. (S.10) when its errors are known.

When decoherence during τ is negligible, the identity

$$(\Omega_0 X - \delta Z)^2 = (\Omega_0^2 + \delta^2)I$$

gives the propagator

$$U(\tau) = \cos\frac{\Omega_{\mathrm{eff}}\tau}{2} I - i\sin\frac{\Omega_{\mathrm{eff}}\tau}{2}\frac{\Omega_0 X - \delta Z}{\Omega_{\mathrm{eff}}}.$$

Here $\Omega_{\text{eff}} = \sqrt{\Omega_0^2 + \delta^2}$. Starting from $|0\rangle$ therefore gives

$$P_\uparrow(\omega, \tau) = \frac{\Omega_0^2}{\Omega_0^2 + \delta^2} \sin^2\left(\frac{\sqrt{\Omega_0^2 + \delta^2}}{2}\tau\right). \tag{N.2}$$

Repeat the full preparation–drive–readout cycle at each chosen frequency and duration, as in Section S.6.

Scanning frequency to measure the spin splitting. Keep the holding voltages, B, pulse amplitude and τ fixed while scanning ω over a range in which Ω_0 can be treated as constant. Choose τ so that $0 < \Omega_0\tau \ll \pi$, so that the resonant response remains far from the later Rabi maxima and nodes. Under this condition, Eq. (N.2), with $\delta = \omega_{\text{res}} - \omega$, has a central maximum at $\delta = 0$. Measure $P_\uparrow$ at each ω; the location of that maximum gives ω_{res}, which obeys

$$\hbar\omega_{\text{res}} = E_Z = g_{\text{e}}\mu_B B. \tag{N.3}$$

It therefore gives E_Z; for known B, it also gives $g_{\text{e}} = \hbar\omega_{\text{res}}/(\mu_B B)$ [37].

Figure N.1 shows one example calculated directly from Eq. (N.2). It uses the fixed pulse area $\Omega_0\tau = 0.5$ and plots the measured probability against the dimensionless frequency offset $(\omega - \omega_{\text{res}})\tau$. The central peak occurs at zero offset, so its horizontal position identifies ω_{res}.

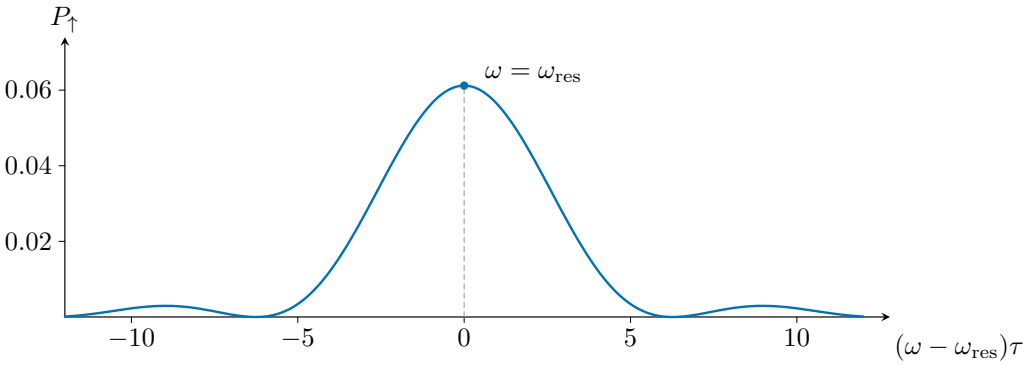


Figure N.1: Example frequency scan from the exact two-level result in Eq. (N.2), for $\Omega_0\tau = 0.5$ and negligible decoherence. The vertical axis is the actual spin-up probability, not a normalized line shape. The dashed line marks the resonant angular frequency. This is a theoretical example, not experimental data.

Scanning duration to measure the rotation rate. Now fix $\omega = \omega_{\text{res}}$ and keep the amplitude constant while varying τ. At $\delta = 0$,

$$P_\uparrow(\omega_{\text{res}}, \tau) = \sin^2\left(\frac{\Omega_0\tau}{2}\right), \qquad \tau_\pi = \frac{\pi}{\Omega_0}. \tag{N.4}$$

The oscillation period is $2\pi/\Omega_0$, and the first maximum gives the spin-flip time τ_π. At each duration, count the up-spin outcomes in repeated trials and fit the probability-versus-duration curve to determine Ω_0.

The measured rotation rate includes the device response and the attenuation between generator and gate for the applied pulse. Repeating at nearby frequencies checks the detuning dependence: the oscillation becomes faster but its maximum transfer becomes smaller away from resonance. Contrast decay during driving can also contain relaxation, fluctuations and pulse errors.

N.1.2 Implementing single-spin gates with calibrated pulses

The preceding scan calibrates the constant strength Ω_0. To turn that calibrated evolution into a gate, use the rectangular drive envelope

$$\Omega(t) = \begin{cases} \Omega_0, & 0 < t < t_p, \\ 0, & \text{otherwise.} \end{cases}$$

Retain the holding voltages and amplitude used in the calibration, choose the transverse axis through the drive phase, and choose the pulse duration t_p to obtain the required rotation angle.

Set $\omega = \omega_{\text{res}} = E_Z/\hbar$, so $\delta = 0$. The constant Hamiltonian gives

$$U(t_p) = U_\phi(\vartheta) = \exp\left[-\frac{i\vartheta}{2}(\cos\phi X - \sin\phi Y)\right], \qquad \vartheta = \Omega_0 t_p.$$

For a desired angle $\theta > 0$, choose

target rotation	ϕ	t_p
$R_x(\theta)$	0	θ/Ω_0
$R_y(\theta)$	$-\pi/2$	θ/Ω_0
$R_x(-\theta)$	π	θ/Ω_0
$R_y(-\theta)$	$+\pi/2$	θ/Ω_0

Thus $R_x(\pi/2)$ uses phase zero and duration $\pi/(2\Omega_0)$, while $R_x(\pi)$ uses the same phase and duration π/Ω_0. The holding voltages and applied amplitude remain those used in the measurement of Ω_0. Successive pulses share the same oscillating phase reference.

Executing the pulse and checking its angle. To test the prescribed x rotation, prepare down, move to the holding voltages, and apply the rectangular pulse just specified. Then turn the drive off, move to the fixed readout voltages, and repeat this full sequence. The up-spin fraction, corrected for known assignment errors, estimates

$$P_1 = |\langle 1|R_x(\theta)|0\rangle|^2 = \sin^2(\theta/2).$$

For the $\pi/2$ and π pulses this predicts $1/2$ and 1. If the holding voltages or delivered amplitude have changed since the Rabi measurement, first update Ω_0 with a duration scan.

Including fixed pulse edges. A finite pulse also rotates the spin while its amplitude rises and falls. At resonance and fixed phase, the angle remains the integral of $\Omega(t)$. Consider a pulse with a constant plateau $\Omega_0 > 0$ of adjustable duration t_{h} and fixed edges. If changing t_{h} leaves the edge waveforms unchanged, write their combined area as ϑ_{edge}. The full angle and the ideal probability measured from $|0\rangle$ are then

$$\vartheta(t_{\text{h}}) = \vartheta_{\text{edge}} + \Omega_0 t_{\text{h}}, \qquad P_1(t_{\text{h}}) = \sin^2\left(\frac{\vartheta_{\text{edge}} + \Omega_0 t_{\text{h}}}{2}\right). \tag{N.5}$$

Keep the programmed amplitude, phase and edges fixed, vary only the plateau duration, and measure P_1 using the same preparation and readout as above. The oscillation period determines Ω_0, while its phase offset determines ϑ_{edge} modulo 2π. For a target angle $\vartheta^\star$, choose

$$t_{\text{h}} = \frac{\vartheta^\star - \vartheta_{\text{edge}} + 2\pi m}{\Omega_0}, \qquad m \in \mathbb{Z}, \tag{N.6}$$

with a branch that gives an allowed positive hold time. Repeat this area calibration when the applied amplitude or edge waveform is changed. These relations use the same negligible-decoherence limit as the rectangular-pulse calculation; a pulse whose phase or detuning changes must be treated with its full Hamiltonian.

This pulse prescription fixes the ideal driven rotation. The next subsection corrects reproducible angle and axis errors before the operation is tested on several inputs in Appendix T.

N.1.3 Correcting systematic pulse errors

The resonance and Rabi scans provide initial settings for the drive frequency, amplitude and duration. We now amplify small, reproducible discrepancies between the intended and actual pulse. An amplitude or duration error changes the rotation angle; a frequency error adds a Z component to the rotation axis; and a relative-phase error rotates the intended transverse axis within the xy plane. The three tests below determine which control must be corrected.

Detecting whether a pulse rotates too far or not far enough. Start with a nominal $R_x(\pi)$ pulse. A single measurement near its maximum transfer is insensitive to the sign of a small angle error. Repeating the pulse lets the error accumulate; an extra $\pi/2$ pulse places the final probability on a slope where its sign is visible.

If a nominal $R_x(\pi)$ is actually $R_x(\pi+\eta)$ and other errors are negligible, m repetitions give

$$P_1(m) = \sin^2\!\left[\frac{m(\pi+\eta)}{2}\right]. \tag{N.7}$$

Thus an error hidden in one pulse becomes a visible modulation as m increases. Scan the pulse amplitude or duration around the initial estimate, record this dependence, and find the setting consistent with $\eta = 0$ for several m. Population data of this form alone do not determine the sign of η.

To obtain a signed correction, take even m and append an $R_x(\pi/2)$ analysis pulse before readout. The ideal repeated π pulses now give $P_1 = 1/2$, on the slope rather than the extremum of the oscillation:

$$P_1^{\mathrm{quad}}(m) = \sin^2\left(\frac{m(\pi+\eta)+\pi/2}{2}\right) = \frac{1+\sin(m\eta)}{2} \simeq \frac{1}{2} + \frac{m\eta}{2}. \tag{N.8}$$

In the range $|m\eta| \ll 1$, a measured deviation from $1/2$ gives $\eta \simeq 2(P_1^{\mathrm{quad}} - 1/2)/m$, including its sign. An analysis-pulse angle error adds an intercept independent of m; measuring several even m separates that intercept from the slope. If the adjustable control is the rectangular amplitude a_0 and its measured angle slope is $s_a = d\vartheta/da_0$, change a_0 by $-\eta/s_a$. Repeating the sequence verifies the correction. All probabilities here refer to assignment-corrected counts.

Correcting the drive frequency from an axis error. Even a correct pulse area fails to give a pure x rotation if $\omega \neq E_Z/\hbar$. The term $-\hbar\delta Z/2$ tilts its axis towards z. A nominal π pulse then leaves a transverse spin component whose sign reveals the sign of the detuning. For a rectangular pulse with constant Rabi frequency Ω_0 and phase zero, define $\Omega_{\mathrm{eff}} = \sqrt{\Omega_0^2 + \delta^2}$. The propagator is

$$U(t) = \cos\frac{\Omega_{\mathrm{eff}} t}{2} I - i \sin\frac{\Omega_{\mathrm{eff}} t}{2} \frac{\Omega_0 X - \delta Z}{\Omega_{\mathrm{eff}}}.$$

Starting in $|0\rangle$, its output satisfies

$$\begin{aligned} P_1(t) &= \frac{\Omega_0^2}{\Omega_0^2+\delta^2}\sin^2\frac{\Omega_{\text{eff}}t}{2}, \\ \langle X\rangle(t) &= -\frac{\Omega_0\delta}{\Omega_0^2+\delta^2}[1-\cos(\Omega_{\text{eff}}t)]. \end{aligned} \tag{N.9}$$

At the nominal π time π/Ω_0 and small detuning, $\langle X\rangle \simeq -2\delta/\Omega_0$. Unlike the population deficit, this is linear in the detuning and reveals its sign. Measure it by adding $R_y(-\pi/2)$ before logical-basis readout, then update the reference frequency by $\omega \mapsto \omega+\delta$. The analysis pulse must have independently established phase and small axis error. The resonance scan supplies the initial frequency setting, and the signed axis-error signal above supplies its local correction. The Ramsey fringe-phase slope in Section U.1.1 independently shows any residual mean detuning during a wait. Amplitude and frequency updates can be alternated until both signed signals vanish within the chosen accuracy.

Checking the relative transverse axes of two pulses. A population measurement after one $\pi/2$ pulse cannot distinguish an x rotation from a y rotation: both give $P_1 = 1/2$ from $|0\rangle$. A second pulse makes their relative axis observable through interference. Apply $R_x(\pi/2)$ followed immediately by a second $\pi/2$ pulse with phase ϕ. Using Eq. (L.3) at $\delta = 0$ gives

$$P_1(\phi) = |\langle 1|U_\phi(\pi/2)R_x(\pi/2)|0\rangle|^2 = \frac{1+\cos\phi}{2}. \tag{N.10}$$

Scan the programmed relative phase and find its offset from this curve. The same phase reference must be maintained between pulses. This establishes relative axes; the choice of what is called the x axis fixes their common phase convention.

These tests return corrections to duration or amplitude, reference frequency, and relative phase, respectively. After a correction, repeat the associated test at the same holding voltages. Random changes between repetitions are not fixed pulse missettings. Their effect during a wait is measured in Section U.1, while their effect during a gate is treated in Appendix U.

N.2 Two-qubit gates for single-spin qubits

During an exchange pulse, the two basis states with opposite spins are $|01\rangle = |{\downarrow\uparrow}\rangle$ and $|10\rangle = |{\uparrow\downarrow}\rangle$. After removing a scalar energy, their dynamics is governed by

$$H_0(t) = \begin{pmatrix} -\Delta E_Z/2 & J(t)/2 \\ J(t)/2 & +\Delta E_Z/2 \end{pmatrix}, \qquad \Delta E_Z = E_{Z,1} - E_{Z,2}. \tag{N.11}$$

The diagonal term is the difference between the two local Zeeman splittings. The off-diagonal term converts $|01\rangle$ into $|10\rangle$ and vice versa; it is the part used to exchange the two spin states. A SWAP-type gate therefore requires a controlled $J(t)$ and either negligible ΔE_Z during the pulse or an explicit treatment of its simultaneous evolution.

The coupling $J(t)$ is produced by lowering the interdot barrier. This increases tunneling, and virtual double occupation produces the exchange Hamiltonian

$$H_{\text{ex}}(t) = J(t)\,\mathbf{s}_1\cdot\mathbf{s}_2, \qquad \mathbf{s}_i = \boldsymbol{\sigma}_i/2. \tag{N.12}$$

Exchange changes the relative phase of the singlet and triplet components of a two-spin state. We first estimate its scale from a virtual-tunneling model, then calibrate the effective J from the measured spin oscillation. The calibrated pulse area sets SWAP and $\sqrt{\text{SWAP}}$ gates. A composite sequence combines the same interaction with local single-spin rotations to obtain a diagonal entangling gate.

N.2.1 Estimating the exchange scale from virtual tunneling

The device model gives an initial estimate of the exchange coupling that can be used to choose a safe range of barrier amplitudes and pulse durations for the first experiment. Lowering the interdot barrier increases the tunneling amplitude t_c. The higher-energy $(2,0)$ and $(0,2)$ configurations then contribute virtually: in a single-orbital model they lower the singlet energy but not the triplet energy. Let $|S_{11}\rangle$ be the separated singlet and let $|S_{20}\rangle$ and $|S_{02}\rangle$ be the doubly occupied singlets. Define their positive energy costs relative to $|S_{11}\rangle$ by

$$\Delta_{20} = E(S_{20}) - E(S_{11}), \qquad \Delta_{02} = E(S_{02}) - E(S_{11}).$$

Single-electron tunneling couples $|S_{11}\rangle$ to each doubly occupied singlet with matrix-element magnitude $\sqrt{2}t_c$. For $\Delta_{20}, \Delta_{02} \gg |t_c|$, second-order perturbation theory therefore gives

$$\delta E_S = -\frac{2t_c^2}{\Delta_{20}} - \frac{2t_c^2}{\Delta_{02}}.$$

The corresponding triplets cannot put two electrons into the same single-dot orbital, so they have no such second-order shift in this single-orbital model. The singlet–triplet splitting is consequently

$$J = E_T - E_S \simeq 2t_c^2 \left(\frac{1}{\Delta_{20}} + \frac{1}{\Delta_{02}}\right), \qquad H_{\text{ex}} = J\,\mathbf{s}_1 \cdot \mathbf{s}_2, \qquad \mathbf{s}_i = \boldsymbol{\sigma}_i/2.$$

This expression estimates the scale and voltage dependence of J within the single-orbital, second-order model. The value used to set a gate is instead obtained from the exchange-oscillation experiment below. This is the virtual-charge mechanism introduced more briefly in Section 4.3.

N.2.2 Calibrating exchange from spin oscillations

Experimentally, J is changed by applying voltages to the electrode above the interdot barrier. A voltage on that electrode also shifts the two dot energies, so two compensating plunger-voltage changes are applied at the same time. Collect the physical electrode voltages as $\mathbf{V} = (V_1, V_2, V_{\text{B}})^T$ and write the coordinated pulse as

$$\mathbf{V}(t) = \mathbf{V}^0 + \mathbf{n}v(t), \qquad \mathbf{n} = (n_1, n_2, 1)^T, \qquad G\mathbf{n} = 0. \tag{N.13}$$

Here $v(t)$ is the applied barrier-voltage increment, while $n_1 v(t)$ and $n_2 v(t)$ are the simultaneous plunger corrections. The condition $G\mathbf{n} = 0$ keeps the two addition energies fixed to first order, as derived in Appendix F.4.

Choose the sign of v so that increasing its pulse amplitude lowers the potential barrier between the dots. The orbital overlap and t_c then increase, so the estimate in Section N.2.1 predicts a larger J. This prediction gives only the direction and scale of the change: the function $J(v)$ is determined experimentally by the duration scan below. During every pulse, keep the two reservoir barriers closed so that the electron number remains fixed; ramp from $v = 0$ to a chosen amplitude, hold, and return to $v = 0$ before readout.

Evolution of two opposite spins. Figure N.2 divides the pulse into a rise of duration t_{r}, a flat hold of duration t_h, and a fall of duration t_{f}. During the hold, $v(t) = v^\star$ and $J(t) = J_{\text{h}}(v^\star)$

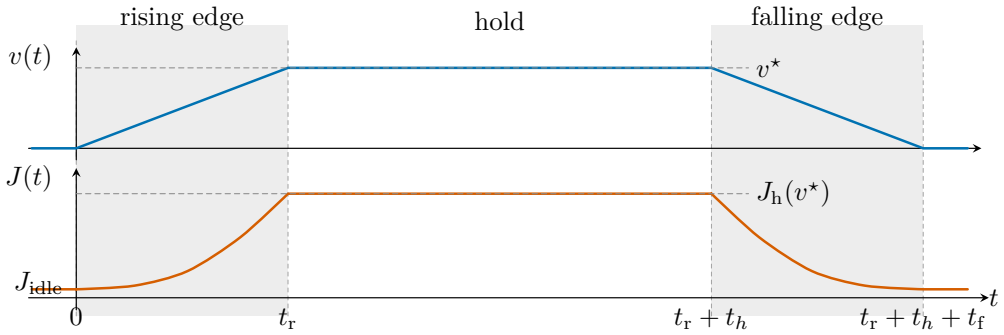


Figure N.2: A compensated barrier pulse and the resulting exchange coupling. The programmed coordinate $v(t)$ rises from zero to $v^\star$, remains there for the hold time t_h, and returns to zero. All physical gate voltages follow $\mathbf{V}(t) = \mathbf{V}^0 + \mathbf{n}v(t)$. The exchange response $J(t)$ need not have linear edges because J is generally a nonlinear function of voltage. The shaded rising and falling intervals together produce the edge phase θ_{edge}; the flat interval contributes $J_{\mathrm{h}}t_h/\hbar$.

are constant. The complete exchange angle is

$$\begin{aligned}\theta &= \frac{1}{\hbar}\int_0^{t_{\mathrm{r}}+t_h+t_{\mathrm{f}}} J(t)\,dt \\ &= \underbrace{\frac{1}{\hbar}\left[\int_0^{t_{\mathrm{r}}} J(t)\,dt + \int_{t_{\mathrm{r}}+t_h}^{t_{\mathrm{r}}+t_h+t_{\mathrm{f}}} J(t)\,dt\right]}_{\theta_{\mathrm{edge}}} + \frac{J_{\mathrm{h}}(v^\star)t_h}{\hbar}.\end{aligned} \tag{N.14}$$

Thus *edge* denotes both nonflat parts of the pulse together, not an additional Hamiltonian term. The singlet and triplets have exchange energies $-3J/4$ and $J/4$, so θ is their accumulated relative phase.

For two single-spin qubits, prepare $|01\rangle = |{\downarrow\uparrow}\rangle = (|T_0\rangle - |S\rangle)/\sqrt{2}$. The ideal exchange formula below assumes that ΔE_Z is negligible during the pulse or that its known evolution is canceled by a refocusing sequence. This is the condition stated in the introductory gate discussion of Section 6.1; it is repeated here because it determines the measured oscillation. Under this condition, exchange produces

$$|01\rangle \longmapsto e^{i\theta/4}\left[\cos(\theta/2)|01\rangle - i\sin(\theta/2)|10\rangle\right].$$

For this test input, the exchanged-state probability is

$$P_{10} = \sin^2(\theta/2). \tag{N.15}$$

Measuring the coupling from the oscillation. Choose one barrier-pulse amplitude and keep its rising and falling edges fixed, so θ_{edge} is the same in every trial. Vary only the flat hold time t_h. If ΔE_Z is negligible, Eqs. (N.14) and (N.15) predict

$$P_{10}(t_h) = \sin^2\!\left(\frac{\theta_{\mathrm{edge}}}{2} + \frac{J_{\mathrm{h}}t_h}{2\hbar}\right). \tag{N.16}$$

The barrier pulse therefore moves the state repeatedly between $|01\rangle$ and $|10\rangle$. Increasing t_h does not change the Hamiltonian during the hold; it only lets the same Hamiltonian act for longer.

For each chosen t_h, initialize both spins down and flip the second spin using Section N.1.2 to prepare $|01\rangle$. Apply the barrier pulse, return to the weak-exchange setting, and read the two spins separately using Section S.1. Repeat the full sequence many times. The assignment-corrected fraction of $|10\rangle$ outcomes estimates $P_{10}(t_h)$. Repeating this procedure for a set of hold times produces the exchange-oscillation curve.

Fit that curve to Eq. (N.16). Its horizontal period T_{ex} satisfies

$$\frac{J_{\mathrm{h}} T_{\mathrm{ex}}}{2\hbar} = \pi, \qquad J_{\mathrm{h}} = \frac{2\pi\hbar}{T_{\mathrm{ex}}} = \frac{h}{T_{\mathrm{ex}}}. \tag{N.17}$$

The horizontal offset determines θ_{edge} modulo 2π; it does not change the period. A measured contrast below one can arise from preparation and readout errors, dephasing, or a nonzero ΔE_Z, but it does not by itself change the ideal period-to-J_{h} conversion when ΔE_Z is negligible.

If ΔE_Z cannot be neglected, the flat-hold Hamiltonian is the matrix in Eq. (N.11). Since

$$H_0^2 = \frac{J_{\mathrm{h}}^2 + \Delta E_Z^2}{4} I,$$

an ideal rectangular hold starting from $|01\rangle$ gives

$$P_{10}(t_h) = \frac{J_{\mathrm{h}}^2}{J_{\mathrm{h}}^2 + \Delta E_Z^2} \sin^2\left(\frac{\sqrt{J_{\mathrm{h}}^2 + \Delta E_Z^2}}{2\hbar} t_h \right). \tag{N.18}$$

The Zeeman difference therefore both reduces the maximum transfer and increases the oscillation frequency. Use an independently measured ΔE_Z when fitting this expression; otherwise an imperfect readout contrast can be confused with the prefactor above. Fixed noninstantaneous pulse edges can instead be included by propagating the full measured waveform.

Finally, repeat the duration scan at several points $\mathbf{V}(v) = \mathbf{V}^0 + \mathbf{n}v$ on the compensated voltage path. The fitted value at each point gives the required response $J_{\mathrm{h}}(v)$.

N.2.3 Choosing the amplitude and duration of a partial-SWAP pulse

The preceding subsection supplies two measured quantities for each barrier-pulse amplitude v: the flat-hold coupling $J_{\mathrm{h}}(v)$ and the phase $\theta_{\mathrm{edge}}(v)$ accumulated on the fixed rising and falling edges. We now convert a target exchange gate into the two pulse settings that the control system needs: an amplitude $v^\star$ and a hold time t_h.

Assume first that ΔE_Z is negligible or refocused. The complete pulse then implements

$$U_{\mathrm{ex}}(\theta) = \exp[-i\theta\, \mathbf{s}_1 \cdot \mathbf{s}_2], \qquad \theta(v, t_h) = \theta_{\mathrm{edge}}(v) + \frac{J_{\mathrm{h}}(v) t_h}{\hbar}. \tag{N.19}$$

The family is called partial-SWAP because varying θ continuously changes how much of $|01\rangle$ is transferred to $|10\rangle$. The choices $\theta^\star = \pi$ and $\theta^\star = \pi/2$ give SWAP and $\sqrt{\mathrm{SWAP}}$, respectively, up to a common phase.

Choose an amplitude $v^\star$ inside the range already tested by the duration scans. The corresponding hold time is

$$t_h = \frac{\hbar[\theta^\star - \theta_{\mathrm{edge}}(v^\star) + 2\pi m]}{J_{\mathrm{h}}(v^\star)}, \qquad m \in \mathbb{Z}, \tag{N.20}$$

where the integer m selects a branch with an allowed positive duration. If $v^\star$ lies between measured amplitudes, interpolate both measured functions $J_{\mathrm{h}}(v)$ and $\theta_{\mathrm{edge}}(v)$; do not extrapolate beyond the calibrated voltage interval.

Prepare $|01\rangle$ and apply the selected pulse. The expected population check is

$$P_{10} = \sin^2(\theta^\star/2),$$

which equals 1 for SWAP and 1/2 for $\sqrt{\text{SWAP}}$. This check verifies the exchange area for that input. The multi-input tests in Appendix T are still required to verify the complete gate, including its relative phases. If ΔE_Z is appreciable and not refocused, the pulse implements evolution under the full H_0 rather than a pure partial-SWAP; its amplitude and duration must then be chosen from that full propagator.

N.2.4 Constructing CZ from exchange and local rotations

A single isotropic exchange pulse contains X_AX_B, Y_AY_B and Z_AZ_B with the same coefficient. It therefore produces a partial SWAP rather than a diagonal controlled-phase gate. Local z rotations can reverse the first two products without reversing the third, allowing two exchange pulses to retain only the Z_AZ_B interaction.

Let

$$G = \exp\left[-\frac{i\pi}{8}(X_AX_B + Y_AY_B + Z_AZ_B)\right]$$

be the exchange gate with area $\theta = \pi/2$, calibrated by the oscillation measurement above. Conjugation by Z_A reverses X_AX_B and Y_AY_B but leaves Z_AZ_B unchanged. Since the three Pauli products commute,

$$GZ_AGZ_A = \exp(-i\pi Z_AZ_B/4). \tag{N.21}$$

This is the exchange construction introduced in Eq. (6.9), now written in terms of the pulses calibrated in Section N.1.2. The remaining local phases are removed by

$$\text{CZ} = e^{-i\pi/4}R_{z,A}(-\pi/2)R_{z,B}(-\pi/2)\exp(-i\pi Z_AZ_B/4). \tag{N.22}$$

The exchange area and the local rotations are first set by their individual calibrations, but the conditional phase of the complete sequence must also be measured. Prepare qubit A in $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ and qubit B in $|b\rangle$, with $b = 0$ and 1 in separate repetitions. If the complete diagonal sequence gives phases χ_{ab}, the relative phase observed on A is $d_b = \chi_{1b} - \chi_{0b}$. Analysis rotations before logical-basis readout determine both $\cos d_b$ and $\sin d_b$. The nonlocal phase is

$$\Theta = d_1 - d_0 = \chi_{11} - \chi_{10} - \chi_{01} + \chi_{00}. \tag{N.23}$$

Adjust the exchange hold durations until $\Theta = \pi$ modulo 2π, then update the two local z rotations from the separately measured single-qubit phases. This measurement calibrates the complete composite gate rather than assuming that the phases of its component pulses add ideally.

O Singlet–Triplet Qubit Control

A singlet–triplet qubit stores its logical state in the singlet and unpolarized triplet of two electrons. Exchange separates their energies, while a Zeeman-energy difference mixes them. Two such qubits can also acquire a conditional phase through the Coulomb interaction between their state-dependent charge distributions. This appendix develops the one-qubit rotations and this capacitive two-qubit gate, including the residual interaction during nominally local operations.

Logical-state preparation and spin-to-charge readout are described in Appendices R and S.

O.1 Single-qubit gates for singlet–triplet qubits

The logical Bloch sphere has two control mechanisms. A difference between the two local Zeeman splittings mixes S and T_0 and therefore rotates about the logical x axis. Exchange separates their energies and rotates about the logical z axis. The gradient usually remains present when exchange is turned on, so an exchange pulse generally rotates about a tilted axis rather than exactly about z. We will measure the two terms, construct x and z rotations from them, and then combine those rotations into an arbitrary one-qubit gate.

O.1.1 Calibrating the two terms in the logical Hamiltonian

Use $|0_{\mathrm{ST}}\rangle = |S\rangle$ and $|1_{\mathrm{ST}}\rangle = |T_0\rangle$, and define

$$Z_{\mathrm{ST}} = |S\rangle\langle S| - |T_0\rangle\langle T_0|, \qquad X_{\mathrm{ST}} = |S\rangle\langle T_0| + |T_0\rangle\langle S|.$$

At a fixed voltage setting, the interdot barrier determines the exchange splitting J, while the difference $\Delta E_Z = E_{Z,1} - E_{Z,2}$ between the local Zeeman splittings gives the transverse term. The logical Hamiltonian is

$$H_{\mathrm{ST}} = -\frac{J}{2} Z_{\mathrm{ST}} + \frac{\Delta E_Z}{2} X_{\mathrm{ST}}. \tag{O.1}$$

Holding this setting for a time τ makes the two coefficients constant. With $E = \sqrt{J^2 + \Delta E_Z^2}$, the corresponding propagator is

$$U_{\mathrm{ST}}(\tau) = \cos\frac{E\tau}{2\hbar} I - i \sin\frac{E\tau}{2\hbar} \frac{\Delta E_Z X_{\mathrm{ST}} - J Z_{\mathrm{ST}}}{E}. \tag{O.2}$$

Thus one voltage setting specifies the rotation axis $(\Delta E_Z, 0, -J)$, and the hold time specifies the rotation angle $E\tau/\hbar$. Gate construction requires both coefficients to be measured.

To measure $|\Delta E_Z|$, prepare the separated singlet using Section R.2, set J negligible, and hold for a variable time τ. Equation (O.2) then gives

$$P_{T_0|S}(\tau) = \sin^2\left(\frac{\Delta E_Z \tau}{2\hbar}\right). \tag{O.3}$$

Use the pair readout of Section S.2 after every hold. If the measured oscillation period is T_Δ, then $|\Delta E_Z| = h/T_\Delta$.

Measuring J requires an input with both S and T_0 components, because a pure singlet is an eigenstate of the exchange term. At $J \simeq 0$, use the calibrated gradient to apply $R_x(\pi/2)$ to $|S\rangle$. Pulse to the voltage at which J is to be measured, hold for a variable time τ, return to the gradient setting, and apply $R_x(-\pi/2)$ before pair readout. If the hardware cannot reverse the gradient, the inverse is the equivalent positive rotation $R_x(3\pi/2)$. When the gradient is negligible during the exchange hold, the singlet-return probability is

$$P_S(\tau) = \frac{1 + \cos\theta_J}{2}, \qquad \theta_J = \frac{1}{\hbar}\int J(t)\, dt. \tag{O.4}$$

For a constant plateau, the oscillation period is $T_J = h/J$. Repeating the measurement at different compensated barrier voltages determines $J(V)$. If the gradient is appreciable during the hold, fit the measured return probability to

$$|\langle S| R_x(-\pi/2) U_{\mathrm{ST}}(\tau) R_x(\pi/2) |S\rangle|^2$$

using the independently measured ΔE_Z; the fitted parameter is then J. Propagators for the rising and falling edges can be included in the same product.

O.1.2 Constructing logical X and Z rotations

With $J \simeq 0$, Eq. (O.1) reduces to $H_{\mathrm{ST}} = \Delta E_Z X_{\mathrm{ST}}/2$. A hold of duration τ_x therefore implements

$$R_x(\theta_x) = \exp\!\left(-\frac{i\theta_x}{2} X_{\mathrm{ST}}\right), \qquad \theta_x = \frac{\Delta E_Z \tau_x}{\hbar}. \tag{O.5}$$

This supplies the first logical rotation axis.

For a target $R_z(-\theta_z)$, an exchange-dominated pulse suggests the duration $\tau_z = \hbar\theta_z/J$. The pulse is only a pure z rotation when the gradient term can be neglected. Starting from S, the unwanted $S \to T_0$ probability during the actual tilted rotation is

$$P_{T_0|S}(\tau_z) = \frac{\Delta E_Z^2}{J^2 + \Delta E_Z^2} \sin^2\!\left(\frac{E\tau_z}{2\hbar}\right). \tag{O.6}$$

This is a logical rotation error, not population outside the $\{S, T_0\}$ space. Requiring it to remain below q for every duration gives the sufficient condition

$$J \geq |\Delta E_Z| \sqrt{q^{-1} - 1}.$$

Even when this bound is satisfied, the actual rotation angle is $E\tau_z/\hbar$, so the duration is refined by measuring the phase of a superposition rather than by using $J\tau_z/\hbar$ alone.

If the accessible J does not satisfy this condition, a longer pulse does not help: it increases the angle but leaves the axis tilted. Rotations about two accessible axes can instead synthesize the required z rotation. For a concrete construction, take $\Delta E_Z = \Delta > 0$ and suppose the controls can set $J = 0$ or $J = \Delta$. The first setting gives $R_x(\theta)$ in time $\hbar\theta/\Delta$. The second setting has the axis $\mathbf{n} = (1, 0, -1)/\sqrt{2}$ and gives $A = R_{\mathbf{n}}(\pi)$ in time $\pi\hbar/(\sqrt{2}\Delta)$. Since this π rotation maps X_{ST} to $-Z_{\mathrm{ST}}$,

$$AR_x(\theta)A = -R_z(-\theta). \tag{O.7}$$

The minus sign is a global phase. The three voltage holds $J = \Delta$, $J = 0$, and $J = \Delta$, with the durations just given, therefore implement the target z rotation in this ideal model. For finite switching edges or voltage-dependent ΔE_Z, use the measured Hamiltonian throughout the waveform and multiply the corresponding propagators before adjusting the three durations.

O.1.3 Constructing an arbitrary logical one-qubit gate

The calibrated x and z rotations complete the single-qubit control problem. Up to a global phase, any target unitary on the $\{S, T_0\}$ space can be written as

$$U = R_z(\alpha)R_x(\beta)R_z(\gamma). \tag{O.8}$$

The rightmost operation acts first. Implement $R_x(\beta)$ at the $J \simeq 0$ setting using Eq. (O.5), and implement each z rotation either with a calibrated exchange-dominated hold or with the composite construction in Eq. (O.7). This sequence turns the measured functions $J(V)$ and $\Delta E_Z(V)$ into the voltage settings and durations of a specified logical gate.

O.2 Two-qubit gates for singlet–triplet qubits

Suppose a pulse does not transfer population between the logical basis states $|ab\rangle = |a\rangle_A |b\rangle_B$. In the ordered basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$, its unitary is diagonal. Direct factorization gives

$$\begin{aligned} &\mathrm{diag}\!\left(e^{i\chi_{00}}, e^{i\chi_{01}}, e^{i\chi_{10}}, e^{i\chi_{11}}\right) \\ &\quad = e^{i\chi_{00}} \left[\mathrm{diag}\!\left(1, e^{i(\chi_{10}-\chi_{00})}\right) \otimes \mathrm{diag}\!\left(1, e^{i(\chi_{01}-\chi_{00})}\right)\right] \\ &\qquad \times \mathrm{diag}\!\left(1, 1, 1, e^{i(\chi_{11}-\chi_{10}-\chi_{01}+\chi_{00})}\right). \end{aligned} \tag{O.9}$$

Denote the phases in the first local factor, the second local factor and the final factor by

$$\alpha = \chi_{10} - \chi_{00}, \qquad \beta = \chi_{01} - \chi_{00}, \qquad \Theta = \chi_{11} - \chi_{10} - \chi_{01} + \chi_{00}.$$

The factor $e^{i\chi_{00}}$ is a global phase. The square brackets are a tensor product of one gate acting on A and one gate acting on B, so they contain only single-qubit operations. The final four-dimensional diagonal matrix cannot be written as a product of one-qubit phase gates unless $\Theta = 0$ modulo 2π. It is therefore the entangling part of the evolution; after removing the local factors, $\Theta = \pi$ modulo 2π gives CZ.

This decomposition also identifies what must be measured. If B is held in $|b\rangle$, the relative phase acquired by A is $d_b = \chi_{1b} - \chi_{0b}$. Hence $d_0 = \alpha$, $d_1 = \alpha + \Theta$, and

$$\Theta = d_1 - d_0.$$

We now need an interaction that makes the logical energy of one qubit depend on the logical state of the other. Consider two singlet–triplet qubits A and B, each containing two electrons, and keep tunneling between the two qubits negligible. They still interact electrostatically: moving charge within A changes the Coulomb energy experienced by the electrons in B. Because a singlet can acquire double occupation while a triplet remains separated, this interaction can produce the required state-dependent energy and hence a conditional phase.

O.2.1 Projecting the electrostatic interaction onto the logical basis

We begin with one double dot before restricting it to the logical states. Retain the charge states $S(1,1)$ and $S(0,2)$ and the separated triplet $T_0(1,1)$. Taking the energy of $S(1,1)$ as zero gives

$$H_{\mathrm{dot}} = \delta_c |S(0,2)\rangle\langle S(0,2)| + E_T |T_0(1,1)\rangle\langle T_0(1,1)| + \sqrt{2}t_c \left[|S(1,1)\rangle\langle S(0,2)| + \mathrm{h.c.} \right]. \quad \text{(O.10)}$$

The tunnel term conserves spin. It connects the two singlets, while there is no low-energy $T_0(0,2)$ state because two electrons in the same lowest orbital must form a spin singlet.

The two qubits have copies H_A and H_B of this Hamiltonian, with their own values of δ_c, t_c and E_T. They also interact through the ordinary Coulomb interaction. Let $\varphi_{A,i}(\mathbf{r})$ and $\varphi_{B,j}(\mathbf{r})$ be the localized orbitals on their respective dots, and let $\hat{n}_{A,i}$ and $\hat{n}_{B,j}$ be the corresponding occupation operators. Neglecting tunneling and orbital overlap between the two qubits gives the density–density form

$$H_C^{AB} = \sum_{i,j=1}^{2} W_{ij} \hat{n}_{A,i} \hat{n}_{B,j}, \qquad W_{ij} = \int d^3r \, d^3r' \, |\varphi_{A,i}(\mathbf{r})|^2 V_C(\mathbf{r},\mathbf{r}') |\varphi_{B,j}(\mathbf{r}')|^2. \quad \text{(O.11)}$$

Here V_C is the electron–electron electrostatic interaction energy, and W_{ij} is the Coulomb energy of one electron in dot i of A and one in dot j of B. The Hamiltonian to be projected is

$$H = H_A + H_B + H_C^{AB}. \quad \text{(O.12)}$$

We next choose the logical basis of one double dot. Diagonalizing the singlet block of Eq. (O.10) gives its lower eigenstate

$$|0\rangle = |\widetilde{S}\rangle = \sqrt{1-p}\,|S(1,1)\rangle - \sqrt{p}\,|S(0,2)\rangle, \qquad |1\rangle = |\widetilde{T}_0\rangle \simeq |T_0(1,1)\rangle. \quad \text{(O.13)}$$
$$p = \frac{1}{2}\left(1 - \frac{\delta_c}{\sqrt{\delta_c^2 + 8t_c^2}}\right),$$

The singlet has mean dot occupations $(1-p, 1+p)$, whereas the triplet has $(1,1)$. Thus the logical states differ mainly in spin at the separated setting $p \simeq 0$, but acquire different charge distributions when tunneling makes p nonzero. This difference comes from spin-conserving tunneling together with the Pauli restriction, not from the electrostatic interaction itself.

Within the logical space of one double dot, the two occupation operators are

$$\begin{aligned} \hat{n}_1 &= (1-p)|0\rangle\langle 0| + |1\rangle\langle 1| = \left(1 - \frac{p}{2}\right) I - \frac{p}{2} Z, \\ \hat{n}_2 &= (1+p)|0\rangle\langle 0| + |1\rangle\langle 1| = \left(1 + \frac{p}{2}\right) I + \frac{p}{2} Z, \end{aligned} \tag{O.14}$$

Here $Z = |0\rangle\langle 0| - |1\rangle\langle 1|$. For qubits A and B, use the corresponding expressions with (p, Z) replaced by (p_A, Z_A) and (p_B, Z_B).

Before collecting the coefficients, separate a conditional energy shift from those that act on only one logical qubit. Let $E_{00}, E_{01}, E_{10}, E_{11}$ denote the diagonal energies of the projected Coulomb interaction in the ordered logical basis $(|00\rangle, |01\rangle, |10\rangle, |11\rangle)$, where $|0\rangle$ is the charge-hybridized singlet and $|1\rangle$ the separated triplet. Define $P_{S,q} = |0_q\rangle\langle 0_q| = (I + Z_q)/2$ for $q = A, B$. Directly matching the four diagonal entries gives

$$\begin{aligned} \mathrm{diag}(E_{00}, E_{01}, E_{10}, E_{11}) = E_{11} I &+ (E_{01} - E_{11}) P_{S,A} \\ &+ (E_{10} - E_{11}) P_{S,B} + \kappa P_{S,A} P_{S,B}, \\ \kappa &= E_{00} - E_{01} - E_{10} + E_{11}. \end{aligned} \tag{O.15}$$

The first term shifts all states equally; the next two shift the singlet energy in only one qubit. They therefore contribute global and local phases. Expanding $P_{S,A} P_{S,B} = (I + Z_A + Z_B + Z_A Z_B)/4$ shows that the conditional energy supplies a $Z_A Z_B$ coefficient $\kappa/4$ as well as additional global and local shifts.

Substituting the projected occupation operators into Eqs. (O.11) and (O.12), and collecting the I, Z_A, Z_B and $Z_A Z_B$ terms, gives

$$\begin{aligned} H_{\mathrm{eff}} &= C I + \frac{h_A}{2} Z_A + \frac{h_B}{2} Z_B + \frac{\kappa}{4} Z_A Z_B, \\ \kappa &= p_A p_B (W_{11} - W_{12} - W_{21} + W_{22}). \end{aligned} \tag{O.16}$$

Here C contains all state-independent energies, while h_A and h_B contain the isolated-qubit splittings and their Coulomb shifts. The result treats the interqubit Coulomb interaction as too weak to substantially change the separate-qubit charge eigenstates [16]. The Hamiltonian remains diagonal while magnetic-gradient or other terms that mix S and T_0 have negligible or refocused net action during the phase pulse; restoring such terms would add X_A or X_B components to H_{eff}.

The voltage pulse controls $p_A p_B$ and therefore $\kappa(t)$. Start at the separated setting, where $p_A \simeq p_B \simeq 0$ and $\kappa \simeq 0$. Ramp towards the two singlet charge anticrossings, hold with $p_A p_B > 0$, and reverse the ramps. If each branch follows its instantaneous charge eigenstate and returns to the same logical basis state, Eq. (O.16) gives the conditional phase

$$\Theta = -\frac{1}{\hbar} \int \kappa(t)\, dt. \tag{O.17}$$

A CZ gate requires $\Theta = \pi$ modulo 2π. For a constant hold, $|\kappa|\tau/\hbar = \pi$ is an initial duration estimate; the ramps also contribute phase.

O.2.2 Measuring the conditional phase and obtaining CZ

The pulse of the preceding subsection leaves each logical basis state in place but gives it a phase χ_{ab}. A basis-state population measurement cannot reveal these phases. A superposition can: fix B in $|b\rangle$ and prepare A in $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. The full two-qubit state then evolves as

$$\begin{aligned} |\Psi_b^{\text{in}}\rangle &= |+\rangle_A |b\rangle_B, \\ |\Psi_b^{\text{out}}\rangle &= \frac{e^{i\chi_{0b}}}{\sqrt{2}}(|0\rangle_A + e^{id_b}|1\rangle_A)|b\rangle_B, \qquad d_b = \chi_{1b} - \chi_{0b}. \end{aligned} \tag{O.18}$$

Because the pulse is diagonal, B remains in $|b\rangle$. The relative phase between the $|0\rangle_A|b\rangle_B$ and $|1\rangle_A|b\rangle_B$ amplitudes is d_b. The probabilities of the two logical-basis outcomes on A remain $1/2$, but the probability of the $+1$ outcome when measuring X_A is $(1 + \cos d_b)/2$. Fixing B first in $|0\rangle$ and then in $|1\rangle$ gives the relative phases d_0 and d_1. Their difference is exactly the phase that remains after the single-qubit factors in Eq. (O.9) are removed:

$$\Theta = d_1 - d_0 = \chi_{11} - \chi_{10} - \chi_{01} + \chi_{00}. \tag{O.19}$$

In particular, a phase common to both choices of b is a local phase on A and does not contribute to Θ.

To determine each d_b, measure the interference in two complementary bases. The output in Eq. (O.18) has $\langle X_A\rangle_b = \cos d_b$ and $\langle Y_A\rangle_b = \sin d_b$. Logical-basis readout measures Z_A; preceding it with $R_y(-\pi/2)$ gives the X_A component, and preceding it with $R_x(\pi/2)$ gives the Y_A component. Consequently,

$$d_b = \operatorname{atan2}(\langle Y_A\rangle_b, \langle X_A\rangle_b) \pmod{2\pi}.$$

Repeat this measurement with B in $|0\rangle$ and $|1\rangle$ while varying the complete pulse duration. Following the phase continuously resolves its 2π branches. The duration is chosen where the measured difference Θ reaches π modulo 2π; this includes the phase accumulated during the ramps as well as the hold.

At that duration, $d_0 = \alpha$ gives the phase of the local factor on A. To find the local factor on B, instead prepare B in $|+\rangle$ and fix A in $|0\rangle$. Its measured relative phase is $\beta = \chi_{01} - \chi_{00}$. With the convention $R_z(\phi) = e^{-i\phi Z/2}$, applying $R_{z,A}(-\alpha)R_{z,B}(-\beta)$ after the interaction cancels these two factors. The remaining gate is CZ, up to a global phase, when $\Theta = \pi$ modulo 2π. This inference assumes the pulse is diagonal in the logical basis, so basis-state population transfer must also be checked.

O.2.3 Canceling local phases with an echoed entangling pulse

A controlled-phase pulse generally rotates each qubit separately as well as entangling them. Section O.2.2 removes the measured local phases with subsequent rotations. If those phases also have nearly constant but unknown offsets during each trial, a fixed correction cannot remove them in every trial. We instead reverse their accumulation halfway through the interaction, without reversing the desired two-qubit term.

For an idle spin, the echo sequence of Section U.1.1 cancels a different error. A central π pulse reverses the sign of Z, so a nearly constant unwanted Zeeman offset contributes opposite phases in the two halves. The cancellation also removes a desired phase generated by that same Z term. An idle-memory echo therefore cannot simply be inserted into a z gate without recalculating the intended operation.

For a two-qubit diagonal interaction, echo can instead remove local phases while retaining entanglement. Suppose the Hamiltonian during the interaction is

$$H = \frac{a}{2}Z_A + \frac{b}{2}Z_B + \frac{\kappa}{4}Z_A Z_B,$$

where a, b, κ are constant energies during the sequence. Simultaneous π rotations about x act, up to global phase, as $P = X_A X_B$. Conjugation changes Z_A and Z_B to their negatives but leaves $Z_A Z_B$ unchanged. Thus

$$\begin{aligned} PHP &= -\frac{a}{2}Z_A - \frac{b}{2}Z_B + \frac{\kappa}{4}Z_A Z_B, \\ U_{\text{echo}} &= Pe^{-iH\tau/(2\hbar)}Pe^{-iH\tau/(2\hbar)} = e^{-i\kappa\tau Z_A Z_B/(4\hbar)}. \end{aligned} \tag{O.20}$$

The last equality follows because all three Z products commute. The two π rotations on each qubit occur at the middle and end, so the logical basis is restored. Local quasistatic fluctuations in a, b cancel in the same way, whereas a fluctuation in κ does not. This construction assumes that the π pulses are short compared with the interaction and error time scales; their actual durations and errors enter the subsequent conditional-phase calibration.

To use the sequence, split the intended interaction time into two equal halves, apply simultaneous x flips at the middle and end, and choose the total time from the measured κ. The retained conditional phase is still $-\kappa\tau/\hbar$. The conditional-phase measurement then tests the complete sequence, including its finite-duration flips. Echo has canceled the local terms; it has not corrected an error in the coupling κ.

O.2.4 Parallel single-qubit gates in the presence of capacitive coupling

The same electrostatic interaction matters when the intended operation is two separate one-qubit gates. An exchange pulse on one singlet–triplet qubit changes the charge admixture of its singlet and hence its logical energy splitting. If both qubits receive such pulses at once, both singlets can have different charge distributions from their respective triplets. The Coulomb energy then depends on the logical states of *both* qubits. Applying the voltage pulses simultaneously does not, by itself, make their joint operation a tensor product of the two separately calibrated gates.

For exchange-dominated pulses whose charge branches return to their logical states and whose S–T_0 mixing is negligible throughout, the instantaneous logical Hamiltonian is the diagonal Hamiltonian of Eq. (O.16). Its coefficients can vary throughout both pulse edges and holds. All its terms commute at different times, so, up to a global phase,

$$\begin{aligned} U_{\parallel} &= R_{z,A}(\phi_A)R_{z,B}(\phi_B)\exp\left(-\frac{i\Phi_{AB}}{4}Z_A Z_B\right), \\ \phi_q &= \frac{1}{\hbar}\int h_q(t)\,dt, \qquad \Phi_{AB} = \frac{1}{\hbar}\int \kappa(t)\,dt = -\Theta. \end{aligned} \tag{O.21}$$

The h_q include the Coulomb-induced local shifts, so even the one-qubit angles must be calibrated for the simultaneous waveform. The last factor is an additional two-qubit operation. It has no entangling part when $\Phi_{AB} = 0$ modulo 2π; otherwise local z corrections alone cannot turn $U_{\parallel}$ into the desired product of one-qubit gates. This statement concerns the complete pulse area, not just the value of κ at the hold point. For gates with a finite error allowance, the phase need only be close enough to a multiple of 2π.

There is a control tradeoff, but no rule that useful exchange requires a large κ. In the charge model of Eq. (O.10), take $\delta_c \gg t_c > 0$ and neglect any other contribution to the singlet–triplet

splitting. The lower singlet energy shift and double-occupation probability then give

$$J_{\mathrm{tun}} \simeq \frac{2t_c^2}{\delta_c}, \qquad p \simeq \frac{2t_c^2}{\delta_c^2} = \frac{J_{\mathrm{tun}}}{\delta_c}. \tag{O.22}$$

At fixed t_c, moving farther from the charge anticrossing reduces both p and the exchange generated by virtual tunneling. If the barrier and detuning can be changed together, the same J_{tun} can instead be obtained at a larger δ_c and a smaller p, until the accessible tunnel coupling or other device constraints limit that choice. Since this model gives $\kappa = p_A p_B(W_{11} - W_{12} - W_{21} + W_{22})$, specified values of J_A and J_B alone do not fix the unwanted conditional phase. Parking B with $p_B \simeq 0$ suppresses κ while A is pulsed, but also reduces the tunneling-generated exchange of B.

For a proposed pair of simultaneous z gates, the relevant question is therefore how much Θ their actual joint waveform adds. The conditional-phase comparison of Section O.2.2 can be applied to that waveform; one can then choose operating points with smaller charge admixture, compensate the two-qubit evolution, or schedule the gates separately if the extra phase is too large. When transverse X_A or X_B terms act appreciably during a pulse, they do not commute with $Z_A Z_B$, and the simple phase integral in Eq. (O.21) must be replaced by the full two-qubit evolution. Experiments on GaAs singlet–triplet arrays have measured state-dependent capacitive shifts during simultaneous operation and have also demonstrated parallel exchange oscillations [40, 41]. Their coupling strengths are device dependent and do not set a value for a silicon array.

P Exchange-Only Qubit Control

Pair exchange acts along different logical axes within a three-spin exchange-only qubit. A coupling between two encoded blocks additionally allows amplitude to leave their individual code spaces, so an encoded two-qubit gate must control the complete six-spin evolution. This appendix gives the single-pair rotations, their calibration and the conditions on a complete two-qubit sequence.

The logical basis is defined in Eq. (3.4); preparation and pair readout are developed in Appendices R and S.

P.1 Single-qubit gates for exchange-only qubits

The two interdot barriers provide rotations about different logical axes. We first identify those axes and the conditions under which a single-pair pulse has a simple propagator. Pair readout then calibrates the two pulse areas in different ways. Finally, we ask whether simultaneous exchange can produce a direct logical x rotation over the complete voltage waveform.

P.1.1 Logical Hamiltonian and single-pair rotations

In the three-spin basis of Eq. (3.4), logical 0 has pair 1–2 in a singlet and logical 1 has that pair in a triplet. Exchange on 1–2 therefore produces a relative phase in this basis. Exchange on 2–3 is diagonal in a different pair-spin basis, so it both mixes and shifts the chosen logical states. To establish the restriction to this two-dimensional space, consider the physical Hamiltonian in Eq. (6.21). Each pair exchange is $\mathbf{s}_i \cdot \mathbf{s}_j = [(\mathbf{s}_i + \mathbf{s}_j)^2 - 3I/2]/2$, so it commutes with all components of the pair's total spin and with the remaining spin. It therefore commutes with $\mathbf{s}_{\mathrm{tot}}^2$ and $s_{\mathrm{tot},z}$. Since the code spans the full $(S, m) = (1/2, +1/2)$ sector, arbitrary time-dependent exchange pulses stay within it. Additional nonuniform magnetic fields need not preserve this space.

Define $P_{\mathrm{E}} = |0_{\mathrm{E}}\rangle\langle 0_{\mathrm{E}}| + |1_{\mathrm{E}}\rangle\langle 1_{\mathrm{E}}|$. Then $(I - P_{\mathrm{E}})H_{\mathrm{E}}P_{\mathrm{E}} = 0$. Restricting $P_{\mathrm{E}}H_{\mathrm{E}}P_{\mathrm{E}}$ means computing its matrix elements in the encoded basis. Using $\mathbf{s}_i \cdot \mathbf{s}_j = \mathrm{SWAP}_{ij}/2 - I/4$ gives

$$\begin{aligned} P_{\mathrm{E}}\mathbf{s}_1 \cdot \mathbf{s}_2 P_{\mathrm{E}}|_{\mathrm{E}} &= \begin{pmatrix} -3/4 & 0 \\ 0 & 1/4 \end{pmatrix} = -\frac{1}{4}I - \frac{1}{2}Z_{\mathrm{E}}, \\ P_{\mathrm{E}}\mathbf{s}_2 \cdot \mathbf{s}_3 P_{\mathrm{E}}|_{\mathrm{E}} &= \begin{pmatrix} 0 & \sqrt{3}/4 \\ \sqrt{3}/4 & -1/2 \end{pmatrix} = -\frac{1}{4}I + \frac{1}{4}Z_{\mathrm{E}} + \frac{\sqrt{3}}{4}X_{\mathrm{E}}. \end{aligned} \tag{P.1}$$

These are the matrices of the physical exchanges acting within the code, with $X_{\mathrm{E}}, Z_{\mathrm{E}}$ the encoded Pauli operators. The two compensated interdot barriers control J_{12} and J_{23}. Combining the matrices with these coefficients and dropping the common energy $-(J_{12} + J_{23})I/4$ gives

$$H_{\mathrm{E,q}}(t) = \frac{\sqrt{3}J_{23}(t)}{4}X_{\mathrm{E}} + \left[-\frac{J_{12}(t)}{2} + \frac{J_{23}(t)}{4}\right]Z_{\mathrm{E}}. \tag{P.2}$$

The following construction assumes selective exchange pulses. To realize $U_{12,\mathrm{E}}$, pulse the 1–2 barrier so that $J_{12}(t)$ is active while $J_{23}(t)$ is negligible. To realize $U_{23,\mathrm{E}}$, pulse the 2–3 barrier with $J_{12}(t)$ negligible. Having two barriers does not by itself establish that the inactive exchange can be suppressed; this is a condition to check at the operating point. For either single-pair pulse, define its exchange area by

$$\theta_{ij} = \frac{1}{\hbar}\int J_{ij}(t)\,dt.$$

To derive the ideal propagators, set the inactive coupling to zero in Eq. (P.2). A 1–2 pulse then has $H_{\mathrm{E,q}}(t) = -J_{12}(t)Z_{\mathrm{E}}/2$, whereas a 2–3 pulse has $H_{\mathrm{E,q}}(t) = J_{23}(t)(\sqrt{3}X_{\mathrm{E}} + Z_{\mathrm{E}})/4$. Integrating each of these single-pair Hamiltonians gives, up to the global phase of the omitted identity term,

$$\begin{aligned} U_{12,\mathrm{E}}(\theta_{12}) &= \exp(i\theta_{12}Z_{\mathrm{E}}/2), \\ U_{23,\mathrm{E}}(\theta_{23}) &= \exp\left[-\frac{i\theta_{23}}{4}(\sqrt{3}X_{\mathrm{E}} + Z_{\mathrm{E}})\right]. \end{aligned} \tag{P.3}$$

If the inactive coupling is appreciable, the actual pulse is not described by these single-pair propagators; its evolution must be calculated from Eq. (P.2) with both couplings. These are the two nonparallel logical rotations summarized earlier in Eq. (6.23); they are written here because the following measurements calibrate their two pulse areas.

P.1.2 Calibrating the two pair-exchange pulse areas

Pair readout distinguishes singlet and triplet character on dots 1–2. A 2–3 pulse mixes those logical states, so its area appears directly as a population oscillation. A 1–2 pulse changes only their relative phase and needs a preparation and analysis rotation before the same readout can reveal it.

Measuring the 2–3 coupling. Prepare $|0_{\mathrm{E}}\rangle$ using Section R.3, suppress J_{12}, and pulse only the 2–3 barrier. For a constant $J_{23} = J$,

$$H_{23}^{\mathrm{E}} = \frac{J}{4}Z_{\mathrm{E}} + \frac{\sqrt{3}J}{4}X_{\mathrm{E}}, \qquad P_{1_{\mathrm{E}}}(\tau) = \frac{3}{4}\sin^2\left(\frac{J\tau}{2\hbar}\right). \tag{P.4}$$

The oscillation reaches 3/4 rather than 1 because this rotation axis is tilted relative to the preparation and pair-readout axis. Vary the hold time, return to the pair-readout setting on dots 1–2, and count pair-triplet outcomes using Section S.3. The period gives J/h. Repeat at

different compensated barrier voltages to determine J_{23} along the control path. Pair-triplet leakage states can produce the same blocked record, so this interpretation assumes that the evolution remains in the code space.

For a pulse with variable coupling but $J_{12} = 0$, the same probability is $3\sin^2(\theta_{23}/2)/4$, where $\theta_{23} = \hbar^{-1}\int J_{23}(t)\,dt$. A duration scan with fixed edges calibrates this complete pulse area, including its edge contribution.

Measuring the 1–2 phase. A 1–2 pulse only adds phase to the initial $|0_\mathrm{E}\rangle$, so first apply $A = U_{23,\mathrm{E}}(\beta)$ with $\cos\beta = -1/3$. The expression above gives $P_{1_\mathrm{E}} = 3(1-\cos\beta)/8 = 1/2$: $A|0_\mathrm{E}\rangle$ is an equal-weight superposition. Insert the 1–2 pulse and then apply $A^\dagger$ before pair readout. Using the pair propagators in Eq. (P.3),

$$P_{0_\mathrm{E}} = \left|\langle 0_\mathrm{E}|A^\dagger e^{i\theta_{12}Z_\mathrm{E}/2}A|0_\mathrm{E}\rangle\right|^2 = \frac{1+\cos\theta_{12}}{2}. \tag{P.5}$$

The inverse rotation can be supplied by a positive 2–3 pulse of area $2\pi-\beta$, up to global phase. Vary the 1–2 hold time between these fixed preparation and analysis pulses to calibrate θ_{12} from the return oscillation. Residual couplings during either pulse must be included in the two-axis Hamiltonian above.

P.1.3 Testing a direct logical x rotation

First consider constant simultaneous couplings J_{12} and J_{23} during a hold of duration τ. With $J_{23} = 0$, the hold gives a pure z rotation $R_z(-\theta_{12})$. With $J_{12} = J_{23}/2$, the Z_E coefficient cancels and the hold gives a pure x rotation:

$$H_{\mathrm{E,q}} = \frac{\sqrt{3}J_{23}}{4}X_\mathrm{E}, \qquad \vartheta_x = \frac{\sqrt{3}J_{23}\tau}{2\hbar}. \tag{P.6}$$

For arbitrary constant values of the two couplings, pair readout after the hold gives

$$P_{1_\mathrm{E}}(\tau) = \frac{3J_{23}^2}{3J_{23}^2+(J_{23}-2J_{12})^2}\sin^2\left(\frac{\tau\sqrt{3J_{23}^2+(J_{23}-2J_{12})^2}}{4\hbar}\right). \tag{P.7}$$

At fixed J_{23}, scanning J_{12} and τ identifies the constant setting with vanishing z component by its maximum oscillation contrast; the period then determines the x-rotation rate. The two barrier voltages must be calibrated when applied together. Their separately measured responses only suggest an initial setting, and readout contrast and decay must be accounted for when interpreting the measured amplitude.

A real voltage pulse can have time-dependent couplings on its edges. The constant-hold result does not by itself show that the complete pulse is a pure R_x. One sufficient condition is $J_{12}(t) = J_{23}(t)/2$ throughout the active waveform, with other logical terms negligible; then $\vartheta_x = (\sqrt{3}/2\hbar)\int J_{23}(t)\,dt$. If the ratio changes, the complete evolution must be evaluated and calibrated as a gate. When the available waveform does not produce the desired R_x, compose it from the two calibrated nonparallel pair rotations instead.

These pulse-area and axis measurements characterize coherent gates. An encoded-qubit relaxation measurement asks a separate question: prepare an excited logical state and measure its population after a wait, as in Section U.1.2. Extra leakage channels may require tracking more than two populations.

P.2 Two-qubit gates for exchange-only qubits

Let qubit A occupy spins $(1, 2, 3)$ and qubit B occupy $(4, 5, 6)$. Exchange within each triple dot produces the one-qubit rotations of the preceding section. A two-qubit gate additionally uses the boundary coupling J_{34} between the two blocks. That coupling can entangle them, but it does not preserve each block's code space by itself. The control problem is therefore to combine intra-block and inter-block exchange pulses so that the final state returns to both code spaces while retaining a nonlocal logical operation.

P.2.1 Why one boundary-exchange pulse is insufficient

For a linear six-dot array, the physical Hamiltonian is

$$H_6(t) = \sum_{i=1}^{5} J_{i,i+1}(t)\, \mathbf{s}_i \cdot \mathbf{s}_{i+1}. \tag{P.8}$$

The boundary term $J_{34}\mathbf{s}_3 \cdot \mathbf{s}_4$ transfers spin angular momentum between the blocks. If $\mathbf{S}_A = \mathbf{s}_1 + \mathbf{s}_2 + \mathbf{s}_3$ and $\mathbf{S}_B = \mathbf{s}_4 + \mathbf{s}_5 + \mathbf{s}_6$, then

$$[\mathbf{s}_3 \cdot \mathbf{s}_4, \mathbf{S}_A^2] \neq 0, \qquad [\mathbf{s}_3 \cdot \mathbf{s}_4, \mathbf{S}_B^2] \neq 0.$$

A single boundary pulse can therefore move amplitude from an $S = 1/2$ encoded block into an $S = 3/2$ state. It may entangle the six physical spins without defining a unitary gate on the two logical qubits.

P.2.2 Conditions on a complete exchange sequence

A candidate encoded gate is an ordered product

$$U_6 = U_{i_M j_M}(\theta_M) \cdots U_{i_1 j_1}(\theta_1), \qquad U_{ij}(\theta) = e^{-i\theta\, \mathbf{s}_i \cdot \mathbf{s}_j}, \qquad \theta = \frac{1}{\hbar} \int J_{ij}(t)\, dt. \tag{P.9}$$

The order matters because exchanges on overlapping pairs do not commute. Let P_{AB} project onto the product of the two exchange-only code spaces. A valid encoded gate must satisfy

$$(I - P_{AB}) U_6 P_{AB} = 0, \qquad P_{AB} U_6 P_{AB}|_{\text{code}} = e^{i\chi} U_{\text{logical}}. \tag{P.10}$$

The first condition allows intermediate leakage but requires its amplitude to cancel at the end. The second condition fixes the logical action after that return. For an encoded CNOT, $U_{\text{logical}} = \text{CNOT}_{A \to B}$, reproducing the target stated in Eq. (6.28).

P.2.3 Calibrating and testing the complete sequence

Each pair pulse is first calibrated through its measured exchange area, but those separate calibrations do not establish the six-spin gate. Program the ordered pulse list, prepare each of the four logical basis inputs, and apply the complete sequence before the exchange-only readout of both blocks. The final code-space population tests the first condition above; the logical outcomes test the basis-state action of the second. Superposition inputs and analysis rotations are additionally required to determine the relative phases and distinguish the target gate from a truth table with incorrect coherent phases.

Exact leakage-free exchange-only CNOT sequences supply a concrete pulse ordering and set of exchange areas [14]. Their individual areas are initial settings, not experimental facts: the full sequence must be retested after the pair responses, switching edges and residual simultaneous couplings are included. A single calibrated J_{34} pulse cannot replace this complete sequence.

Q From Tunneling Amplitudes to Rate Equations

Section 5 uses reservoir entry and escape rates to describe loading and spin-selective readout. This appendix derives those rates from tunneling amplitudes and states the conditions under which occupation probabilities obey a master equation.

Q.1 From tunneling amplitudes to a master equation

An electron entering from the source and subsequently leaving into the drain produces the charge cycle

$$\underbrace{|N-1,0\rangle}_{|A\rangle} \xrightarrow{\text{source}\to\text{dot}} \underbrace{|N,0\rangle}_{|B\rangle} \xrightarrow{\text{dot}\to\text{drain}} |A\rangle,$$

where 0 denotes a ground state. Quantum mechanics assigns amplitudes to these tunneling processes. A rate equation instead evolves the probabilities of occupying A and B, treating entry and exit as random events. We establish the connection between these descriptions below.

Quantum amplitudes and the Green function. During an interval with fixed electrode voltages, write the Hamiltonian of the dot and its reservoirs as

$$H_{\text{tot}} = H_{\text{dot}} + H_{\text{R}} + H_{\text{T}}, \qquad U(\tau) = e^{-iH_{\text{tot}}\tau/\hbar}, \tag{Q.1}$$

where H_{T} transfers electrons between the dot and the reservoirs. Let $|i\rangle, |j\rangle$ be dot states and $|r\rangle, |s\rangle$ be many-electron states of the reservoirs; $|i,r\rangle = |i\rangle \otimes |r\rangle$. The transition amplitude is

$$\mathcal{A}_{js,ir}(\tau) = \langle j,s|U(\tau)|i,r\rangle. \tag{Q.2}$$

It is a matrix element of the quantum propagator. With the following normalization, the retarded Green operator for the full Schrödinger equation is

$$G^R(\tau) = -\frac{i}{\hbar}\theta(\tau)U(\tau), \qquad (i\hbar\partial_\tau - H_{\text{tot}})G^R(\tau) = \delta(\tau)\mathbb{I}, \tag{Q.3}$$

where θ is the step function. Thus the amplitude is related to a matrix element of G^R, rather than being a probability.

Probabilities and the addition of channels. Prepare the dot in $|i\rangle$, initially uncorrelated with reservoirs whose density matrix is $\rho_{\text{R}} = \sum_r w_r |r\rangle\langle r|$, with $w_r \geq 0$ and $\sum_r w_r = 1$. If only the final dot state is measured, the conditional probability is

$$P_{j\leftarrow i}(\tau) = \sum_{r,s} w_r \left|\mathcal{A}_{js,ir}(\tau)\right|^2. \tag{Q.4}$$

The sum over r averages the initial reservoir mixture; the sum over s includes its unobserved, orthogonal final states.

For the two entry processes in Appendix I, the final dot states $|N,0\rangle$ and $|N,1\rangle$ are orthogonal. Suppressing the reservoir labels for a moment, their contributions satisfy

$$\||a_0|N,0\rangle + a_1|N,1\rangle\|^2 = |a_0|^2 + |a_1|^2. \tag{Q.5}$$

The same addition holds after summing over reservoir states. Consequently, the probability of entry into either state is the sum of the two entry probabilities. This follows from quantum orthogonality, even if the final superposition is coherent. By contrast, two coherent contributions $a^{(1)}$ and $a^{(2)}$ to the *same full final state* give $|a^{(1)} + a^{(2)}|^2$, including an interference term.

From a transition probability to a rate. To describe repeated tunneling by rates, assume weak coupling to large reservoirs whose distributions remain approximately fixed. Let τ_{R} be the decay time of their correlation functions. We consider intervals long enough for reservoir memory to decay but short enough that a tunneling event is unlikely:

$$\tau_{\mathrm{R}} \ll \Delta t \ll \Gamma_{\max}^{-1}, \qquad P_{j\leftarrow i}(\Delta t) \simeq \Gamma_{j\leftarrow i}\Delta t \quad (j \neq i), \tag{Q.6}$$

where $\Gamma_{\max}$ is the largest total escape rate from a retained dot state. This defines a conditional rate: the dot is already in i, and $\Gamma_{j\leftarrow i}$ gives its probability per unit time of making the transition to j. Its unit is s^{-1}, not energy or velocity.

The linear dependence arises in weak-coupling perturbation theory after summing over the reservoir continuum. To see the connection, write $H_{\mathrm{dot}}|i\rangle = E_i|i\rangle$, $H_{\mathrm{R}}|r\rangle = \epsilon_r|r\rangle$, and define $T_{js,ir} = \langle j, s|H_{\mathrm{T}}|i, r\rangle$ and $\Delta E_{js,ir} = E_j + \epsilon_s - E_i - \epsilon_r$. First-order time-dependent perturbation theory gives the interaction-picture amplitude and its probability:

$$\begin{aligned} \mathcal{A}^{(1)}_{js,ir}(\tau) &= -\frac{iT_{js,ir}}{\hbar}\int_0^\tau e^{i\Delta E_{js,ir}u/\hbar}\,du, \\ |\mathcal{A}^{(1)}_{js,ir}(\tau)|^2 &= \frac{4|T_{js,ir}|^2\sin^2(\Delta E_{js,ir}\tau/2\hbar)}{(\Delta E_{js,ir})^2}. \end{aligned} \tag{Q.7}$$

Under a continuum energy integral, the long-time kernel obeys

$$\frac{4\sin^2(\Delta E\tau/2\hbar)}{\tau\Delta E^2} \longrightarrow \frac{2\pi}{\hbar}\delta(\Delta E).$$

Substituting into Eq. (Q.4) gives Fermi's golden rule:

$$\Gamma_{j\leftarrow i} = \frac{2\pi}{\hbar}\sum_{r,s} w_r|T_{js,ir}|^2\delta(E_j + \epsilon_s - E_i - \epsilon_r). \tag{Q.8}$$

The delta function enforces conservation of the total energy. The reservoir sum includes its available occupied and empty states, giving the Fermi factors in Eq. (G.7). Here the long-time form of the perturbative kernel is used within the interval in Eq. (Q.6), before appreciable depletion of the initial state. It is not the exact $\tau \to 0$ limit: at very short times the perturbative probability above is proportional to τ^2.

In the positive-bias regime where entry occurs from the source and exit into the drain, set $\Gamma_{\mathrm{in}} = \Gamma_{B\leftarrow A}$ and $\Gamma_{\mathrm{out}} = \Gamma_{A\leftarrow B}$. If two orthogonal final states can be loaded, probability addition in Eq. (Q.5) gives

$$\Gamma_{\mathrm{in}} = \Gamma_{\mathrm{S0}} + \Gamma_{\mathrm{S1}}. \tag{Q.9}$$

Each term is itself calculated from quantum amplitudes; adding rates does not replace addition of amplitudes for coherent paths to the same final state.

The master equation for occupation probabilities. A closed equation for occupations requires a further condition: coherences between the retained dot states must not feed back appreciably into their populations. For separated levels, rapidly oscillating coherence terms can often be averaged out on the same coarse time scale. Together with weak coupling and short reservoir memory, this gives the population description used here [42]. If coherences or reservoir memory remain important, these occupation probabilities alone do not provide a closed description.

Let $p(t)$ be the probability of B and $1 - p(t)$ that of A. Over one coarse time step,

$$\begin{aligned} p(t+\Delta t) \simeq\ & \underbrace{p(t)(1-\Gamma_{\mathrm{out}}\Delta t)}_{\text{already in } B \text{ and remains there}} \\ &+ \underbrace{[1-p(t)]\Gamma_{\mathrm{in}}\Delta t}_{\text{starts in } A \text{ and enters } B}. \end{aligned} \tag{Q.10}$$

Subtracting $p(t)$ and dividing by Δt yields the differential equation for the coarse-grained evolution,

$$\dot{p} = \Gamma_{\text{in}}(1-p) - \Gamma_{\text{out}}p. \tag{Q.11}$$

This is the two-state *Pauli master equation*, and reproduces Eq. (G.4) with $\Gamma_{\text{in}} = \Gamma_{\text{S}}$ and $\Gamma_{\text{out}} = \Gamma_{\text{D}}$. For more dot states, with occupation probabilities p_i, the same gain-minus-loss argument gives

$$\dot{p}_j = \sum_{i\neq j} \Gamma_{j\leftarrow i} p_i - p_j \sum_{i\neq j} \Gamma_{i\leftarrow j}. \tag{Q.12}$$

Orthogonal-channel addition alone does not justify merging their final-state occupations. The two-state treatment of excited-state transport in Appendix I also uses fast relaxation to the N-electron ground state; otherwise the excited states need their own p_i.

Waiting times and current. For a constant total entry rate, the probability $q(t)$ that no electron has yet entered, starting in A, obeys $q(t+\Delta t) \simeq q(t)(1-\Gamma_{\text{in}}\Delta t)$. Thus

$$q(t) = e^{-\Gamma_{\text{in}}t}, \qquad \langle \tau_{\text{in}} \rangle = \int_0^\infty q(t)\, dt = \Gamma_{\text{in}}^{-1}. \tag{Q.13}$$

The corresponding mean waiting time for exit, starting in B, is Γ_{out}^{-1}. These are conditional waiting times. In a running transport cycle, the ensemble-averaged numbers of entries and exits per unit time are instead $\Gamma_{\text{in}}(1-p)$ and $\Gamma_{\text{out}}p$. In the stationary state these are equal, and the current convention of Appendix G gives

$$p_{\text{st}} = \frac{\Gamma_{\text{in}}}{\Gamma_{\text{in}}+\Gamma_{\text{out}}}, \qquad I = e\Gamma_{\text{out}}p_{\text{st}} = e\frac{\Gamma_{\text{in}}\Gamma_{\text{out}}}{\Gamma_{\text{in}}+\Gamma_{\text{out}}}. \tag{Q.14}$$

The rates are therefore quantum-mechanically determined inputs to a classical stochastic approximation for the occupations; neither a rate nor an occupation probability is a quantum propagation amplitude.

R Preparing Spin-Qubit States

This appendix develops the preparation conditions used in Section 5: the selected charge window, the relaxation or loading time, and the transfer into the encoded state.

Preparing a qubit input requires fixing three things: the number of electrons, their spin state and, for an encoded qubit, how they are distributed among the dots. Loading selects the charge sector; relaxation or spin-selective loading selects the spin state; an isolated charge-transfer ramp can then redistribute the electrons without exchanging them with a reservoir. We use these processes to prepare $|\downarrow\rangle$ for a single-spin qubit, a separated singlet for a singlet–triplet qubit, and a pair singlet with the chosen third-spin state for an exchange-only qubit.

For each encoding below, we first construct that preparation, then use measured rates or gaps to choose its duration. The output is a state at the holding voltages, ready for a gate. The loading window, the hold time, and the separation speed are different controls and will be determined from different physical conditions.

R.1 A single-spin qubit

We want one electron in the lower Zeeman state. Keep a static field B along z, with the convention $E_Z = g_{\text{e}}\mu_B B > 0$, where g_{e} is the electron g factor. In the one-electron spin space,

$$H_Z = \frac{E_Z}{2}(|\uparrow\rangle\langle\uparrow| - |\downarrow\rangle\langle\downarrow|). \tag{R.1}$$

Thus $|0\rangle = |\downarrow\rangle$ has lower energy. The protocol first loads an electron, of either spin, and then lets it release any spin excitation energy to a cold environment. Loading fixes the electron number; energy relaxation fixes the spin.

R.1.1 Selecting the loading voltages

Let V_{P} control mainly the dot well and V_{B} mainly the barrier to one reservoir. Keep the reservoir contact voltage fixed, so its chemical potential μ_{R} stays fixed. Retain one orbital–valley state with two spin states and assume the omitted levels are inaccessible during loading. A minimal charge model is

$$H_{\mathrm{d}} = \epsilon_d(\mathbf{V})(n_\uparrow + n_\downarrow) + \frac{E_Z}{2}(n_\uparrow - n_\downarrow) + U n_\uparrow n_\downarrow,$$

where n_σ is the occupation of spin state σ and $U > 0$ is the double-occupation energy cost. With the empty-dot energy set to zero, the energies are

$$E_\downarrow = \epsilon_d - \frac{E_Z}{2}, \qquad E_\uparrow = \epsilon_d + \frac{E_Z}{2}, \qquad E_2 = 2\epsilon_d + U.$$

The first-electron addition energies are $\mu_\sigma = E_\sigma$. The second-electron costs are $E_2 - E_\uparrow = \epsilon_d + U - E_Z/2$ and $E_2 - E_\downarrow = \epsilon_d + U + E_Z/2$.

First choose an emptying setting $\mathbf{V}^{\mathrm{E}}$ with $\epsilon_d - E_Z/2 > \mu_{\mathrm{R}}$ and a barrier that permits tunneling. An electron can then leave, and the sensor should report an empty dot. Next lower the dot energies to a loading setting $\mathbf{V}^{\mathrm{L}}$ satisfying

$$\epsilon_d + \frac{E_Z}{2} < \mu_{\mathrm{R}} < \epsilon_d + U - \frac{E_Z}{2}. \tag{R.2}$$

Both first-electron spin states can load; even the cheaper second-electron process cannot. This window requires $U > E_Z$ in the model. Choose margins larger than the reservoir thermal energy $k_B T_e$ and the tunnel broadening, and keep the first-electron excited orbital/valley thresholds above μ_{R}. The more general condition for excluding a second electron is that every accessible second-electron addition cost exceed μ_{R}, as in Eq. (I.5).

The loading pulse is the voltage change $\mathbf{V}^{\mathrm{E}} \to \mathbf{V}^{\mathrm{L}}$, followed by a hold of duration t_L. For example, if the measured local response with the barrier voltage fixed is $d\epsilon_d = -e\alpha_{\mathrm{P}}\, dV_{\mathrm{P}}$ with $\alpha_{\mathrm{P}} > 0$, increasing V_{P} lowers both spin levels. The required increment is $\Delta V_{\mathrm{P}} = -\Delta\epsilon_d/(e\alpha_{\mathrm{P}})$. If the barrier is also changed, use its energy response in the same voltage calculation. The charge-transition positions and energy measurements in Section G.1.4 and Appendix I locate the window; there is no device-independent loading voltage.

R.1.2 Loading one electron and retaining it

At $\mathbf{V}^{\mathrm{L}}$, keep the reservoir barrier open enough for an electron to enter. For weak tunneling, the total rate out of the empty state is

$$\Gamma_{\mathrm{in}} = \sum_{\sigma=\uparrow,\downarrow} \Gamma_\sigma^{(0)} f_{\mathrm{R}}(\mu_\sigma), \qquad f_{\mathrm{R}}(E) = \frac{1}{1 + e^{(E-\mu_{\mathrm{R}})/(k_B T_e)}}.$$

Here $\Gamma_\sigma^{(0)}$ is the loading rate for that spin when the reservoir states at the required energy are occupied. Lowering the barrier increases their wave-function overlap with the dot and hence the tunnel rate. The Fermi factor accounts for whether the reservoir can supply an electron at

that energy. Under the loading-window conditions, unloading is suppressed and the empty-dot probability obeys

$$\dot{p}_{\text{empty}} = -\Gamma_{\text{in}} p_{\text{empty}}, \qquad p_{\text{empty}}(t_L) = e^{-\Gamma_{\text{in}} t_L}.$$

Use the charge waiting-time measurement at this loading setting to determine Γ_{in}: the measured fraction of empty records after a wait t_L follows the exponential above. The preparation uses that measured rate to choose its loading time. A target empty-dot probability q_{empty} requires $t_L \geq \ln(1/q_{\text{empty}})/\Gamma_{\text{in}}$.

After loading, raise the reservoir barrier to suppress electron exchange for the remaining sequence. Compensate its shift of the dot energy with the plunger, using the response method of Appendix F.4. Keep the dot in the one-electron region while closing the barrier; do not return to the emptying setting. Verify retention by varying the hold time and checking that the charge remains one.

R.1.3 Removing the spin excitation

Under H_Z alone, an up spin remains an up spin: its phase changes but its population does not. Raising the tunnel barrier isolates the electron from particle exchange, not from the crystal. Weak spin–orbit coupling admixes orbital or valley excitations with opposite spin into the Zeeman states. A lattice displacement can then have a nonzero matrix element between these mixed states. An upper spin state can become a lower one while emitting a phonon of energy E_Z. A phonon is one quantum of lattice vibration; such spin relaxation has been observed in silicon dots [31].

An effective model for this energy exchange is

$$H_{\text{env}} = \sum_k \hbar\omega_k b_k^\dagger b_k, \qquad H_{\text{int}} = \sum_k \left(c_k |\downarrow\rangle\langle\uparrow| b_k^\dagger + c_k^* |\uparrow\rangle\langle\downarrow| b_k \right).$$

The operator $b_k^\dagger$ creates a phonon in mode k; c_k is an effective coupling energy that includes the spin admixture. The two terms describe emission and absorption near $\hbar\omega_k = E_Z$. This resonant, weak-coupling model explains the relaxation process; its coupling strengths need not be calculated microscopically to measure the relaxation time.

For a thermal phonon environment at temperature T, the emission and absorption factors are $\bar{n}+1$ and $\bar{n}$, with $\bar{n} = [e^{E_Z/(k_B T)} - 1]^{-1}$. Consequently,

$$\frac{\gamma_\uparrow}{\gamma_\downarrow} = e^{-E_Z/(k_B T)}, \qquad \dot{p}_\uparrow = -\gamma_\downarrow p_\uparrow + \gamma_\uparrow (1 - p_\uparrow).$$

The rates are taken constant at the chosen hold voltages. Writing $T_1 = (\gamma_\downarrow + \gamma_\uparrow)^{-1}$ gives

$$p_\uparrow(t) = p_\uparrow^{\text{eq}} + [p_\uparrow(0) - p_\uparrow^{\text{eq}}] e^{-t/T_1}, \qquad p_\uparrow^{\text{eq}} = \frac{1}{1 + e^{E_Z/(k_B T)}}. \tag{R.3}$$

Here t starts after loading and isolation. At $k_B T \ll E_Z$, the environment can accept the excitation energy but rarely supplies it, so the final state is predominantly down.

R.1.4 Choosing the wait and completing the preparation

The relaxation model explains why a wait prepares the lower state. Its measured parameters now determine how long that wait must be. Use the waiting-time measurement of Section U.1.2 at these same hold voltages to determine T_1 and the long-time population. To reach $p_\uparrow(t) \leq q$ when $p_\uparrow(0) > q > p_\uparrow^{\text{eq}}$, choose

$$t \geq T_1 \log \frac{p_\uparrow(0) - p_\uparrow^{\text{eq}}}{q - p_\uparrow^{\text{eq}}}.$$

A longer wait cannot reach a target below the equilibrium population. With $p_\uparrow$ conditioned on successful loading, the total failure probability is $p_{\text{empty}} + (1 - p_{\text{empty}})p_\uparrow$. The complete sequence is therefore: empty at $\mathbf{V}^{\text{E}}$, load at $\mathbf{V}^{\text{L}}$ for t_L, close the reservoir barrier while retaining one electron, and wait at the holding voltages for the time just calculated. It produces one predominantly down-spin electron. The loading rate controls the probability of a missing electron; T_1 and the equilibrium population control the remaining spin error.

R.2 A singlet–triplet qubit

The logical zero is the singlet $|S\rangle = (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)/\sqrt{2}$ with one electron in each dot. We prepare it by first placing both electrons in the same lowest orbital–valley state of dot 2. Their spatial state is then symmetric, so fermionic antisymmetry requires a spin singlet. Separating this pair without rotating its spins produces the desired $(1, 1)$ singlet.

R.2.1 Loading the pair singlet

Begin with dot 1 empty and one electron in dot 2, prepared as in Section R.1. Let its energy be $E_{01,\downarrow}$. The possible final two-electron states include the singlet S_{02} and triplets that require a different orbital or valley state. Use a magnetic field and confinement for which the singlet remains the lowest two-electron state. The relevant addition energies are

$$\mu_{2S} = E(S_{02}) - E_{01,\downarrow}, \qquad \mu_{2T,\min} = \min_m E(T_{02,m}) - E_{01,\downarrow}.$$

Here m labels the triplet spin component. Lower the right-dot potential with its plunger, and keep its reservoir barrier open, until

$$\mu_{2S} < \mu_{\text{R}} < \mu_{2T,\min}.$$

Also keep the third-electron addition energy above μ_{R}. With thermal margins, an incoming electron can complete the lowest-orbital singlet but cannot load an excited triplet or a third electron. The charge sensor verifies that two electrons have loaded; a loading-time scan sets the hold duration by the same exponential waiting probability as for one electron. Then close the reservoir barrier while retaining $(0, 2)$. The excitation measurement supplies the singlet–triplet energy interval needed to choose this loading window.

R.2.2 Separating the electrons by adiabatic evolution

Keep the pair isolated from the reservoirs and allow interdot tunneling. We want to move one electron from dot 2 to dot 1 while retaining the singlet spin state. Changing the relative plunger voltages raises the energy of S_{02} relative to S_{11}. With tunneling, their crossing becomes an avoided crossing. If the voltages change sufficiently slowly, the system follows the lower instantaneous eigenstate as it changes from both electrons in dot 2 to one electron in each dot. This is adiabatic charge transfer.

First consider the singlet space alone. In the basis $(|S_{11}\rangle, |S_{02}\rangle)$, use

$$H_S(t) = \begin{pmatrix} 0 & g \\ g & \delta_c(t) \end{pmatrix}, \qquad g = \sqrt{2}\, t_c, \quad \delta_c = E(S_{02}) - E(S_{11}). \tag{R.4}$$

The factor $\sqrt{2}$ comes from the two spin-conserving tunneling terms that connect the singlets. The voltage response $d\delta_c = \sum_j (\partial\delta_c/\partial V_j) dV_j$ sets a ramp from $\delta_c^{\text{i}} < 0$ to $\delta_c^{\text{f}} > 0$; the interdot barrier controls g. Choose endpoints with $|\delta_c^{\text{i},f}| \gg g$ to make their charge states nearly pure, and compensate the barrier response if g is to stay fixed.

R.2.3 Choosing the speed for adiabatic charge transfer

Let $|-(t)\rangle$ and $|+(t)\rangle$ be the instantaneous lower and upper eigenstates of $H_S(t)$. Their energies and gap are

$$E_\pm = \frac{\delta_c \pm \sqrt{\delta_c^2 + 4g^2}}{2}, \qquad \Delta_c = E_+ - E_- = \sqrt{\delta_c^2 + 4g^2}.$$

At the chosen negative-detuning endpoint, $|-\rangle$ is close to $|S_{02}\rangle$; at the positive-detuning endpoint, it is close to $|S_{11}\rangle$. Adiabatic evolution therefore gives

$$|\psi(t_\mathrm{i})\rangle \simeq |S_{02}\rangle \quad \longrightarrow \quad |\psi(t_\mathrm{f})\rangle \simeq e^{i\gamma}|S_{11}\rangle,$$

where γ is an overall phase. The speed must suppress transitions from $|-\rangle$ to $|+\rangle$ during this evolution. For fixed $g > 0$, the usual local adiabatic criterion is

$$\frac{\hbar|\langle +|\dot{H}_S|-\rangle|}{(E_+ - E_-)^2} = \frac{\hbar g|\dot{\delta}_c|}{(\delta_c^2 + 4g^2)^{3/2}} \ll 1. \tag{R.5}$$

Here $\dot{H}_S = \mathrm{diag}(0, \dot{\delta}_c)$ gives $|\langle +|\dot{H}_S|-\rangle| = g|\dot{\delta}_c|/\Delta_c$. The gap is smallest at $\delta_c = 0$, where it equals $2g$; the condition there is $\hbar|\dot{\delta}_c| \ll 8g^2$. Farther from the avoided crossing, the larger gap permits a faster change of δ_c while maintaining charge adiabaticity.

R.2.4 Preserving the singlet during the transfer

The adiabatic condition above concerns the two singlet charge states. We also need the spin to remain a singlet. If the two dots have different Zeeman splittings, define $\Delta E_Z = E_{Z,1} - E_{Z,2}$. Within (S, T_0) their difference supplies the term $\Delta E_Z X_\mathrm{ST}/2$. For $|\Delta E_Z| \ll g$, its mixing is small near the charge anticrossing, where the lower singlet is separated from T_0 by an energy of order g. After the pulse reaches the separated region where $J \ll |\Delta E_Z|$, a time t_low there rotates S toward T_0 with probability $\sin^2(\Delta E_Z t_\mathrm{low}/2\hbar)$. To keep that unwanted probability below $q_s \ll 1$, require $|\Delta E_Z| t_\mathrm{low}/(2\hbar) \lesssim \sqrt{q_s}$. A suitable ramp is therefore slow enough near the charge avoided crossing to separate the electrons adiabatically, but spends little time in the low-exchange region where the gradient rotates their spin. These requirements concern different energy scales: we preserve adiabaticity for the charge transfer while limiting unwanted singlet–triplet evolution. The period of singlet–triplet oscillations at negligible exchange determines $|\Delta E_Z|$.

R.2.5 Completing and testing the preparation

The tunnel-coupling measurement and the voltage response of δ_c supply the inputs for the ramp. Use Eq. (R.5) to choose the crossing speed. The measured ΔE_Z limits the low-exchange exposure. Test these choices by separating and returning a singlet while scanning the ramp durations. Charge readout at the separated point tests incomplete transfer; singlet return tests spin preservation together with the return ramp. The return probability alone does not assign an error to each ramp separately. The preparation itself ends at the separated $(1, 1)$ setting; the return ramp is a test carried out in separate repetitions.

R.3 An exchange-only qubit

For the fixed-$m = +1/2$ encoding, the target is $|0_\mathrm{E}\rangle = |S_{12}\rangle|\uparrow\rangle_3$. The pair singlet has total spin zero, so the third spin supplies the required total z component. Prepare dot 3 with one electron and retain it while loading the singlet on dots 1–2. The pair then follows $(0, 2, 1) \to (1, 1, 1)$.

Raise the 2–3 barrier to make t_{23}, and hence J_{23}, negligible during this preparation. Otherwise $J_{23}\mathbf{s}_2 \cdot \mathbf{s}_3$ can change the pair spin while the singlet is being prepared. Compensate the barrier's energy shifts to retain the chosen occupations. Apply the singlet-loading window and separation ramp of Section R.2, measuring the pair energies with dot 3 occupied because its Coulomb interaction shifts them.

R.3.1 Preparing the specified third spin

The pair protocol alone has not yet prepared the fixed-m target: dot 3 must also be up. One route is to load and relax it into $|\downarrow\rangle_3$, then apply a calibrated single-spin π pulse with J_{23} suppressed. This route requires a separately addressable spin drive; the construction of its π pulse is given in Section N.1.2. Spin-selective readout of dot 3 tests the resulting polarization. If its up-spin probability is p_3, an otherwise ideal pair preparation gives

$$\rho_{\text{prep}} = |S_{12}\rangle\langle S_{12}| \otimes [p_3|\uparrow\rangle\langle\uparrow| + (1-p_3)|\downarrow\rangle\langle\downarrow|]_3 \,. \tag{R.6}$$

The population in the intended fixed-$m = +1/2$ sector is p_3.

R.3.2 What changes if the third spin is not polarized?

The fixed-m preparation just described is complete only when p_3 is sufficiently close to one. Leaving the third spin unpolarized defines a different preparation, using the subsystem encoding explained in Section 5.3, and its gate sequences must respect that choice [13, 15]. Within the total-spin-1/2 subspace, isotropic exchange has the form $H_{\text{logical}} \otimes I_m$: it rotates the encoded qubit identically in the two m sectors. The state above then contains a pure logical zero with an unprepared m degree of freedom. A magnetic-field gradient need not have this form, so the unpolarized-subsystem interpretation also requires testing the non-exchange terms during the gate.

S Converting Spin States into Readout Probabilities

Spin readout requires the possible spin states to produce distinguishable charge records. This appendix develops the conversion, timing, and assignment of those records for the encodings in Section 5, then relates repeated outcomes to logical-state probabilities. For an array, the sensor responses must also distinguish which qubits produced the charge signals. The final section develops the separation of several outcomes and the recovery of their joint probabilities, supporting the readout requirements in Section 8.

The sensor is a separate conducting device whose current changes when nearby target electrons move. For a sensor dot, the Coulomb term $\sum_i U_{is} N_i n_s$ shifts its addition energy when the target occupations N_i change. At fixed sensor source and drain voltages, the local current response is

$$\delta I_s \simeq \frac{\partial I_s}{\partial \mu_s} \sum_i U_{is} \delta N_i. \tag{S.1}$$

Here n_s is the sensor occupation, μ_s its addition energy and U_{is} the interaction energy with target dot i. The sensor is placed on a sloping part of its current curve, as in Section G.2.

To read a spin, we must first make the two possible spin states produce different target charge records. An electron can leave for one single-spin state, or a pair can enter a doubly occupied dot only when it is a singlet. We choose the voltage and duration of that conversion, then infer the spin outcome from the resulting sensor trace.

S.1 A single-spin qubit

The spin state is initially stored in an occupied dot isolated from its reservoir. To read it, raise the dot potential and reopen the reservoir barrier so that an up electron can leave but a down electron cannot. With the notation of Section R.1, the readout setting $\mathbf{V}^{\mathrm{R}}$ must satisfy

$$\mu_{\downarrow} = \epsilon_d - \frac{E_Z}{2} < \mu_{\mathrm{R}} < \mu_{\uparrow} = \epsilon_d + \frac{E_Z}{2}, \qquad \mu_{\mathrm{R}} - \frac{E_Z}{2} < \epsilon_d < \mu_{\mathrm{R}} + \frac{E_Z}{2}. \tag{S.2}$$

Keep the second-electron addition threshold above the reservoir. The plunger sets ϵ_d and the barrier sets the tunnel rates; use their joint voltage response to reach this condition.

At zero temperature, the reservoir has empty states at $\mu_{\uparrow}$ but occupied states at $\mu_{\downarrow}$. An up electron can therefore escape. Once the dot is empty, a down electron can enter, giving the charge sequence

$$\uparrow: \quad N = 1 \longrightarrow 0 \longrightarrow 1, \qquad \downarrow: \quad N = 1.$$

The information is the temporary empty interval, not the final electron number. A nearby sensor has different mean signals s_0 and s_1 for an empty and occupied target. A simple record model is

$$s(t) = s_0 + (s_1 - s_0)N(t) + \xi(t), \tag{S.3}$$

where $\xi(t)$ is detector noise. Thus the up-spin record can contain an excursion to s_0, while the down-spin record stays near s_1. Resolved empty and occupied intervals in charge reference traces at these readout voltages determine s_0 and s_1; using the same voltages also fixes the direct gate response of the sensor.

To locate the readout interval, measure the two first-electron thresholds relative to μ_{R}, then compare records after down-spin preparation and after a spin-flip pulse that prepares up. Vary the readout plunger voltage while keeping the reservoir contact fixed. Choose a setting between the thresholds, with margins against thermal activation. This window distinguishes an already present spin through its ability to leave [11].

The basic readout is now specified: move the isolated electron to $\mathbf{V}^{\mathrm{R}}$, allow spin-selective tunneling for a time t_r, and record whether a temporary empty interval occurs. An excursion reports up; no excursion reports down in the ideal conversion. The finite-time and relaxation errors in that rule determine the choice of t_r in Section S.4.

S.2 A singlet–triplet qubit

The two logical states S_{11} and $T_{0,11}$ initially have the same charge configuration. To distinguish them, lower the energy of double occupation on dot 2 while keeping the reservoirs closed. Spin-conserving tunneling connects S_{11} to S_{02}. A triplet cannot place both electrons in the same lowest orbital–valley state, because its symmetric spin state requires an antisymmetric spatial state. Its lowest doubly occupied state is higher in energy. Choose the detuning so that

$$E(S_{02}) < E(S_{11}), \qquad E(T_{02,m}) > E(T_{0,11}) \quad \text{for relevant triplet final states.} \tag{S.4}$$

Both inequalities refer to the same readout voltages. Together with small spin mixing during conversion, they allow $S_{11} \to S_{02}$ while keeping $T_{0,11}$ in $(1,1)$. This is the spin-dependent obstruction to charge transfer called Pauli spin blockade.

The voltage pulse reverses the singlet-separation ramp of Section R.2. Follow the singlet charge anticrossing into $(0,2)$ without waiting long enough for substantial singlet–triplet mixing or relaxation. At the endpoint, a sensor with local response $s = s_{\mathrm{bg}} + \kappa_1 N_1 + \kappa_2 N_2 + \xi$ has mean levels

$$s_{11} = s_{\mathrm{bg}} + \kappa_1 + \kappa_2, \qquad s_{02} = s_{\mathrm{bg}} + 2\kappa_2, \qquad s_{02} - s_{11} = \kappa_2 - \kappa_1.$$

The coefficients κ_i are signal changes per electron on dot i. The sensor must respond differently to the two dots. If the converted state persists, averaging the plateau signal distinguishes singlet from triplet.

To set the readout voltage, locate the $(1,1)$–$(0,2)$ boundary and scan across it using singlet and triplet reference preparations. The singlet reference comes from double-occupation loading. For a T_0 reference, apply the gradient-induced x rotation of Eq. (6.13) to a separated singlet: when exchange is negligible, a wait $\pi\hbar/|\Delta E_Z|$ interchanges S and T_0. Determine $|\Delta E_Z|$ from the period of $P_{T_0}(t) = \sin^2(\Delta E_Z t/2\hbar)$ while varying this wait; the charge-conversion contrast sets the observed oscillation amplitude. Compare the two charge-transfer probabilities as functions of the readout voltage. Choose the interval in which the singlet transfers but the triplet remains blocked. For the coherent conversion used here, select the ramp speed from the charge-following and spin-preservation conditions in Section R.2. After the ramp, record the charge plateau before appreciable triplet unblocking.

The resulting map is $S_{11} \mapsto (0,2)$ and $T_{0,11} \mapsto (1,1)$. The ramp performs the conversion; the subsequent plateau supplies the sensor record. A hold that instead relies on irreversible singlet transfer is a different implementation, treated separately below when choosing its readout duration.

S.3 An exchange-only qubit

Read the spin of the pair on dots 1–2 while dot 3 remains occupied. For logical 0, the pair is a singlet and can undergo $(1,1,1) \to (0,2,1)$; for logical 1, it is a triplet and remains blocked in $(1,1,1)$. Suppress J_{23} during conversion so that exchange with dot 3 does not change the pair being measured. Use the pair-readout energy window of Eq. (S.4), adding the fixed occupation of dot 3 to each charge state and measuring the energies with that electron present. The sensor contrast is now $s_{021} - s_{111} = \kappa_2 - \kappa_1$; the unchanged third-dot contribution cancels. The same voltage and duration tests then apply. Within the logical space, pair singlet and pair triplet are the two outcomes even though logical 1 need not be a product of a definite spin on dot 3 and a definite triplet component.

This readout distinguishes pair singlet from pair triplet, but does not determine the third spin. In particular, every state in Eq. (R.6) produces the same pair-singlet result, regardless of p_3. Testing preparation in the fixed-$m = +1/2$ encoding therefore also requires the separate polarization measurement described in Section R.3.

S.4 Choosing a duration that preserves the spin contrast

The conversion conditions allow the desired charge motion, but do not guarantee that it happens before recording ends. Waiting longer can recover slow desired events while also admitting unwanted ones. We now hold the readout voltages fixed and choose the recording time from the probabilities of these competing events.

S.4.1 A single spin: escape before relaxation

For an up spin, the useful event is escape to the reservoir. Relaxation to down before escape destroys that indication. For a down spin, thermally activated escape gives a false indication. The time should allow the first process and suppress the other two.

For a reservoir at temperature T, a weak spin-conserving tunnel contact with bare rate $\Gamma_s^{(0)}$ for spin $s \in \{\uparrow, \downarrow\}$ gives

$$\Gamma_{\mathrm{out},s} = \Gamma_s^{(0)}[1 - f(\mu_s)], \qquad f(E) = \frac{1}{1 + e^{(E-\mu_\mathrm{R})/(k_\mathrm{B}T)}}. \tag{S.5}$$

Raising both dot addition energies increases the desired up-spin exit rate, but eventually also allows a down spin to escape. The voltage scan therefore selects the rates, not just the ordering of three energies.

The readout duration must allow the desired exit before spin relaxation. Write $\Gamma_u = \Gamma_{\mathrm{out},\uparrow}$ and $\Gamma_d = \Gamma_{\mathrm{out},\downarrow}$. At $E_Z \gg k_B T$, upward spin excitation is negligible and the downward relaxation rate is $\gamma_1 \simeq T_1^{-1}$ at the readout setting. An initial up spin survives without either exit or relaxation until time u with probability $e^{-(\Gamma_u+\gamma_1)u}$. Its desired exit probability is therefore

$$\begin{aligned} P_{\mathrm{exit}}(t_r) &= \int_0^{t_r} \Gamma_u e^{-(\Gamma_u+\gamma_1)u}\, du \\ &= \frac{\Gamma_u}{\Gamma_u+\gamma_1}\left[1 - e^{-(\Gamma_u+\gamma_1)t_r}\right]. \end{aligned} \tag{S.6}$$

This is the competing-rate model of Eq. (5.6) for the present voltage setting. To choose a duration, consider the regime $\Gamma_d t_r \ll 1$, in which an up spin that relaxes before escaping is normally missed. A down-spin reference produces a false exit with probability $1 - e^{-\Gamma_d t_r}$, whereas the desired up-spin exit probability is the expression above. For equally likely input spins, the conversion error in this approximation is

$$\epsilon_{\mathrm{conv}}(t_r) = \frac{1}{2}\left[1 - \frac{\Gamma_u}{k}(1 - e^{-k t_r}) + 1 - e^{-\Gamma_d t_r}\right], \qquad k = \Gamma_u + \gamma_1. \tag{S.7}$$

The first contribution decreases with time; the second increases. For $\Gamma_u > \Gamma_d > 0$, their balance gives

$$\frac{d\epsilon_{\mathrm{conv}}}{dt_r} = 0 \quad \Longrightarrow \quad t_r^\star = \frac{\ln(\Gamma_u/\Gamma_d)}{k - \Gamma_d}. \tag{S.8}$$

Check $\Gamma_d t_r^\star \ll 1$ before using this estimate. For example, the illustrative rates $(\Gamma_u, \Gamma_d, \gamma_1) = (10^4, 10, 100)\,\mathrm{s}^{-1}$ give $t_r^\star \simeq 0.685\,\mathrm{ms}$ and about 0.89% conversion error. Increasing the wait alone does not remove the competition with relaxation.

Repeated up/down reference traces determine the exit probabilities versus time. A separate relaxation measurement fixes γ_1; the up-spin curve then determines Γ_u, and the down-spin curve determines Γ_d. Repeat at nearby readout voltages and barrier settings, and compare the resulting minimum errors. Finally test the same settings with the actual sensor decision rule. These formulas assume that an exit event is resolved; an electron that leaves and refills too quickly to be detected adds a separate missed-event contribution.

S.4.2 A pair readout implemented by irreversible transfer

In the coherent protocol above, a ramp carries S_{11} into S_{02}. An alternative is to move to a fixed detuning where S_{02} is lower, then let tunneling and energy dissipation complete the transfer. An initially blocked triplet may also change spin and transfer later. The two inputs consequently produce different charge-transfer probabilities at intermediate times.

Here the clock starts with the pair still in $(1,1)$ at the conversion setting. This is not a rate description of the preceding coherent ramp. When the fixed-setting records support an incoherent, one-way model, write

$$S_{11} \xrightarrow{\Gamma_S} S_{02}, \qquad T_{0,11} \xrightarrow{\gamma_T} S_{11} \xrightarrow{\Gamma_S} S_{02}.$$

Here Γ_S is the singlet transfer rate including energy dissipation, and γ_T describes spin-changing processes that unblock a triplet. Assume that the converted charge remains detectable and that reverse processes are negligible at this endpoint. Starting from S gives

$$p_{02|S}(t) = 1 - e^{-\Gamma_S t}.$$

Starting from T_0, the state must first unblock. Integrating over its unblocking time gives, for $\Gamma_S \neq \gamma_T$,

$$\begin{aligned} p_{02|T_0}(t) &= \int_0^t \gamma_T e^{-\gamma_T u}[1 - e^{-\Gamma_S(t-u)}]\, du \\ &= 1 - \frac{\Gamma_S e^{-\gamma_T t} - \gamma_T e^{-\Gamma_S t}}{\Gamma_S - \gamma_T}. \end{aligned} \tag{S.9}$$

A singlet that has not yet transferred and a triplet that has already unblocked both give a wrong charge label. For equally likely inputs, their conversion error at a chosen time is

$$\epsilon_{\text{pair}}(t) = \frac{1}{2}[1 - p_{02|S}(t) + p_{02|T_0}(t)].$$

Thus choosing t to maximize $p_{02|S} - p_{02|T_0}$ minimizes this error. That difference is large in the interval $\Gamma_S^{-1} \ll t \ll \gamma_T^{-1}$, when such an interval exists. Reference time scans determine whether this separation of rates is actually available. The fitted rates characterize this particular conversion setting. Moving the voltage or changing the ramp requires repeating the conversion test.

For either pair-conversion implementation, the subsequent sensor average must resolve the two levels before the charge contrast is lost. Its duration is tested on the actual traces. A measured irreversible transfer rate cannot be substituted for the coherent ramp condition.

S.5 Assigning 0 or 1 to the measured record

The two charge configurations produce different mean sensor levels, but noise makes the observed records overlap. We must specify which feature of a record counts as logical 0 or 1. The conversion and its duration are already fixed. We now determine how to extract an outcome from its noisy sensor trace; this step does not change the physical spin-to-charge mapping.

S.5.1 Choosing the feature to classify

For pair readout, average over a fixed plateau interval of length t_{a} to obtain $\bar{s}$. The normalized statistic

$$x = \frac{\bar{s} - s_{02}}{s_{11} - s_{02}}$$

is near zero for a converted singlet and near one for a blocked triplet. For the three-dot encoding, use s_{021} and s_{111}.

Single-spin readout requires a different statistic because the empty interval is temporary. After averaging samples over the detector response time, one simple choice is

$$x = \max_{0<t<t_r} \frac{s(t) - s_1}{s_0 - s_1}.$$

An empty-dot excursion moves this statistic toward one; an occupied-dot record stays near zero. The reference traces must use the same sampling, averaging and readout duration, because all three affect the chance of a noise excursion or a missed short event. For either statistic, a threshold x_c reports logical 1 when $x > x_c$.

S.5.2 Choosing a decision threshold

Reference preparations of 0 and 1 give two distributions of x. Count their misclassified records:

$$\epsilon_0 = P(x > x_c \mid 0), \qquad \epsilon_1 = P(x \le x_c \mid 1).$$

For equally weighted reference states, choose x_c to minimize $(\epsilon_0 + \epsilon_1)/2$. Keep the voltage pulse, record duration and decision rule the same in the subsequent experiment.

The threshold can be selected directly from the two measured distributions. To see analytically how their separation controls the error, suppose a plateau-averaged signal has Gaussian conditional distributions with means $m_0 < m_1$ and the same standard deviation σ. At a proposed threshold,

$$\epsilon_0 = \int_{x_c}^{\infty} \frac{e^{-(x-m_0)^2/(2\sigma^2)}}{\sqrt{2\pi}\sigma}\, dx, \qquad \epsilon_1 = \int_{-\infty}^{x_c} \frac{e^{-(x-m_1)^2/(2\sigma^2)}}{\sqrt{2\pi}\sigma}\, dx.$$

Differentiating their sum equates the two probability densities, giving $x_c = (m_0 + m_1)/2$. The error for either state is then

$$\epsilon_0 = \epsilon_1 = \frac{1}{2}\,\mathrm{erfc}\left(\frac{m_1 - m_0}{2\sqrt{2}\,\sigma}\right).$$

Here $\mathrm{erfc}(z) = (2/\sqrt{\pi}) \int_z^{\infty} e^{-u^2} du$. This example connects histogram separation to assignment error. For non-Gaussian records, the same threshold scan can be performed directly by counting the two tails; the Gaussian model is not needed.

S.5.3 Recovering an outcome probability from repeated records

The decision rule produces a count of 1 outcomes. If its conditional errors are independently known, a state with true logical-1 probability p produces the reported probability

$$p_{\mathrm{obs}} = \epsilon_0(1 - p) + (1 - \epsilon_1)p, \qquad p = \frac{p_{\mathrm{obs}} - \epsilon_0}{1 - \epsilon_0 - \epsilon_1}. \tag{S.10}$$

The second relation requires nonzero readout contrast. If n repeated trials give the fraction $\hat{p}_{\mathrm{obs}}$ of outcome 1, and the assignment probabilities are treated as known, the variance of the corrected estimator is

$$\mathrm{Var}(\hat{p}) = \frac{p_{\mathrm{obs}}(1 - p_{\mathrm{obs}})}{n(1 - \epsilon_0 - \epsilon_1)^2}. \tag{S.11}$$

The inverse map removes a known bias but amplifies statistical uncertainty when the two records are difficult to distinguish. It cannot recover information from zero-contrast readout. An uncertain reference preparation gives a mixture of the two distributions, so its histogram alone does not separately determine preparation error and readout error. Initial calibration can use well-understood loading and relaxation limits; precision claims require the reference-state errors to be established as well. This distinction matters when deciding what a measured gate probability actually verifies.

S.6 From repeated readout to a spin probability

A single readout assigns one outcome to one preparation. To measure an outcome probability after a control pulse, repeat the whole sequence: prepare the same state, isolate the electrons, apply the same control, and then perform the same charge conversion and sensor measurement. Reading one spin repeatedly would instead probe a state already changed by the first readout.

Let ρ_0 be the prepared spin density matrix and $U(\tau)$ the coherent evolution during a control interval of duration τ. For ideal readout with outcome projector M_a,

$$P_a(\tau) = \mathrm{Tr}\Big[M_a U(\tau)\rho_0 U^\dagger(\tau)\Big]. \tag{S.12}$$

The spin-to-charge conversion fixes the measured projector. For single-spin readout, $M_\uparrow = |\uparrow\rangle\langle\uparrow|$ and $M_\downarrow = |\downarrow\rangle\langle\downarrow|$. For singlet–triplet readout, the converted outcome has $M_S = |S\rangle\langle S|$ within the logical space. Pair readout of an exchange-only qubit similarly distinguishes its logical pair-singlet and pair-triplet states, but does not separately identify every leakage state.

A common timing signal starts the voltage sequence and sensor recording. For each chosen τ, classify the records from K independent trials. The fraction k_a/K of assigned outcomes estimates the observed probability; the assignment map in Eq. (S.10) connects it to P_a. Keep the preparation and readout settings fixed while changing the intervening control. Its Hamiltonian and propagator determine how the measured probability varies.

S.7 Separating several readout outcomes

A sensor records a charge-dependent signal, not a qubit label. For sensor ℓ, a linear model for its mean signal is

$$\bar{s}_\ell = s_{\ell,0} + \sum_i c_{\ell i}\langle N_i\rangle, \tag{S.13}$$

where N_i is the occupation variable after the chosen spin-to-charge conversion and $c_{\ell i}$ is the sensor response to it. If one sensor responds equally to two dots, it measures their total charge but cannot tell 10 from 01. A second sensor, unequal response, or a conversion at a different time can supply that distinction.

For example, subtract the backgrounds from two sensor signals and suppose their response is symmetric:

$$\begin{pmatrix}\bar{s}_1\\ \bar{s}_2\end{pmatrix} = \begin{pmatrix}c & d\\ d & c\end{pmatrix}\begin{pmatrix}\langle N_1\rangle\\ \langle N_2\rangle\end{pmatrix}.$$

The sum and difference of the occupations are

$$\langle N_1\rangle + \langle N_2\rangle = \frac{\bar{s}_1 + \bar{s}_2}{c+d}, \qquad \langle N_1\rangle - \langle N_2\rangle = \frac{\bar{s}_1 - \bar{s}_2}{c-d}. \tag{S.14}$$

This inversion needs $c \neq \pm d$. When c is close to d, the difference estimate strongly amplifies signal noise. An array thus needs sufficiently distinct response patterns, not simply a count of sensors equal to the count of qubits.

These equations determine mean occupations, not the joint probabilities of single-shot outcomes. For example, a distribution with $P_{00} = P_{11} = 1/2$ and one with $P_{01} = P_{10} = 1/2$ both have $\langle N_1\rangle = \langle N_2\rangle = 1/2$, but the first always has even parity and the second always has odd parity. Their identical marginal occupations therefore do not determine their different correlations. To estimate $P_{00}, P_{01}, P_{10}, P_{11}$, first classify the pair of outcomes in each run, then average those classifications. One sensor can in principle classify all four if its levels $s_0, s_0 + c_1, s_0 + c_2, s_0 + c_1 + c_2$ are distinguishable. If $c_1 = c_2$, its 01 and 10 levels coincide. Sequential conversion can separate them in time, provided the later qubit retains its spin state until its readout.

Readout capacity also sets a time cost. If N_m qubits must be measured, at most R_m independent results can be obtained at once, and one readout group takes t_r, then

$$T_{\mathrm{meas}} \geq \left\lceil \frac{N_m}{R_m} \right\rceil t_r.$$

Conversion, reset, and shared-control delays may increase this bound. Qubits awaiting measurement must retain their state for those extra rounds.

T Gate Verification and Fidelity

Section 5.4 explains why logical populations alone do not determine a state or a gate. This appendix develops the additional measurements that reconstruct states and operations, and relates them to tests of fidelity and entanglement.

A population oscillation establishes one aspect of a gate, but it does not fix its action on every input state. A phase error can be invisible on $|0\rangle$ and $|1\rangle$ while changing their superpositions. Three tests provide different information. State and operation reconstruction determine what a gate does to selected inputs; a Bell-state test checks whether a two-qubit gate has created entanglement; random gate sequences test how errors accumulate. They are alternatives for different questions, not three consecutive steps required to execute a gate.

T.1 Measuring a state beyond its 0 and 1 populations

The states $(|0\rangle + |1\rangle)/\sqrt{2}$ and $(|0\rangle + i|1\rangle)/\sqrt{2}$ both give equal 0 and 1 populations. A relative phase error between them is therefore invisible to logical-basis readout. A known rotation before readout converts the phase information into a population difference. For a qubit that remains in its computational space, write

$$\rho = \frac{1}{2}(I + r_x X + r_y Y + r_z Z). \tag{T.1}$$

Logical-basis readout gives $r_z = P_0 - P_1$. To obtain r_x or r_y, rotate that axis to the readout axis before measuring. If R is the analysis rotation, the measured observable on the unrotated state is $R^\dagger Z R$:

R	$R^\dagger Z R$
I	Z
$R_y(-\pi/2)$	X
$R_x(\pi/2)$	Y

For a single spin, Section N.1.2 makes these analysis settings explicit. A resonant rectangular pulse of duration $\pi/(2\Omega_0)$ with phase $+\pi/2$ gives $R_y(-\pi/2)$; the same duration with phase zero gives $R_x(\pi/2)$. Apply the analysis pulse while the electron remains isolated, then switch to the fixed readout voltages. Repeat preparation and the operation being tested for each analysis setting in separate trials. Correct the outcome counts using Eq. (S.10). The differences $P_0 - P_1$ for the three settings give r_z, r_x, r_y, and hence ρ; this is state *tomography*.

For encoded qubits, the rotations act on the logical basis and are generated by the exchange/gradient controls of Section N.2. Once logical x and z rotations are available, the required y rotation follows from

$$R_y(\vartheta) = R_z(\pi/2) R_x(\vartheta) R_z(-\pi/2),$$

with the rightmost pulse applied first. This connects the analysis axes to the same calibrated controls used for the gates.

This procedure reconstructs the state produced in one specified experiment: three analysis settings give the three components of its Bloch vector. It is complete as a state measurement.

T.2 Reconstructing the action of a one-qubit gate

Different operations can give the same output from one input. For example, an ideal z rotation leaves the populations of $|0\rangle$ unchanged whatever its angle, but changes the relative phase of a superposition. To determine the gate rather than just one output, repeat the state

measurement for independent inputs. A trace-preserving linear quantum operation acts on Bloch vectors as

$$\mathbf{r}_{\text{out}} = M\mathbf{r}_{\text{in}} + \mathbf{t}, \tag{T.2}$$

where M is a real 3×3 matrix and $\mathbf{t}$ a real vector. Start each trial from initialized $|0\rangle$. The preparations

input state	$\|0\rangle$	$\|1\rangle$	$\|+\rangle$	$\|+i\rangle$
preparation rotation	I	$R_x(\pi)$	$R_y(\pi/2)$	$R_x(-\pi/2)$

give four independent inputs, where $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ and $|+i\rangle = (|0\rangle + i|1\rangle)/\sqrt{2}$. Apply the same gate to each input and measure its three output components by the analysis rotations above. Denote the output vectors by $\mathbf{r}_0, \mathbf{r}_1, \mathbf{r}_+, \mathbf{r}_{+i}$. Their input vectors are respectively $\hat{\mathbf{z}}, -\hat{\mathbf{z}}, \hat{\mathbf{x}}, \hat{\mathbf{y}}$, so

$$\begin{aligned} \mathbf{t} &= (\mathbf{r}_0 + \mathbf{r}_1)/2, & M\hat{\mathbf{z}} &= (\mathbf{r}_0 - \mathbf{r}_1)/2, \\ M\hat{\mathbf{x}} &= \mathbf{r}_+ - \mathbf{t}, & M\hat{\mathbf{y}} &= \mathbf{r}_{+i} - \mathbf{t}. \end{aligned} \tag{T.3}$$

This determines the affine map from four input states, each measured along three axes. It assumes that preparation and analysis rotations are known. Without that assumption, the data characterize their combined action with the gate rather than isolating the gate error.

T.2.1 Interpreting a reconstructed operation

The measurements have supplied M and $\mathbf{t}$. We now compare their entries with a specific error model to distinguish a coherent phase error from loss of coherence and energy relaxation. For example, remove the intended gate and suppose the remaining operation consists of an unwanted z rotation by χ, random z phases, and relaxation $|1\rangle \to |0\rangle$ with probability γ. The resulting map has the form

$$M = \begin{pmatrix} \lambda\cos\chi & -\lambda\sin\chi & 0 \\ \lambda\sin\chi & \lambda\cos\chi & 0 \\ 0 & 0 & 1-\gamma \end{pmatrix}, \qquad \mathbf{t} = \begin{pmatrix} 0 \\ 0 \\ \gamma \end{pmatrix}. \tag{T.4}$$

To derive the last row, use $\rho'_{11} = (1-\gamma)\rho_{11}$ and $\rho'_{00} = \rho_{00} + \gamma\rho_{11}$. For the first two rows, relaxation reduces ρ_{01} by $\sqrt{1-\gamma}$: its two Kraus operators are $K_0 = |0\rangle\langle 0| + \sqrt{1-\gamma}|1\rangle\langle 1|$ and $K_1 = \sqrt{\gamma}|0\rangle\langle 1|$, and substitution into $\rho' = \sum_a K_a \rho K_a^\dagger$ gives this factor. A phase rotation multiplies the off-diagonal entry by $e^{-i\chi}$. An independent Gaussian phase with variance σ_ϕ^2 supplies the additional factor $e^{-\sigma_\phi^2/2}$, so $\lambda = \sqrt{1-\gamma}e^{-\sigma_\phi^2/2}$ in this model.

The four-state measurements now have a direct interpretation: $\gamma = t_z$ describes motion of the Bloch sphere toward $|0\rangle$, $\lambda = \sqrt{M_{xx}^2 + M_{yx}^2}$ describes transverse contraction, and $\chi = \text{atan2}(M_{yx}, M_{xx})$ gives the residual phase. A phase correction can remove χ; it cannot reverse the contraction or restore the lost excitation. An observed map that lacks this structure requires a different error model, rather than forcing all deviations into one rotation angle.

T.3 Comparing the measured operation with the target gate

The reconstructed map predicts an output for every input qubit state. To quantify its discrepancy from the requested gate, compare those outputs with the ideal ones before averaging over inputs. This distinguishes an error measure from a parameter such as a Rabi contrast or a readout misclassification rate.

T.3.1 Averaging the output overlap

Let the target unitary U rotate Bloch vectors by the real matrix R_U. For a pure input with unit vector $\mathbf{r}$, the overlap of the measured output with the ideal output is

$$F(\mathbf{r}) = \frac{1 + (R_U\mathbf{r}) \cdot (M\mathbf{r} + \mathbf{t})}{2}.$$

Averaging uniformly over input directions uses $\langle r_i \rangle = 0$ and $\langle r_i r_j \rangle = \delta_{ij}/3$, giving

$$F_{\text{avg}} = \frac{1}{2} + \frac{\text{Tr}(R_U^T M)}{6}. \tag{T.5}$$

This is an average over specified input states, distinct from the noise average that may already be included in the operation. For a coherent angle error η about the correct axis, $\text{Tr}(R_U^T M) = 1 + 2\cos\eta$, so $1 - F_{\text{avg}} = (1 - \cos\eta)/3 \simeq \eta^2/6$.

T.3.2 Accounting for population that leaves the qubit space

The preceding comparison assumed a trace-preserving map within a two-dimensional space. If a pulse also excites an orbital, valley, or excluded spin state, counting only the surviving qubit outcomes omits part of the error. For the computational-space projector P_{q}, determine or bound

$$L(\rho) = 1 - \text{Tr}[P_{\text{q}}\mathcal{E}(\rho)], \tag{T.6}$$

where $\mathcal{E}$ is the physical operation on the full state space. Renormalizing the surviving qubit population hides L. A measurement that labels every triplet as logical 1, for example, cannot distinguish a logical triplet from a leaked spin state. An additional state-selective measurement is needed before reporting leakage separately. The trace-preserving qubit formula above applies when such leakage is negligible.

T.4 Testing whether a two-qubit gate creates entanglement

Use the calibrated conditional-phase interaction to prepare an entangled state. After its local phases are corrected, a CZ gives CNOT by a basis change on qubit B:

$$\text{CNOT}_{A\to B} = (I_A \otimes \mathsf{H}_B)\,\text{CZ}\,(I_A \otimes \mathsf{H}_B), \qquad \mathsf{H} = (X + Z)/\sqrt{2}.$$

The Hadamard H can be implemented, up to global phase, as $R_y(\pi/2)R_z(\pi)$. Initialize both qubits in 0, prepare $|+\rangle_A$ with $R_{y,A}(\pi/2)$, and apply this CNOT sequence. Ideally the output is $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$. However, observing equal probabilities of 00 and 11 does not establish this coherent state: the mixture $(|00\rangle\langle 00| + |11\rangle\langle 11|)/2$ gives exactly those populations. We therefore measure correlations in rotated bases as well as these populations, using the one-qubit analysis pulses already established.

T.4.1 The three measurements required by this test

The Bell-state projector has the Pauli expansion

$$|\Phi^+\rangle\langle\Phi^+| = \frac{I + X_A X_B - Y_A Y_B + Z_A Z_B}{4}.$$

For the measured output ρ, its overlap with the target is therefore

$$F_{\Phi^+} = \frac{1 + \langle X_A X_B \rangle - \langle Y_A Y_B \rangle + \langle Z_A Z_B \rangle}{4}. \tag{T.7}$$

Measure $Z_A Z_B$ by joint logical-basis readout: $\langle Z_A Z_B \rangle = P_{00} + P_{11} - P_{01} - P_{10}$. To measure $X_A X_B$, apply $R_y(-\pi/2)$ to both qubits before the same readout. Applying $R_x(\pi/2)$ to both measures $Y_A Y_B$. The three measurement settings thus determine the displayed fidelity without reconstructing all entries of ρ. If the two assignments are independent, the joint assignment matrix is the tensor product of the individual matrices in Eq. (S.10); otherwise joint reference preparations are needed.

T.4.2 Why the result can establish entanglement

For any normalized product state $|a\rangle|b\rangle$,

$$|\langle \Phi^+ | a, b \rangle|^2 = \frac{|a_0 b_0 + a_1 b_1|^2}{2} \leq \frac{1}{2},$$

by the Cauchy–Schwarz inequality. Convex mixtures of product states obey the same bound. A measured $F_{\Phi^+} > 1/2$, with the measurement uncertainties accounted for, therefore establishes entanglement in this output. The incoherent 00/11 mixture has correlations $(0, 0, 1)$ and gives exactly $F_{\Phi^+} = 1/2$. This tests one important action of the entangling gate; it does not determine the gate on every input state.

T.5 Accumulating error with random gate sequences

To detect small gate errors, apply many gates whose ideal product is known to return the state to its starting point. Individual sequences can amplify or cancel coherent errors differently. Averaging over random sequences reduces dependence on that particular choice.

T.5.1 Constructing and measuring a sequence with known ideal output

For a one-qubit protocol, choose gates from the *Clifford group*: these unitaries permute the Pauli axes, with possible sign changes. For each sequence, draw m independent uniformly random Cliffords $C_1, \ldots, C_m$, and append their ideal inverse

$$C_{\text{inv}} = (C_m \cdots C_1)^\dagger.$$

Compile each Clifford into the calibrated logical $\pi/2$ and π rotations. For example, $C = R_y(\pi/2) R_x(\pi/2)$ is made by an x pulse followed by a y pulse. For a single spin, their phases are zero and $-\pi/2$; both have duration $\pi/(2\Omega_0)$. Compute C_{inv} as one Clifford and compile it using the same set of short implementations, rather than physically reversing all m gates. Thus increasing m increases the tested sequence length without also giving the final inverse a length proportional to m.

Prepare $|0\rangle$, run the sequence, and use the fixed readout procedure to count outcomes 0. Average this return probability over independently drawn sequences at each length m. In the ideal experiment every sequence returns to 0; errors in the implemented pulses reduce the return probability. This is *randomized benchmarking* [43].

T.5.2 Deriving the dependence on sequence length

The measurement supplies a return probability at each length. To infer an error per Clifford from that curve, we need a noise model. The simplest decay model assumes the same trace-preserving noise channel $\mathcal{N}$ after each Clifford, negligible leakage, and noise without memory between gates. We can derive the decay using the Bloch representation $\mathbf{r} \mapsto M_{\mathcal{N}} \mathbf{r} + \mathbf{t}_{\mathcal{N}}$.

Changing the axes by a Clifford rotation R_C gives the noise matrix $R_C^T M_{\mathcal{N}} R_C$ and translation $R_C^T \mathbf{t}_{\mathcal{N}}$.

First average over the four Pauli conjugations. Their Bloch rotations are I, $\mathrm{diag}(1,-1,-1)$, $\mathrm{diag}(-1,1,-1)$ and $\mathrm{diag}(-1,-1,1)$. Every translation component and every off-diagonal matrix entry occurs with opposite signs and cancels. Averaging the remaining diagonal matrix over Clifford permutations of x, y, z replaces all three diagonal entries by their mean. Hence

$$\frac{1}{24}\sum_C R_C^T M_{\mathcal{N}} R_C = pI, \qquad p = \frac{\mathrm{Tr}\, M_{\mathcal{N}}}{3}, \qquad \frac{1}{24}\sum_C R_C^T \mathbf{t}_{\mathcal{N}} = 0. \tag{T.8}$$

There are 24 distinct single-qubit Clifford rotations after discarding global phases.

Why does this average appear in the pulse sequence? Write $D_j = C_j \cdots C_1$. Inserting $D_j D_j^\dagger$ between the errors and using the final ideal inverse puts the sequence's error into the form

$$\mathcal{N} \circ (\mathcal{D}_m^{-1} \circ \mathcal{N} \circ \mathcal{D}_m) \circ \cdots \circ (\mathcal{D}_1^{-1} \circ \mathcal{N} \circ \mathcal{D}_1),$$

where $\mathcal{D}_j(\rho) = D_j \rho D_j^\dagger$, $D_0 = I$, and the leftmost $\mathcal{N}$ is the final-inverse error. The uniformly and independently selected C_j make the cumulative D_j uniformly and independently distributed as well: the map between the two lists is invertible, $C_j = D_j D_{j-1}^\dagger$. Each parenthesis therefore averages to the same isotropic channel [44]. In density-matrix notation, that channel and its repeated action are

$$\mathcal{D}_p(\rho) = p\rho + (1-p)\frac{I}{2}, \qquad \mathcal{D}_p^m(\rho) = p^m\rho + (1-p^m)\frac{I}{2}. \tag{T.9}$$

With perfect preparation and readout, the return probability is $(1+p^m)/2$. Fixed preparation, readout and final-inverse errors change the constants, giving the fit

$$\overline{P}_{\mathrm{return}}(m) = Ap^m + B. \tag{T.10}$$

T.5.3 Extracting and interpreting the error

Fit A, B, p from the length scan. For a pure input, $\langle\psi|\mathcal{D}_p(|\psi\rangle\langle\psi|)|\psi\rangle = (1+p)/2$; the corresponding mean infidelity per Clifford is therefore

$$r_{\mathrm{C}} = \frac{1-p}{2}. \tag{T.11}$$

For three equally spaced lengths $m, m+d, m+2d$, the exact exponential model implies

$$\frac{\overline{P}(m+2d) - \overline{P}(m+d)}{\overline{P}(m+d) - \overline{P}(m)} = p^d.$$

The unknown offset B and amplitude A cancel. This shows how a length scan distinguishes gate accumulation from fixed readout contrast. In data analysis, use many lengths and random sequences in a common fit; the three-point ratio becomes unstable when the differences are comparable to counting noise.

For example, a fixed small rotation error $R_z(\eta)$ after each ideal Clifford has $p = (1 + 2\cos\eta)/3 \simeq 1 - \eta^2/3$. Randomization gives $p^m \simeq e^{-m\eta^2/3}$. Repeating the same unrandomized rotation error instead produces $R_z(m\eta)$ and an equatorial-state error $\sin^2(m\eta/2) \simeq m^2\eta^2/4$. Thus a good mean benchmark does not make a coherent error harmless in every circuit. The signed calibration signals identify such an error; benchmarking summarizes its accumulation under the specified randomization.

The interpretation of r_{C} relies on the noise model. Strong drift, gate dependence or leakage can invalidate this simple exponential description [44]. Also, a Clifford may require several physical pulses, so r_{C} is not automatically the error of each pulse. Tomography identifies how an operation acts on states; benchmarking tests how errors accumulate in a specified gate set.

U Coherence, Relaxation, and Pulse Errors

During a drive-free wait, fluctuations of the spin splitting reduce phase coherence, while energy exchange with the environment changes spin populations. During a gate, voltage fluctuations can change the intended rotation angle, and the pulse can populate states outside the computational space. This appendix develops the corresponding calculations behind Section 7: Ramsey and echo probabilities, population relaxation, and pulse-dependent angle and leakage errors. The conditions for reducing each error depend on the waiting configuration, noise, and pulse shape.

U.1 Correcting and characterizing evolution during a wait

Calibrated $\pi/2$ and π pulses prepare and analyze a spin before and after a drive-free wait. The pulse conventions are those of Section N.1.2. The detuning during the wait contains two physically different parts,

$$\delta(t) = \delta_{\mathrm{r}} + \xi(t).$$

The reproducible part δ_{r} produces a Ramsey-fringe phase drift and can be corrected by changing the reference frequency. The random part $\xi(t)$ reduces the fringe contrast and defines the phase-coherence time. An echo pulse cancels a constant or sufficiently slow contribution during the wait and exposes the remaining phase noise. A separate relaxation experiment measures the loss of upper-state population. The resulting T_2^*, $T_{2,\mathrm{echo}}$ and T_1 belong to the chosen waiting configuration.

U.1.1 Measuring phase coherence with Ramsey and echo

The resonance scan in Section N.1.1 determines ω_{res}. The calibrated rotations of Section N.1.2 use $\omega = \omega_{\mathrm{res}}$, so their nominal detuning is $\delta = \omega_{\mathrm{res}} - \omega = 0$. Ramsey now tests what happens to the spin phase during a drive-free wait. Finite precision in the frequency scan or later drift may leave a residual mean detuning δ_{r}. The spin-transition frequency $\omega_q(t) = E_Z(t)/\hbar$ may also fluctuate, causing different repetitions to accumulate different phases relative to the fixed control reference. The first $\pi/2$ pulse creates a superposition, and the second pulse converts this relative phase into a logical-basis population. The phase of the resulting fringe reveals δ_{r}, while the decay of its contrast defines the phase-coherence time T_2^*.

Use the logical basis $|0\rangle = |\downarrow\rangle$, $|1\rangle = |\uparrow\rangle$ and $Z = |0\rangle\langle 0| - |1\rangle\langle 1|$ from Section 6.1. Let $Y = -i|0\rangle\langle 1| + i|1\rangle\langle 0|$ and $R_y(\theta) = e^{-i\theta Y/2}$. All states and rotations below are expressed in the frame rotating at the drive angular frequency $\omega = \omega_{\mathrm{res}}$. The gates and the waiting evolution must use this same frame. Turning off the transverse drive means setting its envelope to $\Omega(t) = 0$; it does not mean setting $\omega = 0$. The oscillator that generates the two pulses retains the phase reference ωt, so the spin can accumulate a measurable phase relative to that reference while no transverse field is applied.

Equation (L.3) specifies the pulses: choose drive phases $\phi_1 = -\pi/2$ and $\phi_2 = +\pi/2$, with the same amplitude Ω_0 and duration t_p. For $|\delta_{\mathrm{r}}| \ll \Omega_0$, the pulse Hamiltonians and duration are

$$H_1 \simeq \frac{\hbar\Omega_0}{2} Y, \qquad H_2 \simeq -\frac{\hbar\Omega_0}{2} Y, \qquad t_p = \frac{\pi}{2\Omega_0}.$$

They give $R_y(\pi/2)$ and $R_y(-\pi/2)$, respectively. The phase difference of π reverses the drive term; the drive envelope is $\Omega(t) = 0$ during the intervening wait, while the oscillator phase reference continues at angular frequency ω. Treating these pulses as short compared with the

phase-evolution and decoherence times gives the ideal Ramsey sequence

$$|0\rangle \xrightarrow{R_y(\pi/2)} \frac{|0\rangle + |1\rangle}{\sqrt{2}} \xrightarrow{\text{wait } \tau} \frac{|0\rangle + |1\rangle}{\sqrt{2}} \xrightarrow{R_y(-\pi/2)} |0\rangle. \tag{U.1}$$

If a residual mean detuning δ_{r} is present but random fluctuations are absent, the drive-free wait adds the phase $\delta_{\mathrm{r}}\tau$, and the inverse-pulse sequence predicts

$$P_0(\tau) = \frac{1 + \cos(\delta_{\mathrm{r}}\tau)}{2}. \tag{U.2}$$

This residual is the difference between the mean transition frequency during the Ramsey acquisition and the previously chosen reference frequency. It is therefore a correction to the nominally resonant setting, not a second independent determination of ω_{res}.

Use the Rabi measurement in Section N.1.1 to determine t_p at the chosen amplitude. In each trial, prepare $|0\rangle$, apply the first pulse, wait τ with the drive off, and apply the second $\pi/2$ pulse. Let ϕ_a denote an additional phase applied to this second pulse relative to the inverse pulse in Eq. (U.1). Then use the single-spin readout of Section S.1. For each fixed τ, scan ϕ_a through a full period and repeat the complete sequence at every phase to estimate $P_0(\tau, \phi_a)$. Keep the field, waiting gate settings, drive frequency, amplitude and pulse duration fixed. The probability-versus-phase curve is the Ramsey fringe; its amplitude measures the reproducible coherence after the wait. Repeat this phase scan for several values of τ to obtain the coherence decay.

The energy difference between the two spin states may fluctuate during the wait. Write

$$E_Z(t) = \overline{E}_Z + \varepsilon_Z(t), \qquad \frac{\overline{E}_Z}{\hbar} = \omega_{\mathrm{res}} + \delta_{\mathrm{r}},$$

where $\overline{E}_Z$ is the mean splitting during the acquisition, δ_{r} is its residual detuning from the fixed control reference ω_{res}, and $\varepsilon_Z(t)$ is a zero-mean energy fluctuation. Define

$$\xi(t) = \frac{\varepsilon_Z(t)}{\hbar}$$

only to express this energy noise in angular-frequency units. Model $\xi(t)$ as a classical random function that can differ between repetitions. Assume that the fluctuation changes the gap while leaving $|0\rangle$ and $|1\rangle$ as the energy eigenstates, and neglect relaxation and pulse errors. With the drive reference fixed at ω_{res}, the waiting Hamiltonian in the same rotating frame is

$$H_{\mathrm{wait}}(t) = -\frac{\hbar}{2}\big[\delta_{\mathrm{r}} + \xi(t)\big]Z.$$

All these Hamiltonians are proportional to Z, so they commute at different times. The waiting evolution is therefore

$$U_\xi(\tau) = \exp\left\{\frac{iZ}{2}\big[\delta_{\mathrm{r}}\tau + \varphi(\tau)\big]\right\}, \qquad \varphi(\tau) = \int_0^\tau \xi(t)\,dt.$$

For a given function $\xi(t)$ in one repetition, the wait adds the phase $\delta_{\mathrm{r}}\tau + \varphi(\tau)$. The analysis pulse with phase offset ϕ_a is

$$R_{\mathrm{a}}(\phi_a) = \exp\left[-\frac{i\pi}{4}(-\sin\phi_a X - \cos\phi_a Y)\right].$$

It reduces to $R_y(-\pi/2)$ when $\phi_a = 0$. The conditional return probability is

$$\begin{aligned} P_0(\tau, \phi_a \mid \xi) &= |\langle 0|R_{\mathrm{a}}(\phi_a)U_\xi(\tau)R_y(\pi/2)|0\rangle|^2 \\ &= \frac{1 + \cos[\delta_{\mathrm{r}}\tau + \varphi(\tau) - \phi_a]}{2}. \end{aligned}$$

Repeated trials sample different fluctuations as well as the quantum readout outcomes. Their counts estimate the average of this probability. Below, $\langle\cdot\rangle$ denotes a classical average over the fluctuations in those repetitions. Using $\cos x = \mathrm{Re}\, e^{ix}$ gives

$$\begin{aligned} P_0(\tau,\phi_a) &= \langle P_0(\tau,\phi_a \mid \xi)\rangle \\ &= \frac{1}{2}\left[1 + \mathrm{Re}\{e^{i(\delta_\mathrm{r}\tau-\phi_a)} W_R(\tau)\}\right] \\ &= \frac{1}{2}\left[1 + |W_R(\tau)|\cos\big(\phi_a - \delta_\mathrm{r}\tau - \arg W_R(\tau)\big)\right], \\ W_R(\tau) &= \left\langle e^{i\varphi(\tau)}\right\rangle. \end{aligned} \tag{U.3}$$

Thus a sinusoidal fit to the measured phase scan directly gives the contrast $|W_R(\tau)|$ and the fringe phase $\delta_\mathrm{r}\tau + \arg W_R(\tau)$. The contrast describes how averaging the random phase over repeated trials reduces the reproducible coherence. If the fluctuations have zero mean and no systematic phase, the slope of the fringe phase versus τ gives δ_r.

For example, suppose $\xi(t)$ is constant during each repetition but Gaussian distributed between repetitions, with zero mean and standard deviation σ_ξ. Then $\varphi(\tau) = \xi\tau$, and averaging $e^{i\xi\tau}$ over that Gaussian distribution gives

$$W_R(\tau) = e^{-\sigma_\xi^2\tau^2/2} = e^{-(\tau/T_2^*)^2}, \qquad T_2^* = \frac{\sqrt{2}}{\sigma_\xi}. \tag{U.4}$$

Thus Ramsey contrast can decay even when each individual spin undergoes coherent evolution: different repetitions accumulate different phases.

The measured contrasts at different waiting times give the coherence envelope. A Gaussian fit yields T_2^* under the quasistatic Gaussian-noise model above. The fitted fringe phases give the residual mean detuning and reveal any additional mean phase shift not included in the zero-mean noise model.

To remove an approximately constant frequency error within a repetition, insert a π rotation about a transverse axis halfway through the wait. It interchanges the two spin amplitudes, so they accumulate the opposite relative phase during the second half. For ideal pulses, the remaining noise phase is

$$\varphi_E(\tau) = \int_0^{\tau/2} \xi(t)\, dt - \int_{\tau/2}^{\tau} \xi(t)\, dt. \tag{U.5}$$

The residual mean term contributes $\delta_\mathrm{r}\tau/2 - \delta_\mathrm{r}\tau/2 = 0$, and any time-independent part of ξ cancels in the same way. This is a *Hahn echo*; its return contrast depends on $|\langle e^{i\varphi_E}\rangle|$ and is often described by an echo decay time $T_{2,\mathrm{echo}}$. Fluctuations that change during the sequence, spin relaxation, and imperfect pulses still reduce the contrast.

For the echo experiment, repeat the same measurement with an additional calibrated π pulse halfway through the wait. Vary the total waiting time and obtain the return-fringe contrast at each value. Comparing Ramsey and echo contrasts tests how much of the apparent dephasing is removed by refocusing slow frequency fluctuations. These two sequences have been implemented with electrically driven silicon spins [37].

U.1.2 Population relaxation during a wait

After a resonant π pulse, the electron is in the upper spin state. With the drive off it can release energy to its environment and return to the lower state, through the mechanism described in Section R.1. This loss of population does not follow from the unitary waiting Hamiltonian H_wait; it is described by transition rates. We measure how long the upper-state population

survives at the chosen holding voltages. If upward and downward transitions have constant rates $\gamma_\uparrow$ and $\gamma_\downarrow$, the population obeys

$$\dot{P}_\uparrow = -\gamma_\downarrow P_\uparrow + \gamma_\uparrow(1 - P_\uparrow), \qquad P_\uparrow(\tau) = P_\uparrow^{\mathrm{eq}} + [P_\uparrow(0) - P_\uparrow^{\mathrm{eq}}]e^{-\tau/T_1}, \tag{U.6}$$

where $T_1 = (\gamma_\downarrow + \gamma_\uparrow)^{-1}$ and $P_\uparrow^{\mathrm{eq}} = \gamma_\uparrow/(\gamma_\downarrow + \gamma_\uparrow)$. At low temperature the upward rate is small and the upper-state population decays toward zero.

To measure this decay, initialize down, apply the π pulse of Section N.1.2, turn off the drive, wait at a fixed gate setting for time τ, and apply the same readout pulse after every wait. Repeat for several τ values and fit the excited-state probability to Eq. (U.6). The fitted T_1 belongs to the waiting configuration. Keeping the readout configuration and its acquisition window fixed prevents their changes from being confused with a change in the waiting-time decay.

U.2 Reducing sensitivity to voltage fluctuations

Consider exchange evolution that acts as a rotation about one fixed logical axis with angle $\theta = \int J(t)\,dt/\hbar$. Examples are a 1–2 pulse of the exchange-only qubit and an exchange-dominated singlet–triplet z rotation with its gradient negligible during the pulse. Voltage fluctuations change the confinement and the tunnel barrier, so nominally identical pulses can have different exchange energies and accumulated phases. For example, at aligned dot energies, $J = 4t_c^2/(U - V_{12})$: a fluctuating overlap already changes J even if the mean dot energies are held fixed.

U.2.1 Connecting voltage noise to an error in the gate angle

Let the actual voltages be $\mathbf{V}(t)+\delta\mathbf{V}$, where $\mathbf{V}(t)$ is the programmed pulse and the offsets δV_g are nearly constant during one pulse but vary between repetitions. For a pulse about one fixed axis, the random extra energy produces an extra rotation angle, $\delta\theta = \hbar^{-1}\int[J(\mathbf{V} + \delta\mathbf{V}) - J(\mathbf{V})]dt$. To first order,

$$\delta\theta = \sum_g b_g \delta V_g, \qquad b_g = \frac{1}{\hbar}\int_0^\tau \left.\frac{\partial J}{\partial V_g}\right|_{\mathbf{V}(t)} dt. \tag{U.7}$$

Here the programmed waveform $\mathbf{V}(t)$ is fixed, while the offsets are random. With zero-mean offsets and covariance $C_{gh} = \mathbb{E}[\delta V_g \delta V_h]$, their angle variance is

$$\mathbb{E}[(\delta\theta)^2] = \sum_{g,h} b_g C_{gh} b_h.$$

For an input perpendicular to the rotation axis, the overlap with the desired output is $\cos^2(\delta\theta/2)$. Its mean loss is approximately $\mathbb{E}[(\delta\theta)^2]/4$ for small fluctuations. Thus reducing the voltage derivatives that enter b_g directly reduces this contribution to gate error.

For an explicit prediction of the averaged signal, take Gaussian offsets. Then $\delta\theta$ is Gaussian with variance $\sigma_\theta^2 = \mathbf{b}^T C \mathbf{b}$. In a basis where the rotation axis is Z, the error multiplies ρ_{01} by $e^{-i\delta\theta}$ in each trial. Averaging over repetitions gives

$$\mathbb{E}[\rho'_{01}] = e^{-\sigma_\theta^2/2}\rho_{01}, \qquad \mathbb{E}[F_\perp] = \frac{1 + e^{-\sigma_\theta^2/2}}{2}. \tag{U.8}$$

Here the intended rotation has been removed and $F_\perp$ is the overlap for an input perpendicular to the axis. The mean error over all pure states of this logical qubit is smaller, $(1 - e^{-\sigma_\theta^2/2})/3$,

because states on the rotation axis are unaffected. This gives a gate-error prediction from the measured voltage response and a specified noise covariance, rather than only a qualitative preference for a small derivative.

U.2.2 Choosing a less sensitive exchange setting

The derivative formula tells us what to reduce, but does not yet choose a voltage setting. The symmetric two-dot model gives a concrete choice. Holding t_c and $D = U - V_{12}$ fixed,

$$J(\varepsilon) = \frac{4t_c^2 D}{D^2 - \varepsilon^2} = \frac{4t_c^2}{D}\left[1 + \frac{\varepsilon^2}{D^2} + O\left(\frac{\varepsilon^4}{D^4}\right)\right]. \tag{U.9}$$

At $\varepsilon = 0$, a small detuning offset has no first-order effect on J. One can therefore hold the dot energies aligned while changing the barrier to control exchange. This is the principle of symmetric exchange operation [26]. Barrier fluctuations still change t_c, and hence J; the suppressed derivative is specifically the detuning derivative.

Locate this setting experimentally by measuring J on both sides of the aligned-dot point. Use the measured response along the actual compensated voltage path, since a voltage can also change t_c and D. Compare candidate pulses at the same target angle: a smaller instantaneous derivative may be offset by a longer gate duration.

U.2.3 Estimating what remains after the first-order term vanishes

At the symmetric point, detuning noise still changes exchange through the quadratic term. Its residual effect can be estimated with the same fixed-angle calculation. For a constant pulse with $\theta_0 = 4t_c^2\tau/(\hbar D)$ and a small quasistatic detuning offset $\delta\varepsilon$,

$$\delta\theta \simeq \theta_0 \frac{(\delta\varepsilon)^2}{D^2}.$$

If $\delta\varepsilon$ is zero-mean Gaussian with variance σ_ε^2, then $\mathbb{E}[(\delta\varepsilon)^4] = 3\sigma_\varepsilon^4$. Consequently,

$$\mathbb{E}[\delta\theta] = \theta_0 \frac{\sigma_\varepsilon^2}{D^2}, \qquad \mathrm{Var}(\delta\theta) = 2\theta_0^2 \frac{\sigma_\varepsilon^4}{D^4}. \tag{U.10}$$

The mean shift can be included in the angle calibration. After that correction, the leading mean gate infidelity from this noise alone is $\mathrm{Var}(\delta\theta)/6$. It is fourth order in σ_ε/D, instead of the second-order dependence produced by a nonzero linear derivative. This improvement does not apply to a fluctuating barrier: $\delta J/J \simeq 2\delta t_c/t_c$ still gives a first-order angle fluctuation. The working-point choice has removed first-order sensitivity to detuning but not to barrier noise. Compare such settings using pulses with the same requested angle.

U.3 Avoiding population of excluded states

A voltage pulse reshapes the orbital potential as well as changing the intended spin interaction. An electron prepared in the lowest orbital can then acquire amplitude in an excited orbital. For a concrete example, let $|1\rangle$ contain an up spin in the orbital ground state, and let $|\ell\rangle$ have the same spin in the first excited orbital along x. We calculate this leakage channel; the other logical state can have its own channels.

U.3.1 Identifying the transition driven by the voltage pulse

Near a holding voltage V^0, a gate pulse gives $\delta H = \delta V(t)\widehat{W}$ with $\widehat{W} = \partial H/\partial V$. In a harmonic confinement model, a gate force $F_v\delta V$ has $\widehat{W} = -F_v x$, so

$$\Delta = \hbar\omega_x, \qquad \lambda(t) = \delta V(t)\langle\ell|\widehat{W}|1\rangle = -F_v\delta V(t)\sqrt{\frac{\hbar}{2m^*\omega_x}}.$$

Here Δ is the orbital gap and F_v is force per voltage. A uniform scalar shift would have no off-diagonal matrix element; excitation occurs because this gate changes the shape or position of the well. Analogous matrix elements couple to valley or charge excitations when the corresponding potential changes allow them.

Set the energy of $|1\rangle$ to zero. For small pulses, neglect the change of the gap and retain the weak coupling:

$$H(t) = \Delta|\ell\rangle\langle\ell| + \lambda(t)|\ell\rangle\langle 1| + \lambda^*(t)|1\rangle\langle\ell|. \tag{U.11}$$

The pulse begins and ends at the holding voltage, where these states define the retained and excluded spaces. Starting in $|1\rangle$, first-order time-dependent perturbation theory gives

$$a_\ell(\tau) \simeq -\frac{i}{\hbar}\int_0^\tau \lambda(t)e^{i\Delta t/\hbar}\,dt. \tag{U.12}$$

The pulse component at angular frequency $\Delta/\hbar$ drives the unwanted transition. For a constant weak coupling switched on for time τ,

$$|a_\ell(\tau)|^2 \simeq \frac{4|\lambda|^2}{\Delta^2}\sin^2\!\left(\frac{\Delta\tau}{2\hbar}\right).$$

U.3.2 Comparing pulse shapes for the same intended action

The leakage amplitude selects the pulse's Fourier component at $\Delta/\hbar$. We now compare a square and a smooth pulse to see which produces less excitation at that gap. Set $\omega_\ell = \Delta/\hbar$. A square pulse $\lambda_{\rm sq}(t) = \lambda_0$ and a smooth pulse

$$\lambda_{\rm sm}(t) = 2\lambda_0\sin^2(\pi t/\tau) = \lambda_0[1-\cos(2\pi t/\tau)], \qquad 0 \le t \le \tau,$$

have the same time integral. This comparison is appropriate when the desired control and the unwanted coupling are both proportional to the same waveform; otherwise the desired gate must be matched using its own response. Let $a = 2\pi/\tau$. Direct integration of the three exponential terms gives

$$\begin{aligned}\int_0^\tau \lambda_{\rm sq}e^{i\omega_\ell t}\,dt &= \lambda_0\frac{e^{i\omega_\ell\tau}-1}{i\omega_\ell},\\ \int_0^\tau \lambda_{\rm sm}e^{i\omega_\ell t}\,dt &= -\frac{a^2}{\omega_\ell^2-a^2}\,\lambda_0\frac{e^{i\omega_\ell\tau}-1}{i\omega_\ell}.\end{aligned} \tag{U.13}$$

Away from the removable singularities at $\omega_\ell = \pm a$, the smooth-pulse probability is the square-pulse result times $a^4/(\omega_\ell^2-a^2)^2$. For $\omega_\ell\tau \gg 2\pi$, this adds a suppression factor approximately $(2\pi/\omega_\ell\tau)^4$. At $\omega_\ell = a$, the integral must instead be evaluated by its finite limit; there is no suppression guarantee near that frequency.

The smooth pulse and its first derivative vanish at both ends. Those endpoint conditions remove the leading boundary terms when integrating by parts and explain its smaller high-frequency component. The comparison assumes perturbatively small leakage and a fixed unwanted gap. If a voltage excursion brings an excluded state close in energy, the gap must be included as time dependent rather than using this fixed-gap formula.

U.3.3 When the gap changes during a large ramp

The preceding calculation assumed fixed unperturbed states and a weak pulse. A large voltage excursion instead changes the instantaneous eigenstates and their gaps. If the intended operation follows one of those states, the corresponding local adiabatic condition is

$$\frac{\hbar|\langle m(t)|\dot{H}(t)|n(t)\rangle|}{|E_m(t)-E_n(t)|^2} \ll 1, \qquad m \neq n. \tag{U.14}$$

Apply it to the unwanted states being excluded, not automatically to every pair of states in the computational space: the desired gate may intentionally drive transitions within that space. Use the measured excitation gaps to select candidate ramp times, then recalibrate the gate angle for the changed waveform. Longer pulses also accumulate more relaxation and phase noise, so smoothness alone is not a performance criterion.

Changing the envelope also changes the intended gate. For an electrically driven spin, replace the rectangular amplitude a_0 of Section N.1.1 by a shaped envelope $a(t)$. The corresponding Rabi angular frequency now varies:

$$\Omega(t) = \frac{g_e \mu_B b_\perp \chi_x a(t)}{2\hbar}, \qquad H_{\mathrm{rot,q}}(t) = \frac{\hbar\Omega(t)}{2}(\cos\phi X - \sin\phi Y)$$

at zero detuning and fixed phase ϕ. All these Hamiltonians are proportional to the same Pauli operator, so the pulse produces angle

$$\vartheta = \int_0^{t_p} \Omega(t)\, dt.$$

For a rectangular envelope this reduces to $\vartheta = \Omega_0 t_p$; a shaped pulse requires the full integral. Reuse the Rabi or exchange measurement with the complete modified pulse to restore its target angle. If detuning also varies, the Hamiltonians at different times need not commute. The complete propagator is then

$$U(t_p) = \mathcal{T}\exp\left[-\frac{i}{\hbar}\int_0^{t_p} H(t)\, dt\right], \tag{U.15}$$

where $\mathcal{T}$ places later-time operators to the left. This is the time-ordered evolution introduced in Eq. (Pref.3); the integral cannot be replaced by a scalar pulse area when the rotation axis changes in time. Test both the intended gate and unwanted excitation before accepting the new shape.

V Connectivity, Parallel Gates and Spectator Response

Section 8 states the conditions for selecting spins and exchange links. This appendix derives the spectator response and the restrictions on scheduling independent gates.

The available exchange links specify which physical spins can interact directly. Several selected links may share a spin, and a control field may also reach a spin outside the selected set. This appendix examines how physical connectivity, encoded blocks and unwanted couplings constrain operations on an array. Its parallel-gate and spectator calculations give conditions for the intended operations to act independently during a shared interval.

V.1 Connectivity and layout

An exchange gate between electron spins requires tunneling between their dots. The positions of the dots therefore restrict which pairs can interact directly. Represent each dot by a vertex

and each controllable exchange coupling by an edge. For a single-spin array, a useful model is

$$H(t) = \sum_i E_{Z,i}(t) s_{iz} + \sum_{(i,j)\in\mathcal{E}} J_{ij}(t)\mathbf{s}_i \cdot \mathbf{s}_j + H_{\text{drive}}(t), \tag{V.1}$$

where $\mathcal{E}$ is the set of available exchange links, $\mathbf{s}_i = \boldsymbol{\sigma}_i/2$, and H_{drive} contains transverse spin controls. An edge states that J_{ij} can be made useful for a gate; it does not assert that this coupling is always on. Conversely, a pair without an edge has no direct exchange gate in this model.

Consider the four-dot chain in Fig. 8.1, with $\mathcal{E} = \{(1,2),(2,3),(3,4)\}$. To act only on spins 2 and 3, we raise J_{23} while keeping the other two couplings small. If J_{12} is also appreciable, spin 2 interacts with both neighbours and the pulse is a three-spin operation. For a residual coupling approximately constant over a duration τ, the unwanted exchange area is $J_{\text{off}}\tau/\hbar$. It is this area, rather than a binary on/off description, that sets the scale of its possible effect. The on-setting must also keep the selected spins in their intended computational space, as established for the individual gate constructions in Appendices N, O and P.

A physical dot and an encoded qubit need not be the same object. The basic operating blocks considered here have

Encoding	Dots per qubit	Dots in a two-qubit block
Single spin	1	2
Singlet–triplet	2	4
Exchange only	3	6

These counts assume one electron per dot during the spin operation and omit reservoirs, sensors and spare dots. A link inside a three-dot exchange-only block gives a logical one-qubit rotation; the boundary link in Fig. V.1 is one ingredient of a six-spin two-qubit sequence, not a complete logical gate. Two singlet–triplet blocks can instead obtain a conditional phase from their state-dependent charge distributions, as in Section O.2. The graph of usable *logical* gates is obtained only after each complete physical sequence has been specified and shown to return to the logical spaces.

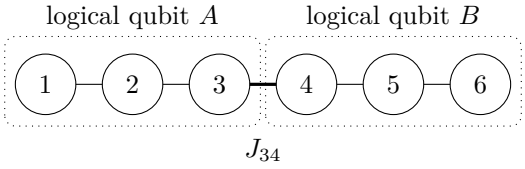


Figure V.1: Two exchange-only qubits formed from six physical dots. Each dotted box groups the three spins of one logical qubit. The thick J_{34} boundary link is a physical exchange coupling, not a complete logical two-qubit gate.

The graph also determines the cost of an operation between non-neighbours. In the four-dot chain of Fig. 8.1, there is no exchange edge $(1,3)$; pulsing one barrier cannot implement a gate on spins 1 and 3. A circuit must route a state along existing edges, move an electron, or use another mediator. The number of edges along a chosen path, not the distance between labels on a drawing, sets the elementary routing count. A two-dimensional layout offers different paths, although interior dots also need control and readout access. A six-dot device may encode six single-spin qubits or two exchange-only qubits, so dot count alone does not describe its connectivity or computational capacity. A six-spin silicon register has demonstrated coordinated preparation, gates and measurement in one array [19].

V.2 Parallel gates and spectator response

A set of gates is independent during one interval only if its complete evolution has the intended action on both the selected spins and the spectators. The exchange graph constrains whether two selected pair gates share a spin. The spatial and frequency response of the drive determines what happens to spins outside the selected set. These are distinct restrictions on parallel operation.

V.2.1 Independent exchange gates on a graph

Consider four occupied dots in a chain. Raising exchange on $(1, 2)$ and $(3, 4)$ couples disjoint spin pairs. Their Hamiltonians act on different tensor factors, so the two pair operations can run during the same interval in the ideal model. Raising exchange on $(1, 2)$ and $(2, 3)$ has a different meaning: spin 2 interacts with both neighbours. With $H_{ij} = J_{ij}\mathbf{s}_i \cdot \mathbf{s}_j$,

$$[H_{12}, H_{34}] = 0, \qquad [H_{12}, H_{23}] \neq 0.$$

For constant simultaneous couplings on the adjacent edges, the evolution is

$$U_{\text{sim}} = e^{-i(H_{12}+H_{23})\tau/\hbar}.$$

It is a three-spin operation, not the product of two independently calibrated pair gates. The same distinction holds for shaped pulses: the shared spin still belongs to both Hamiltonian terms at once. Such a pulse can be useful if its complete three-spin action is designed, but it cannot simply replace two successive pair gates.

For isolated pair gates, a set of edges with no shared vertex is therefore one candidate layer. Figure V.2 shows two such layers on a six-spin chain. The first uses $(1, 2), (3, 4), (5, 6)$; the second uses $(2, 3), (4, 5)$. Together they cover the nearest-neighbour edges while assigning each spin to at most one gate in each interval.

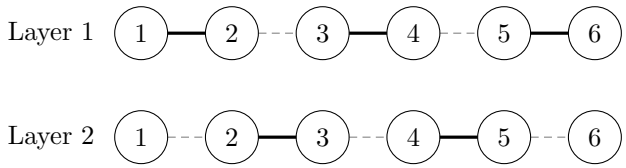


Figure V.2: Two layers of pair-exchange gates on a six-spin chain. Thick edges are scheduled gates; dashed edges are not selected for that layer. Residual exchange on an unselected edge must still be included in the calibrated operation.

The dashed links in the figure are unselected, not mathematically absent. If they retain exchange during a layer, their accumulated coupling changes its operation. The graph rule tells us which pair gates do not compete for the same spin; it does not establish that the voltage waveform actually suppresses every other coupling.

V.2.2 Response of spins outside the scheduled gates

A control pulse can extend beyond the vertex or edge it was meant to address. For example, a microwave field resonant with spin i can also reach a neighbouring spin j. The latter sees both a transverse drive and a frequency mismatch. In the frame rotating with the drive at ω_i, its Hamiltonian is

$$H_j = \frac{\hbar}{2}(-\Delta_j Z_j + \Omega_j X_j), \qquad \Delta_j = \omega_j - \omega_i.$$

Here Ω_j is the amplitude at j, not the programmed amplitude at i. Figure V.3 shows the carrier frequency relative to the two spin resonances and the spectator's effective rotation axis. When that axis is nearly longitudinal, a Z_j eigenstate barely flips; a superposition can nevertheless change phase.

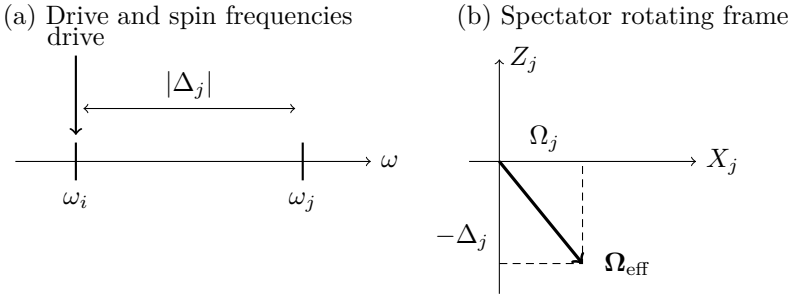


Figure V.3: A drive resonant with i is detuned from spectator j. Panel (b) depicts $\Delta_j > 0$: the spectator's effective rotation vector has transverse component Ω_j and longitudinal component $-\Delta_j$. Its transverse fraction controls possible flips; its changed length also changes the phase accumulated by a superposition.

The magnitude of the effective rotation vector is $\Omega_{\text{eff}} = \sqrt{\Delta_j^2 + \Omega_j^2}$. For a rectangular pulse lasting τ, a spectator initially in $|0\rangle_j$ has flip probability

$$P_{1,j}(\tau) = \frac{\Omega_j^2}{\Omega_j^2 + \Delta_j^2} \sin^2\left(\frac{\sqrt{\Omega_j^2 + \Delta_j^2}\,\tau}{2}\right) \leq \frac{\Omega_j^2}{\Omega_j^2 + \Delta_j^2}. \tag{V.2}$$

The prefactor is the squared transverse fraction of the axis in Fig. V.3. When $|\Delta_j| \gg |\Omega_j|$, this bounds the population transfer even if the pulse duration changes. A zero of the sine gives no net flip at that particular duration, but it is not evidence that the pulse left an arbitrary spectator state unchanged.

The drive also changes the spectator's rotating-frame level separation. For $|\Omega_j/\Delta_j| \ll 1$, $\Omega_{\text{eff}} - |\Delta_j| \simeq \Omega_j^2/(2|\Delta_j|)$. Relative to an otherwise identical wait, the corresponding additional evolution is

$$\delta H_j \simeq -\frac{\hbar\Omega_j^2}{4\Delta_j} Z_j, \qquad U_{j,\text{extra}} \simeq R_z(-\phi_j), \qquad \phi_j = \int \frac{\Omega_j(t)^2}{2\Delta_j}\, dt. \tag{V.3}$$

The integral applies to a slowly switched, far-detuned envelope whose dressed states are followed. A rectangular pulse may also leave the transverse response quantified by Eq. (V.2). Thus a spectator superposition can acquire a phase even when a population test finds few flips.

Increasing the target drive can shorten its gate but also increases the field reaching j. If the measured cross-drive ratio is $\Omega_j = c\Omega_i$ with $c \neq 0$, a rectangular target π pulse takes $\tau_\pi = \pi/|\Omega_i|$. Requiring the spectator-flip envelope to remain below $0 < q < 1$ yields

$$|\Omega_i| \leq \frac{|\Delta_j|}{|c|}\sqrt{\frac{q}{1-q}}, \qquad \tau_\pi \geq \frac{\pi|c|}{|\Delta_j|}\sqrt{\frac{1-q}{q}}. \tag{V.4}$$

At that speed the spectator phase is approximately $\pi c^2|\Omega_i|/(2|\Delta_j|)$ in magnitude, so the flip limit does not replace a phase check. Population readout after the target pulse tests flips; a Ramsey comparison with and without it reveals the extra phase.

The two parts of this section impose different conditions on a proposed layer. Disjoint exchange edges prevent one spin from participating in two intended pair gates. Spectral

separation and cross-drive determine whether an unselected spin remains idle during a microwave pulse. Voltage changes can also shift nearby exchange couplings and spin gaps; disjoint singlet–triplet blocks may acquire a capacitive phase during parallel local gates (Section O.2.4). Shared drive and readout channels further constrain which operations can overlap. A layer is justified as a product of its intended gates, up to calibrated local phases, only when the complete joint waveform has the required action on both targets and spectators.

W Communication between Qubits

This appendix expands the final question in Section 8: how qubits without a direct interaction link can participate in a gate.

A circuit can request a gate between qubits whose vertices share no interaction edge. The missing edge is a physical problem, not merely a change of gate notation. We can bring the two quantum states next to one another with SWAP gates, transport an electron carrying its spin, or couple stationary spins through another quantum system. These choices move different objects and have different conditions for preserving an unknown state.

W.1 Moving a state with SWAP gates

An exchange pulse with the calibrated SWAP area exchanges the spin states at neighbouring occupied dots [Eq. (6.6)]. For example,

$$|\psi\rangle_1|0\rangle_2|0\rangle_3 \xrightarrow{\mathrm{SWAP}_{12}} |0\rangle_1|\psi\rangle_2|0\rangle_3 \xrightarrow{\mathrm{SWAP}_{23}} |0\rangle_1|0\rangle_2|\psi\rangle_3. \tag{W.1}$$

Each dot still holds an electron. The state $|\psi\rangle$, which may be unknown or entangled with other qubits, changes its site because SWAP permutes spin states. The displayed $|0\rangle$ states make the path easy to read; intermediate qubits need not be initialized for the permutation to work. Their own states are permuted as well, and the control program must track the resulting placement.

This permutation builds a gate between distant states from a gate available only on an edge. In the three-site chain let S_{23} be SWAP on sites 2 and 3, and let U_{12} be the desired two-qubit operation on edge $(1, 2)$. Then

$$U_{13} = S_{23}U_{12}S_{23}. \tag{W.2}$$

The rightmost SWAP first moves the state originally at site 3 to site 2; U_{12} acts on that state and the one at site 1; the leftmost SWAP restores both locations. The identity holds on arbitrary three-qubit inputs, so routing neither measures nor copies the transported state.

For a chosen path of d edges, bringing one state beside the other takes $d - 1$ SWAPs; restoring the placement takes another $d - 1$. If all are serial and have duration t_{SWAP}, a routed gate of duration t_U takes

$$T_{\mathrm{route}} = 2(d - 1)t_{\mathrm{SWAP}} + t_U.$$

In an illustrative model where each SWAP independently fails with probability p_S and the central gate fails with probability p_U, the probability of at least one fault is

$$p_{\mathrm{fault}} = 1 - (1 - p_S)^{2(d-1)}(1 - p_U) \simeq 2(d - 1)p_S + p_U.$$

This stochastic estimate is not a rule for adding coherent gate errors or average infidelities. If later operations can use the new placement, the restoring SWAPs may be omitted, provided the program tracks every moved state. For a singlet–triplet or exchange-only encoding, swapping one physical spin generally does not swap the whole logical qubit; the logical-block operation must be constructed and checked separately.

W.2 Moving an electron while preserving its spin

SWAP moves a spin state through occupied sites. Shuttling instead moves the electron that carries the spin into another dot. For one electron in dots A and B, its position is a charge two-level system with $\tau_z = |A\rangle\langle A| - |B\rangle\langle B|$ and $\tau_x = |A\rangle\langle B| + |B\rangle\langle A|$. A minimal model is

$$H(t) = \left[\frac{\varepsilon(t)}{2}\tau_z + t_c\tau_x\right] \otimes I - \frac{1}{2}\left[E_{Z,A}|A\rangle\langle A| + E_{Z,B}|B\rangle\langle B|\right] \otimes Z, \tag{W.3}$$

where $\varepsilon = \epsilon_A - \epsilon_B$, $t_c > 0$ is spin-conserving tunneling, and $Z|0\rangle = |0\rangle$ for $|0\rangle = |\downarrow\rangle$. A sweep from negative to positive ε changes the charge ground state from mostly $|A\rangle$ to mostly $|B\rangle$. Successful transport of the electron does not, by itself, establish that its spin state survived the trip.

First neglect the differential Zeeman energy when finding that charge ground state. Its gap to the excited charge state and its probability of occupying A are

$$\Delta_c = \sqrt{\varepsilon^2 + 4t_c^2}, \qquad n_A = \frac{1}{2}\left(1 - \frac{\varepsilon}{\Delta_c}\right).$$

Thus n_A changes continuously from one to zero and equals one half at the avoided crossing. To follow the ground branch, the local adiabatic condition is

$$\frac{\hbar t_c|\dot{\varepsilon}|}{(\varepsilon^2 + 4t_c^2)^{3/2}} \ll 1. \tag{W.4}$$

It has the form derived for a charge ramp in Eq. (R.5), with the single-electron coupling t_c. The crossing is the restrictive part of a linear sweep because the gap is smallest there.

One way to allocate more time near the crossing is to hold the dimensionless quantity in Eq. (W.4) at a chosen small value η_{ad} over the main part of the ramp. Then $\dot{\varepsilon} = \eta_{\mathrm{ad}}(\varepsilon^2 + 4t_c^2)^{3/2}/(\hbar t_c)$. Integrating between equal and opposite endpoint detunings gives

$$T = \frac{\hbar\varepsilon_{\max}}{2\eta_{\mathrm{ad}} t_c \sqrt{\varepsilon_{\max}^2 + 4t_c^2}}. \tag{W.5}$$

Smooth connections to the endpoints alter the total pulse time. Moreover, η_{ad} is a local condition, not a prediction of the full transfer error. Charge readout after the sweep tests whether the electron reached B.

Even perfect charge following can rotate the spin. Provided $|E_{Z,A} - E_{Z,B}| \ll \Delta_c$ throughout the sweep, projection onto the charge ground state gives

$$H_{\mathrm{spin}}(t) = -\frac{1}{2}E_Z(t)Z, \qquad E_Z(t) = n_A(t)E_{Z,A} + [1 - n_A(t)]E_{Z,B}.$$

The electron therefore samples both local Zeeman energies as it crosses. If spin flips and charge excitation are negligible,

$$|A\rangle \otimes (a|0\rangle + b|1\rangle) \longrightarrow |B\rangle \otimes R_z(-\varphi)(a|0\rangle + b|1\rangle), \qquad \varphi = \frac{1}{\hbar}\int_0^T E_Z(t)\,dt, \tag{W.6}$$

up to a common orbital phase. This phase is written in the laboratory frame. Relative to a rotating control reference of angular frequency ω, the measured phase is instead $\int_0^T [E_Z(t)/\hbar - \omega]\,dt$. A reproducible phase can be included in the control frame; an uncertain phase cannot.

This distinction requires more than an arrival measurement. For a transferred $|+\rangle$, X and Y analyses in the chosen frame yield $\langle X\rangle = \cos\varphi$ and $\langle Y\rangle = -\sin\varphi$. If the phase varies between shots, the combined contrast becomes

$$\sqrt{\langle X\rangle^2 + \langle Y\rangle^2} = |\mathbb{E}[e^{-i\varphi}]|.$$

A fixed phase rotates the vector without reducing its length; phase fluctuations shorten it. Basis-state transfers also test spin flips. Charge arrival, spin preservation conditioned on arrival, and the reproducibility of the accumulated phase are therefore separate properties of a shuttle. Silicon experiments have tested coherent spin transport with these distinct observables [45].

W.3 Coupling through an electromagnetic mode

A third route leaves both electrons at their dots. Each spin couples to the same resonator mode, and an excitation can pass between the spins through that mode. Its field is a quantum degree of freedom: unlike a prescribed microwave drive, it can receive an excitation and later return it.

One mechanism converts the resonator's electric field into spin coupling through a magnetic-field gradient. Let a annihilate a photon of angular frequency ω_c, and let $E_{\mathrm{zpf}}(a + a^\dagger)$ be the electric field along the electron's displacement. In a harmonic dot with orbital frequency $\omega_{\mathrm{orb}} \gg \omega_c, \omega_i$,

$$x \simeq -\frac{eE_{\mathrm{zpf}}}{m^*\omega_{\mathrm{orb}}^2}(a + a^\dagger).$$

A transverse field gradient $b_x = \partial B_x/\partial x$ then turns the displacement into the spin term $g_e\mu_B b_x x X/2$, giving an angular-frequency coupling of magnitude

$$g_s = \frac{|g_e\mu_B b_x eE_{\mathrm{zpf}}|}{2\hbar m^*\omega_{\mathrm{orb}}^2}.$$

Here g_e is the electron Landé factor. This expression explains the route from an electric field to a magnetic spin drive. Actual charge eigenstates can change the coupling, so g_s is determined for the device rather than inferred from this harmonic estimate alone. Spin–photon coupling has been observed in silicon [46].

With measured couplings g_1, g_2, the excitation-conserving model is

$$\frac{H}{\hbar} = \omega_c a^\dagger a + \sum_{i=1}^{2} \omega_i |1\rangle_i\langle 1| + \sum_{i=1}^{2} g_i(a\sigma_i^+ + a^\dagger\sigma_i^-), \qquad \sigma_i^+ = |1\rangle_i\langle 0|. \tag{W.7}$$

The rotating-wave approximation requires $g_i, |\omega_i - \omega_c| \ll \omega_i + \omega_c$. When one spin is resonant with an initially empty mode, an initial $|1, 0_{\mathrm{ph}}\rangle$ oscillates with $P_1(t) = \cos^2(g_i t)$ if losses are negligible during the hold. The mode is then populated during the exchange.

A detuned mode can instead act mainly as a virtual mediator. For equal bare spin frequencies ω_q, put $\Delta = \omega_q - \omega_c$ and assume $|\Delta| \gg g_1, g_2$. In the one-excitation basis $\{|10, 0_{\mathrm{ph}}\rangle, |01, 0_{\mathrm{ph}}\rangle, |00, 1_{\mathrm{ph}}\rangle\}$, after subtracting $\hbar\omega_q$,

$$\frac{H_1}{\hbar} = \begin{pmatrix} 0 & 0 & g_1 \\ 0 & 0 & g_2 \\ g_1 & g_2 & -\Delta \end{pmatrix}.$$

The mode amplitude follows the two spin amplitudes approximately as $c_c = (g_1 c_1 + g_2 c_2)/\Delta$ on their slow time scale. Substitution gives

$$\frac{H_{\mathrm{eff},1}}{\hbar} = \frac{1}{\Delta}\begin{pmatrix} g_1^2 & g_1 g_2 \\ g_1 g_2 & g_2^2 \end{pmatrix}. \tag{W.8}$$

Its diagonal shifts must be included when matching the two spin frequencies; the off-diagonal interaction has magnitude $|g_{\mathrm{med}}| = |g_1 g_2/\Delta|$. With the shifts compensated, an initial $|10\rangle$

evolves to $\cos(g_{\mathrm{med}}t)|10\rangle - i\sin(g_{\mathrm{med}}t)|01\rangle$, apart from a common phase. The equal-weight entangling time is $\pi/(4|g_{\mathrm{med}}|)$, and full excitation transfer takes $\pi/(2|g_{\mathrm{med}}|)$. Increasing $|\Delta|$ reduces the mode occupation, of order $(g_i/\Delta)^2$, while lengthening both times. Whether the remote operation is useful depends on this tradeoff and on spin decoherence and resonator loss over the full gate.

X Inferring and Testing an Effective Model

Sections 2 and 4.3 use addition energies and an exchange coupling to describe the device. This appendix develops a supplementary check: whether a common set of model parameters accounts for the measurements of those quantities.

Charge-transition positions, excitation gaps and charge-distribution curves constrain different terms of an effective Hamiltonian. We now combine such data to estimate its parameters. A separate measurement can then test a prediction that was not used in the fit. The example below fits charge data and predicts an exchange splitting; the spin experiment that tests that prediction is developed in Section N.2.2.

X.1 Relating model parameters to measurements

First choose the retained states and interaction terms, as in Appendix J.1. Collect the unknown parameters in $\boldsymbol{\theta}$. Each measurement then has a prediction $f_k(\boldsymbol{\theta})$, obtained from the corresponding charge-boundary or spectroscopy calculation. Refer all data to matched electrode settings, or include the measured voltage dependence in the model.

Consider the two-dot Hubbard model of Appendix J.1, with one orbital per dot, equal on-site repulsions $U_1 = U_2 = U$, and zero detuning. Write $t_c = |t_{12}|$ and $D = U - V_{12} > 0$, and omit Zeeman and direct-exchange terms. For this example, $\boldsymbol{\theta} = (t_c, D)$. The charge-polarization measurement in Appendix K constrains t_c. Suppose spectroscopy also identifies the two mixed singlet eigenstates of the matrix in Eq. (J.16). Their energy separation is

$$\Delta_{\mathrm{ch}} = \sqrt{D^2 + 16t_c^2}. \tag{X.1}$$

Use these two charge measurements to determine t_c and D, and reserve the spin-oscillation measurement of J for testing the result. This comparison assumes that the same retained-orbital model describes the charge sectors used in the measurements.

X.2 Joint estimation, uniqueness, and uncertainty

Let y_k be the kth measured value and σ_k the standard deviation of its measurement error. For independent Gaussian errors, estimate $\boldsymbol{\theta}$ by minimizing

$$\chi^2(\boldsymbol{\theta}) = \sum_k \frac{[y_k - f_k(\boldsymbol{\theta})]^2}{\sigma_k^2}. \tag{X.2}$$

In the example, y_1 is the measured tunnel coupling and y_2 the identified singlet excitation gap; the predictions are $f_1 = t_c$ and $f_2 = \sqrt{D^2 + 16t_c^2}$. In the ideal limit of negligible uncertainty, these give $t_c = y_1$ and $D = \sqrt{y_2^2 - 16y_1^2}$.

Check which parameters the chosen data actually determine. If only J were measured, its weak-tunneling expression $J \simeq 4t_c^2/D$ would constrain a ratio, leaving many pairs (t_c, D) possible. The two charge measurements resolve that ambiguity, but still determine only $U - V_{12}$, not U and V_{12} separately. To separate those parameters, add charge measurements

whose predicted energies depend on them differently. If the available data cannot distinguish parameter choices within their uncertainty, report the constrained combinations.

Estimate the parameter uncertainty by refitting simulated repetitions generated from the fitted predictions and the measurement-noise model. In each repetition, use the same sampled calibration error for all measurements sharing that calibration; it shifts several inferred energies together. Such correlated errors require their covariance in the fit, or an explicit fit of the calibration parameters together with the device parameters. Keep the resulting joint parameter uncertainty for the prediction test below.

X.3 Testing predictions on independent data

Use the fitted parameters to predict a quantity excluded from the fit, without readjusting them to that measurement. For the two-dot example, Eq. (J.16) predicts

$$J_{\mathrm{pred}} = \frac{\sqrt{D^2 + 16t_c^2} - D}{2}. \tag{X.3}$$

Compare this with J obtained from the two-spin exchange measurement of Section N.2.2. Using the exact Hubbard expression avoids attributing an error in $J \simeq 4t_c^2/D$ to the Hamiltonian itself. Recompute J_{pred} for the joint parameter estimates obtained above to determine its uncertainty, including uncertainty in the operating point. If the measured J had been used to determine D, that same J would no longer be an independent test.

Compare the predicted range with the measured J and its uncertainty. If they disagree, first check the voltage calibration, thermal broadening in the charge fit, and repeated scans for drift. If these checks do not account for a reproducible discrepancy at matched settings, revisit the Hamiltonian assumptions. For example, low-lying orbital or valley states seen in spectroscopy may need to be retained, or direct exchange omitted from the simple model may contribute to J. Include such terms when supported by the data and test the revised model on another independent measurement or voltage setting.

Y Calibration Feedback

The voltage response in Section 2 and the pulse settings in Section 6 refer to a chosen operating point. This supplementary calculation explains how measured errors can update those settings when the operating point drifts.

A charge transition or spin resonance can drift after a gate pulse has been calibrated. Correcting one electrode voltage may also shift another dot or change a tunnel coupling. A feedback update must therefore relate the measured discrepancies to the physical quantities that each voltage changes, and its stopping criterion must account for measurement uncertainty. This appendix develops that update for a two-spin experiment and for a local vector of device parameters.

The experiment uses the preparation and readout methods of Appendices R and S, and the spin gates of Appendix N.

Y.1 Assembling a complete two-spin experiment

Take two dots, each with a reservoir barrier, and a charge sensor. As a concrete target, prepare $|01\rangle$ and apply $\sqrt{\mathrm{SWAP}}$. Under the exchange conditions of Section N.2.2, the intended output is

$$|01\rangle \longmapsto \frac{|01\rangle - i|10\rangle}{\sqrt{2}}$$

up to a common phase. This example requires charge control, initialization, a single-spin flip, exchange, and two spin readouts. We first establish those controls, then assemble their pulses.

Y.1.1 Establishing the charge settings

Put the sensor on a current flank and verify its response by loading and emptying a target dot, using Section G.2. With the reservoir barriers open enough for equilibration, trace target transitions towards depletion and establish the empty reference before counting inward to $(1, 1)$. A featureless image alone does not identify zero occupation: a closed barrier can also hide transitions. The measured transition positions and responses supply the loading and holding voltages of Appendix R and the compensated exchange direction of Appendix F.4.

Y.1.2 Obtaining initial spin controls and refining their references

Load and relax both spins at those settings. The resonance and Rabi methods of Section N.1.1 give the frequency and duration of a spin-flip pulse. An oscillation period can be estimated with imperfect readout contrast. The resulting flip then supplies an up-spin reference to improve the readout voltage and decision rule in Appendix S. Repeat the spin tests with those improved references. This is a mutual refinement of preparation, drive and readout, not an assumption that each was already perfect before the others existed.

Measure $J(v)$ along the compensated path and select the complete exchange pulse with area $\pi/2$, including the ramps, as in Section N.2.2. These measurements have now supplied all pulse parameters for the chosen trial.

Y.1.3 Running the trial with fixed settings

For each repetition, load and relax both spins to $|00\rangle$, retain the electrons by closing the reservoir barriers, and apply $R_{x,2}(\pi)$ to prepare $|01\rangle$ up to a common phase. Apply the compensated exchange pulse, return to weak exchange, and read both spins with the stored spin-to-charge procedures. The ideal population prediction is

$$P_{01} = P_{10} = \frac{1}{2}, \qquad P_{00} = P_{11} = 0.$$

For this test, keep the preparation and readout pulses fixed while varying or correcting the exchange pulse.

Those populations test the transfer but not its relative phase. The correlation measurements of Appendix T, with analysis rotations chosen for the intended output, test its coherence as well. This separates executing the trial from the extra measurements needed to verify the gate.

Store the voltage vectors, ramps, hold times, drive frequency and phase convention, and the readout rule used for this trial. If a later test fails, remeasure the associated parameter: a charge-boundary shift changes the voltage setting, a resonance shift changes the spin-drive reference, and a changed exchange oscillation changes the exchange pulse. The charge-control stages can be automated [47, 48]; the feedback calculation below describes how measured errors can drive those updates.

Y.2 Calculating a voltage correction from measured errors

Suppose the charge map shows that the first dot's addition energy is too low, while an exchange measurement indicates insufficient coupling. Changing one barrier can improve the coupling

while shifting both dot energies. A feedback step uses all measured responses to make these corrections together.

Near the target, let $\mathbf{q}(\mathbf{V})$ contain an equal number of independent measured quantities and adjustable voltages. For example, use the two specified addition-energy offsets relative to the reservoirs and the interdot tunnel coupling, controlled by two plungers and one barrier. Their target values encode the loading or gate condition already chosen above. Perturb one electrode at a time and remeasure the quantities, giving the local response

$$R_{ag} \simeq \frac{q_a(\mathbf{V} + \Delta V_g \mathbf{e}_g) - q_a(\mathbf{V} - \Delta V_g \mathbf{e}_g)}{2\Delta V_g}, \tag{Y.1}$$

where $\mathbf{e}_g$ changes only voltage g. Each measurement uses the charge-transition or tunnel-coupling methods of Appendices H and K, respectively.

Write the current discrepancy as $\mathbf{r} = \mathbf{q}^\star - \mathbf{q}(\mathbf{V})$. For a small change,

$$\mathbf{q}(\mathbf{V} + \delta\mathbf{V}) = \mathbf{q}(\mathbf{V}) + R\delta\mathbf{V} + O(\|\delta\mathbf{V}\|^2).$$

Canceling the first-order discrepancy requires $R\delta\mathbf{V} = \mathbf{r}$. If R is nonsingular, this gives the feedback step

$$\mathbf{V}_{\text{new}} = \mathbf{V} + \eta R^{-1}[\mathbf{q}^\star - \mathbf{q}(\mathbf{V})], \qquad 0 < \eta \leq 1. \tag{Y.2}$$

The factor η limits the step to the measured linear interval and permitted voltages. Substitution into the expansion gives

$$\mathbf{r}_{\text{new}} = (1 - \eta)\mathbf{r} + O(\|\delta\mathbf{V}\|^2).$$

Thus $\eta = 1$ cancels the first-order residual when the response is accurate; smaller steps trade a slower correction for a better local approximation. Remeasure $\mathbf{q}$ after applying the step. If the residual does not decrease, reduce the step or determine R again. A singular R means the selected controls do not give independent local changes of all the requested quantities.

Y.2.1 A simultaneous energy and tunnel-coupling correction

As a worked update, take two addition-energy offsets and $q_3 = \ln(t_c/t_{\text{ref}})$, where t_{ref} is a fixed energy used only to make the logarithm dimensionless. Suppose the measured local responses are

$$\begin{pmatrix} \delta\mu_1/(\mu\text{eV}) \\ \delta\mu_2/(\mu\text{eV}) \\ \delta\ln(t_c/t_{\text{ref}}) \end{pmatrix} = \begin{pmatrix} -100 & -25 & -10 \\ -20 & -80 & -15 \\ 0.05 & 0.04 & 0.50 \end{pmatrix} \begin{pmatrix} \delta V_1/(\text{mV}) \\ \delta V_2/(\text{mV}) \\ \delta V_{\text{B}}/(\text{mV}) \end{pmatrix}. \tag{Y.3}$$

All entries in this displayed, scaled equation are dimensionless. The first two rows use the addition-energy response from Appendix F.4; the last row adds illustrative measurements of the tunnel-coupling response. To request changes $(40, -40, 0.10)$ on the left-hand side, solve the system to obtain

$$(\delta V_1, \delta V_2, \delta V_{\text{B}}) \simeq (-0.572,\ 0.604,\ 0.209)\,\text{mV}.$$

This shifts the two energies oppositely while increasing t_c by a factor $e^{0.10} \simeq 1.105$. Changing only the barrier would also shift the dot energies. Apply this full step only if it stays within the interval used to establish the local response; otherwise multiply all three changes by the same smaller η. Remeasure the two transition positions and the charge-distribution curve to compute the actual new residual.

Y.2.2 The same update for a pulse angle

The signed-angle measurement in Section N.1.3 supplies a scalar error instead of a vector of charge and coupling errors. For a pulse angle, the same first-order calculation gives

$$a_{0,\mathrm{new}} = a_0 + \eta \, \frac{\theta^\star - \theta(a_0)}{d\theta/da_0},$$

where a_0 is the rectangular pulse amplitude, $\theta(a_0)$ comes from the oscillation measurement, and $\theta^\star$ is the target angle. Measure the derivative using nearby amplitudes with the same pulse shape and duration.

Y.3 Deciding whether the correction is resolved and sufficient

A small measured residual can mean that the control is accurate, or that the data are too noisy to resolve its error. We therefore compare the residual with a tolerance tied to the intended operation, and require the measurement uncertainty to be smaller than that tolerance.

Before running either loop, specify tolerances d_a for the measured quantities and require $|q_a - q_a^\star| \le d_a$ on a new measurement, with measurement uncertainty small enough to judge that tolerance. Gate-angle or conditional-phase tolerances are separate from charge-setting tolerances. If a later check fails, repeat the measurement associated with that failure: a shifted charge boundary requires retuning the voltage setting; a shifted spin resonance requires updating the drive reference. Changing a setting also invalidates downstream calibrations whose Hamiltonian parameters changed. This dependency, rather than a universal retuning schedule, determines what must be measured again.

The tolerance should be connected to the operation being requested. For a coherent angle error η_θ about the intended axis, Eq. (T.5) gives $1 - F_{\mathrm{avg}} \simeq \eta_\theta^2/6$. Allocating an error r_{angle} to this mechanism therefore sets the angle tolerance $|\eta_\theta| \lesssim \sqrt{6 r_{\mathrm{angle}}}$. In the signed sequence of Eq. (N.8), n trials near $P_1 = 1/2$ have counting uncertainty approximately $1/(2\sqrt{n})$. The inferred angle uncertainty is then $1/(m\sqrt{n})$ before assignment corrections and other errors. This relation determines how many repetitions can resolve the chosen tolerance. Increasing m helps only while the repeated sequence still obeys the small-error model and retains contrast. The stopping test consequently concerns both the measured residual and whether the measurement is precise enough to resolve it.

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